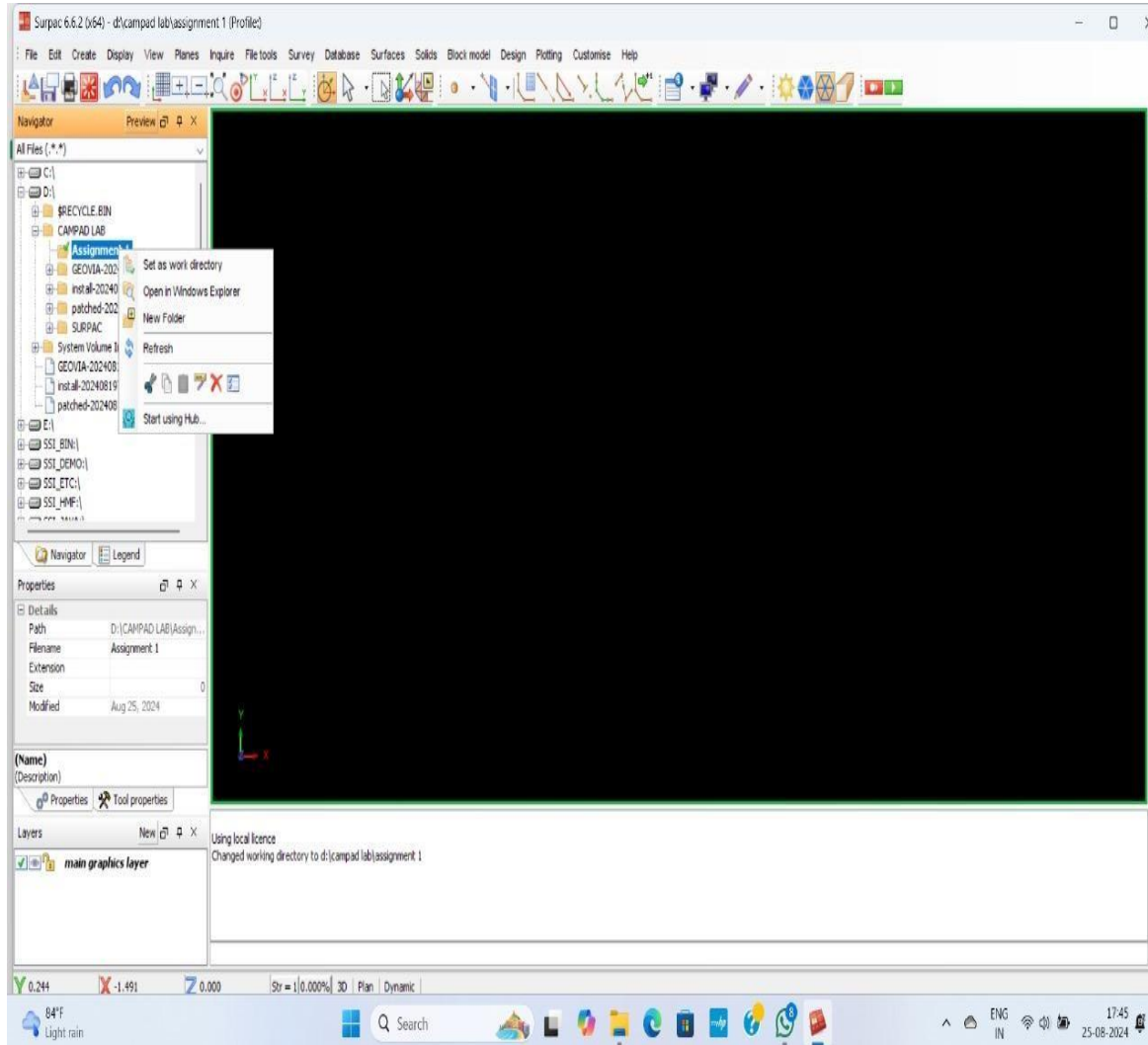


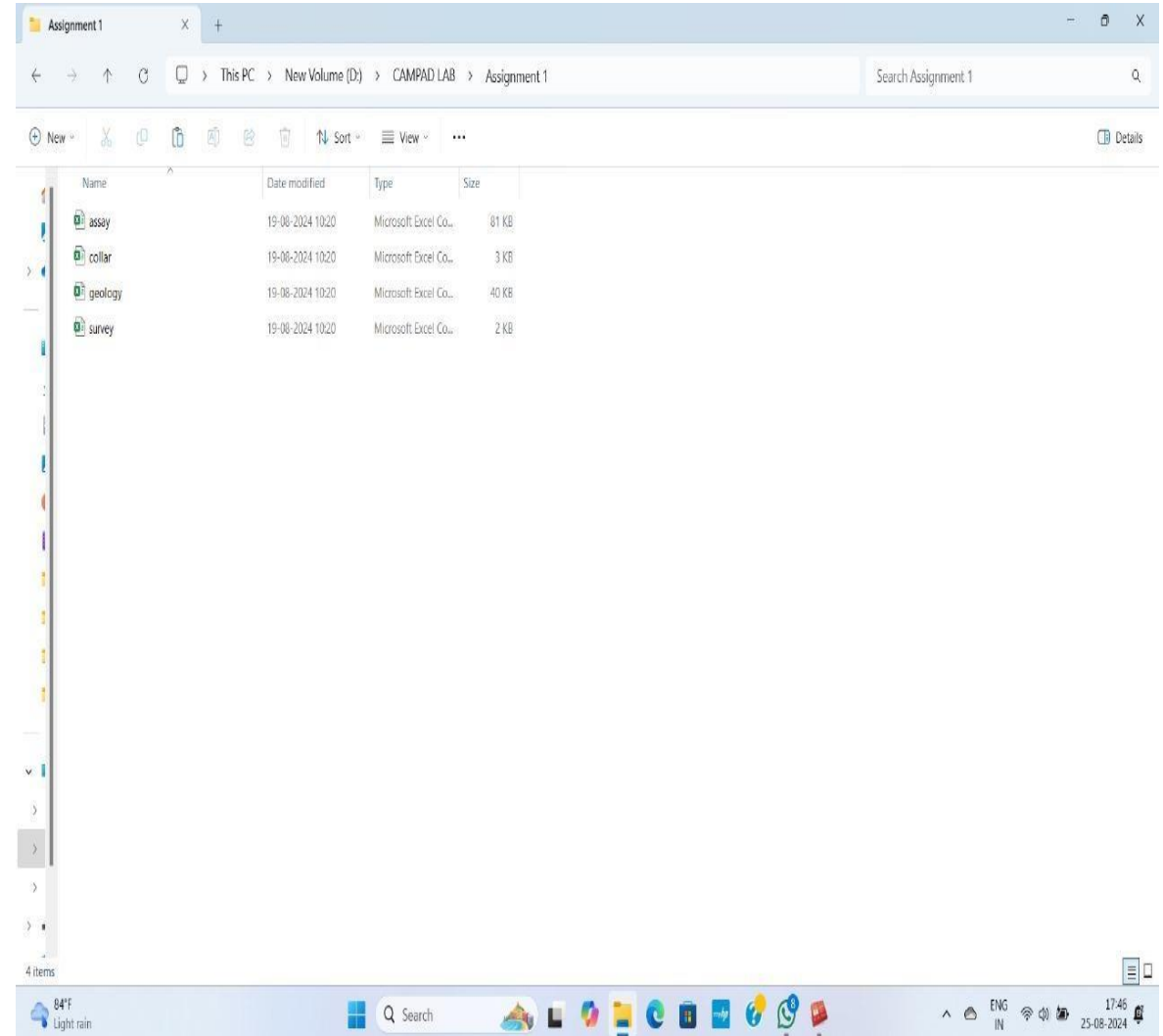
Assignment-01

- Group-13
- 1.Katravath Padma-21JE1103
- 2.Banoth Nikhila-22JE0234
- 3.Barakala Neeha Madhuri-22JE0238

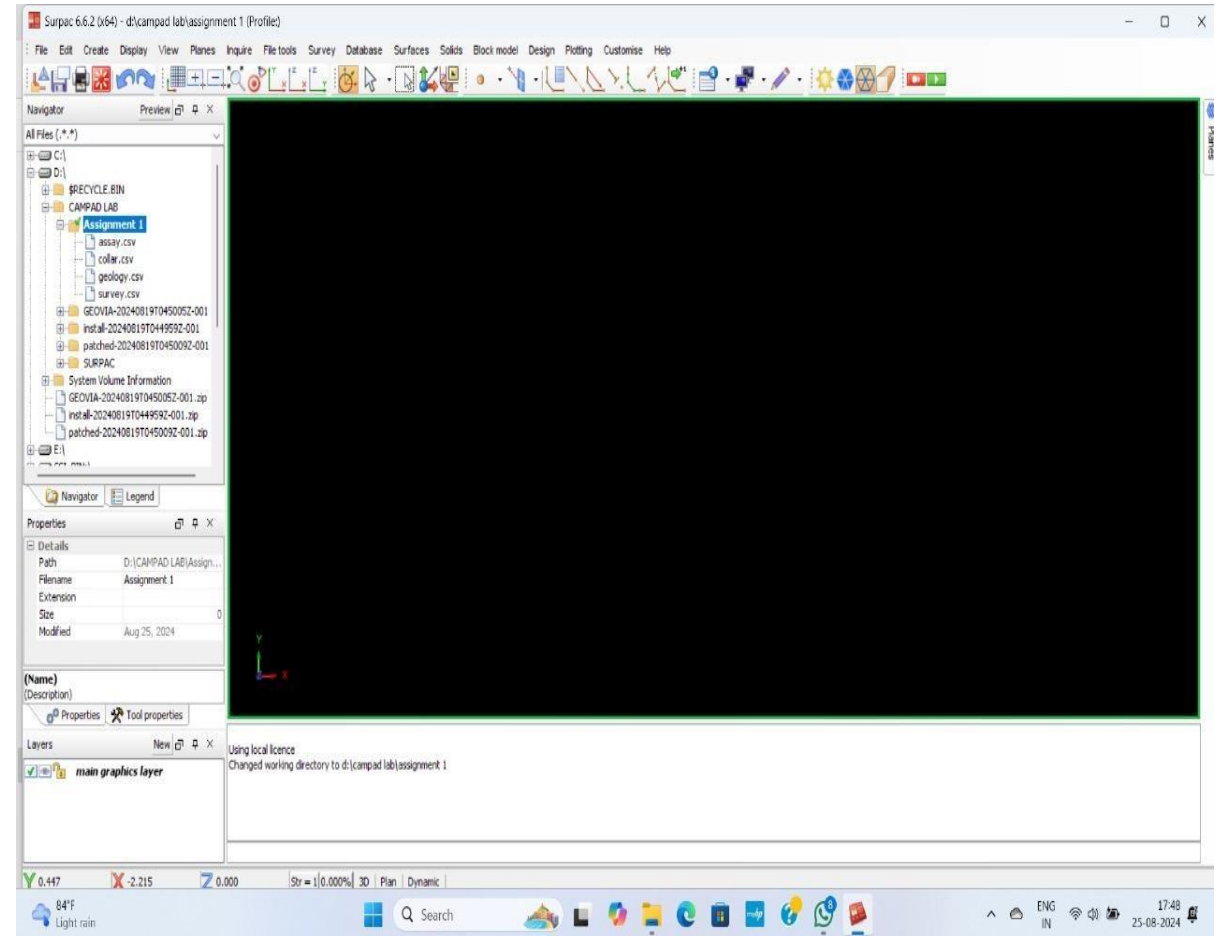
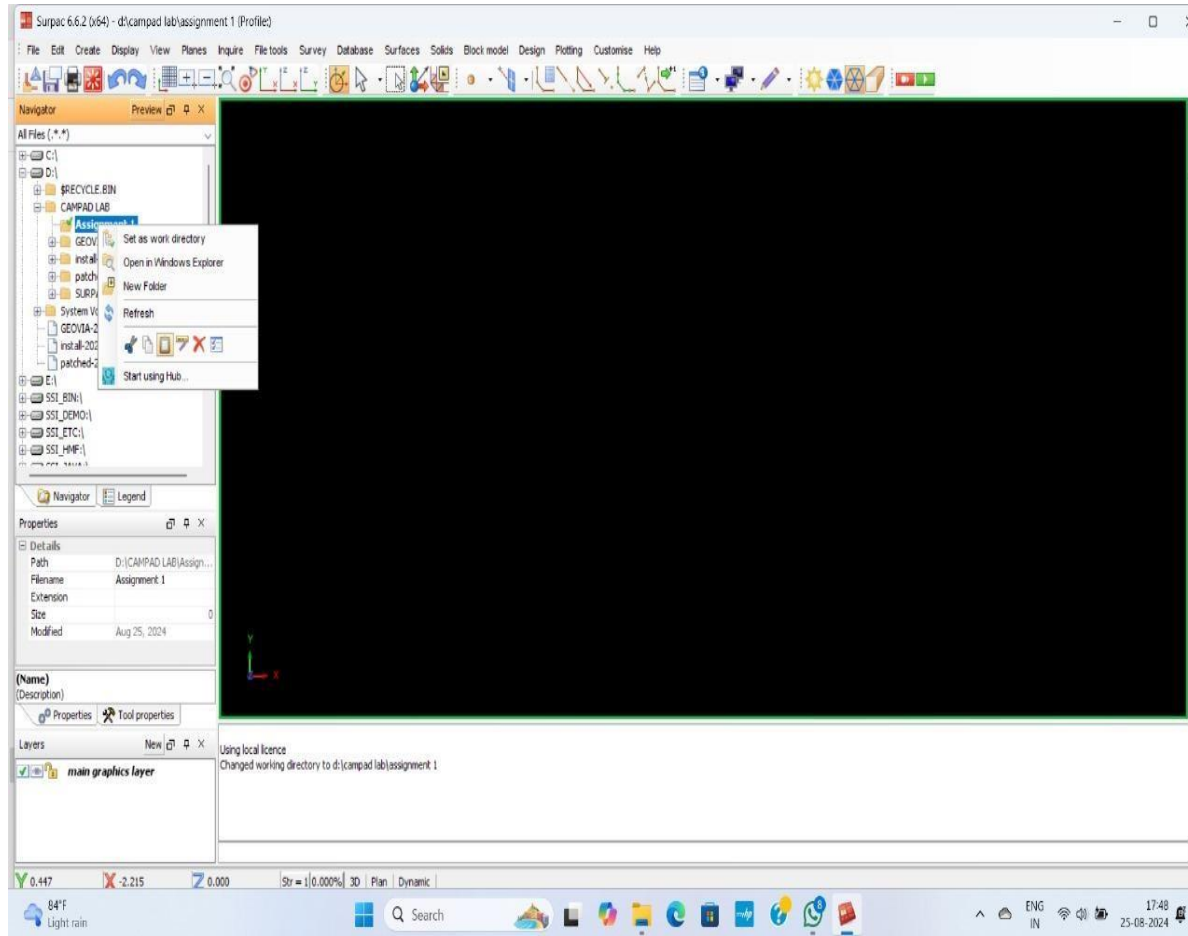
Step 1:-Create a new folder as assignment1 in C drive in your desktop and set it as a work directory



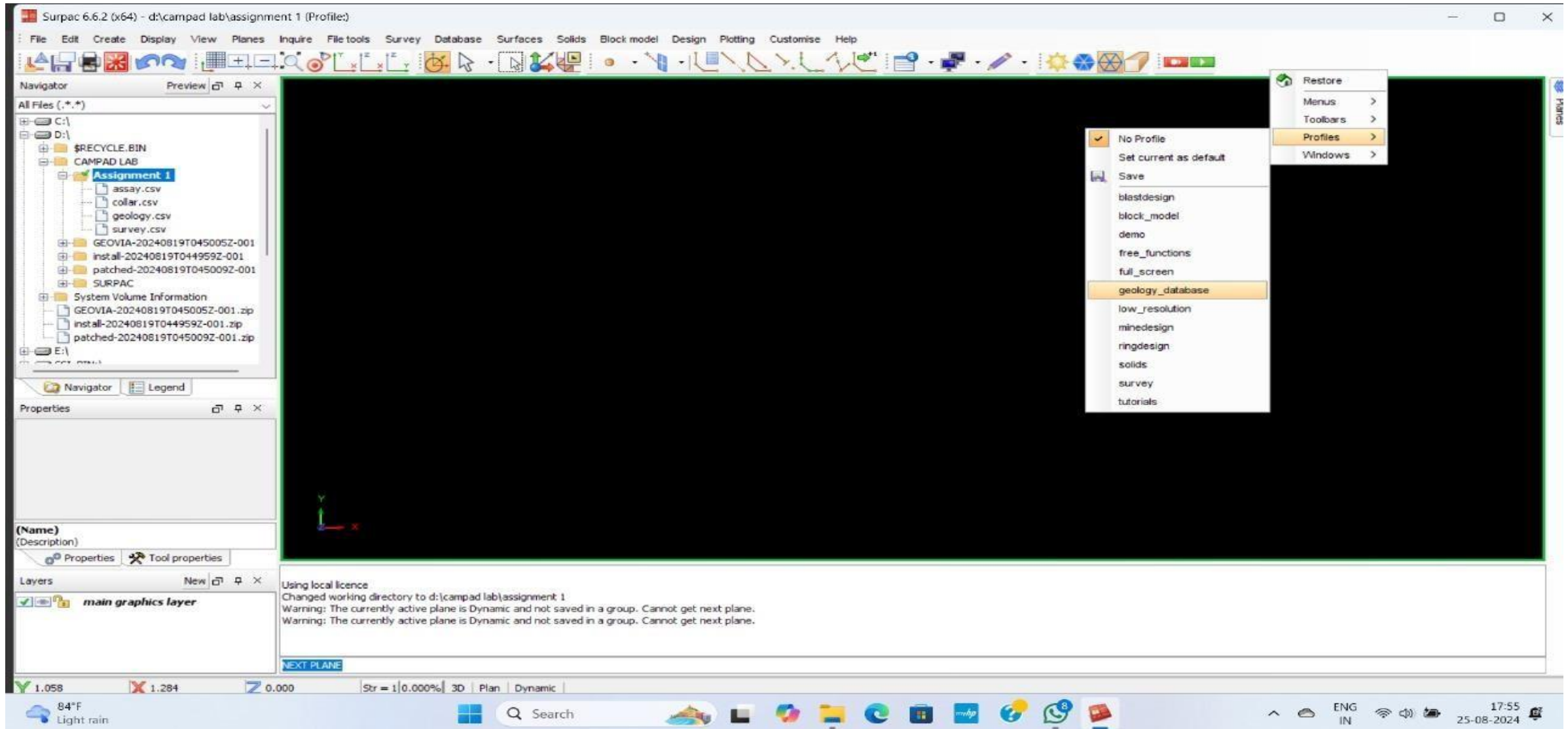
Step 2:-Save assay, collar, geology, and survey in the Assignment 1 folder.



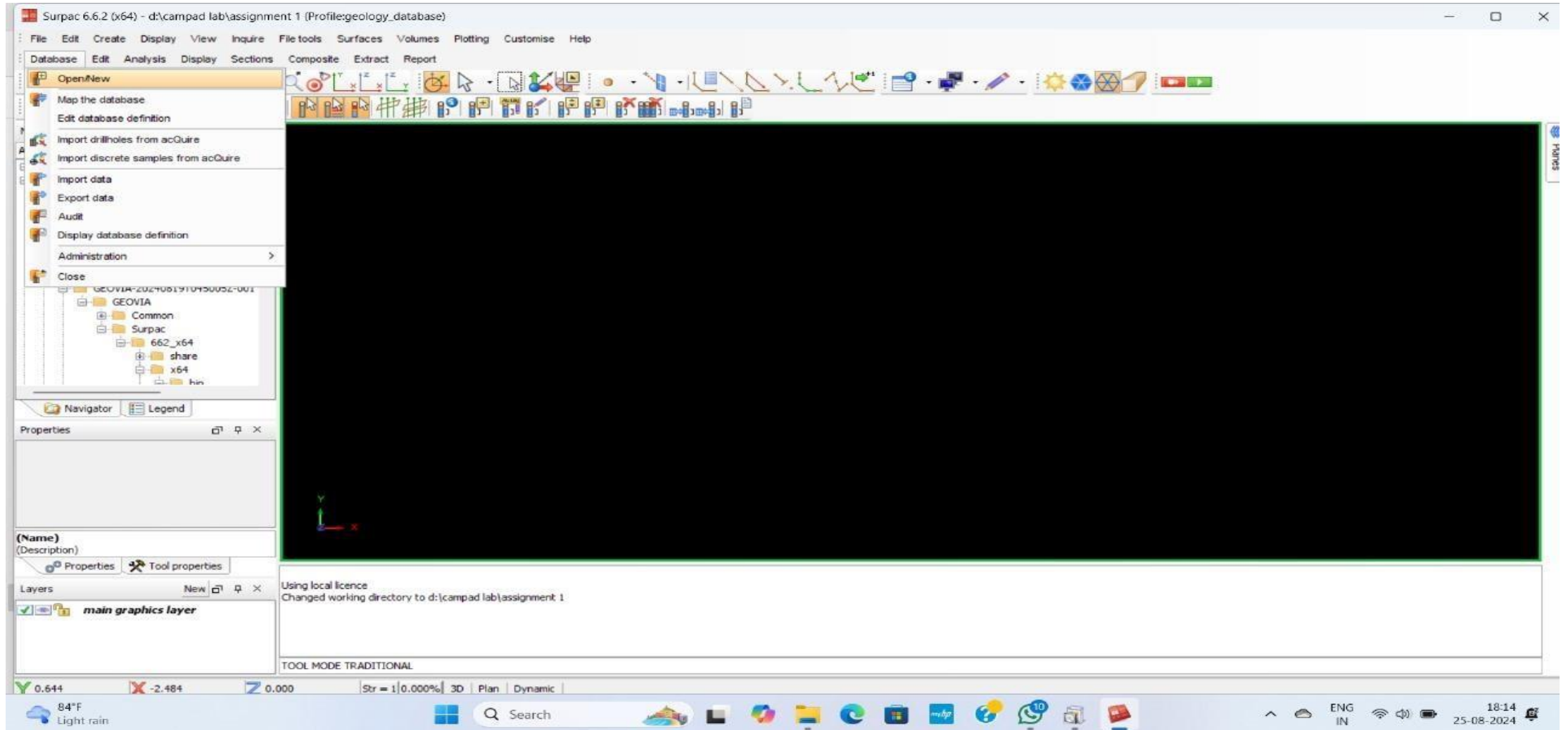
Step 3: Copy those Excel files>> right-click on assignment 1>>paste those Excel files into assignment 1.



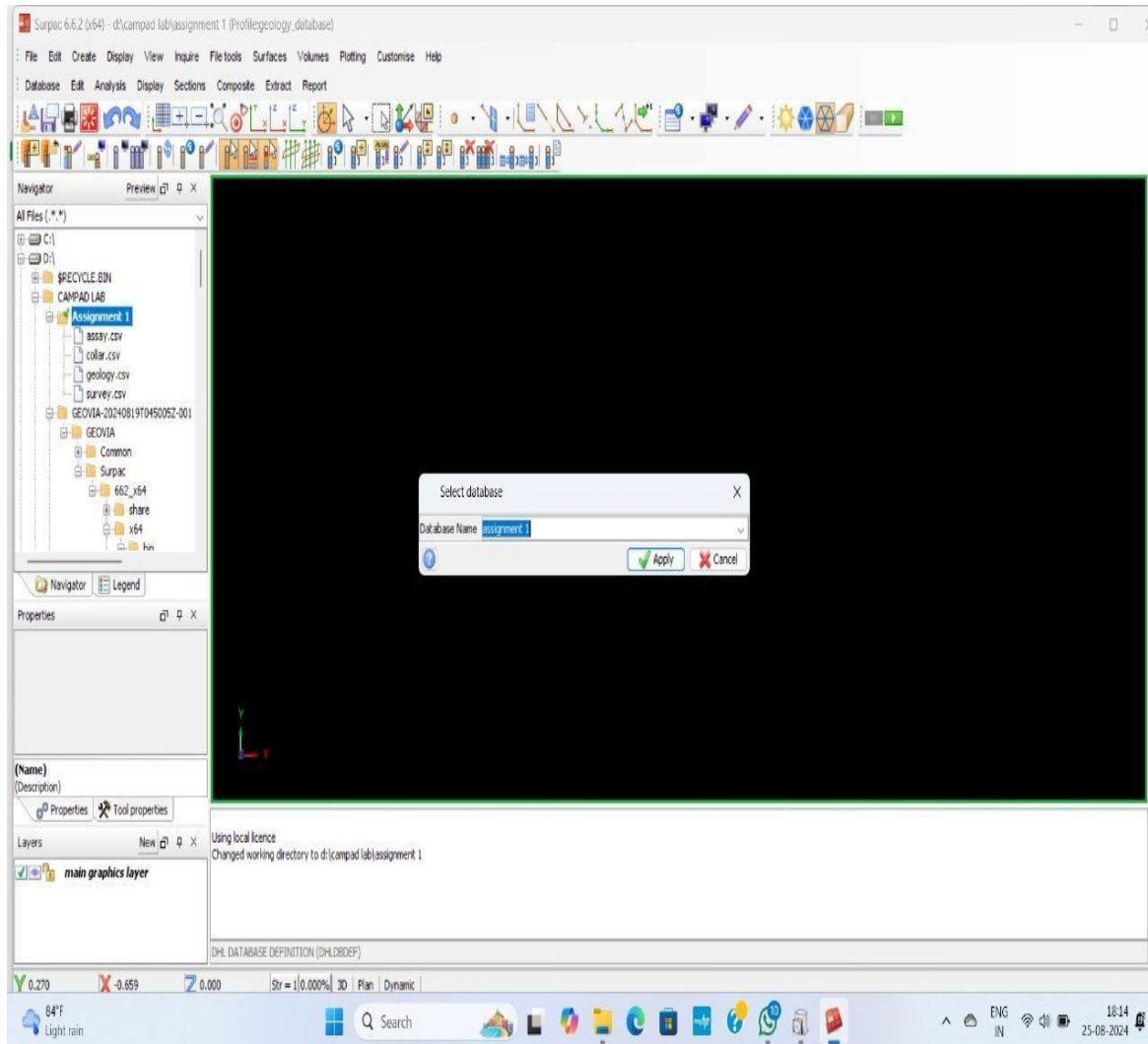
Step 4:- Right click on the top >> Go to profile>> select geology database



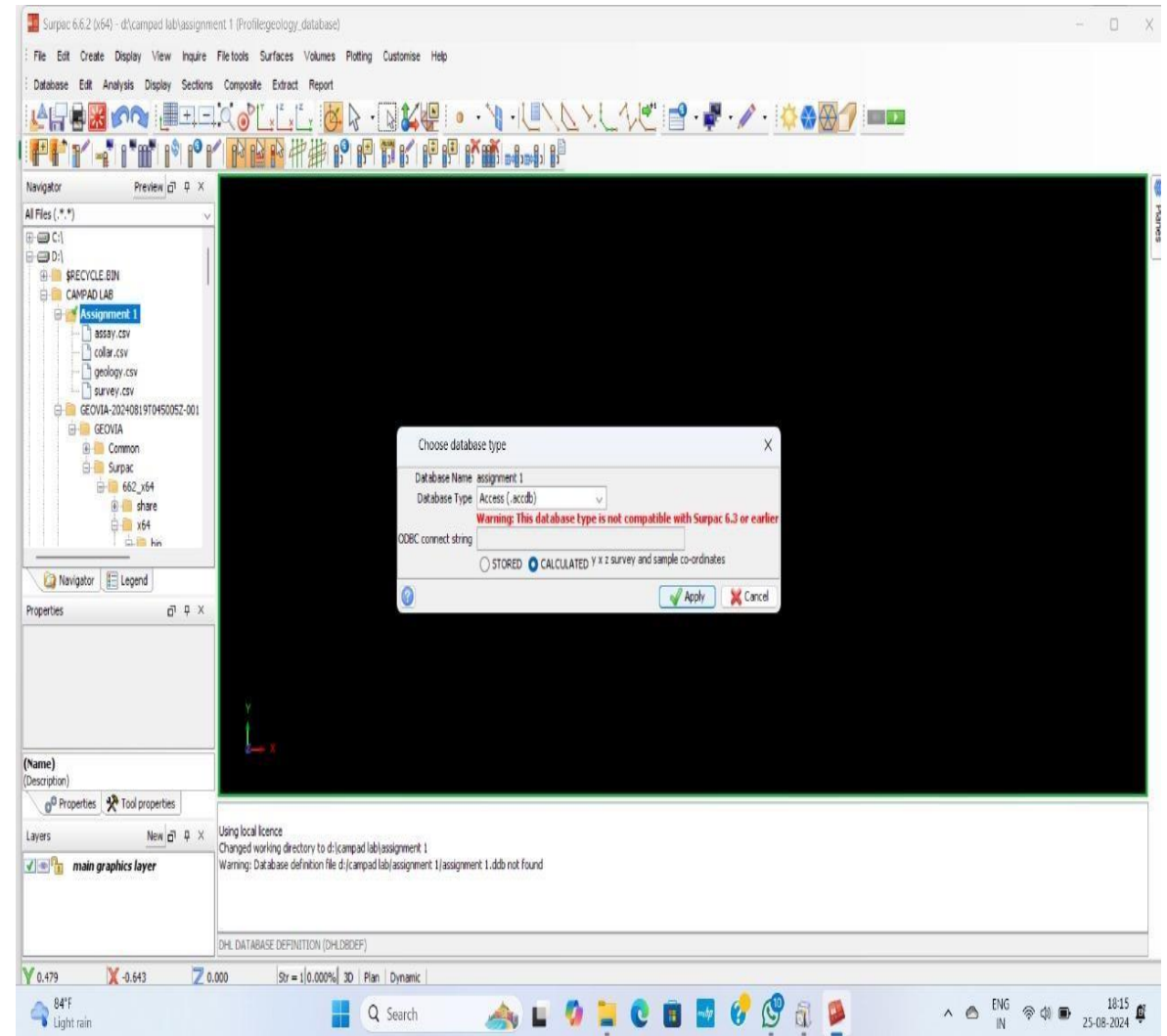
Step 5:- Click database option>> click on open/New



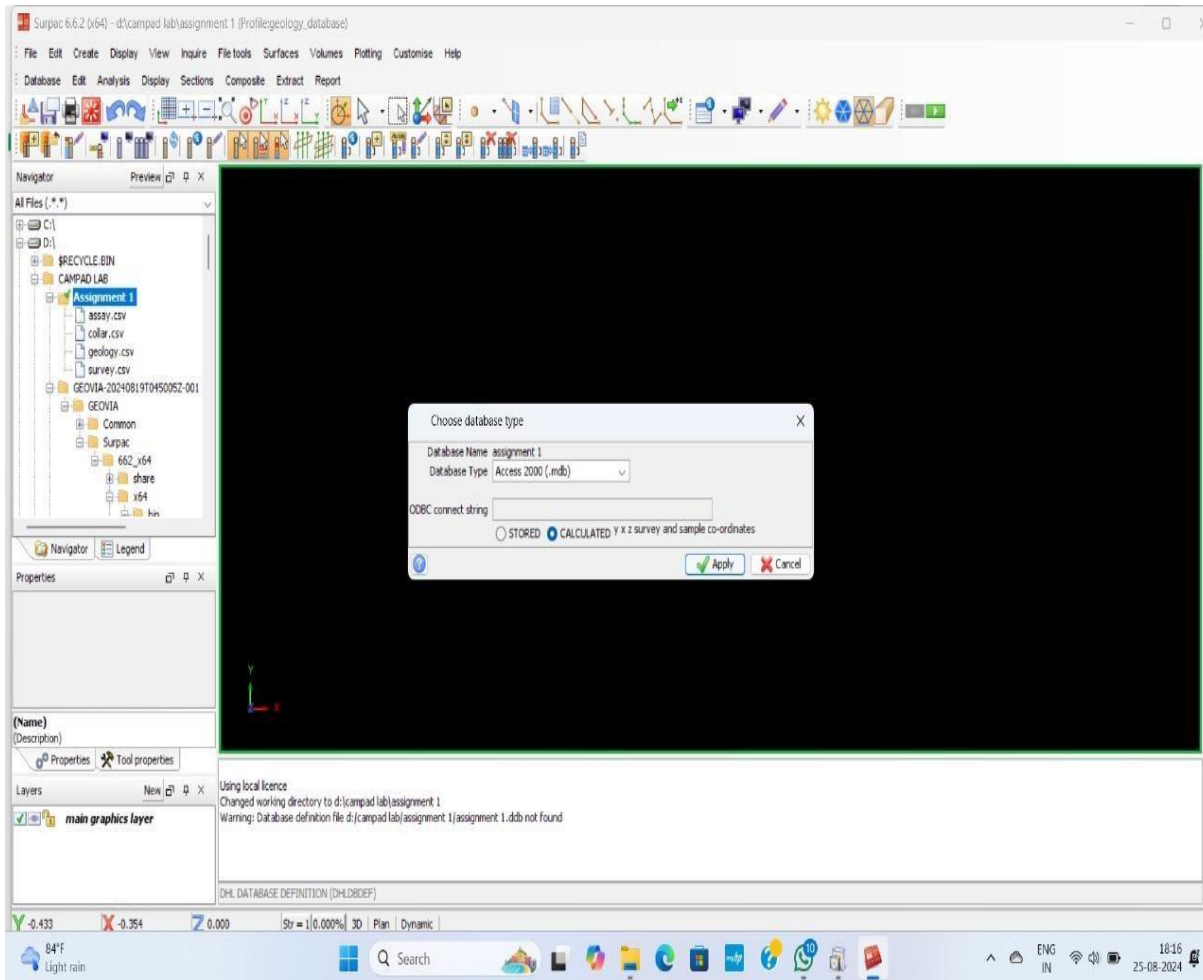
Step 6:- Name the database as assignment1 >>click on apply



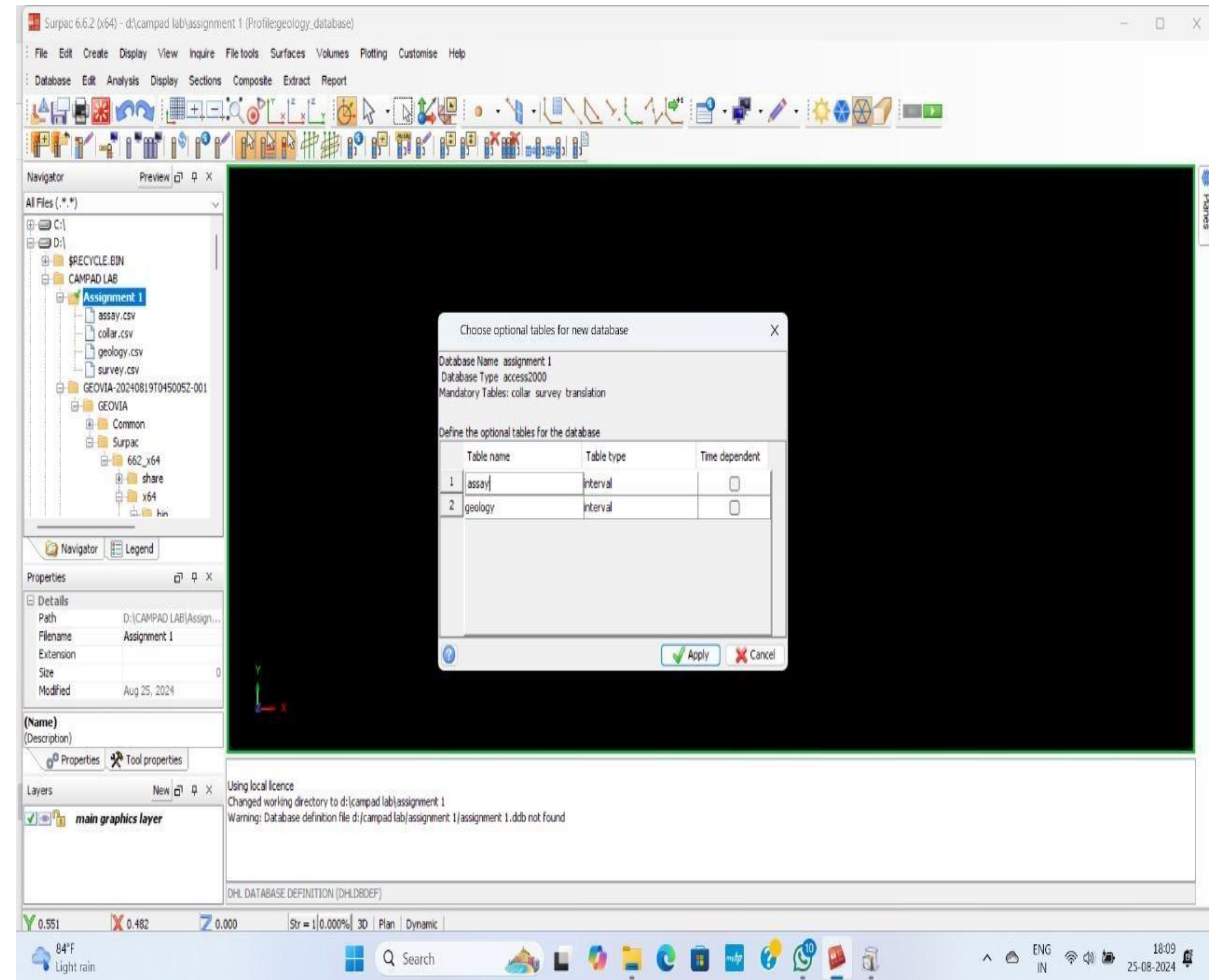
Step 7:- Check whether the database type is showing (.accdb) >> click on apply



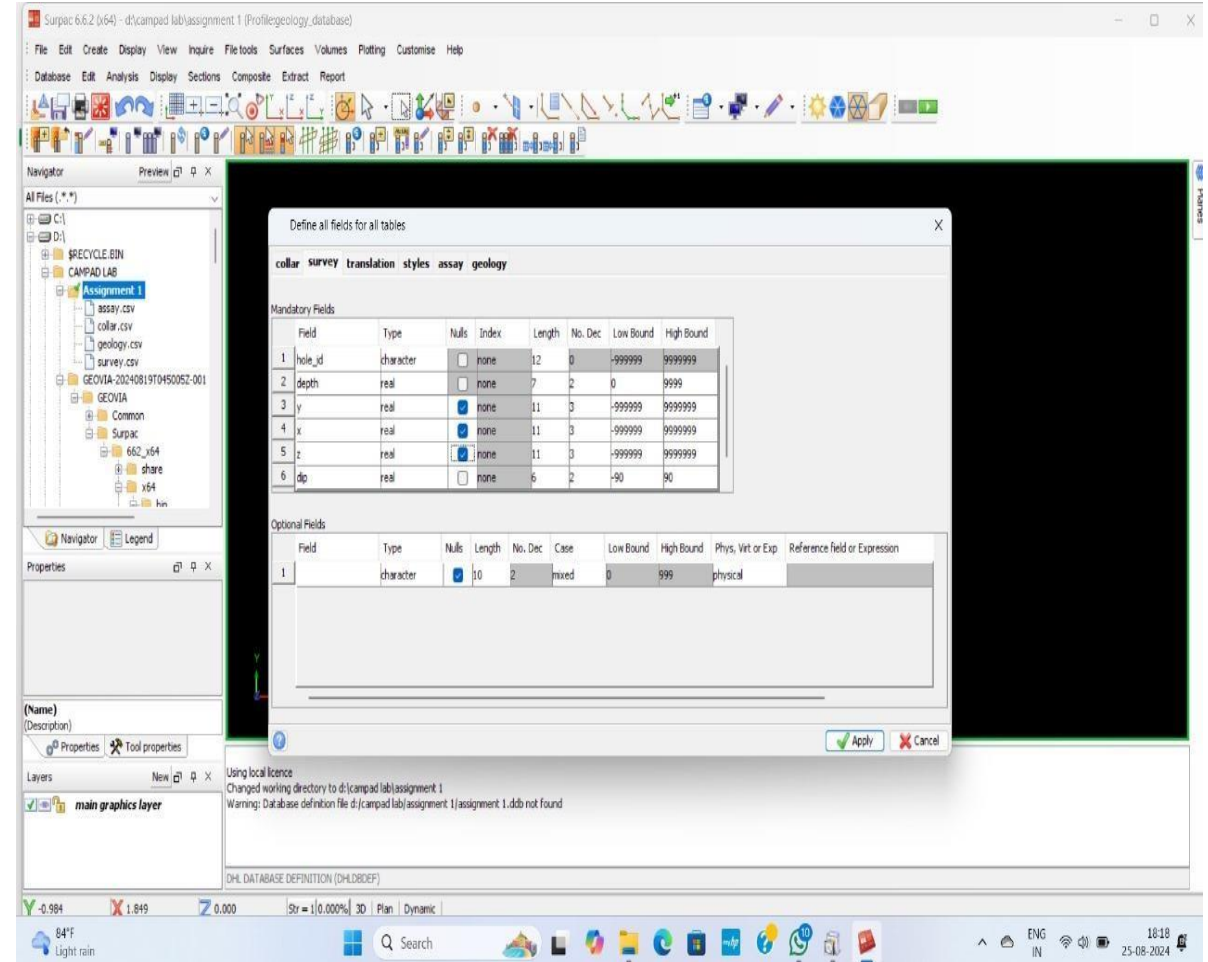
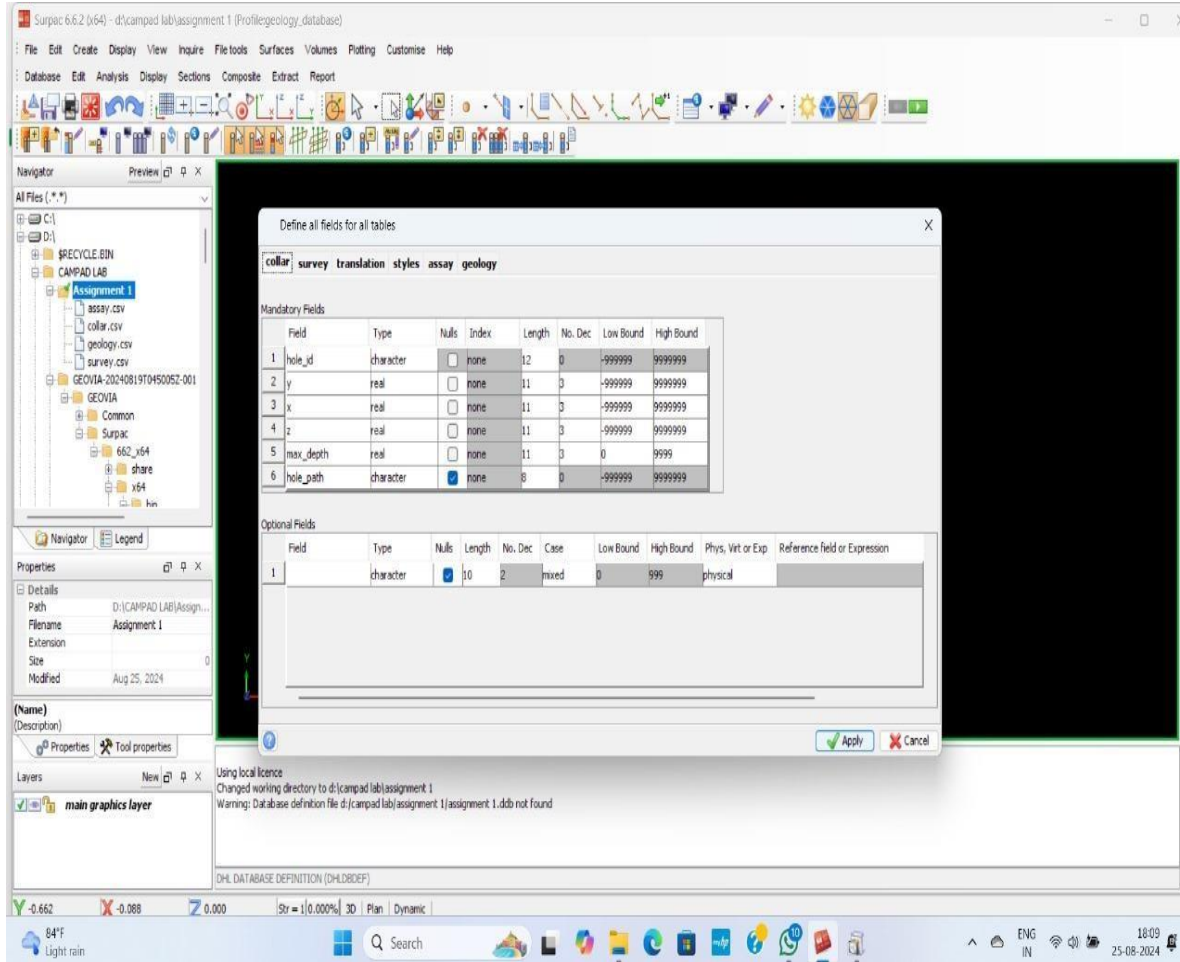
Step 8:- In the next pop up select it as calculated and click on apply



Step 9:- Right-click and choose the add button then add the assay and geology >> Click on Apply.



Step 10:- Check collar, Survey, assay, and geology tables with the Excel sheet and untick the rows that are not required.



Step 11- In the assay section unselect not required rows and add required options in optional fields. Add fe, sio2,al2o3,loi,p. modify integers.

Define all fields for all tables

collar survey translation styles **assay** geology

Mandatory Fields

Field	Type	Nuls	Index	Length	No. Dec	Low Bound	High Bound
1 hole_id	character	☐	none	12	0	-999999	9999999
2 samp_id	character	☒	none	10	0	-999999	9999999
3 depth_from	real	☐	none	7	2	0	9999
4 v_from	real	☒	none	11	3	-999999	9999999
5 x_from	real	☒	none	11	3	-999999	9999999
6 z_from	real	☒	none	11	3	-999999	9999999

Optional Fields

Field	Type	Nuls	Length	No. Dec	Case	Low Bound	High Bound	Phys. Unit or Exp	Reference field or Expression
1 Fe	real	☐	10	2	mixed	0	999	physical	
2 SiO2	real	☐	10	2	mixed	0	999	physical	
3 Al2O3	real	☐	10	2	mixed	0	999	physical	
4 LOI	real	☐	10	2	mixed	0	999	physical	
5 P	real	☐	10	2	mixed	0	999	physical	

Apply Cancel

Step 12:- In the geology section add code. Modify integers. Click on apply.

Define all fields for all tables

collar survey translation styles assay **geology**

Mandatory Fields

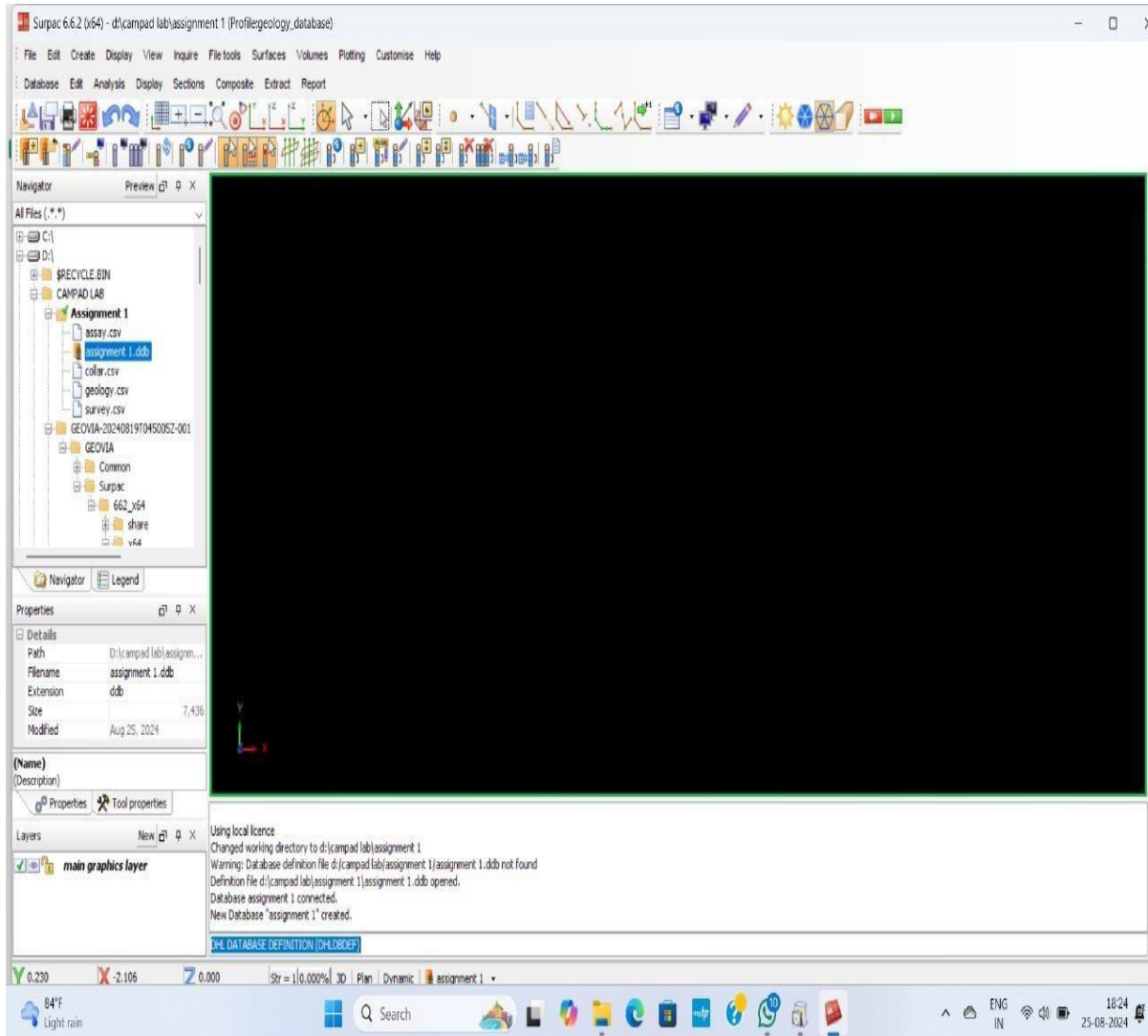
Field	Type	Nuls	Index	Length	No. Dec	Low Bound	High Bound
1 hole_id	character	☐	none	12	0	-999999	9999999
2 samp_id	character	☒	none	10	0	-999999	9999999
3 depth_from	real	☐	none	7	2	0	9999
4 v_from	real	☒	none	11	3	-999999	9999999
5 x_from	real	☒	none	11	3	-999999	9999999
6 z_from	real	☒	none	11	3	-999999	9999999

Optional Fields

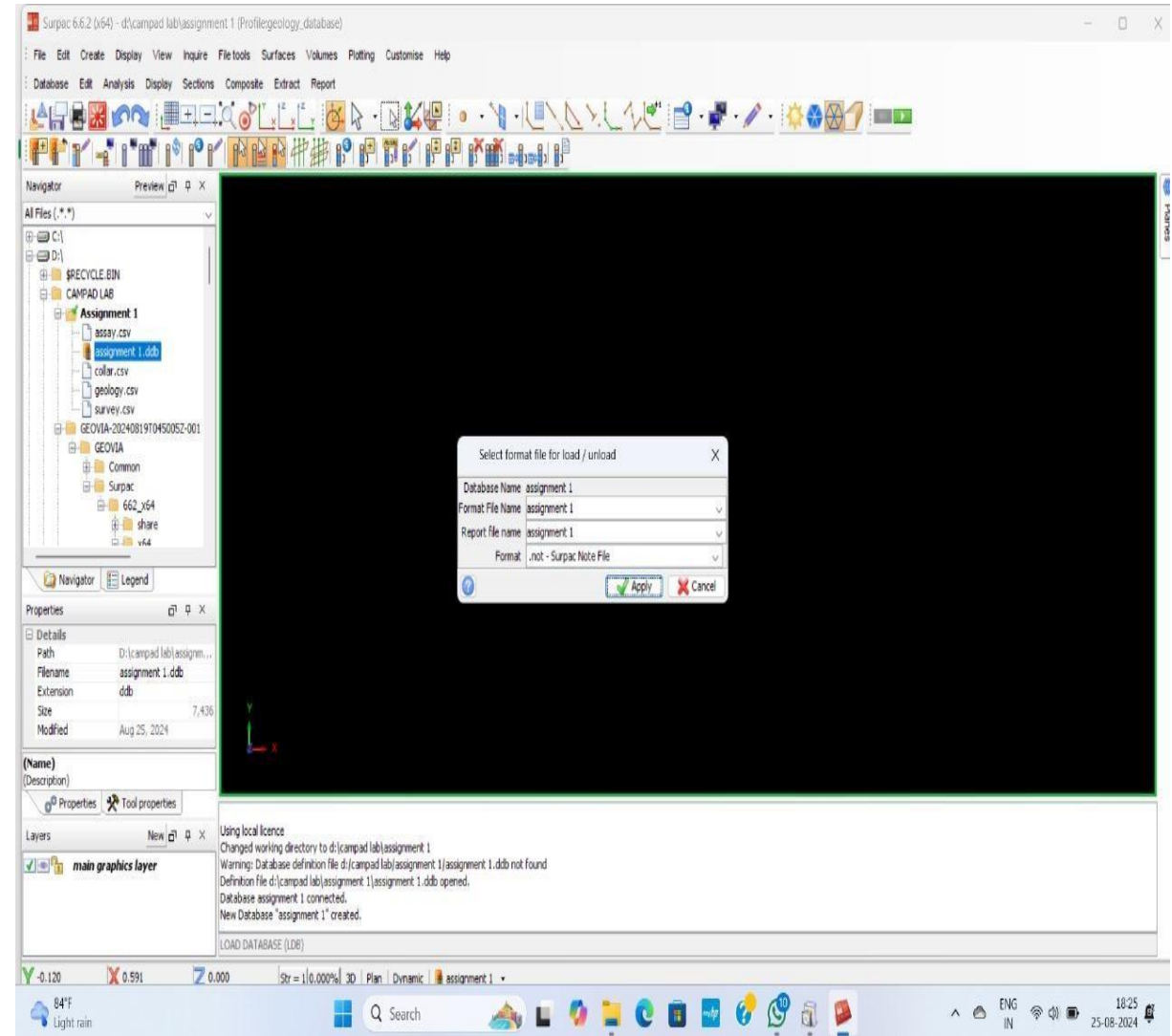
Field	Type	Nuls	Length	No. Dec	Case	Low Bound	High Bound	Phys. Unit or Exp	Reference field or Expression
1 code	real	☐	10	2	mixed	0	999	physical	

Apply Cancel

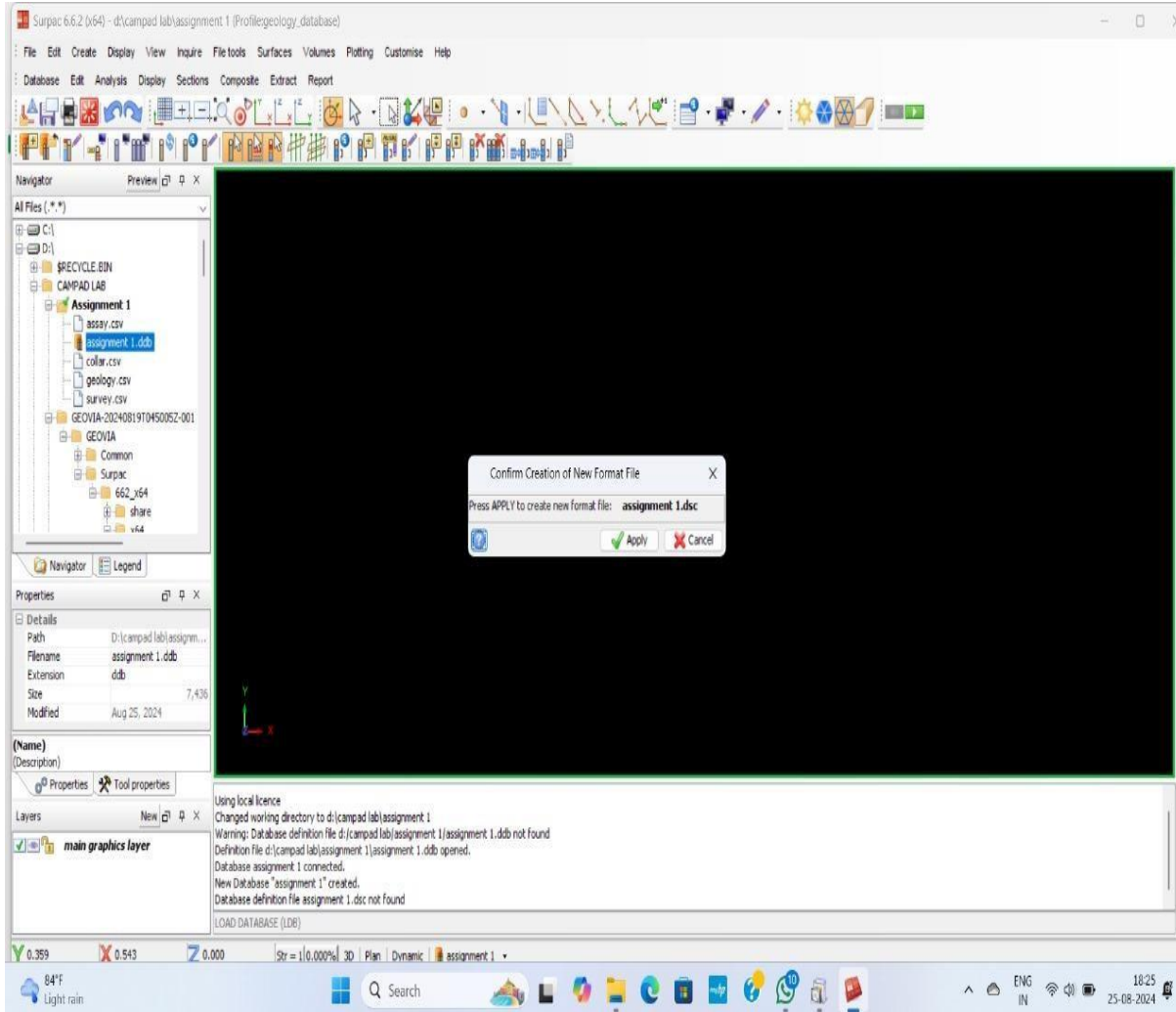
Step 13:- Check whether the new database is created or not in the form of (.ddb)



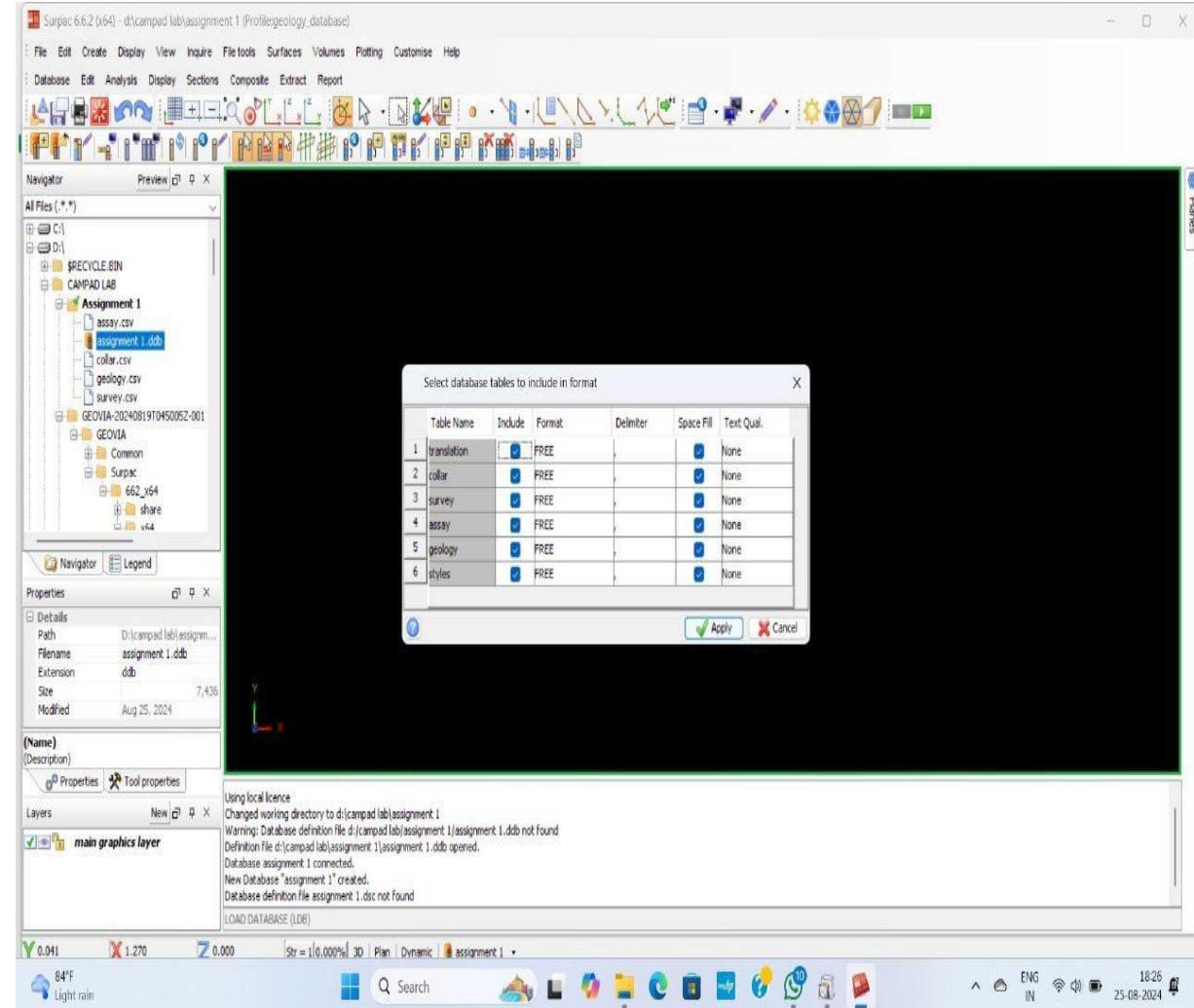
Step 14:- Go to Database >>Click on Import data>>Click on apply



Step 15 :- Click on Apply



Step 16:- Select database labels to include in format



Step 17:- Check whether column numbers is matched with excel sheets or not. Click on apply

Surpac 6.6.2 (x64) - d:\campad lab\assignment 1 (Profile:geology_database)

File Edit Create Display View Inquire File tools Surfaces Volumes Plotting Customise Help

Database Edit Analysis Display Sections Composite Extract Report

Navigator

All Files (*.*)

C:\

D:\

\$RECYCLE.BIN

CAMPAD LAB

Assignment 1

assay.csv

assignment 1.ddb

collar.csv

geology.csv

survey.csv

GEOVIA-20240819T045005Z-001

GEOVIA

Common

Surpac

662_x64

share

v6.6

Properties

Details

Path D:\campad lab\assignm...

Filename assignment 1.ddb

Extension ddb

Size 7,436

Modified Aug 25, 2024

(Name)

(Description)

Properties Tool properties

Layers

New

main graphics layer

Using local licence

Changed working directory to d:\campad lab\assignment 1

Warning: Database definition file d:\campad lab\assignment 1\assignment 1.ddb not found

Definition file d:\campad lab\assignment 1\assignment 1.ddb opened.

Database assignment 1 connected.

New Database "assignment 1" created.

Database definition file assignment 1.dsc not found

LOAD DATABASE (LDB)

0.085 0.486 0.000 Str = 1|0.000% 3D Plan Dynamic assignment 1

84°F Light rain

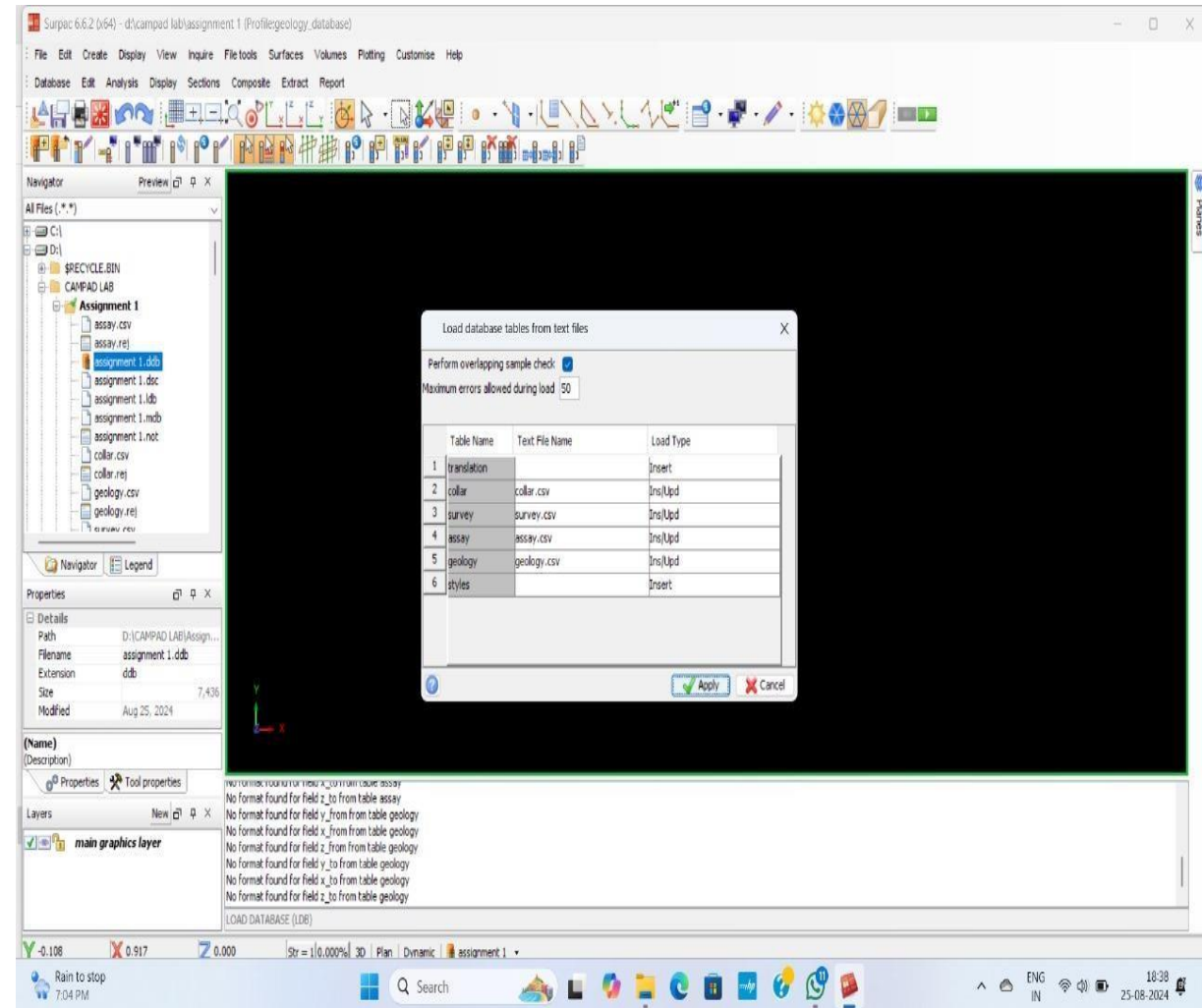
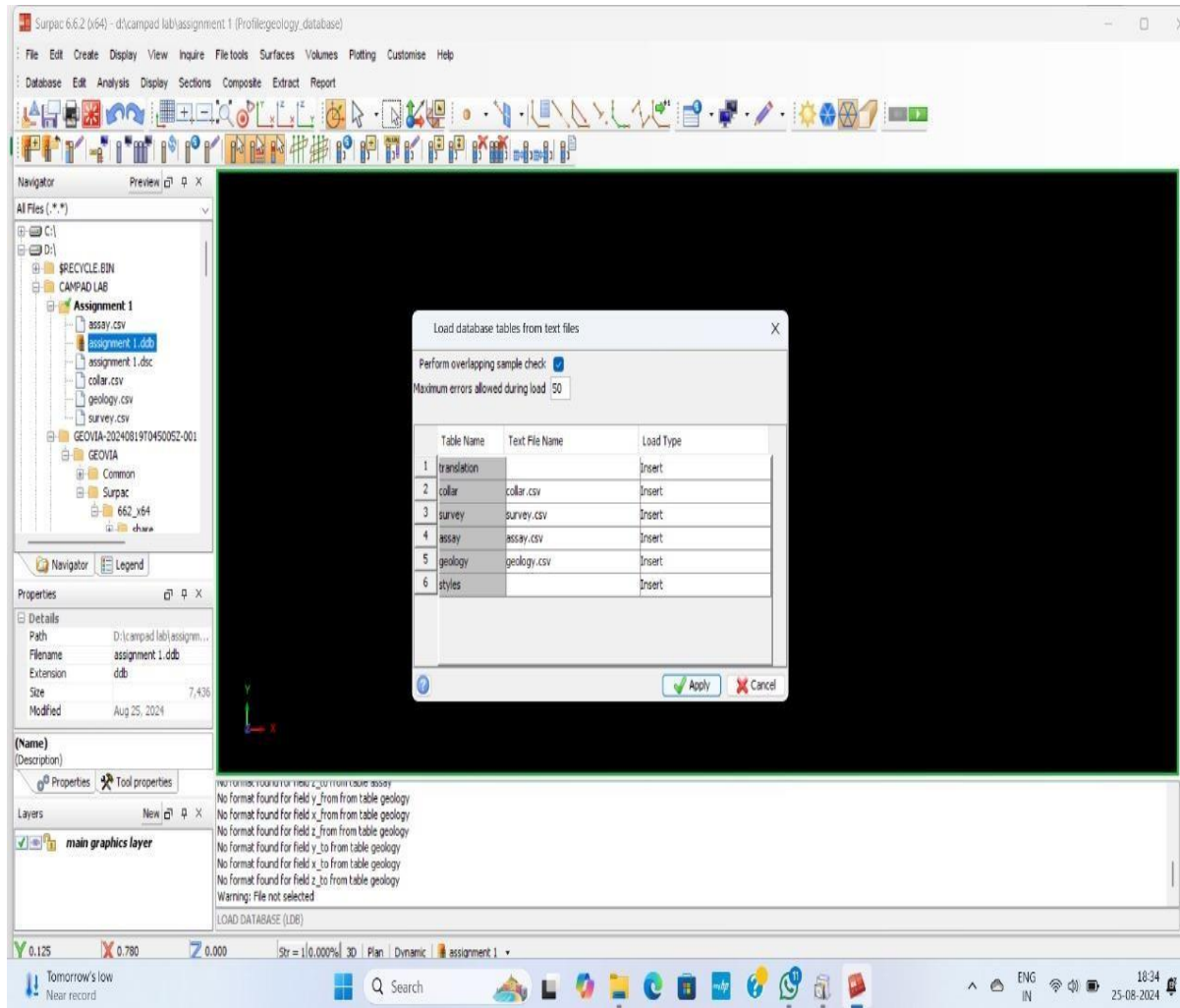
Search

ENG IN

18:27 25-08-2024

	Table Name	Field Name	Include	Column	Length	Format
1	translation	table_name	☒	1	0	FREE
2		field_name	☒	2	0	FREE
3		code	☒	3	0	FREE
4		num_equiv	☒	4	0	FREE
5		description	☒	5	0	FREE
6	collar	hole_id	☒	1	0	FREE
7		y	☒	2	0	FREE
8		x	☒	3	0	FREE
9		z	☒	4	0	FREE
10		max_depth	☒	5	0	FREE
11		hole_path	☒	6	0	FREE

Step 18:- Upload the .csv files in the text file name column. After Uploading files Change the load type column will show Ins/Upd>> Now Click on Apply .



After clicking "Apply," you will receive a report file containing the number of records that were inserted and rejected from your database.

The screenshot displays the Surpac 6.6.2 (x64) software interface. The main window shows a 'Database Management - Database Load Report' for 'assignment 1' dated 'Aug 25, 2024'. The report details the loading of a table from 'collar.csv' and lists several warnings regarding data validity. It concludes that 0 records were inserted and 71 records were updated. The status bar at the bottom indicates 'Completed load for table geology with 1 errors' and 'Completed load for database assignment 1'. A 'LOAD DATABASE (LDB)' button is visible in the bottom right corner of the report window.

Surpac 6.6.2 (x64) - d:\campad lab\assignment 1 (Profile:geology_database)

File Edit Create Display View Inquire File tools Surfaces Volumes Plotting Customise Help

Database Edit Analysis Display Sections Composite Extract Report

Navigator

All Files (*.*)

C:\

D:\

\$RECYCLE.BIN

CAMPAD LAB

Assignment 1

assay.csv

assay rej

assignment 1

assignment 1

assignment 1

assignment 1

assignment 1

collar.csv

collar rej

geology.csv

geology rej

geology rej

Properties

Details

Path D:\CAMPAD LAB\assignment 1

Filename assignment 1

Extension ddb

Size

Modified Aug 25, 2024

(Name)

(Description)

Properties

Tools

Layers

main graphics

Database Management - Database Load Report Aug 25, 2024

Date : 25-Aug-24

Database : assignment 1

Format file : assignment 1.dsc

=====

Loading Table : collar from file collar.csv

Warning: value for field hole_id contains illegal lowercase characters
Warning: "y" is not numeric
Warning: "x" is not numeric
Warning: "z" is not numeric
Warning: "max_depth" is not numeric
Warning: Value for field hole_path not in allowed set
hole_id,y,x,z,max_depth,hole_path

0 records were inserted.
71 records were updated.

Ln 6, Col 24 4,214 characters

100% Windows (CRLF) UTF-8

Completed load for table geology with 1 errors
Completed load for database assignment 1

LOAD DATABASE (LDB)

-0.666 -0.044 0.000 Str = 1|0.000% 3D Plan Dynamic assignment 1

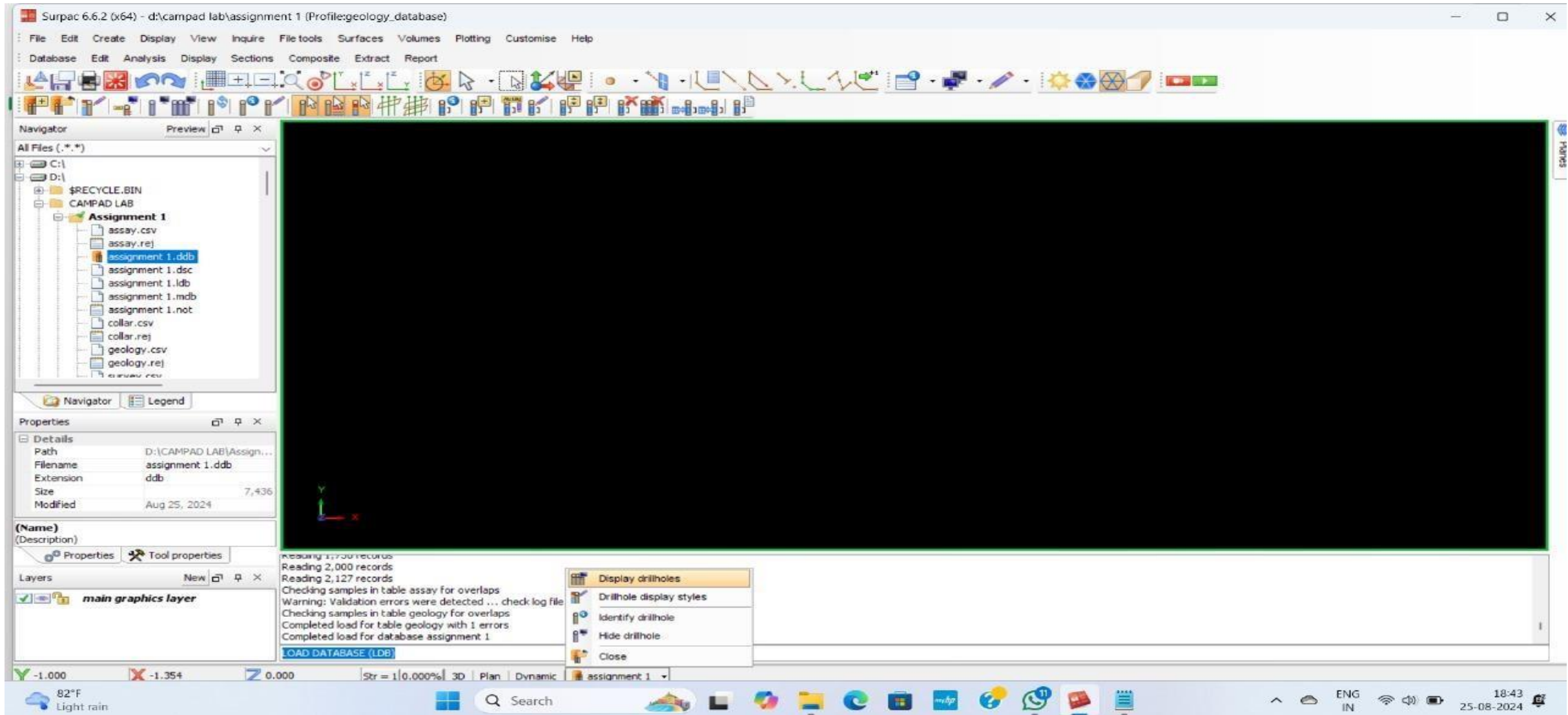
82°F Light rain

Search

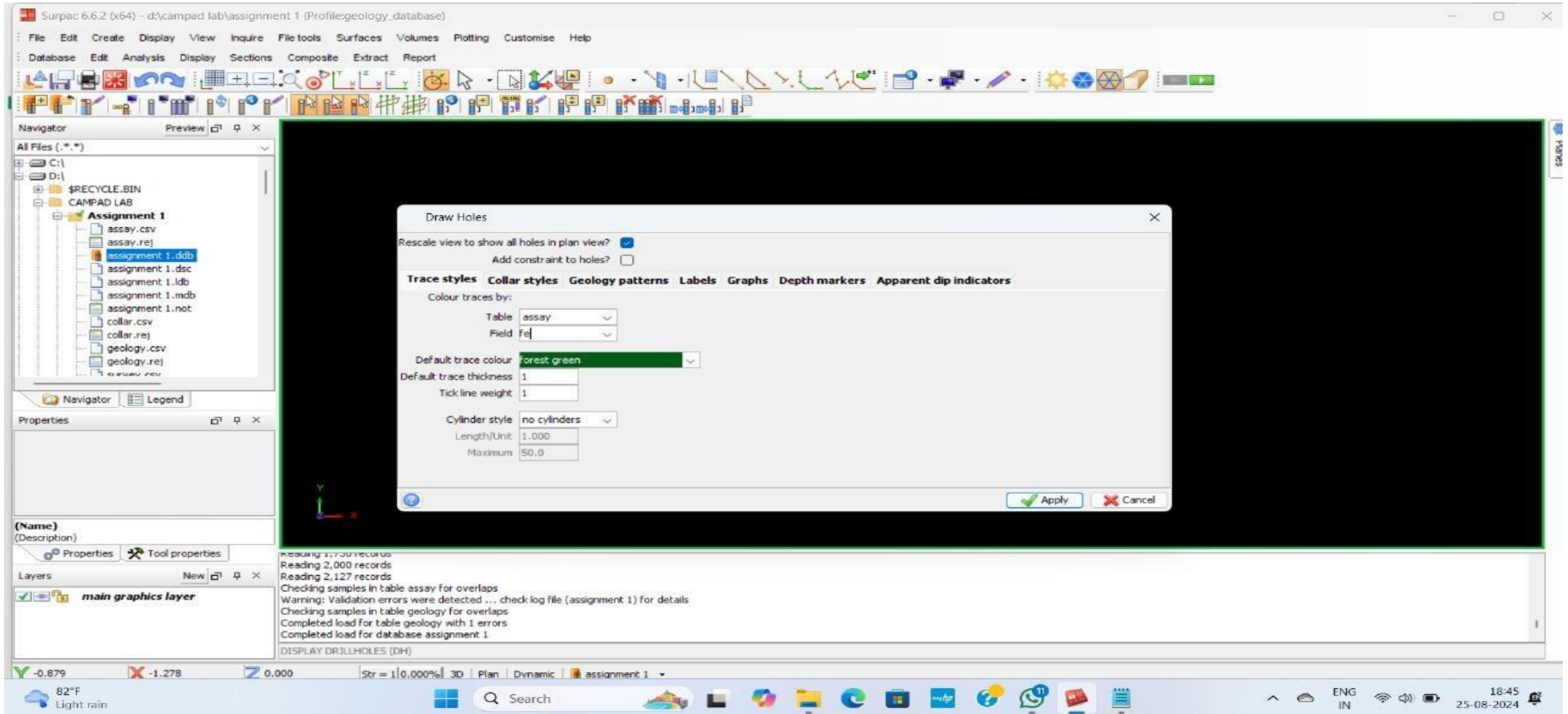
ENG IN

18:40 25-08-2024

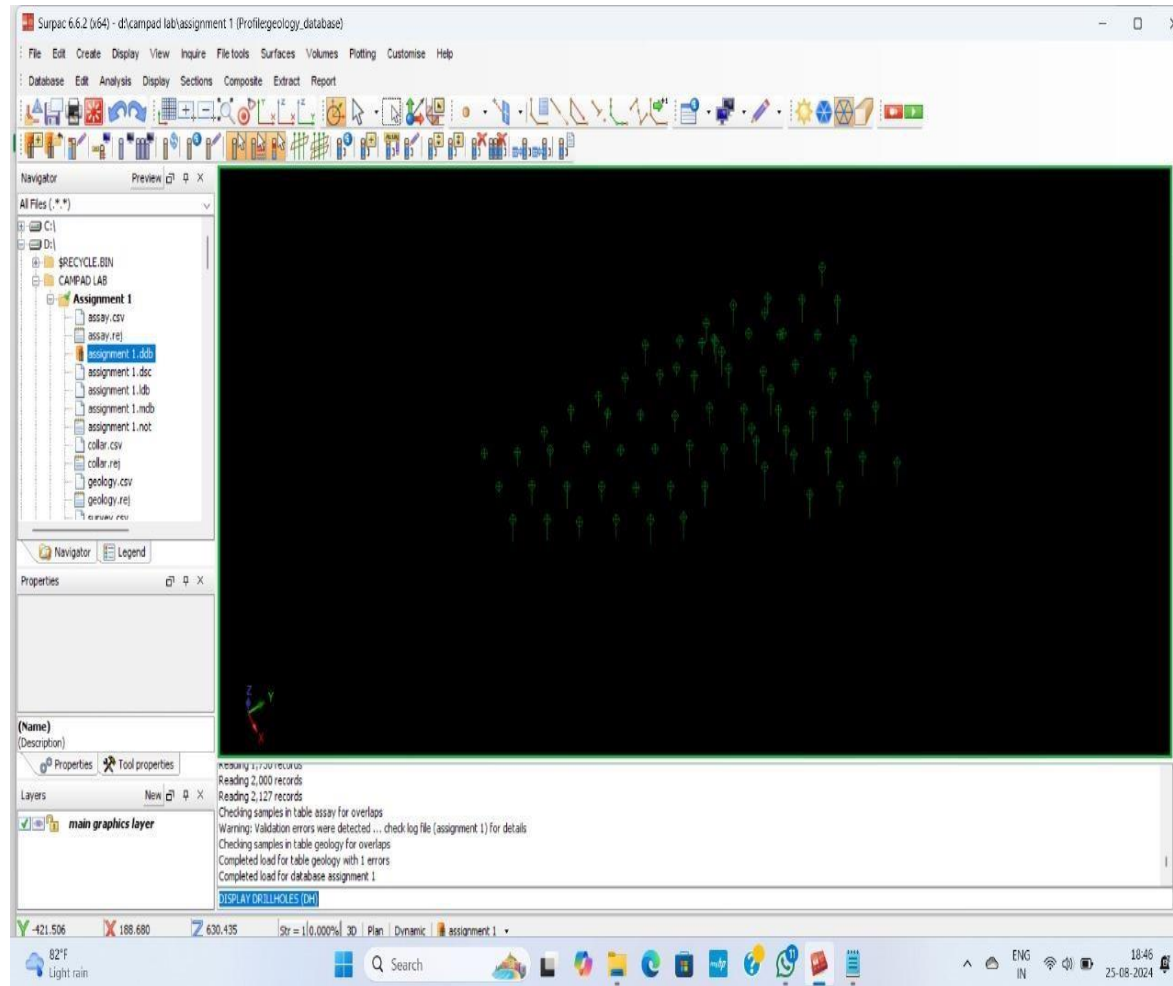
Step 19:- click on assignment1 >> click on display holes.



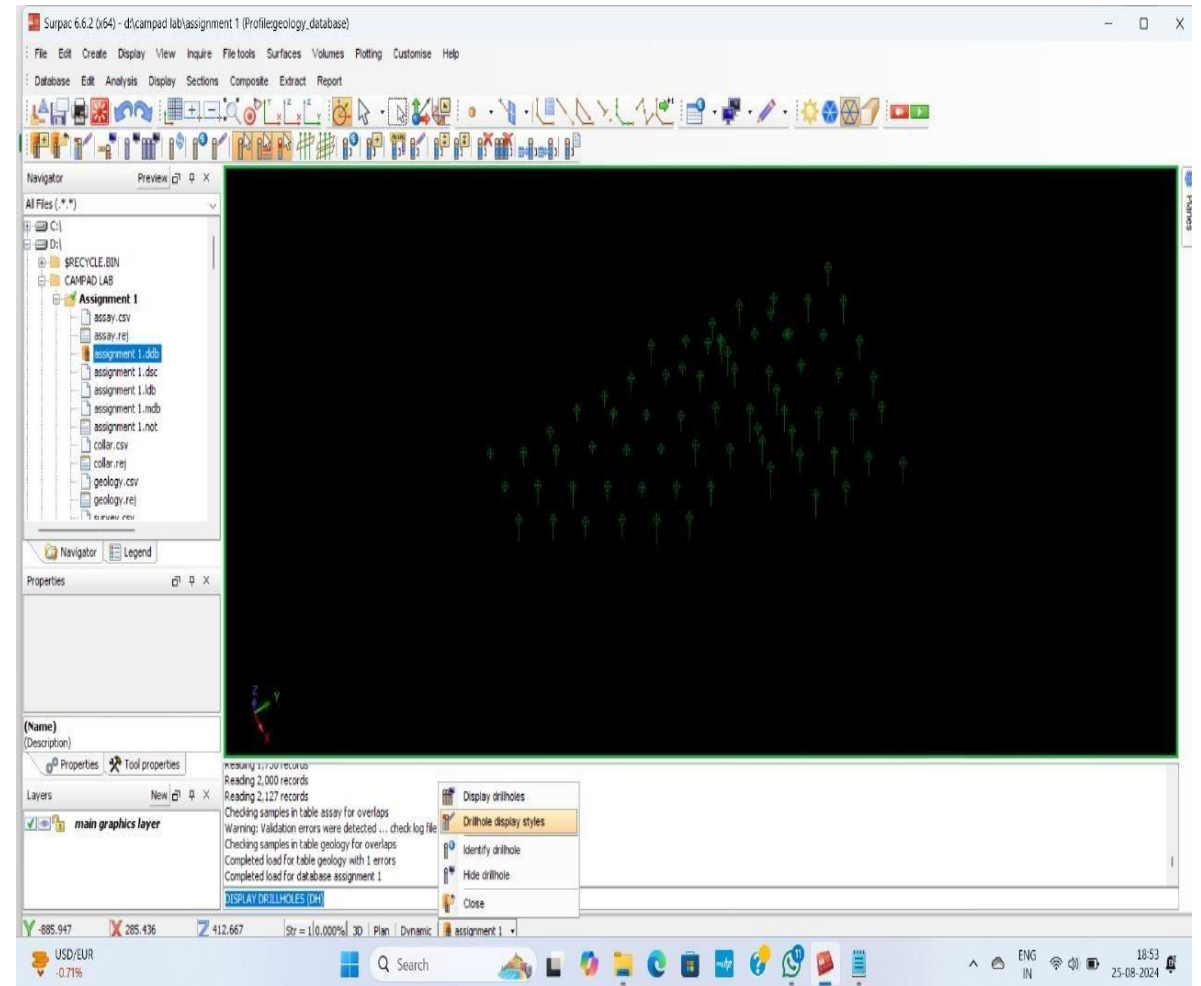
Step 20:- Select any color and click on apply.



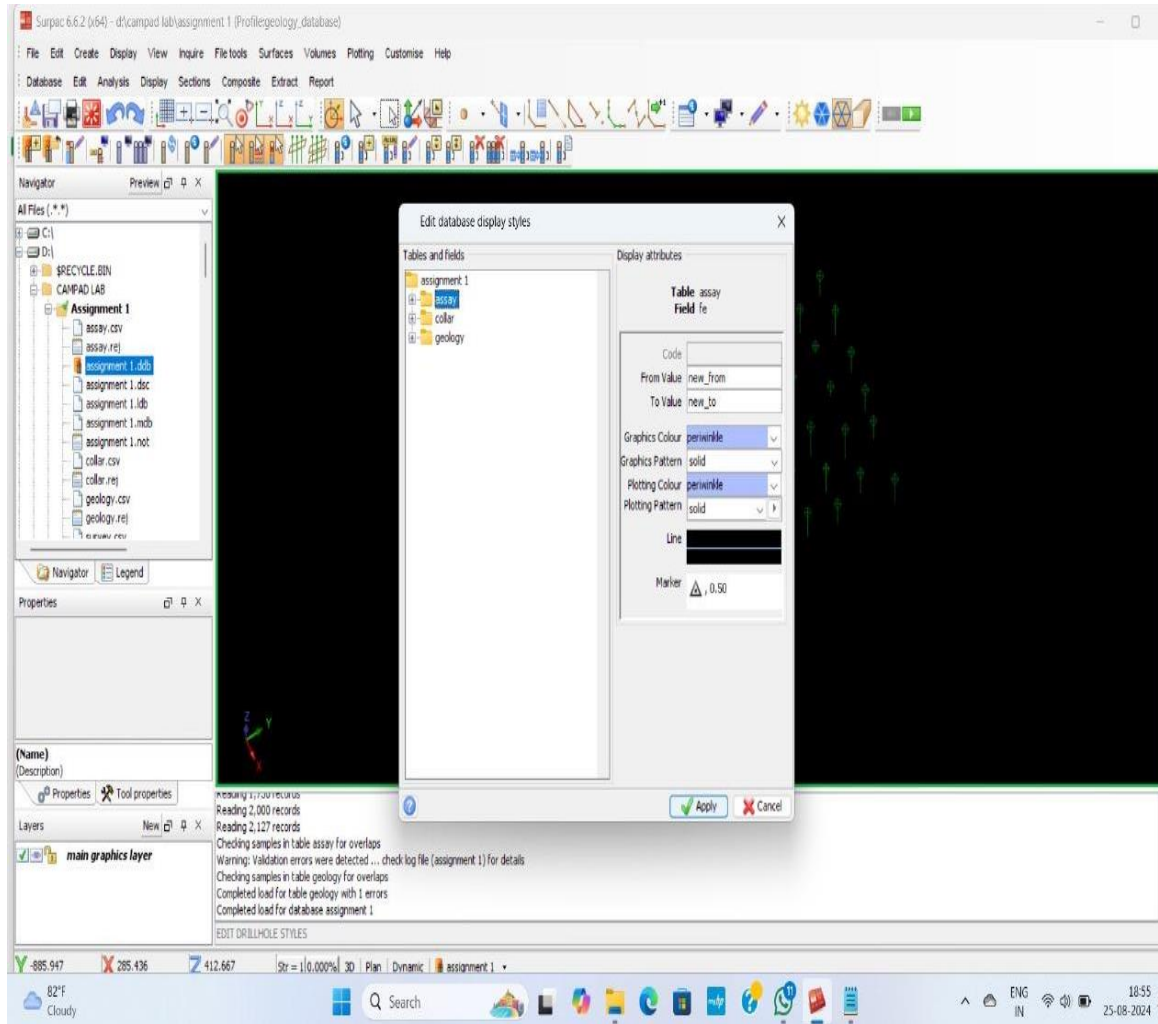
Step 21:- The next window will pop up showing drill holes.



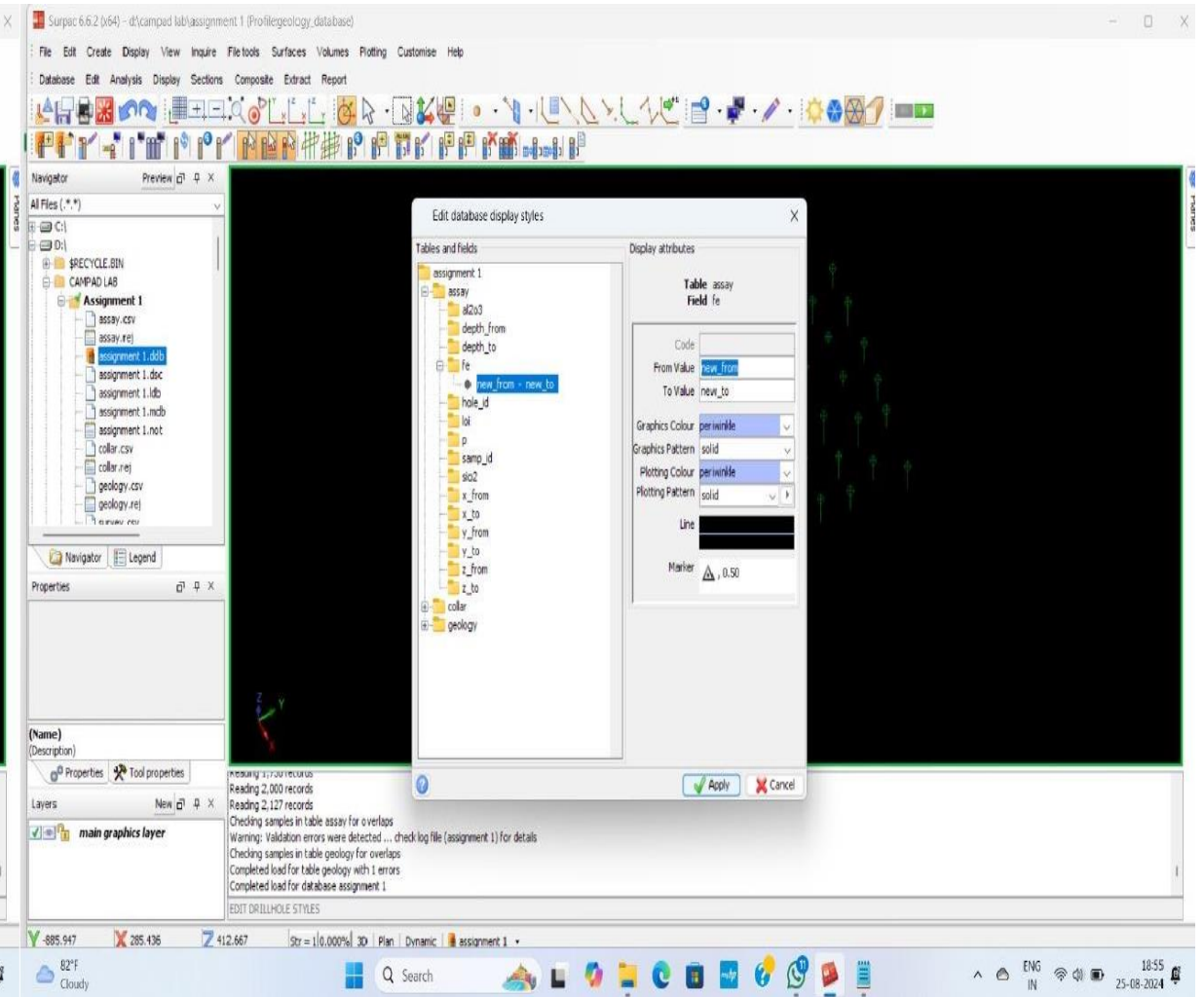
Step 22:- Click on assignment1 >> Click on Drill hole display styles.



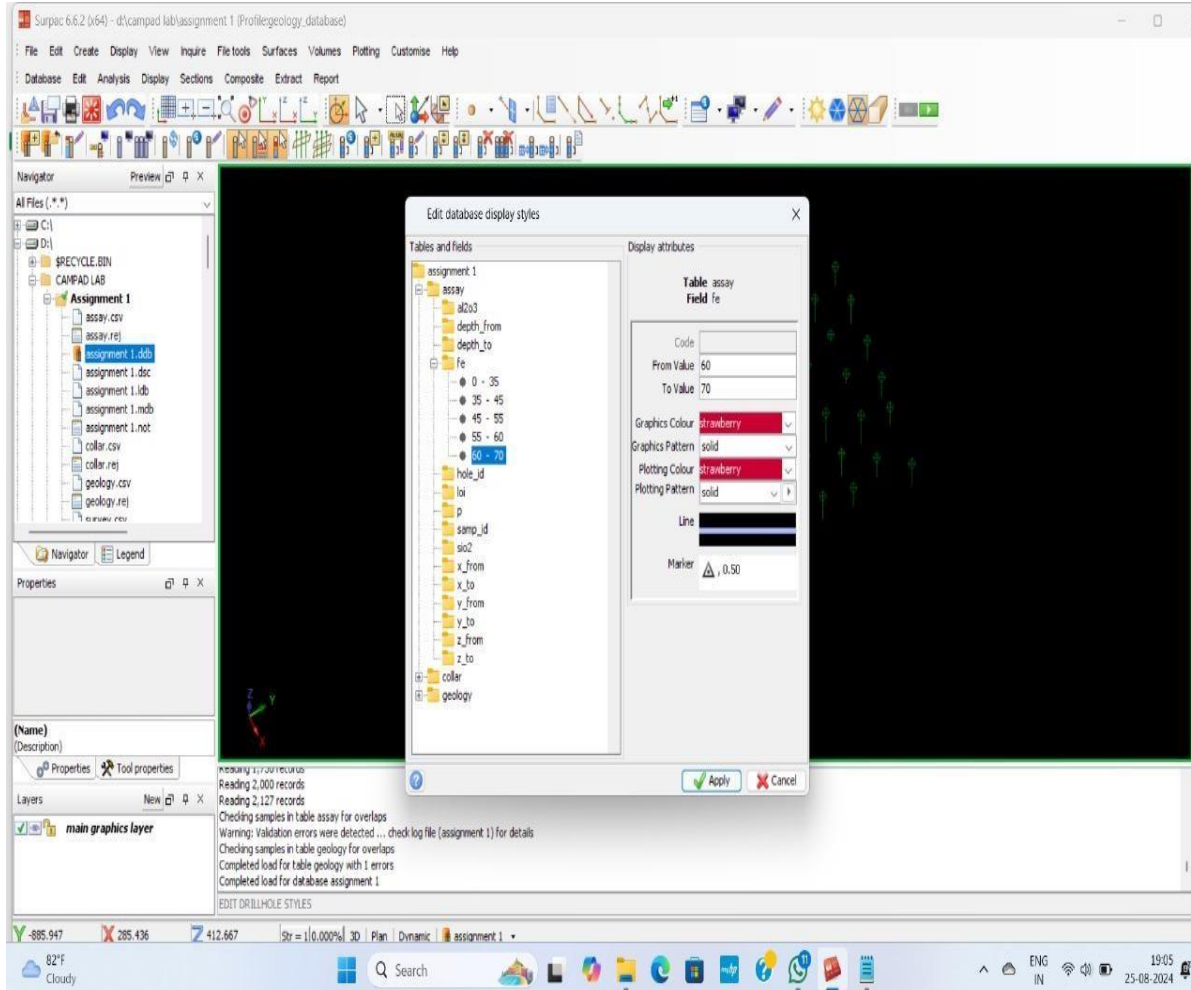
Step 23:- Click on assay.



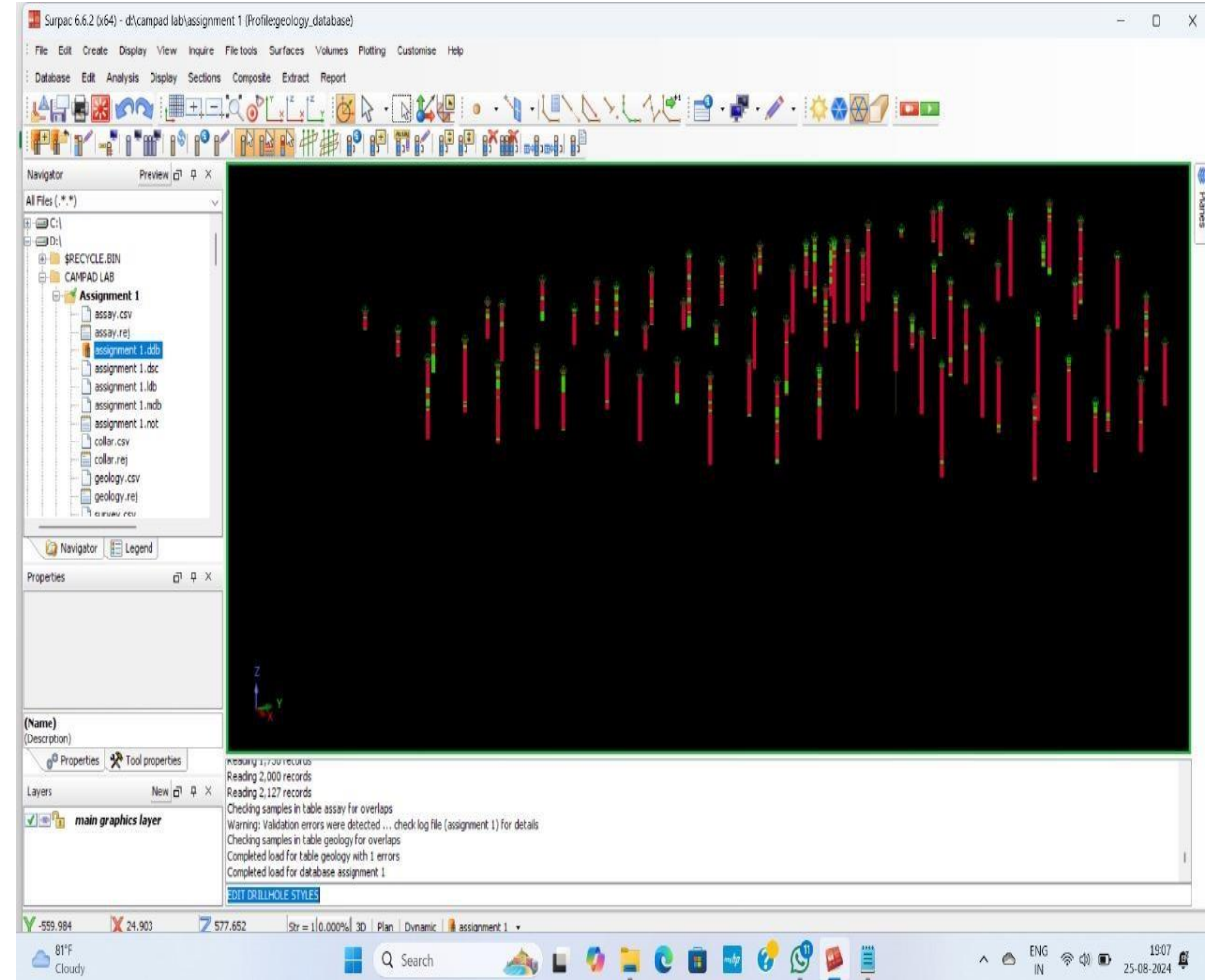
Step 24:- Go to fe>>right click on fe and add new styles.



Step 25:- Mention the ranges for different grades of fe. Here we choose 0-35,35-45,45-55,55-60,60-70 in ascending order of grades

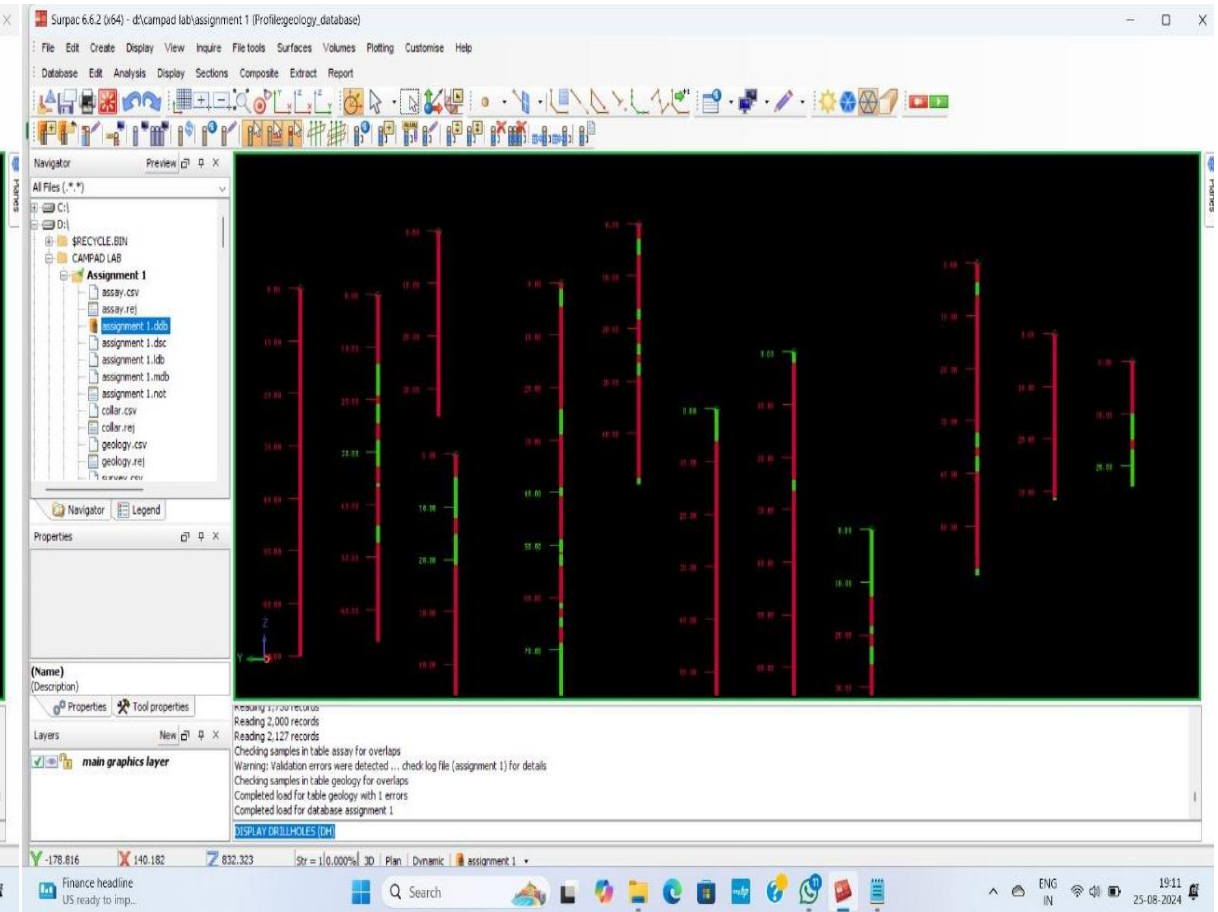
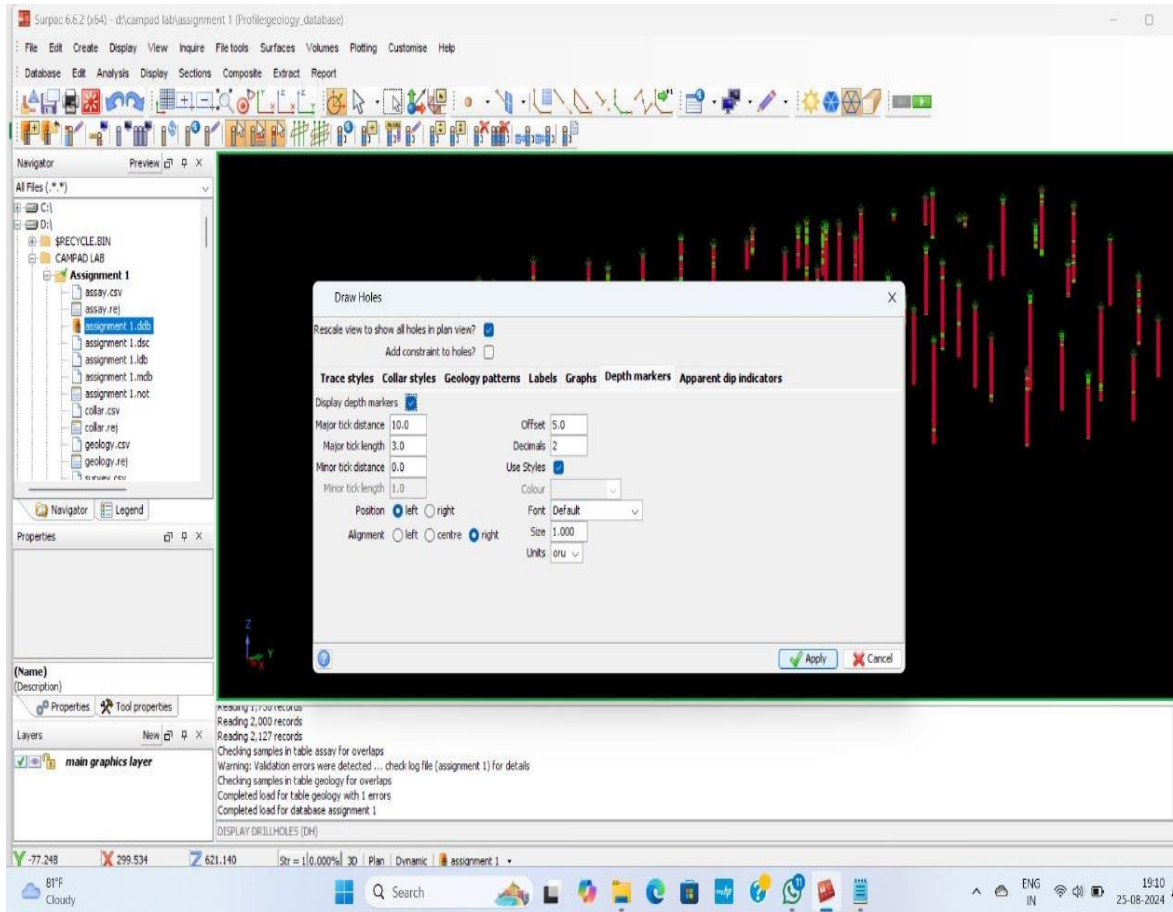


Step 26:- The drilling section displays the various grades of fe as distinct colors.



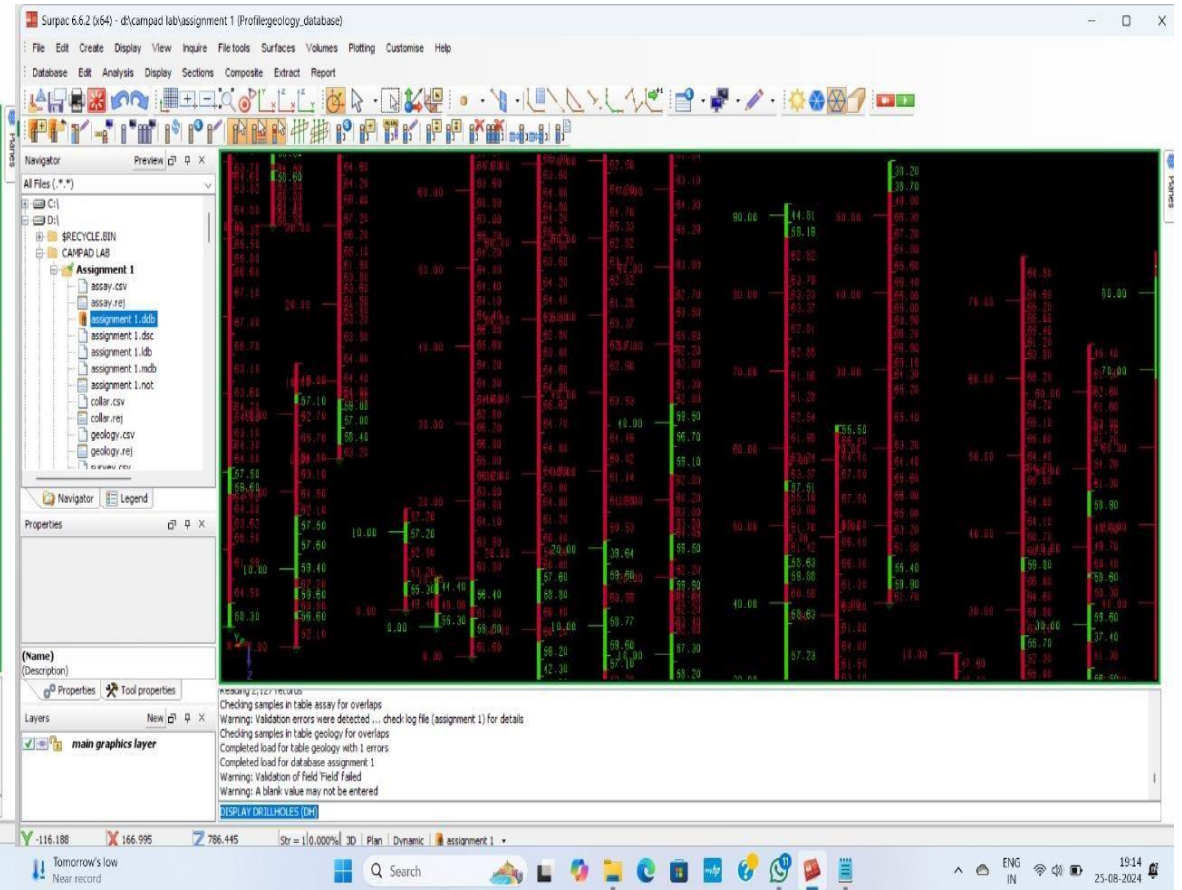
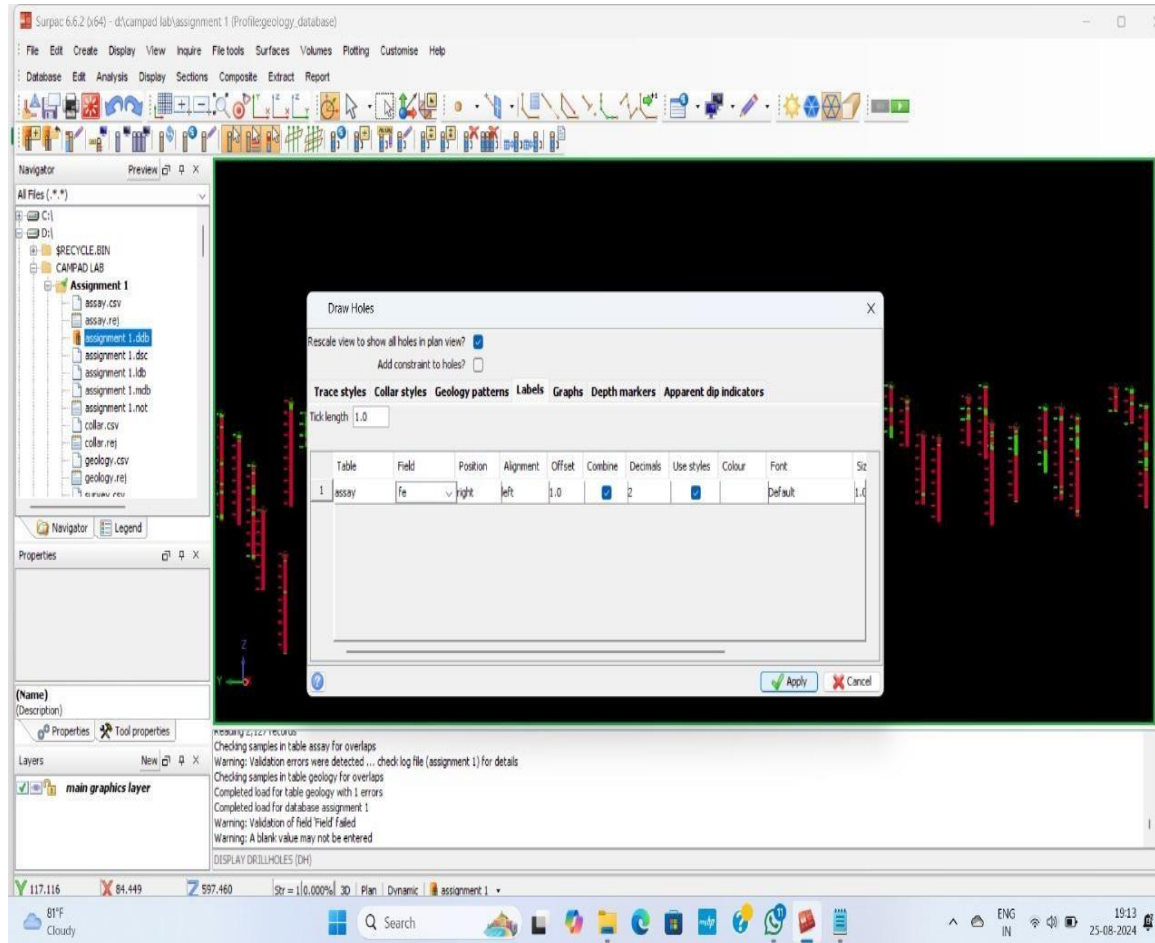
Step 27:- And then again go to display drill holes>>go to depth markers and select the interval of the depth markers as we want.

Step 28:- The various ore grades are indicated based on their depth of presence.



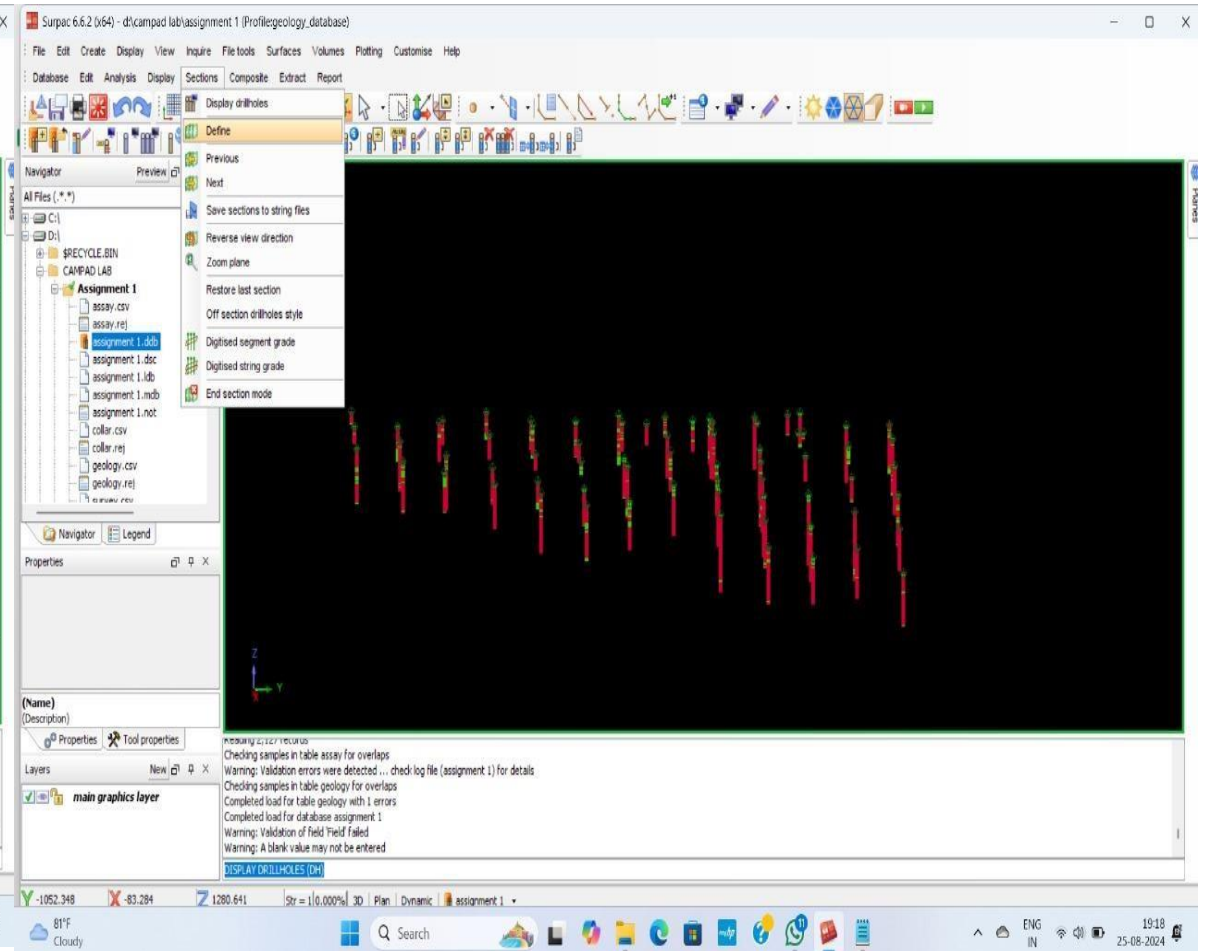
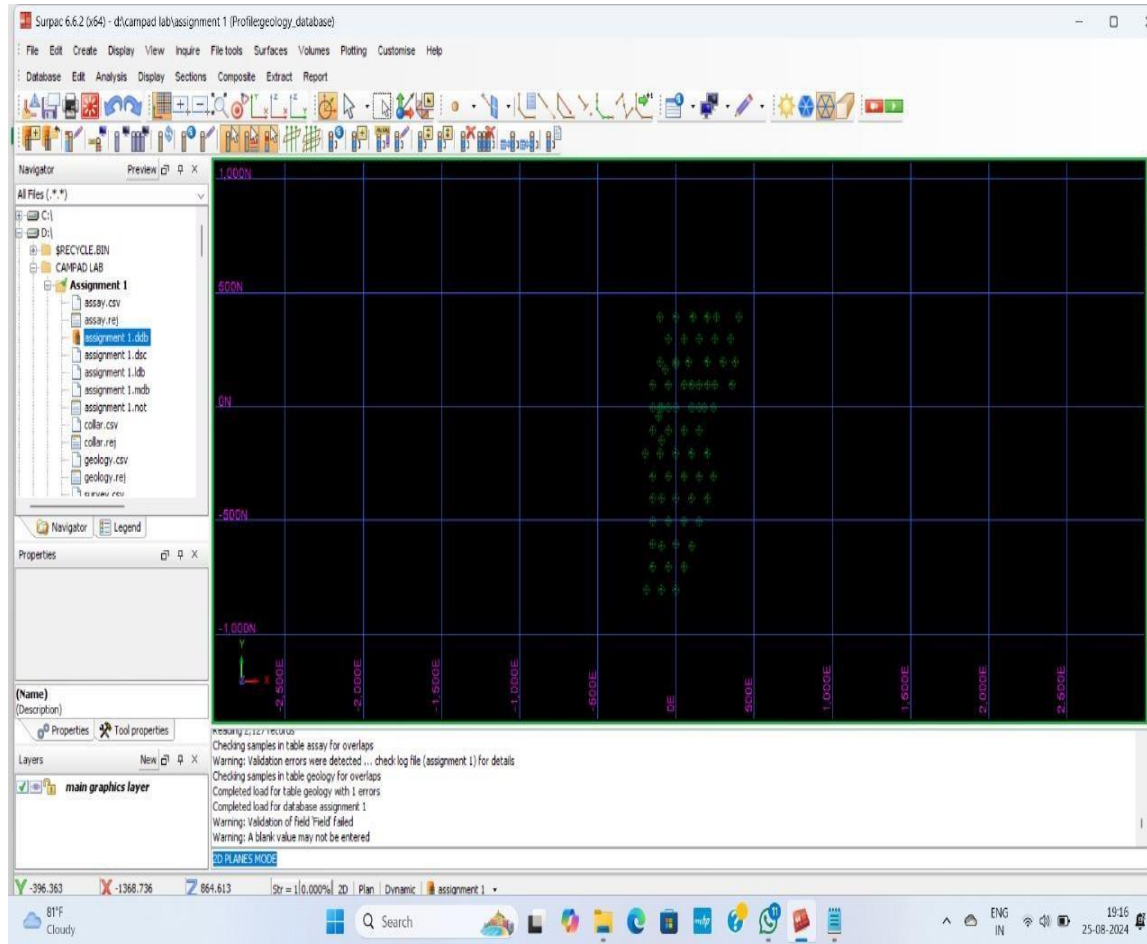
Step 29:- And then again go to display drill holes>>go to labels and select assay>>select fe.

Step 30:- After applying, all the drill holes will display the fields on the both sides as shown in the below figure.

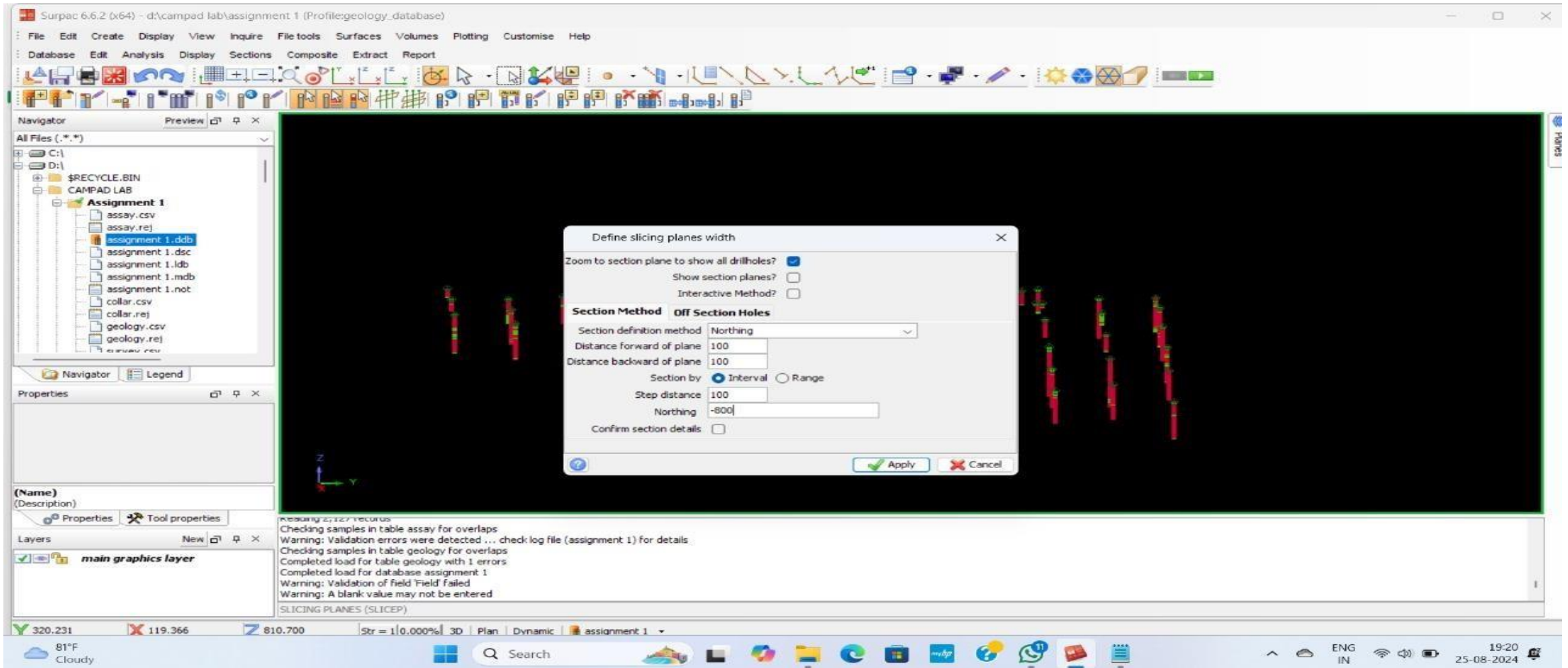


Step 31:- click on the grid.

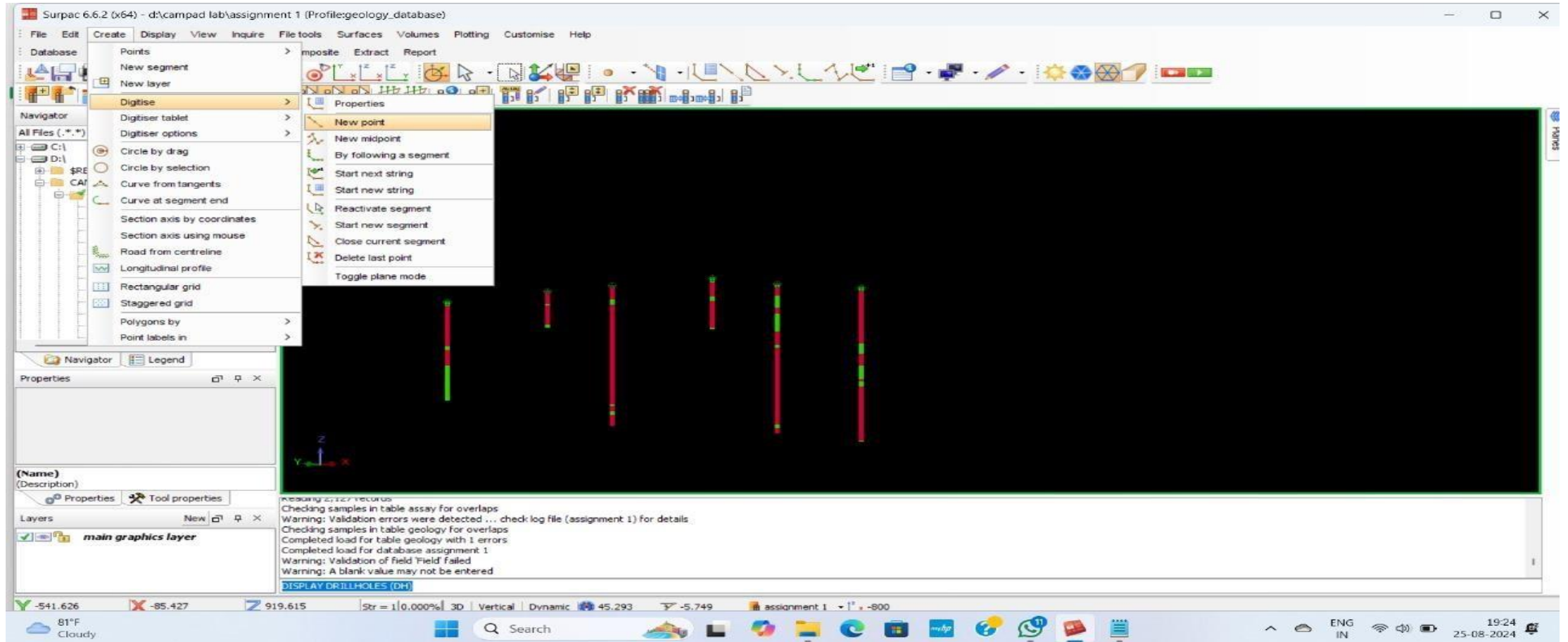
Step 32:- Go to section>>click on define.



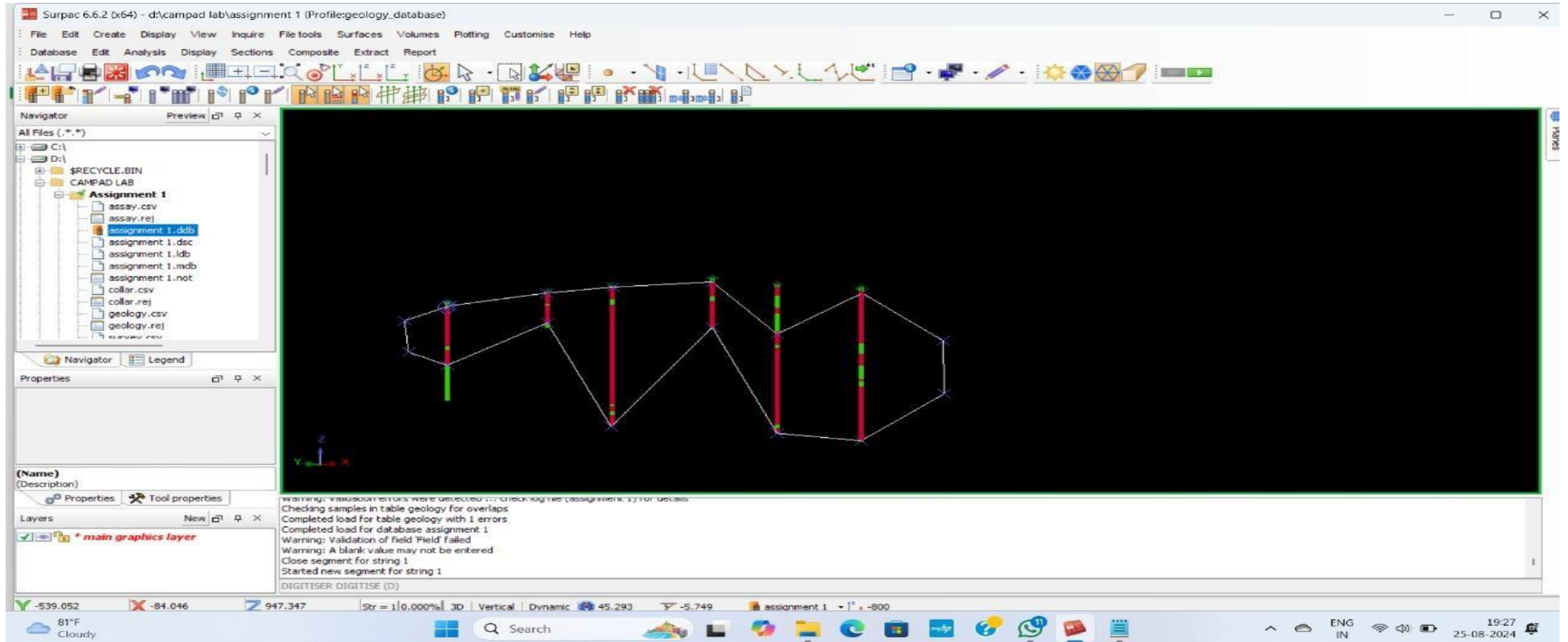
Step 33:- Here the starting boreholes are at -800N and the ending boreholes are at 400N. Here we are using the northing definition method. Depending on the distance between the boreholes, choose the step distance, forward, and backward distance in the plane.



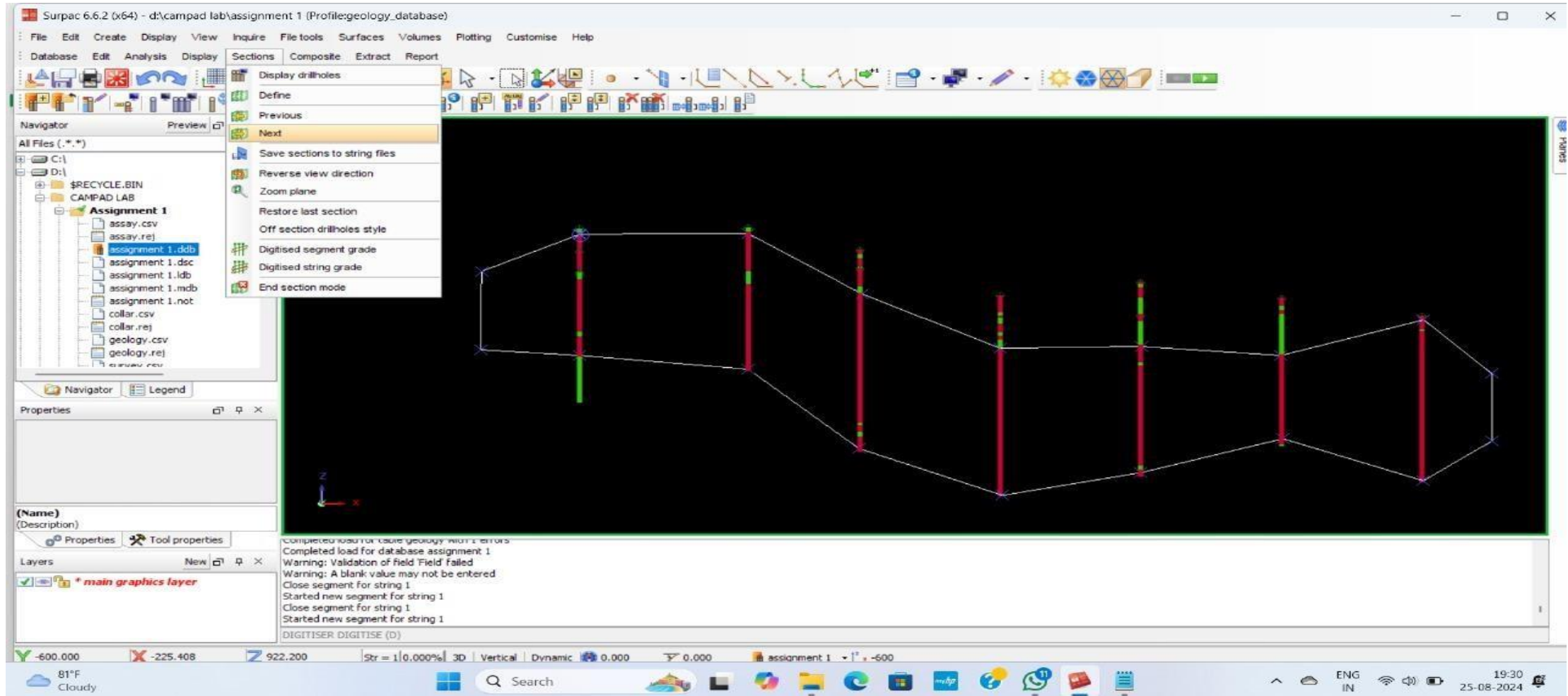
Step 34:- Go to create>>select digitise>> then select new point.



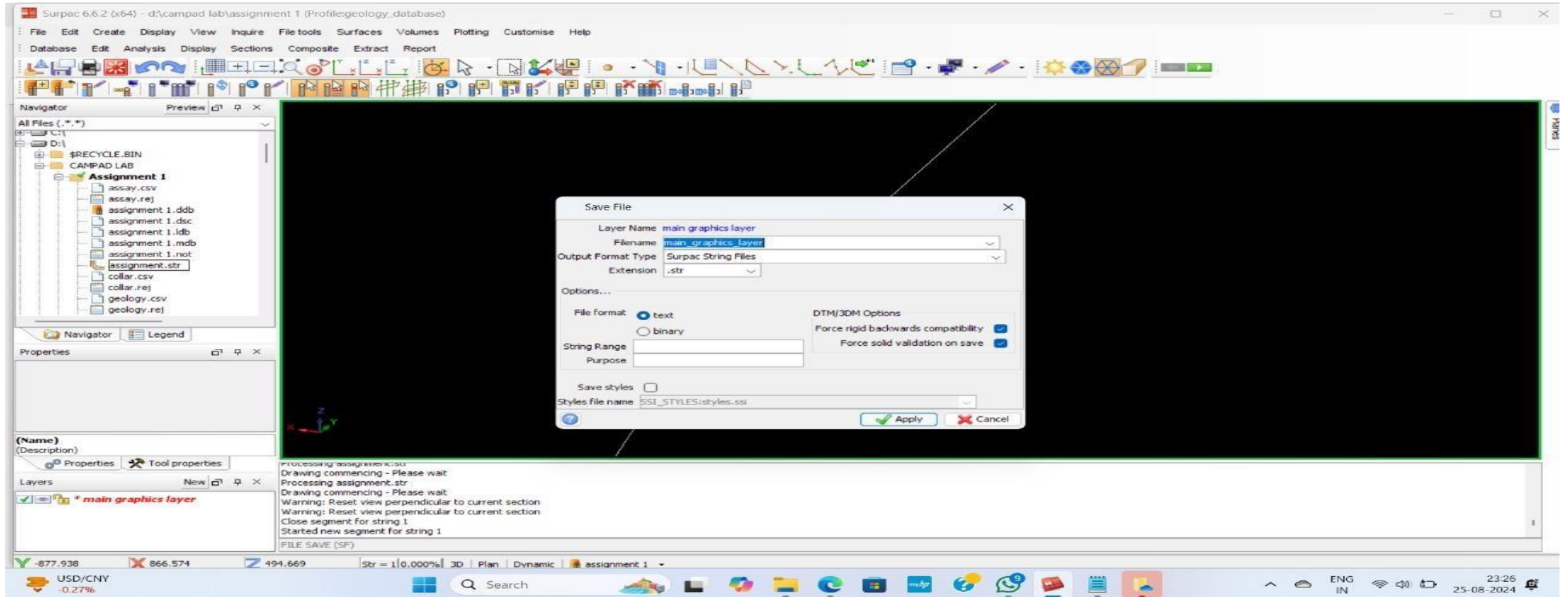
Step 35:- Start joining the lines around the red lines to create a segment of high grade of fe ore, to create a close segment.



Step 36:- Select the sections, then click "Next" to move to the next row of boreholes. Continue to digitise the remaining borehole rows in the same way



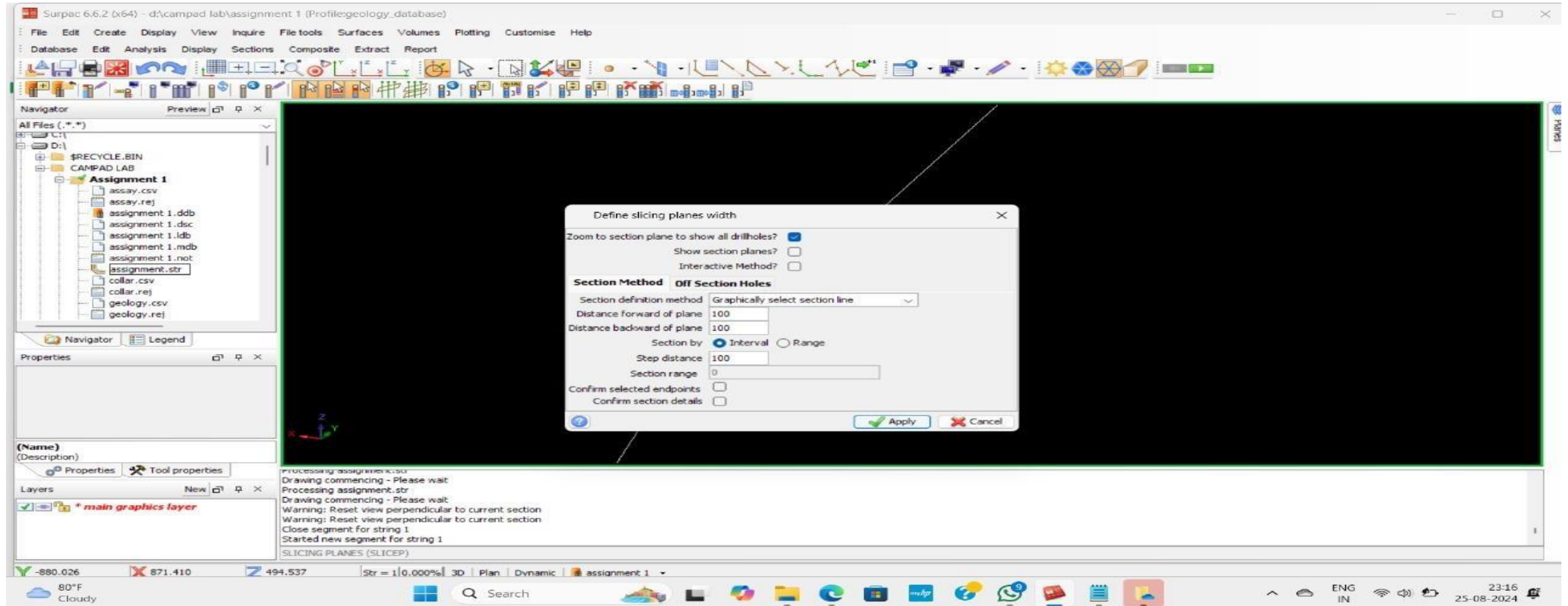
Step 37:- Save the segment as a string file.



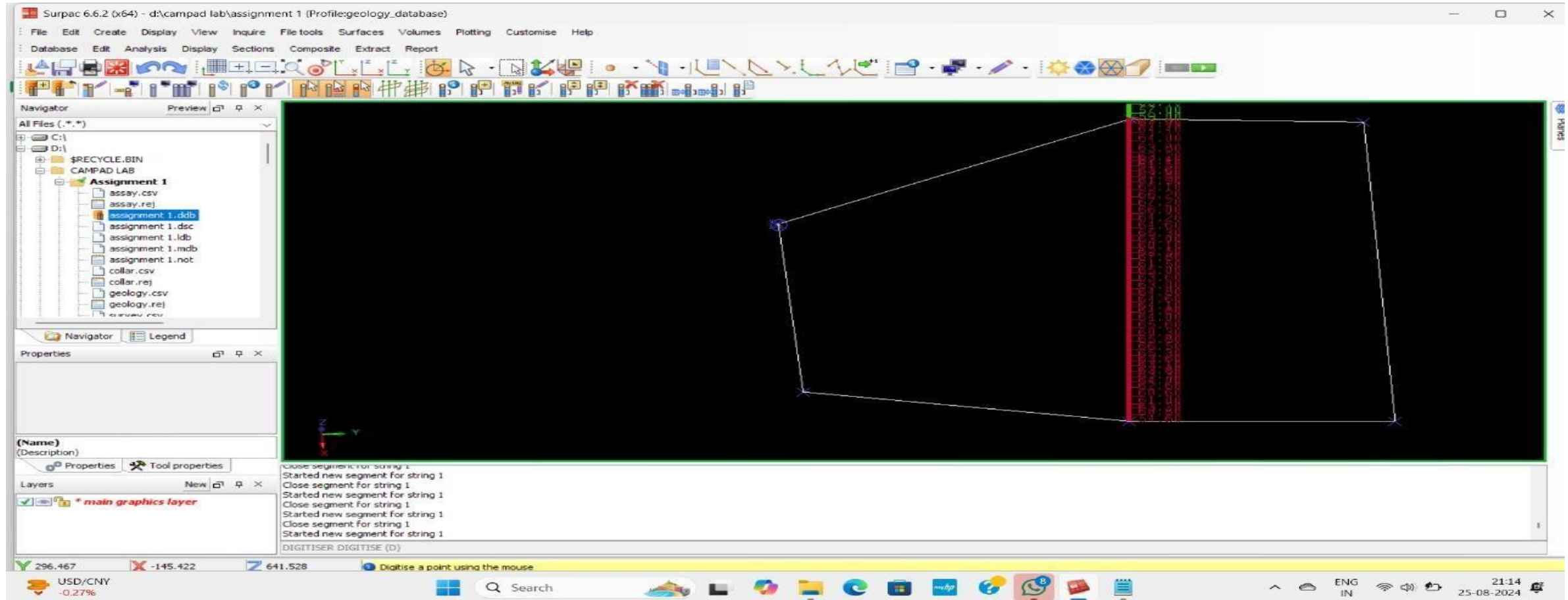
Step 38:- After nothing, this will generate a close segment.



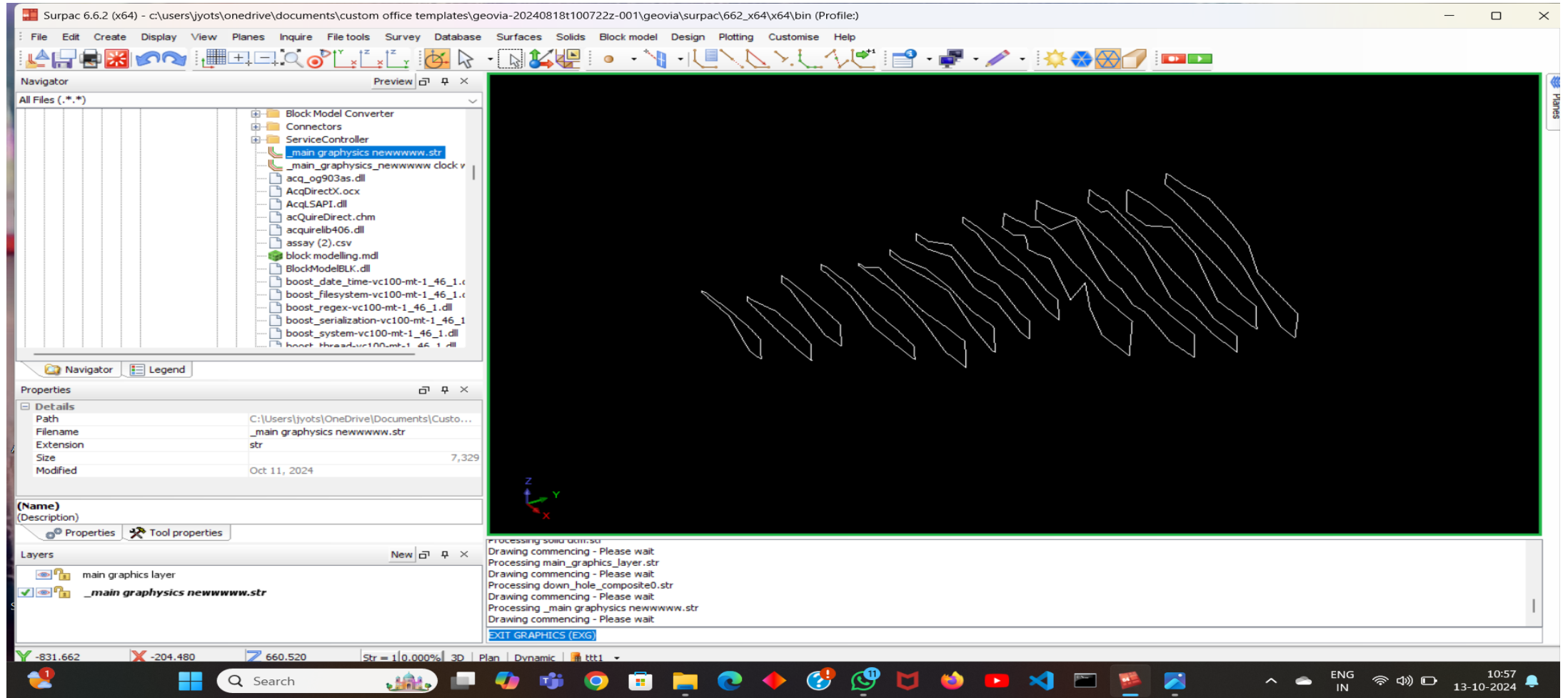
Step 39:- For boreholes that aren't aligned to the north or are located between other boreholes, we need to use the graphical section line method to define them.



Step 40:- left click and drag the line through the borehole/boreholes.

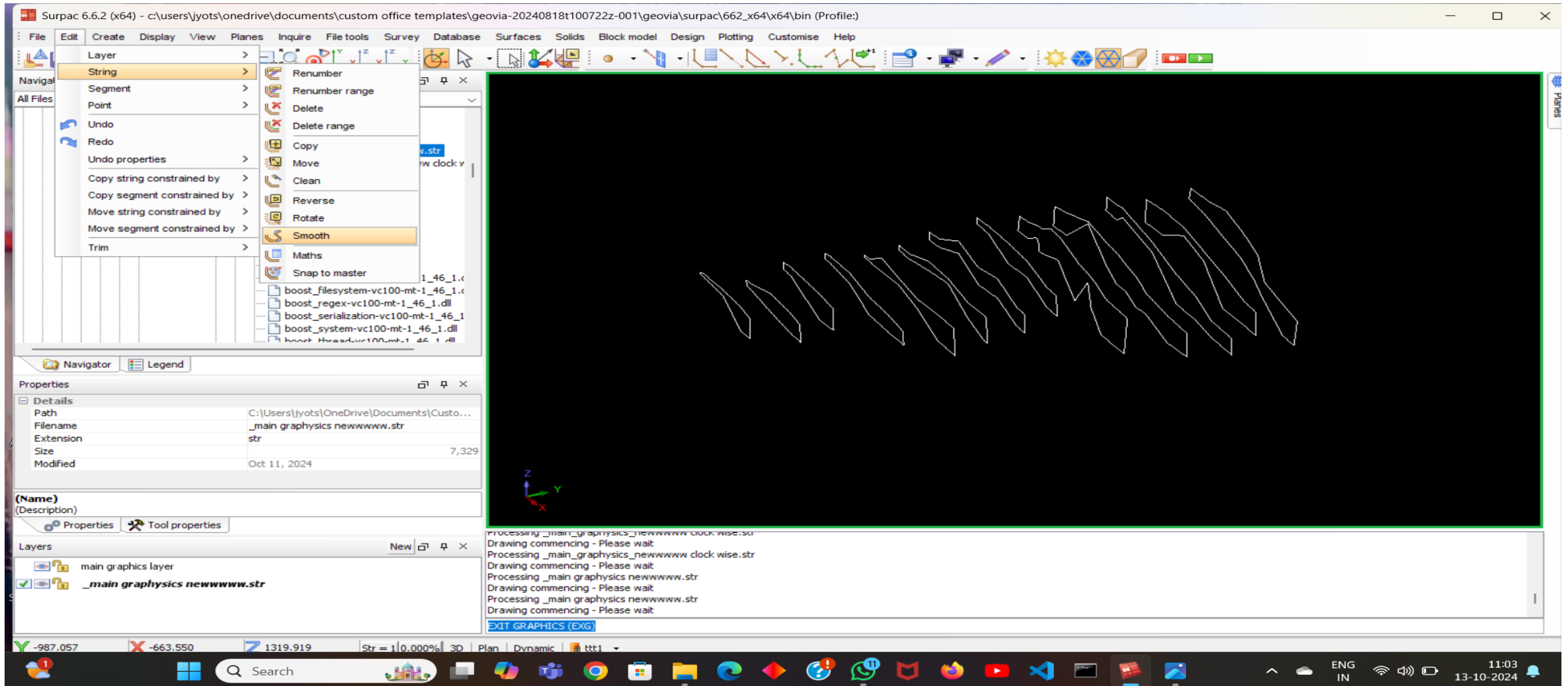


Step 41:- Choose the string files with the segments and drag and drop them to display on the screen.

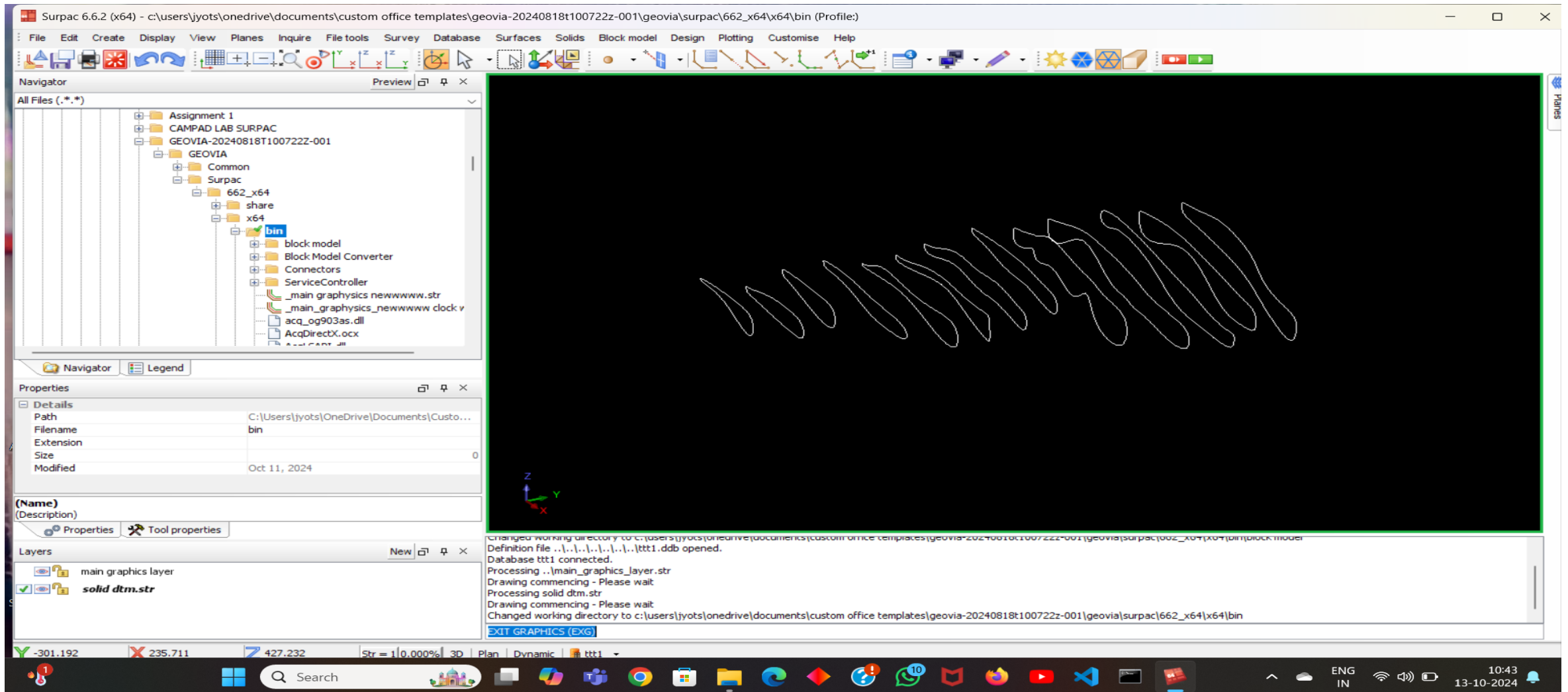


Before Smoothing

Step 42:- Smoothing: ➤ Go to Edit >> String >> Smooth



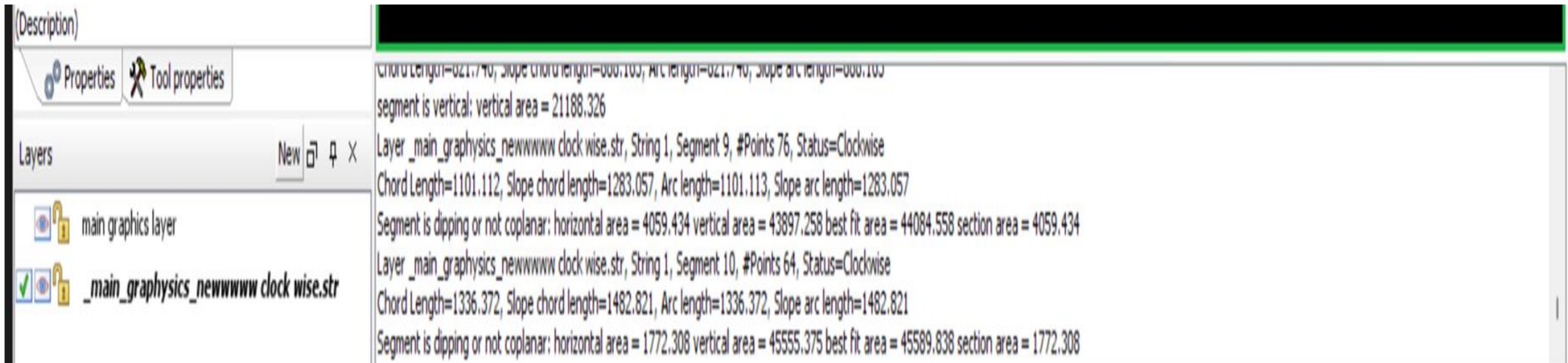
After smoothing sections will look something like the below picture.



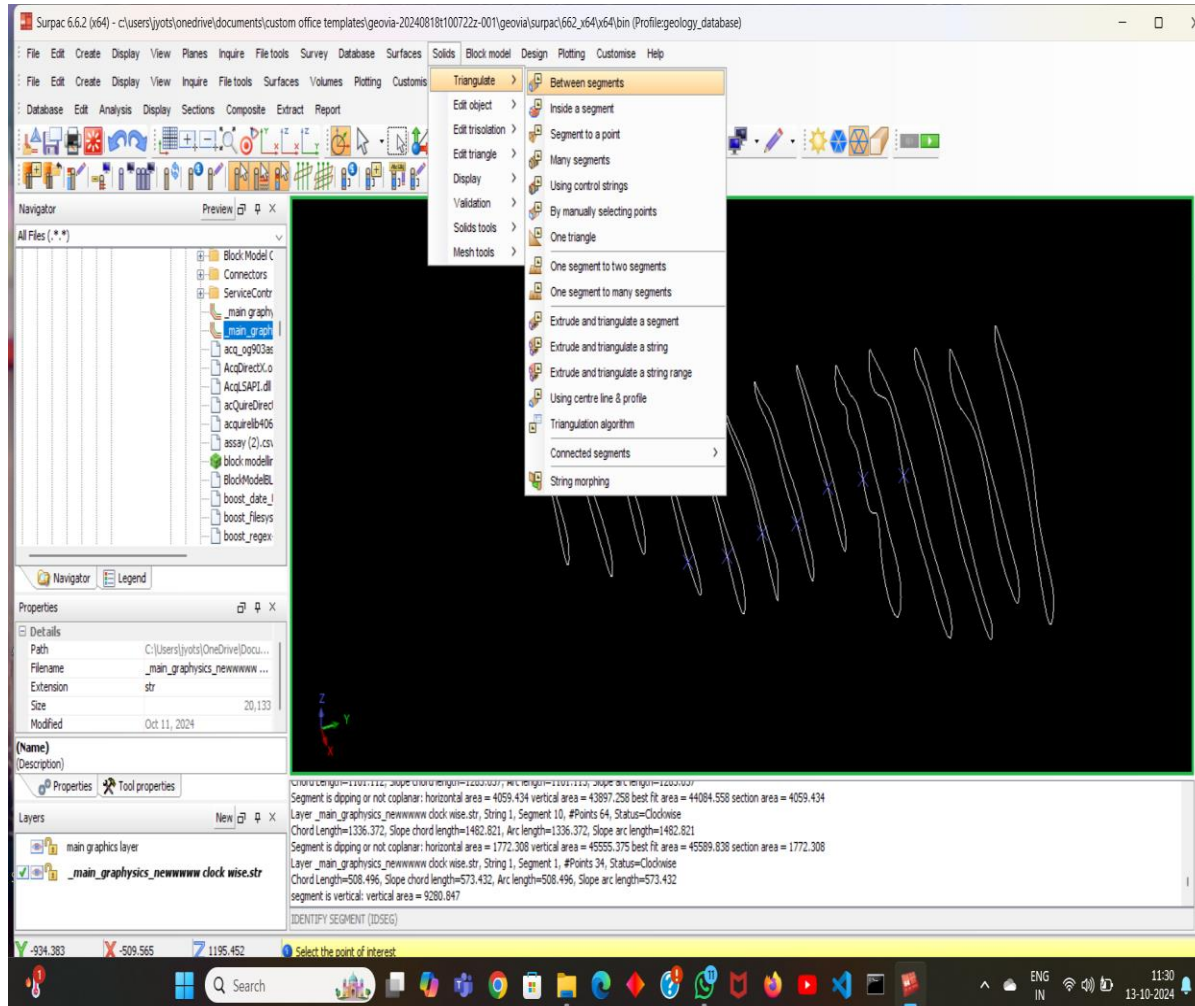
To get a solid all the segments must be in a clockwise direction. If the segment is in anti-clock direction, it will create a void instead of creating a solid.

CHECK orientation of the segment, Go to inquire >> Segment properties >> click on the section.

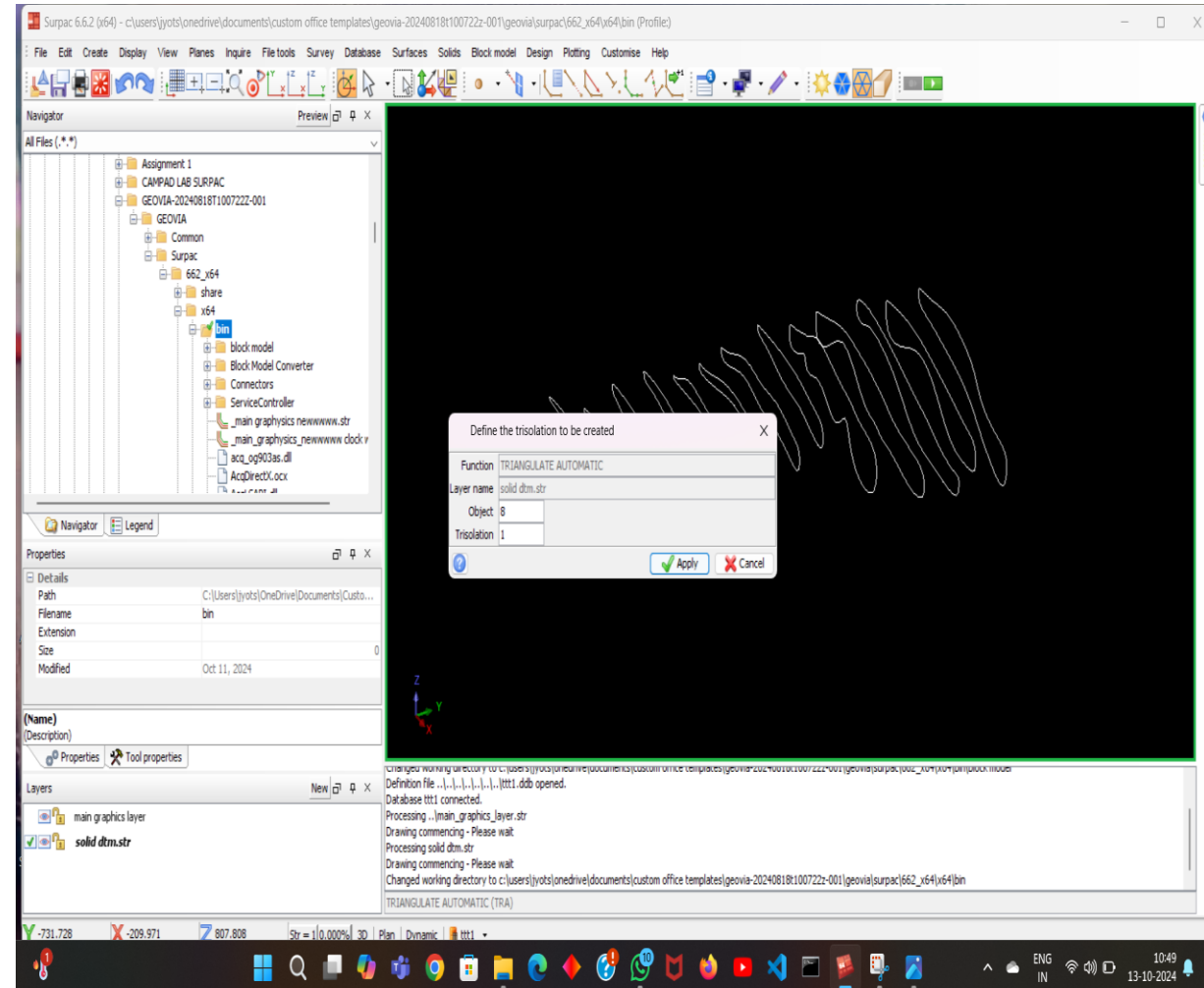
CHANGE the orientation, Go to Edit >> Segment >> Reverse >> Select the segment.



To Create Solid :- Go to Solids >> Triangulate >> Triangulate between segment



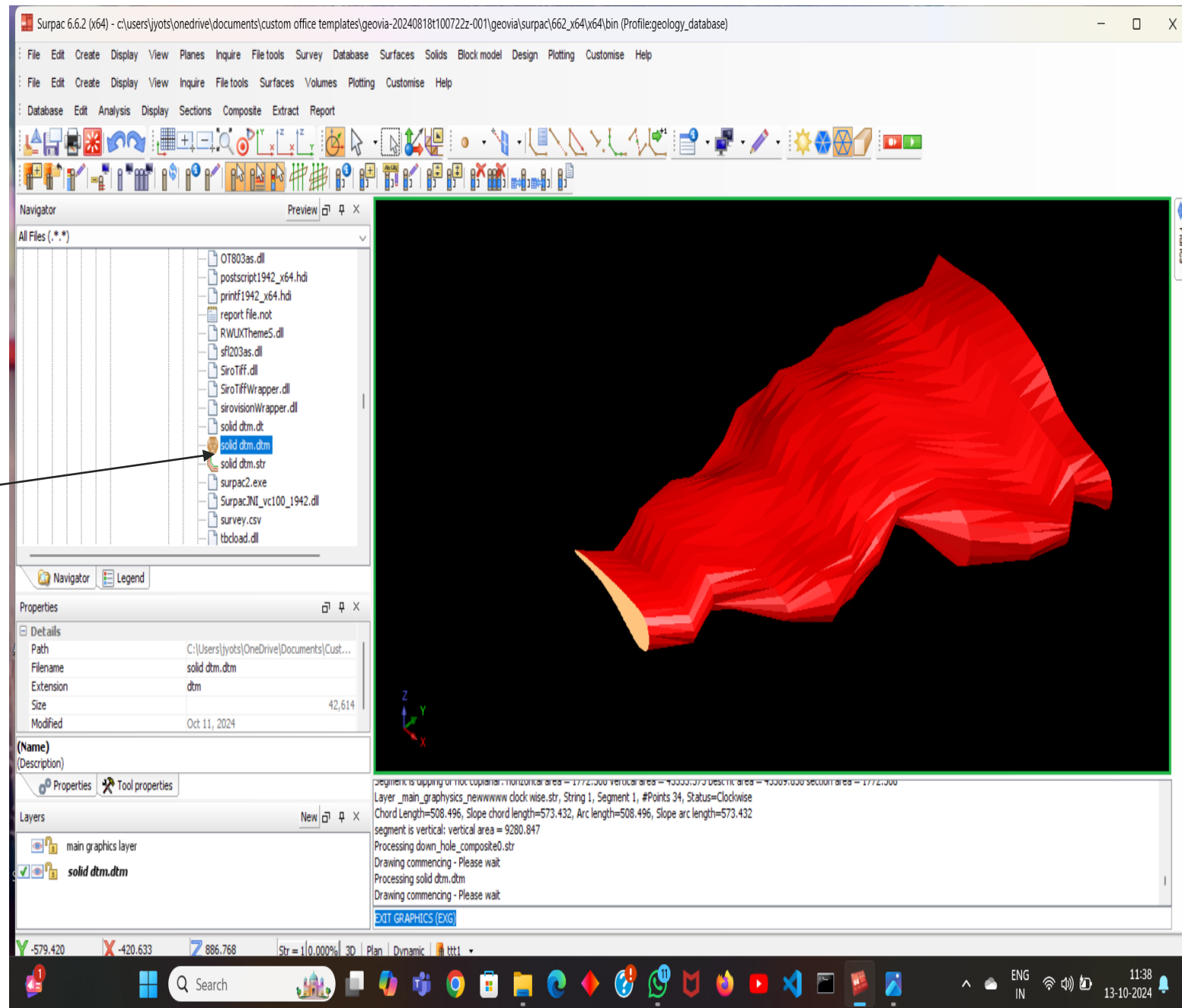
A pop up will Appear to define object ID and Trisolation. Type object ID number as 8 and trisolation as 1.



To close the segments, Go to Solids >> Triangulate >> Triangulate inside segment >> Select the segment.

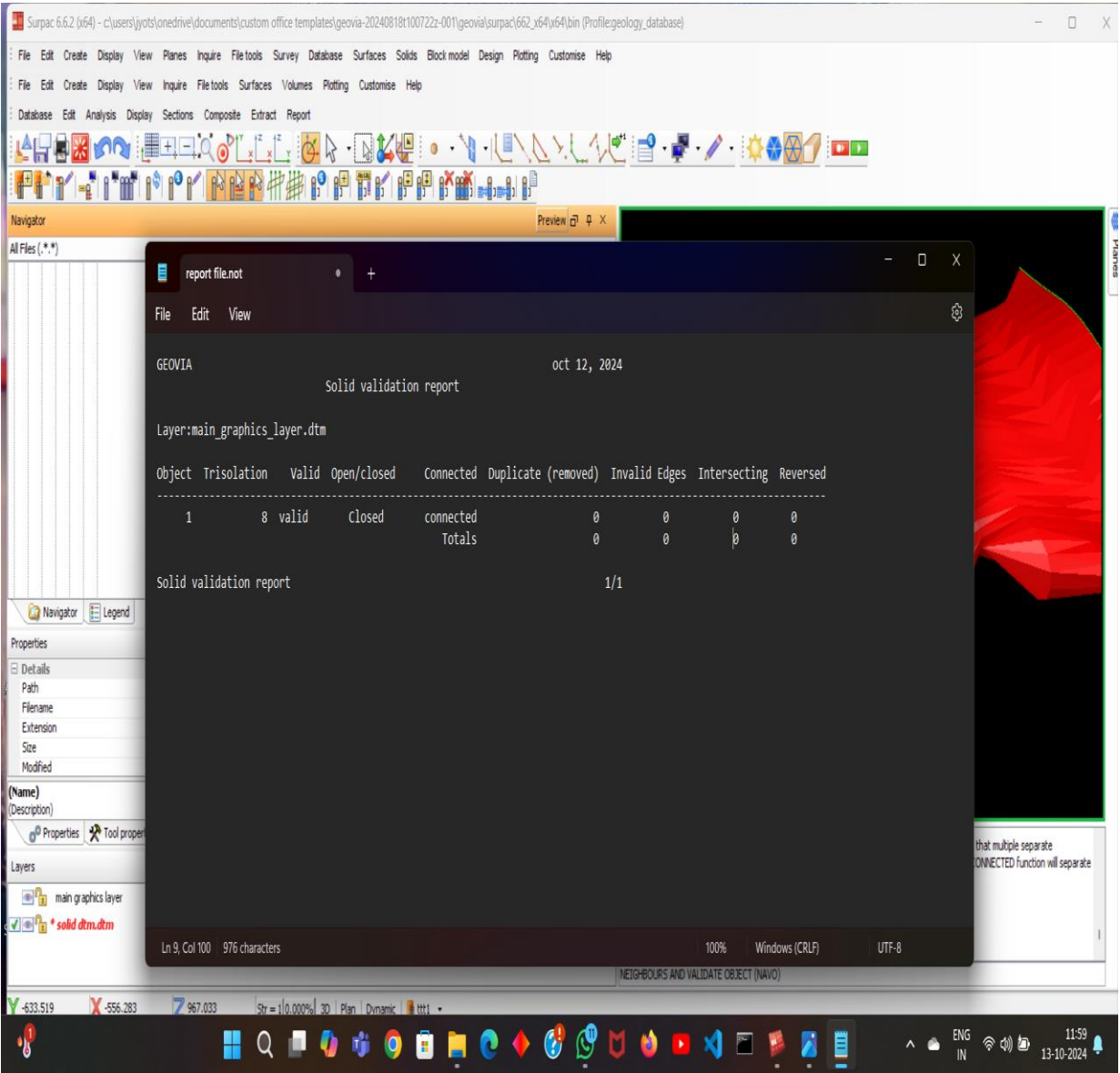
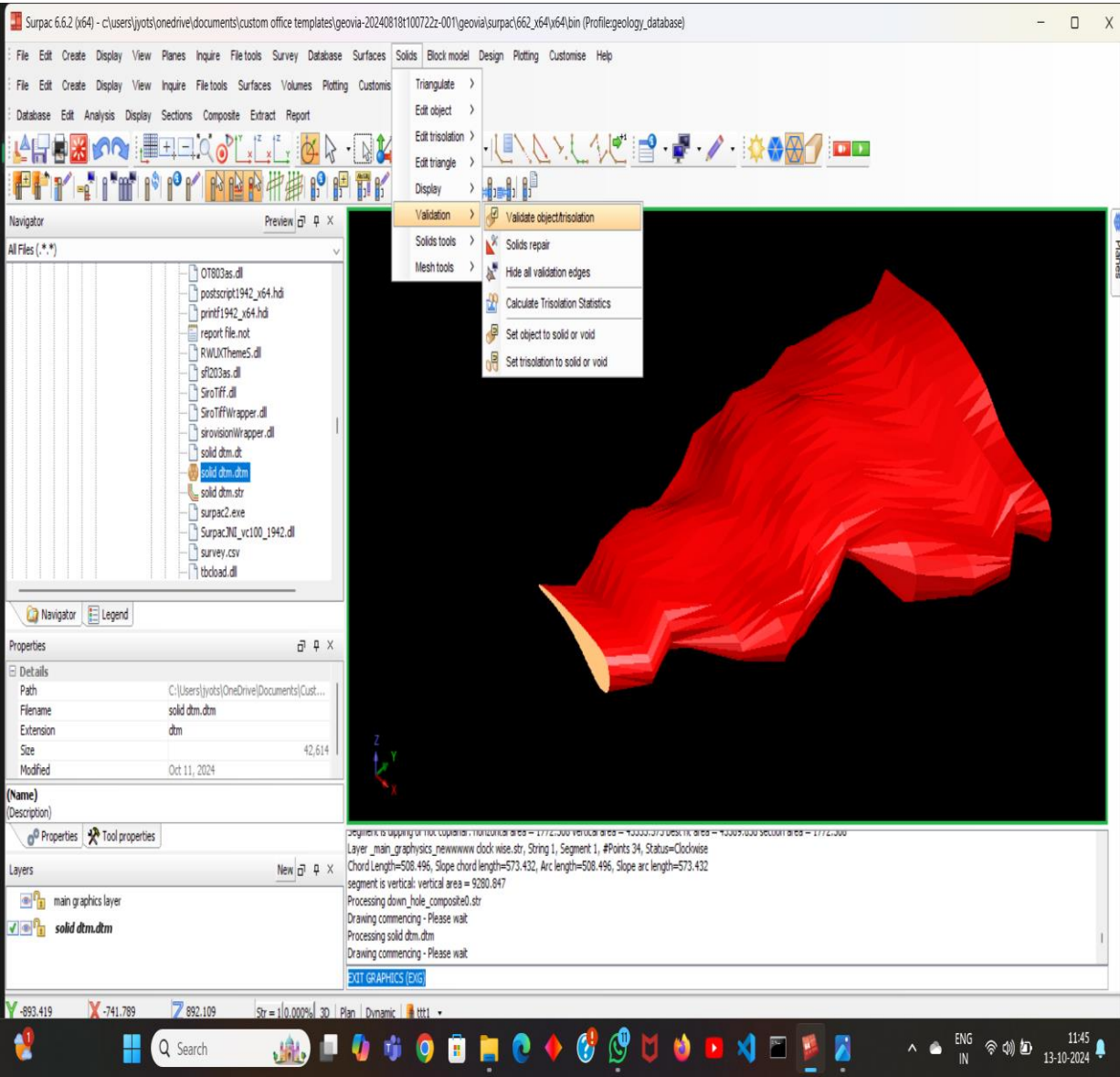
After triangulating all the segments. We have to close the start and end segments to finish the solid dtm.

Save the new layer as solid dtm.

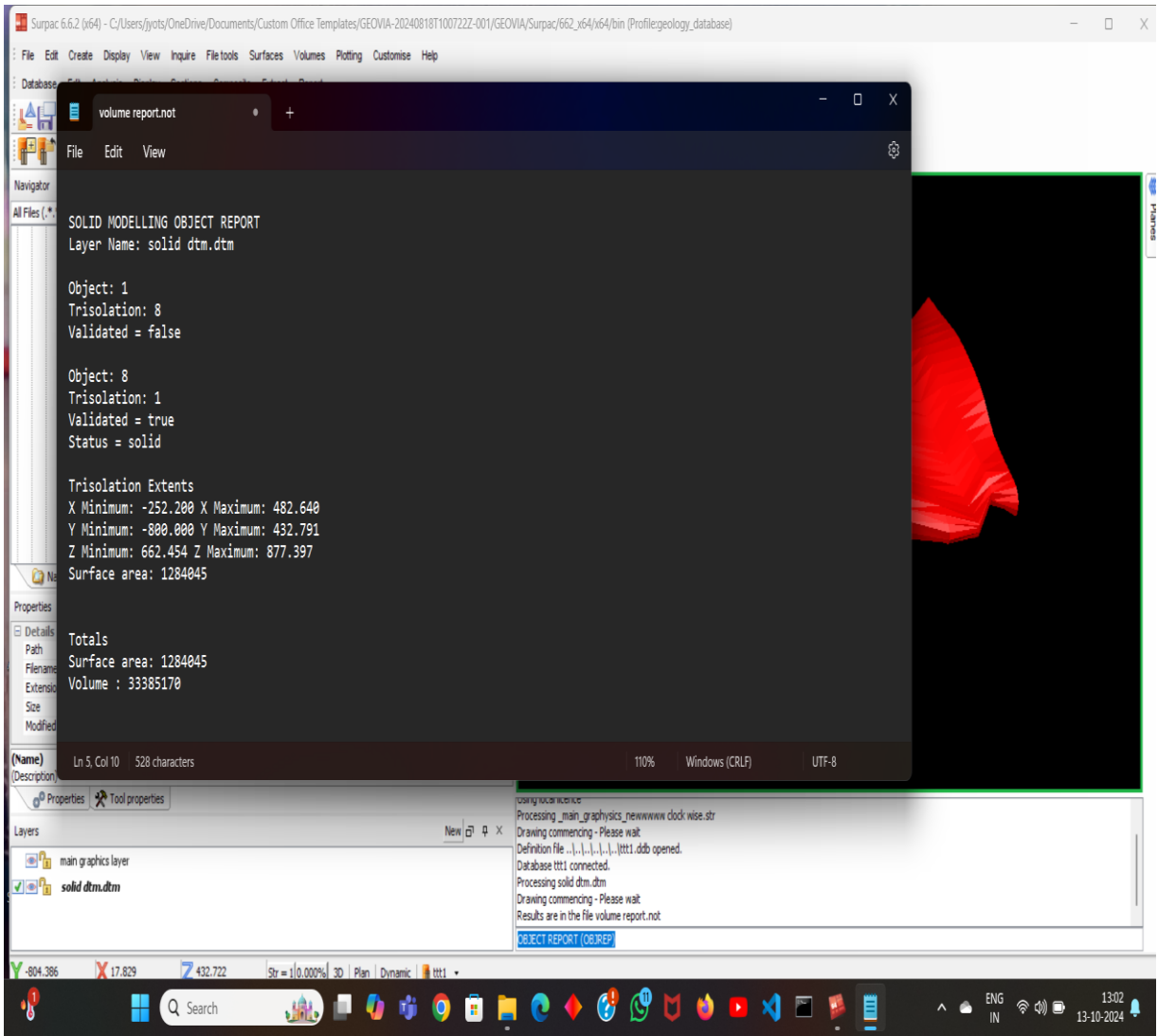
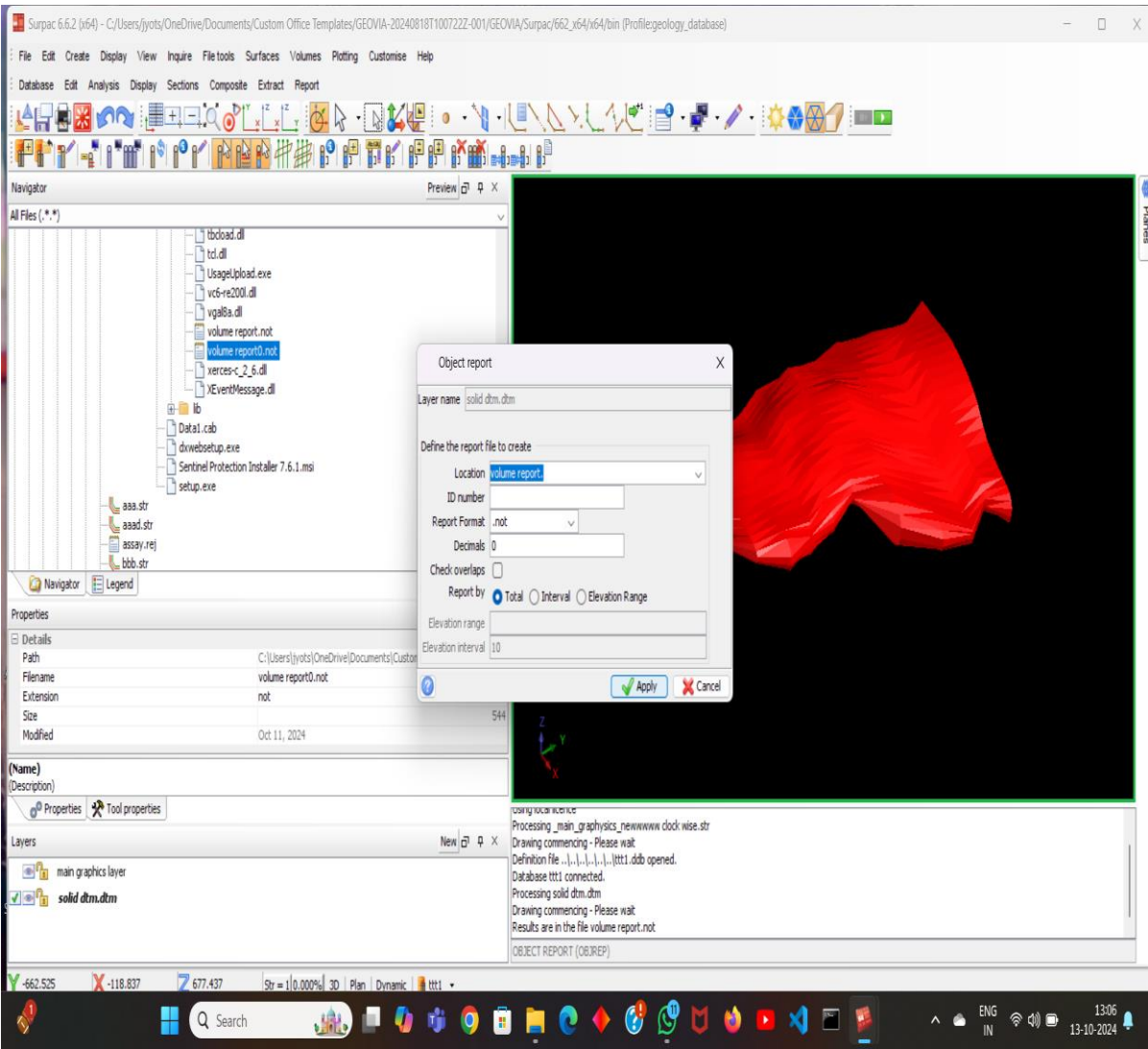


We need to get the solid validation report:- Go to Solids >> validation >> Validate object/trisolation

A report file in .not format will appear on your screen.

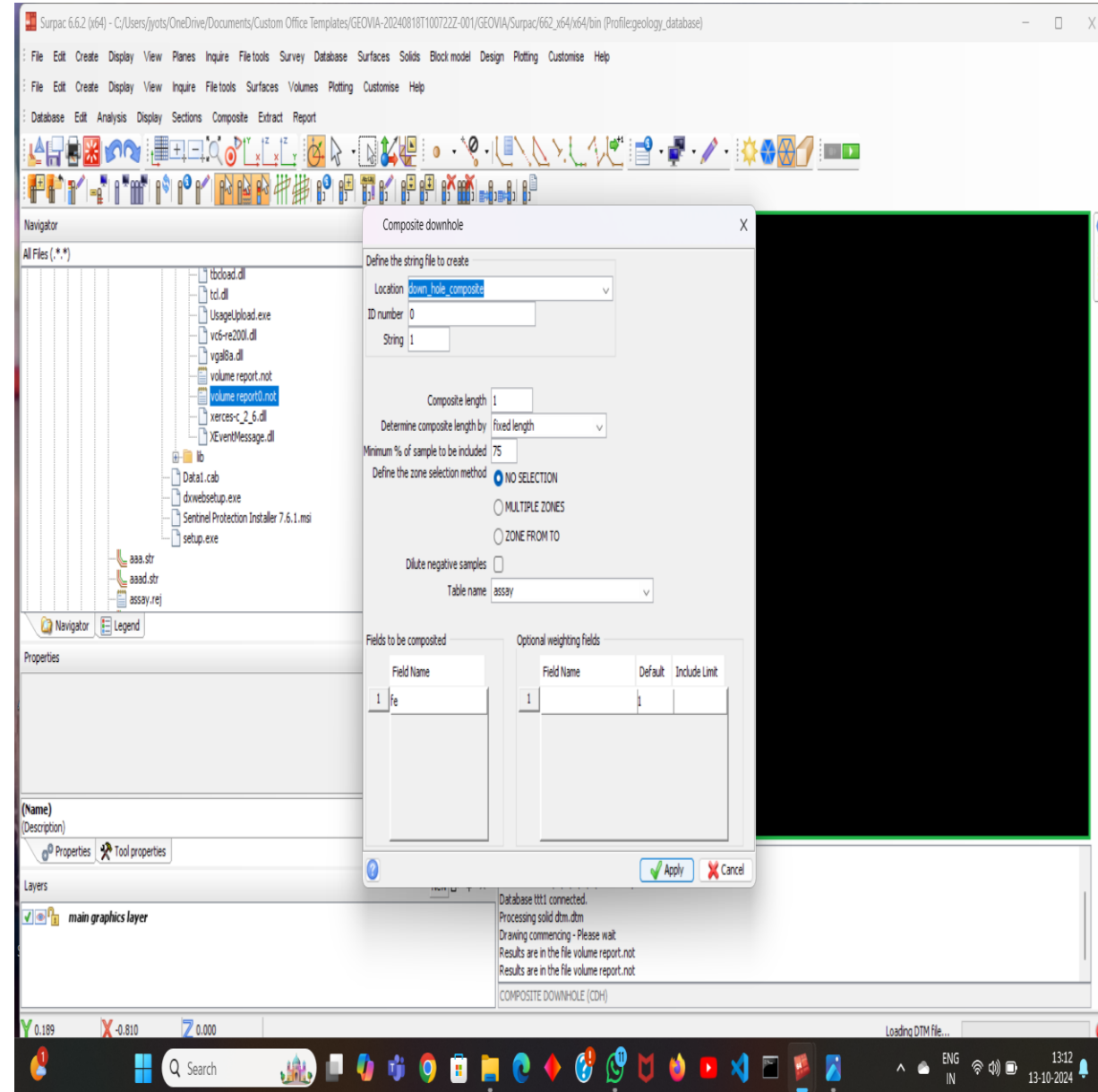
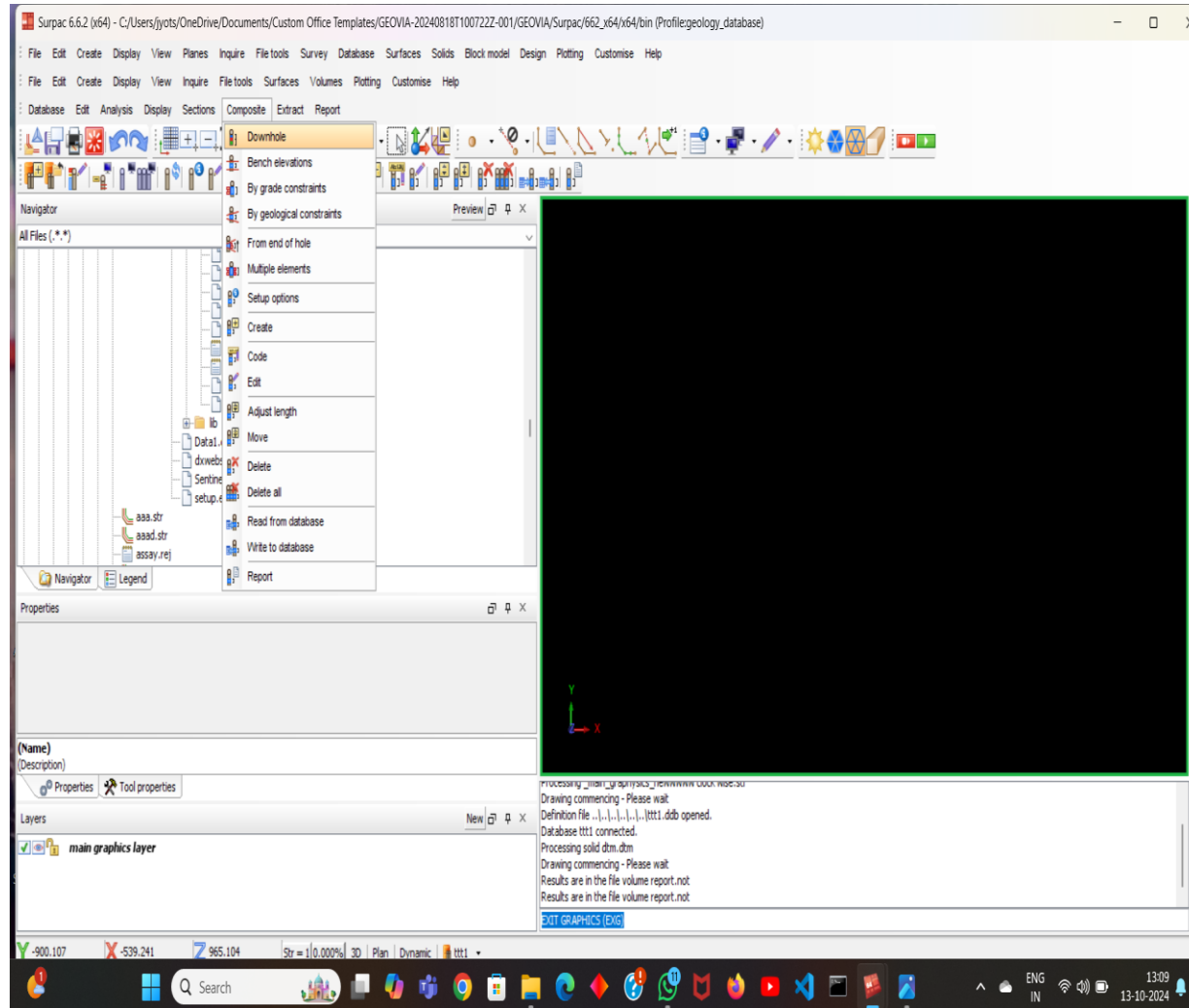


To Calculate Solid Volume Calculation :-
Go to Surfaces >> Volumes >> Report volume of solids >> define the report file name >> Click on Apply



There are different methods of compositing, we implemented downhole Compositing

Go to Composite >> Down hole >> click on Apply



Check in the description box to find out how many raw samples and how many holes were processed. Note the sample data's mean, standard deviation, and variance as well.

One string file is generated and saved

about the load problem with this file can be obtained by turning on the plugins log file in Plugin Preferences, and attempting to reload the file.

Attempting to load file using external application...

File successfully loaded using external application

Results are in the file volume report0.not

Raw sample length statistics: Number of samples = 2126, Total = 4822.26, Minimum = 0.08, Maximum = 12.43, Mean = 2.27, Standard deviation = 1.034, Variance = 1.068

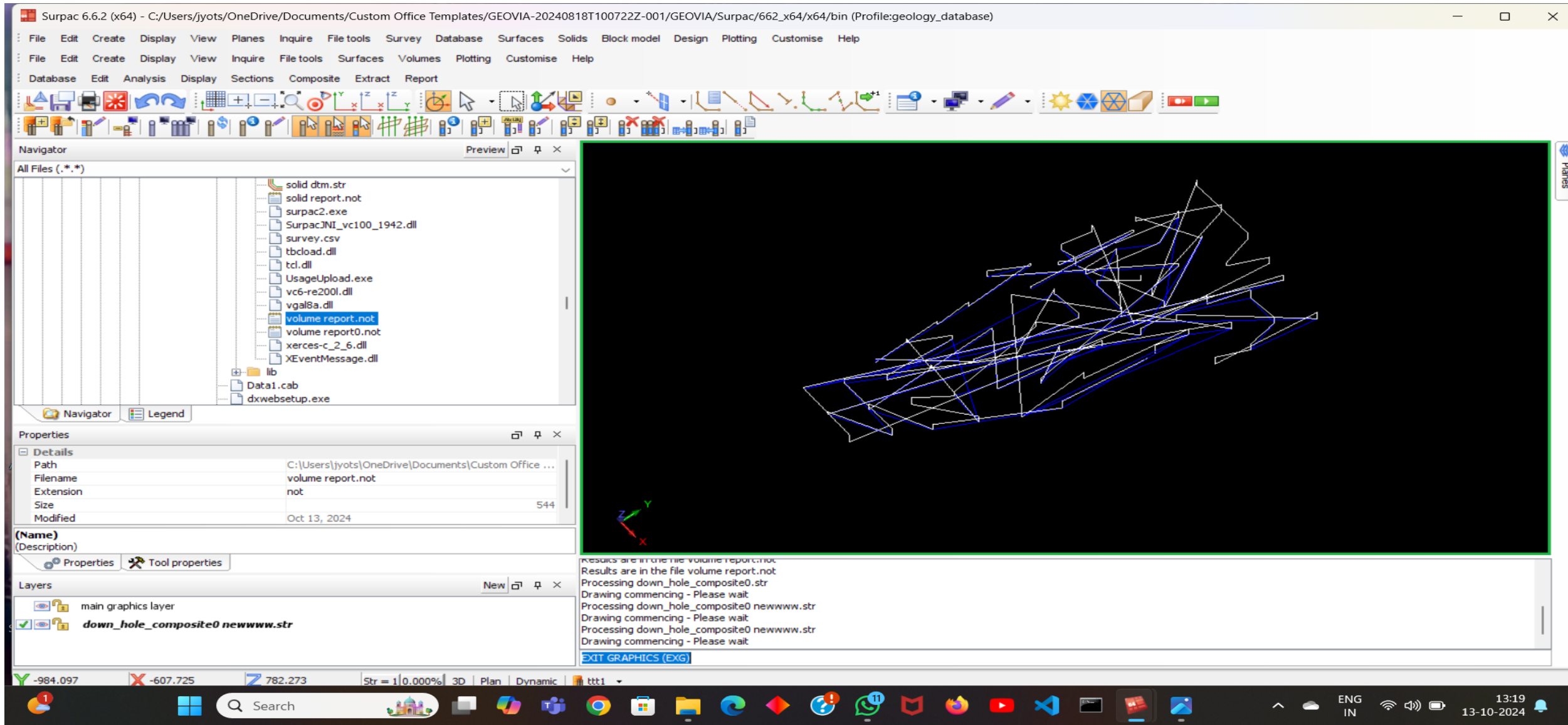
Composite sample length statistics: String 1, Number of samples = 4806, Total = 4802.90, Minimum = 0.75, Maximum = 1.00, Mean = 1.00, Standard deviation = 0.010, Variance = 0.000

Short composite sample length statistics: String 2, Number of samples = 49, Total = 19.36, Minimum = 0.02, Maximum = 0.70, Mean = 0.40, Standard deviation = 0.192, Variance = 0.037

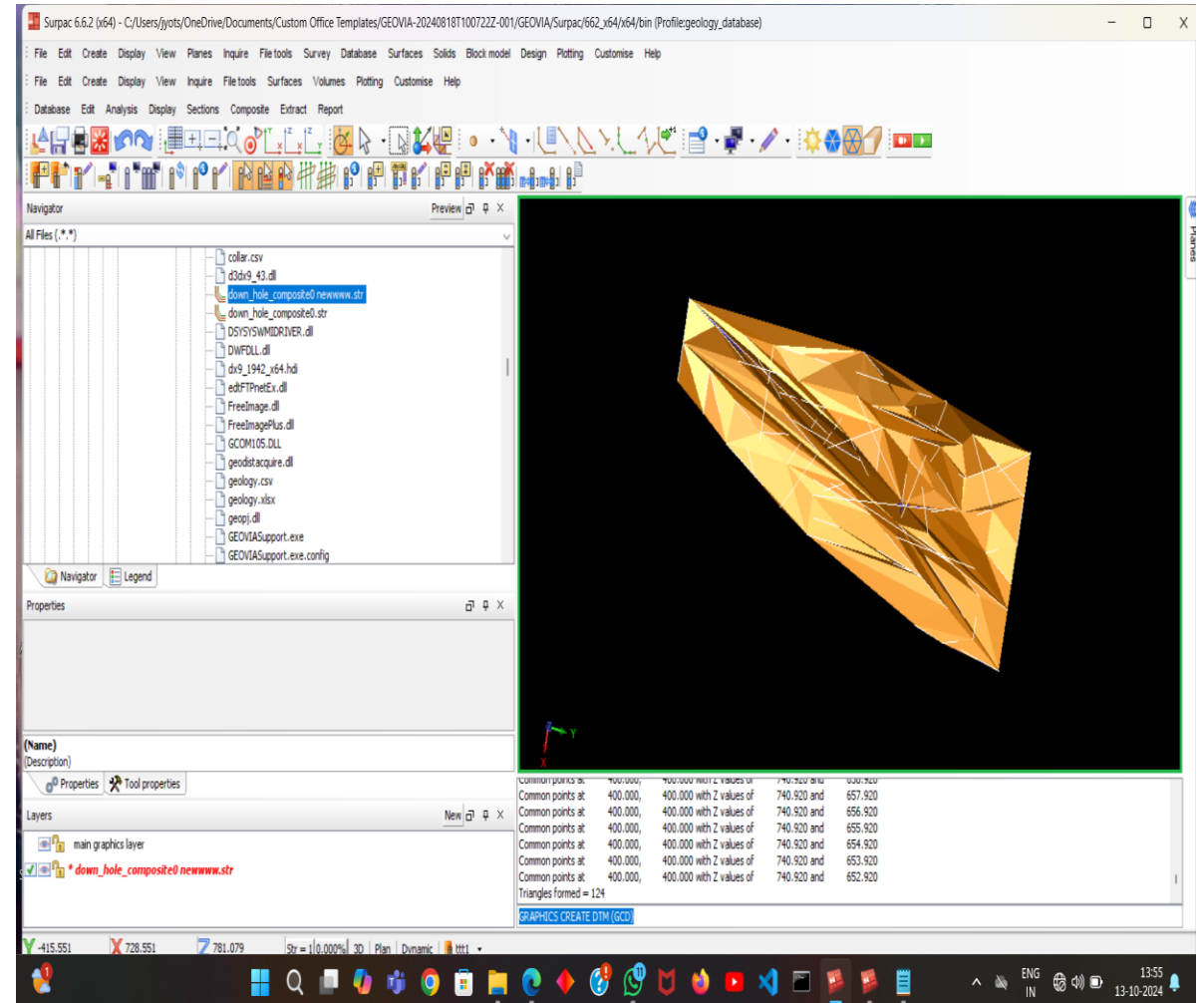
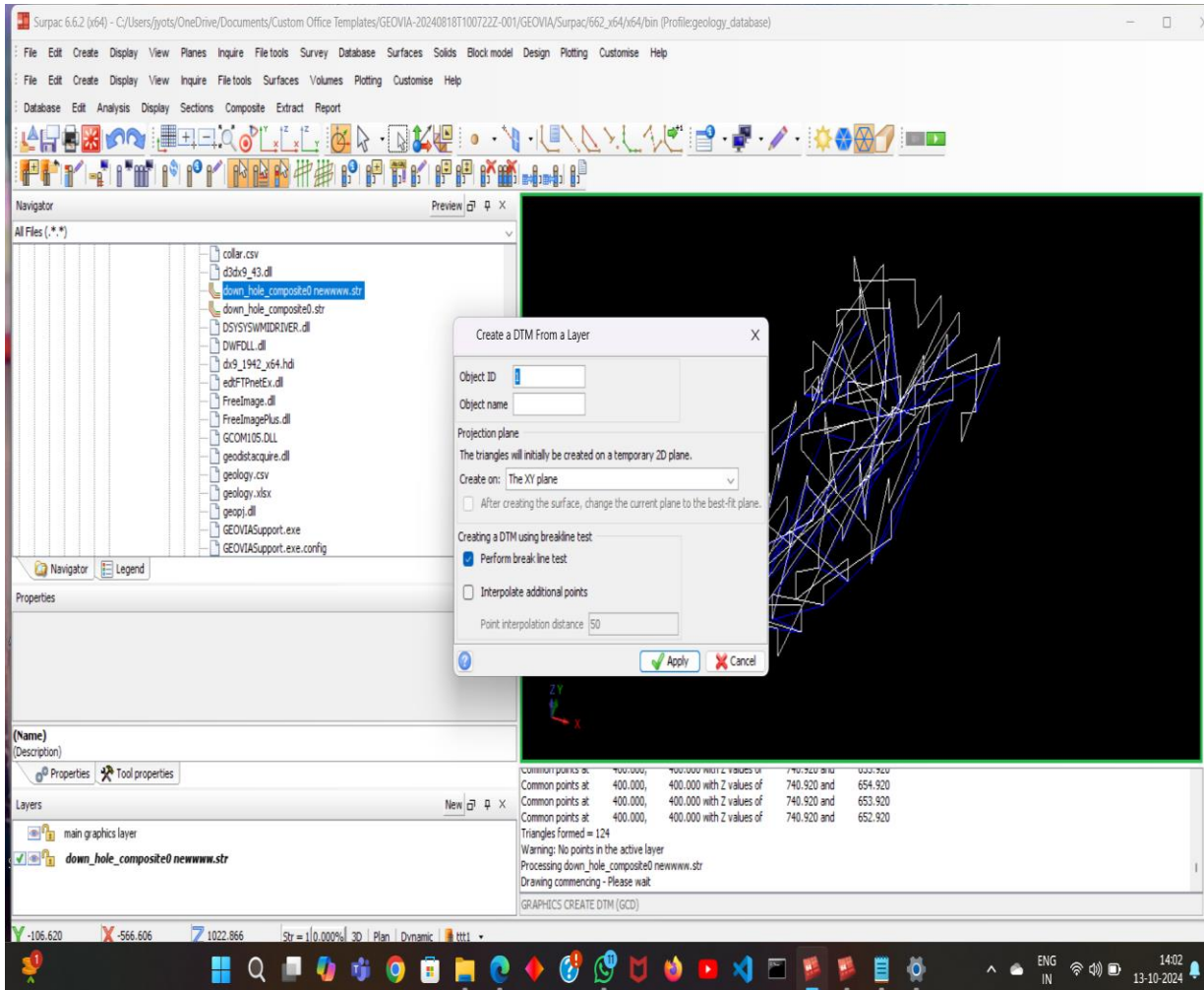
71 holes processed

COMPOSITE DOWNHOLE (CDH)

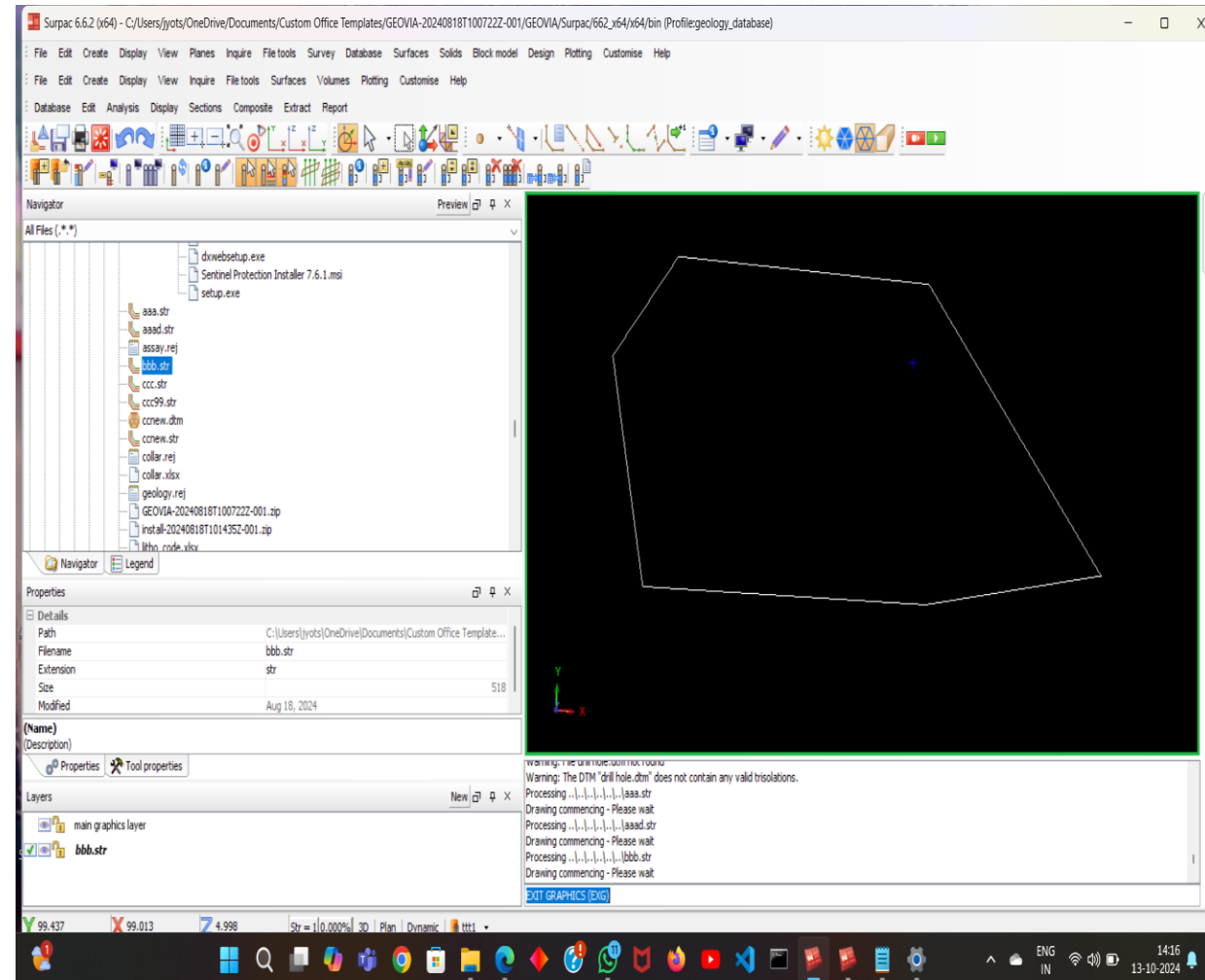
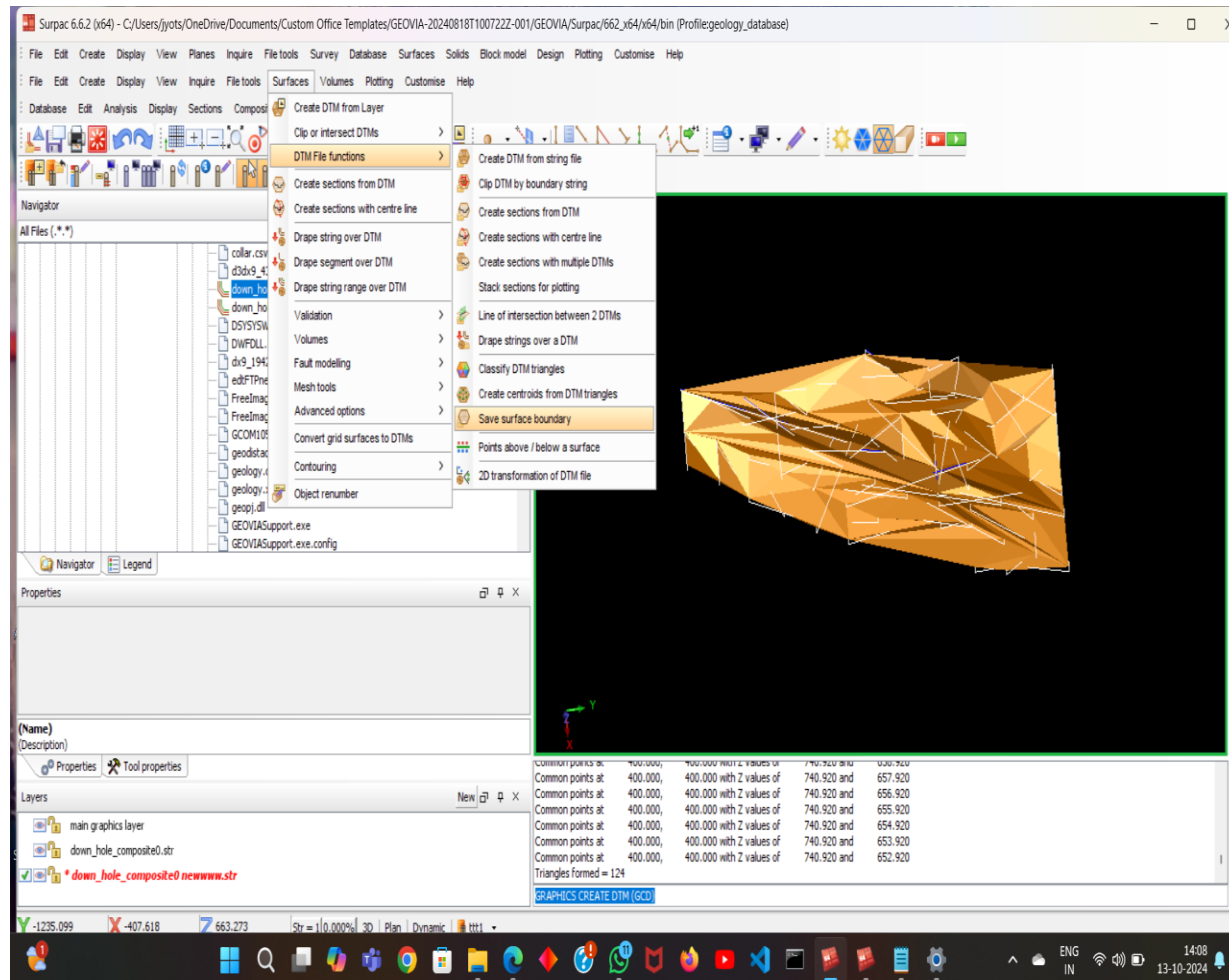
Drag the string file to main graphics layer which was saved in work directory



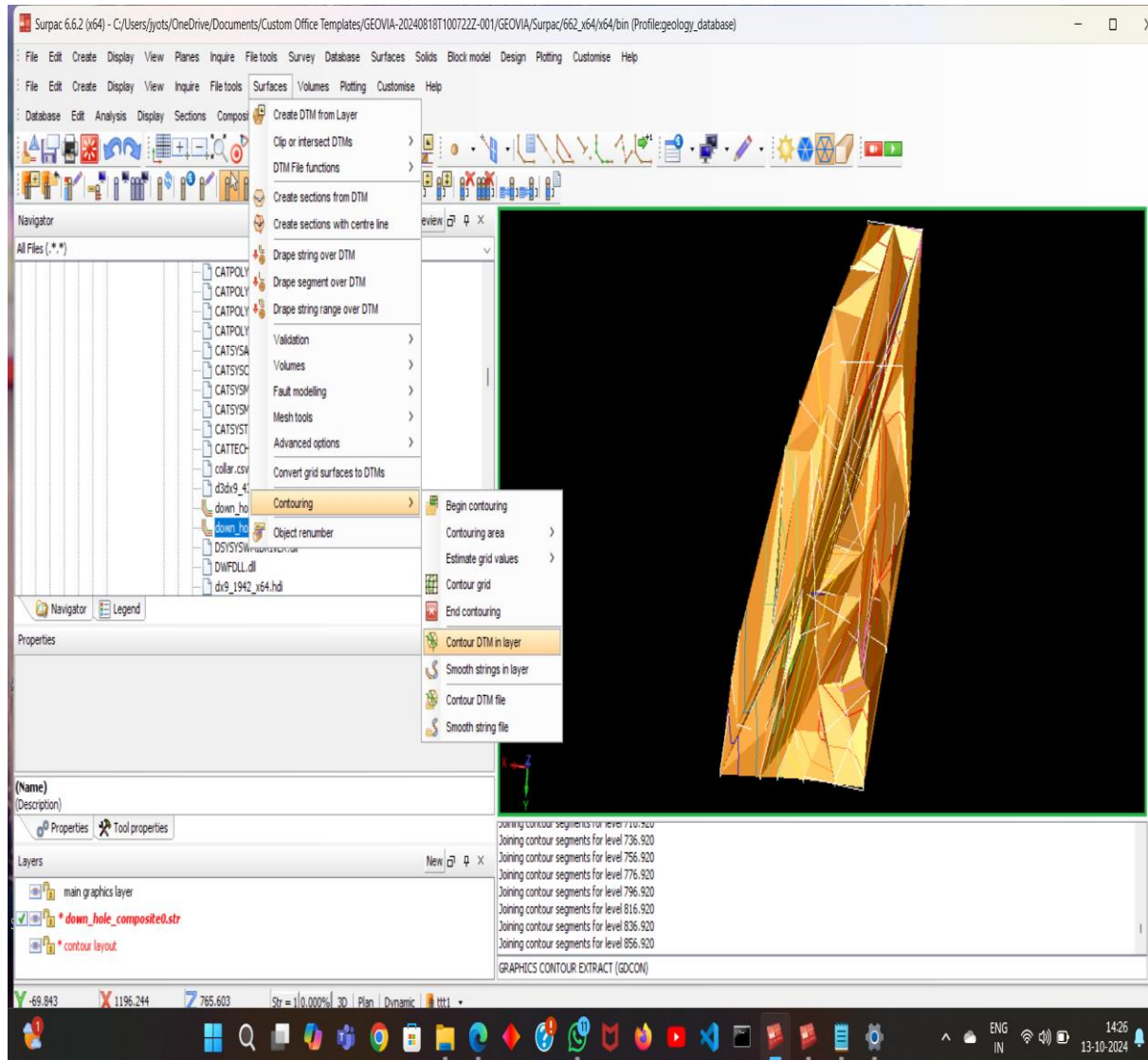
Click on surfaces>> Create DTM from Layer>> Then apply
New Dtm is formed .



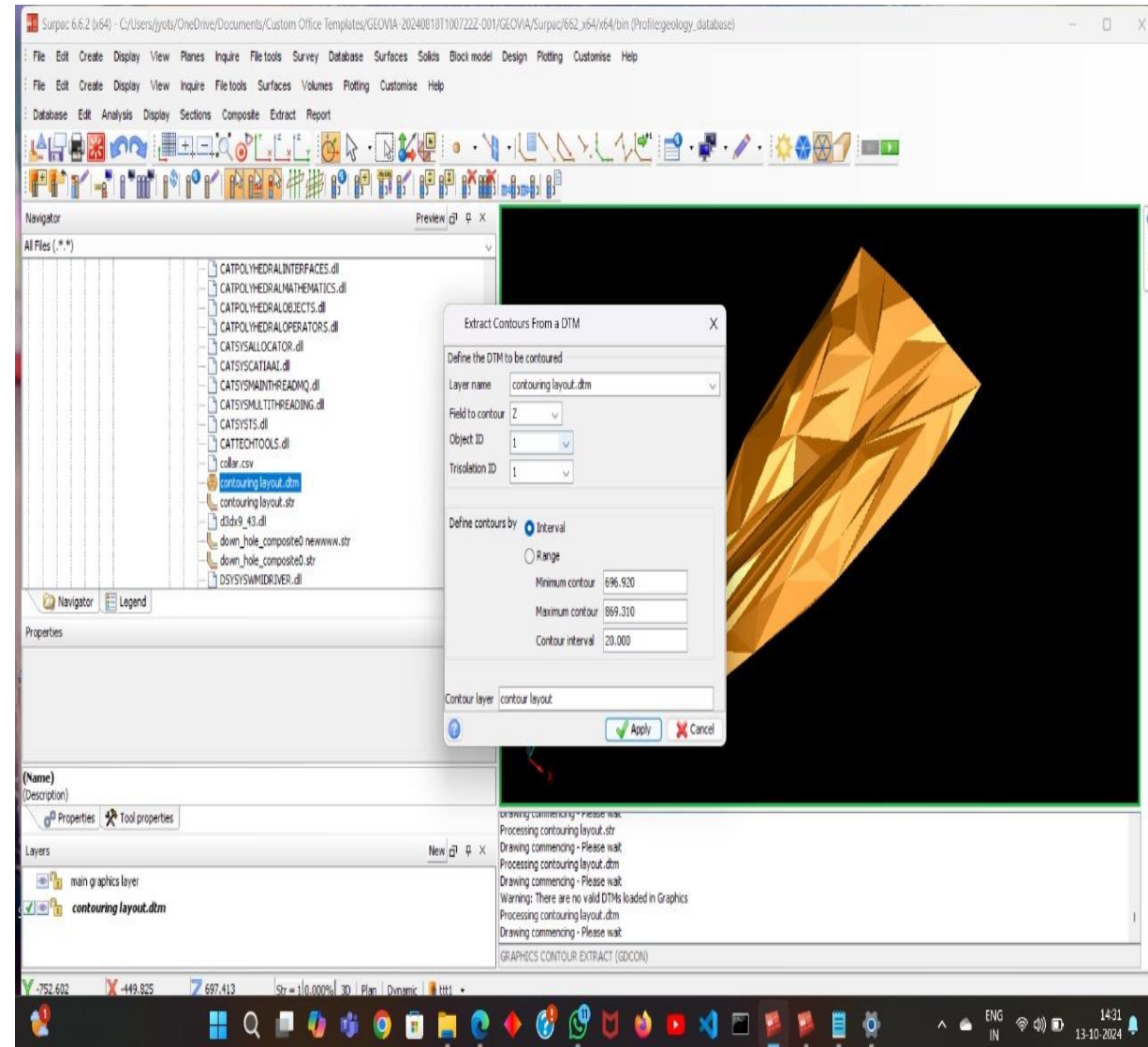
Formation of surface boundary:- click on surfaces>> DTM File functions>> Save surface boundary. In The next Pop up >> Select the Topographical DTM >> Define a name for the boundary >>Click on Apply



CONTOURING:- Go to Surface >> Contouring
>> Contour DTM in layer



Define contour interval and then click on apply

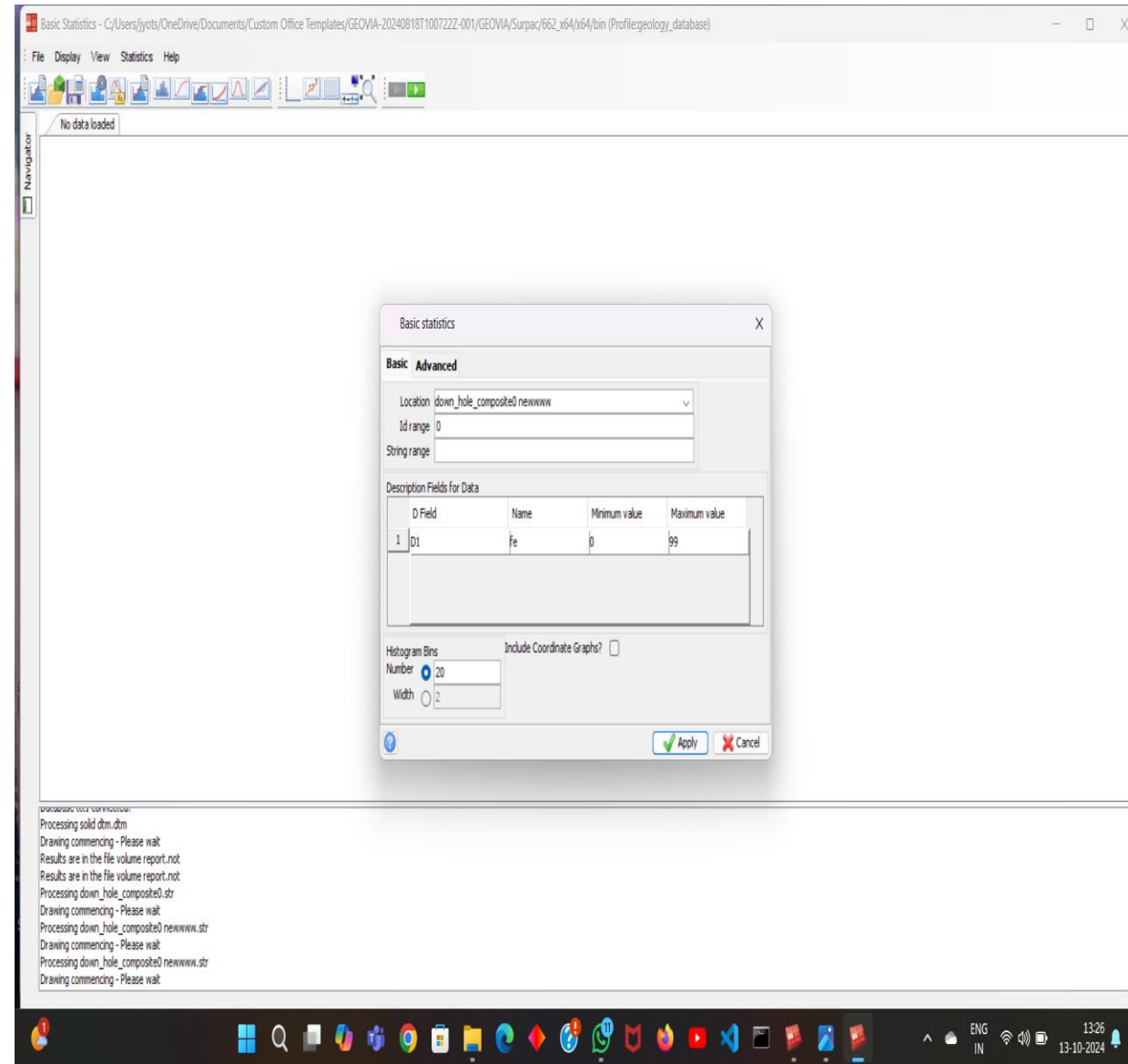
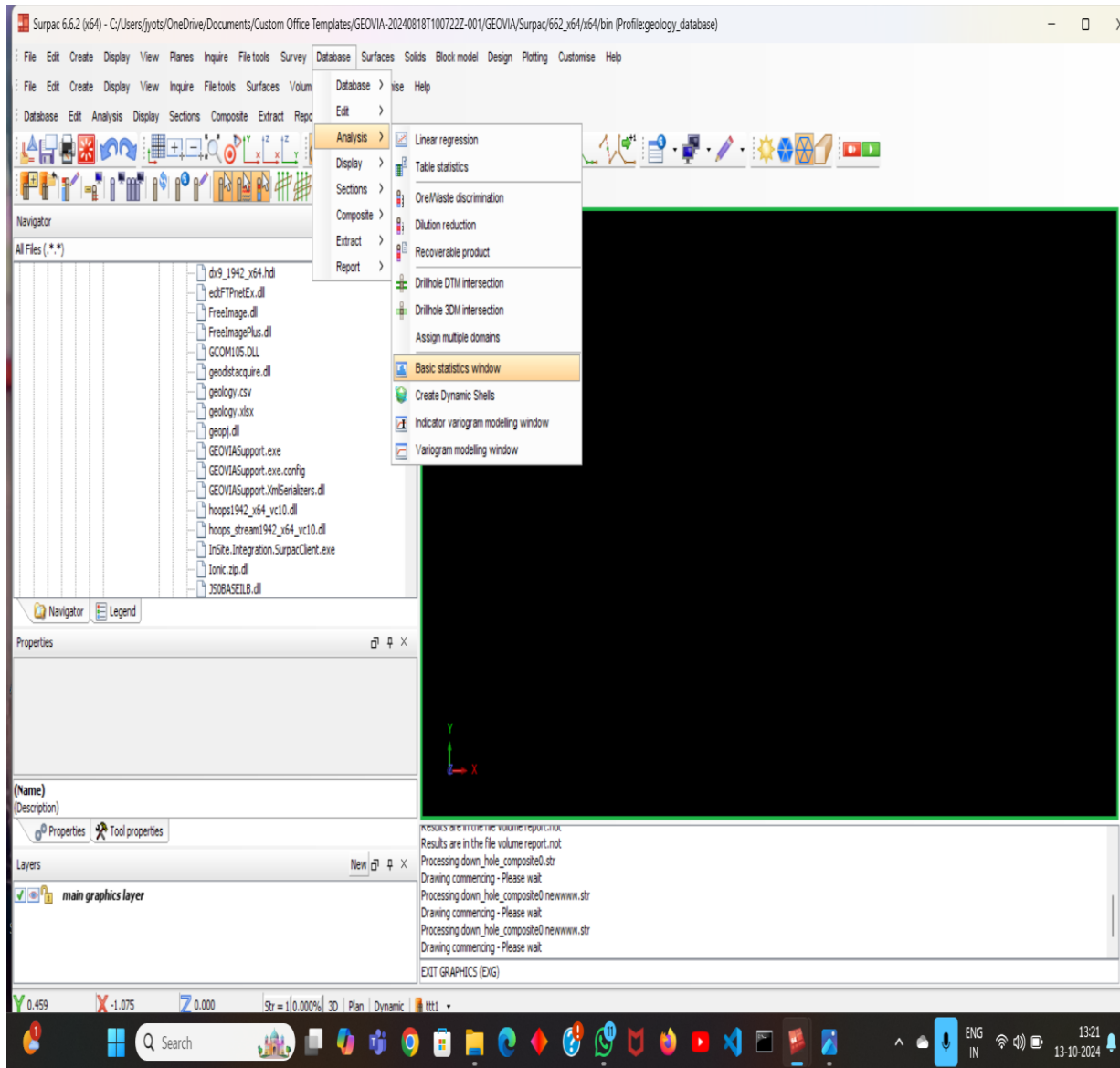


- Type DRWZ in the command box.
- In the next pop up define the contour layer and click on apply. This command will show the elevation (Z) of each contour line.
- Save the new contoured file.
- Then saved file appear like this

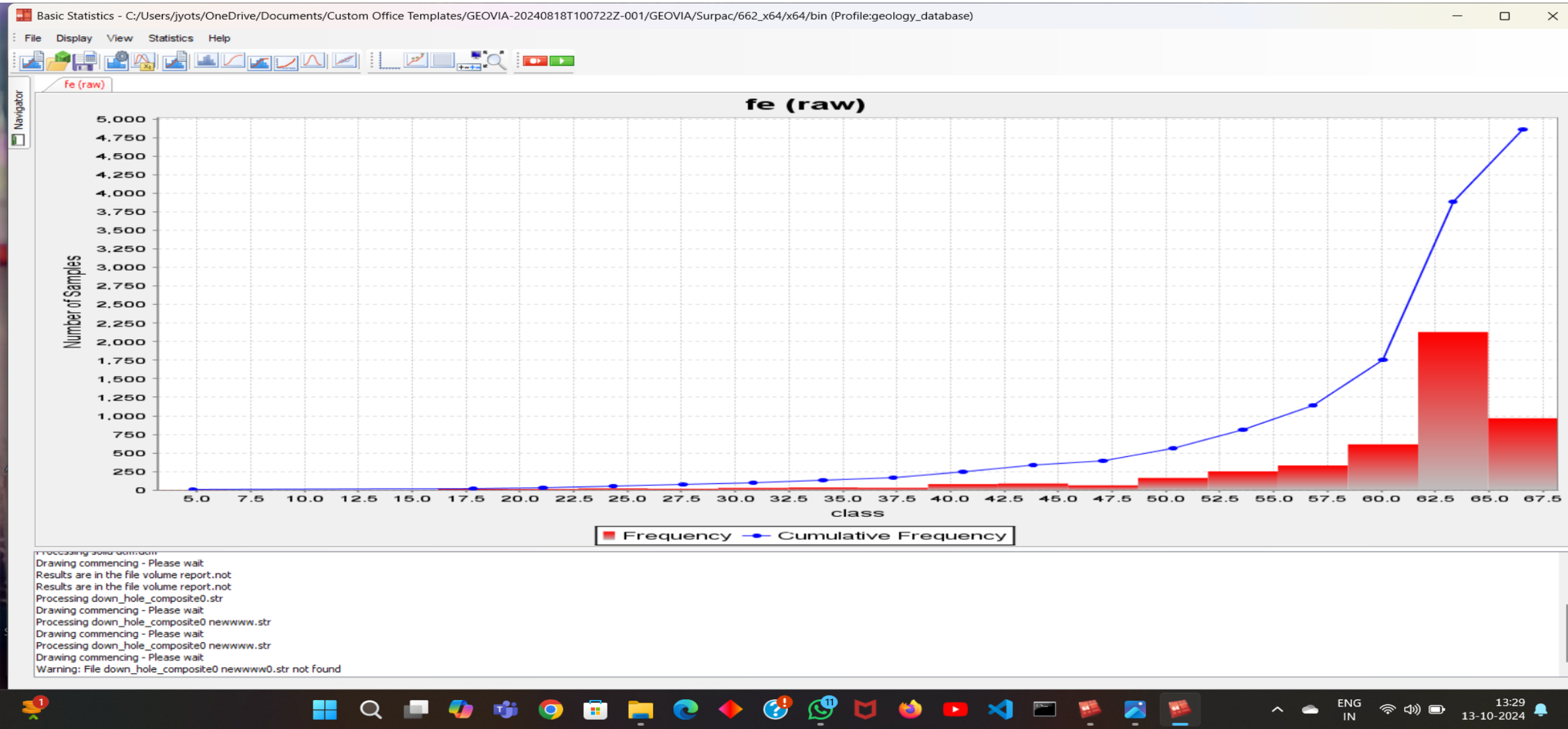


Statistical Analysis:- Go to Database >> Analysis >> Basic statistics window

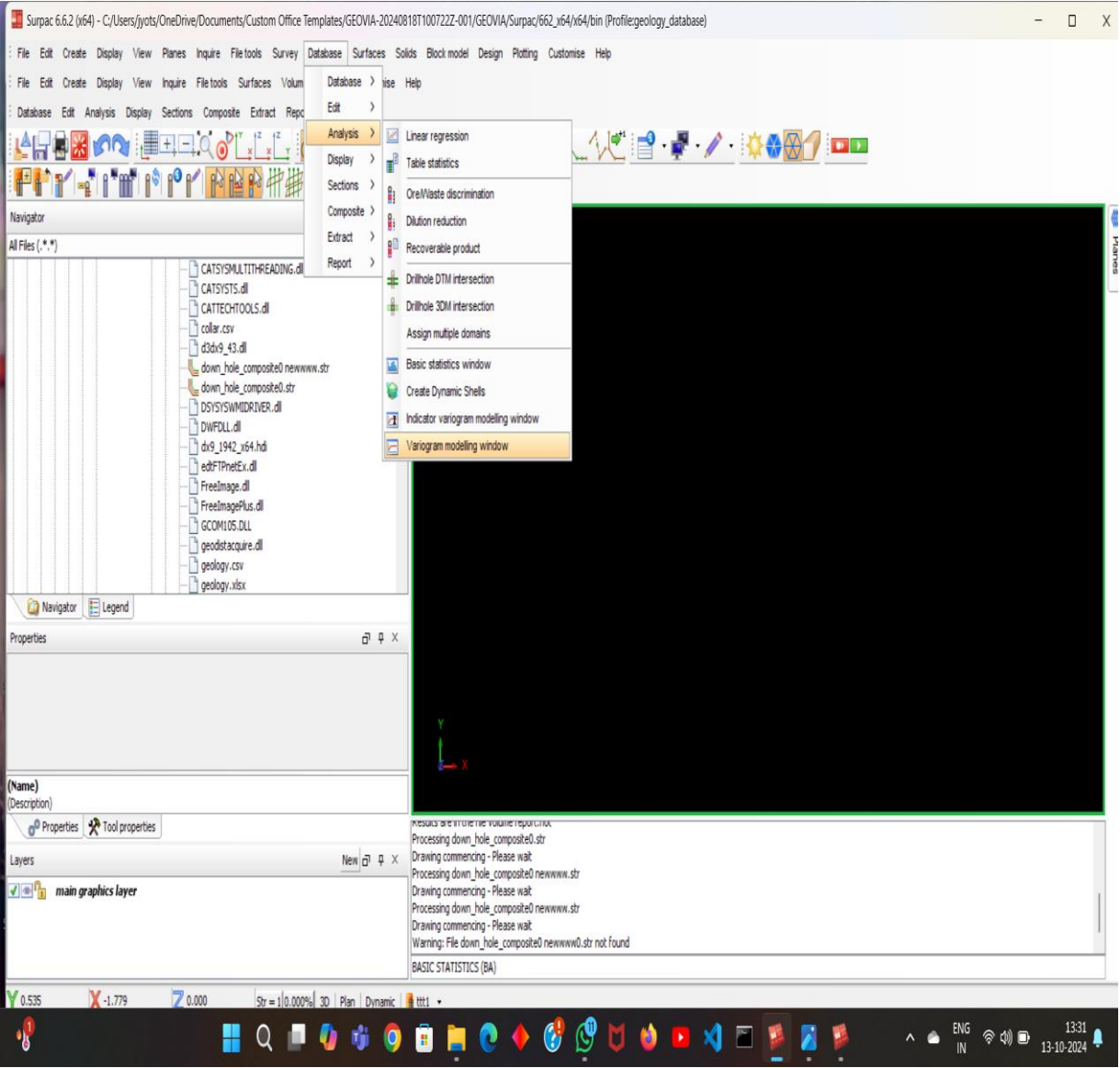
Click on Apply



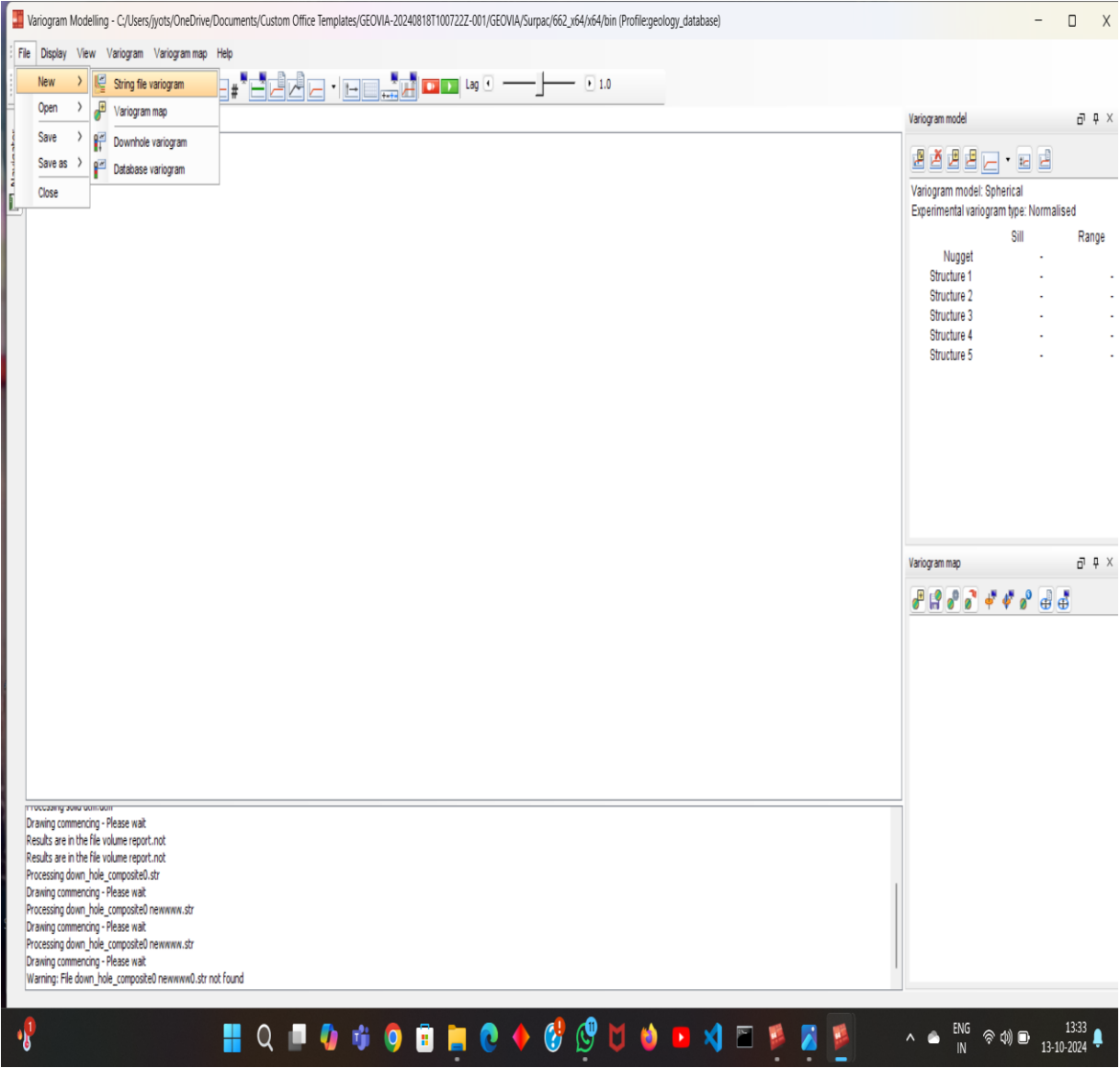
The basic statistical histogram and cumulative frequency curve will display on our screen



Variogram modelling:- Compositing.str file will be used for variogram modelling. Database >> Analysis >> Variogram modelling window



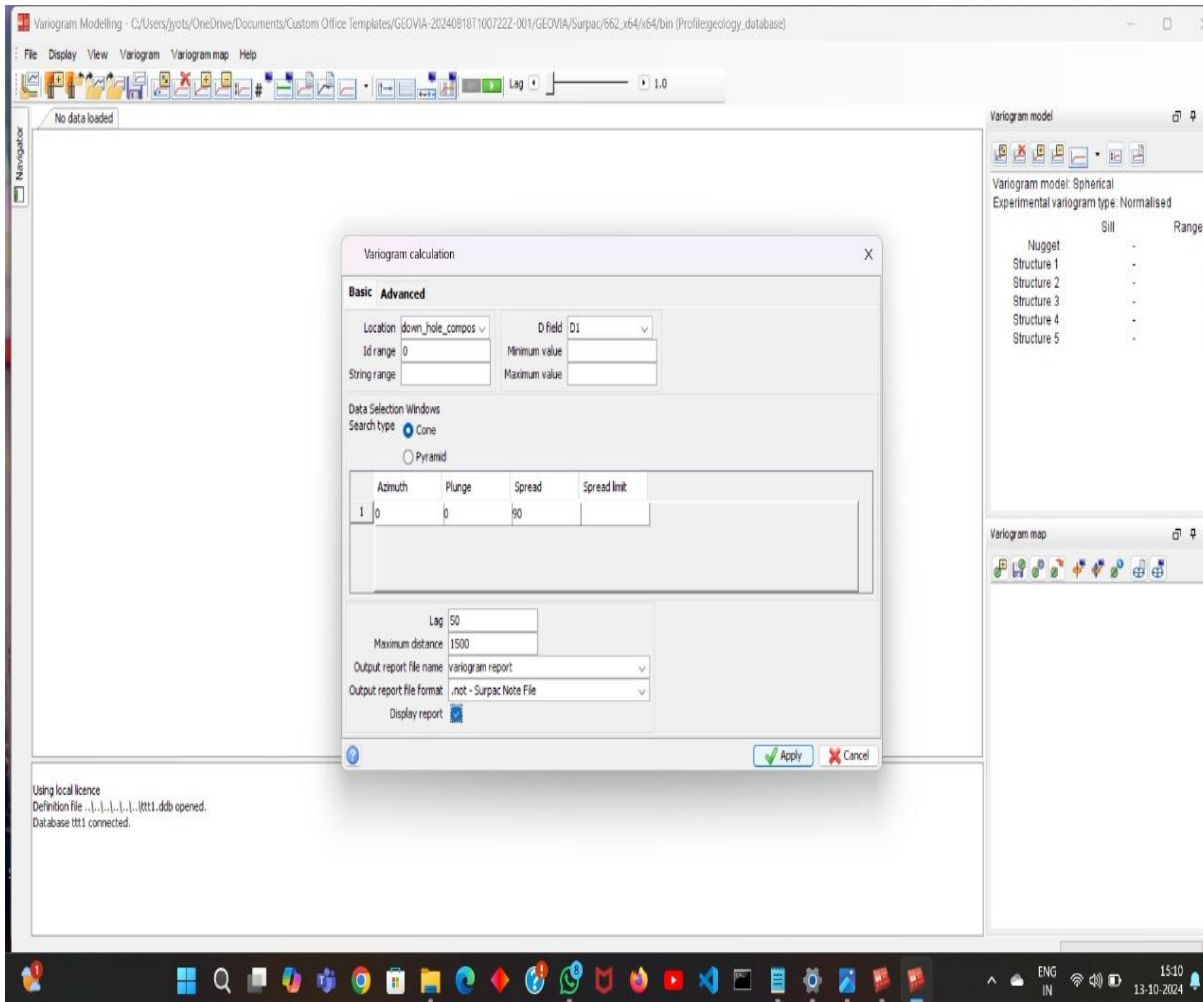
File >> New >> String file variogram



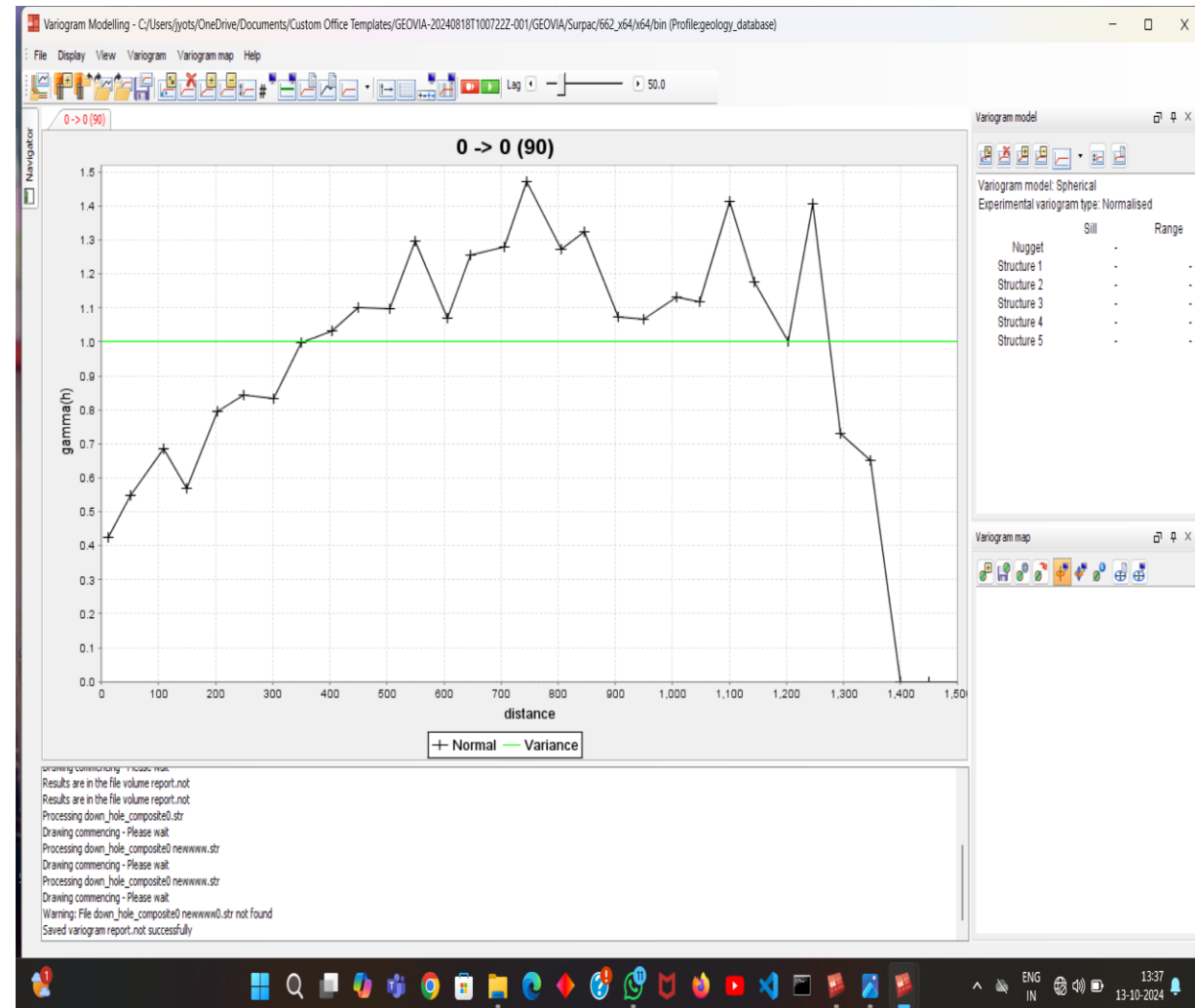
In Location attach compositing.str file

Number of variograms: Choose from the drop-down list

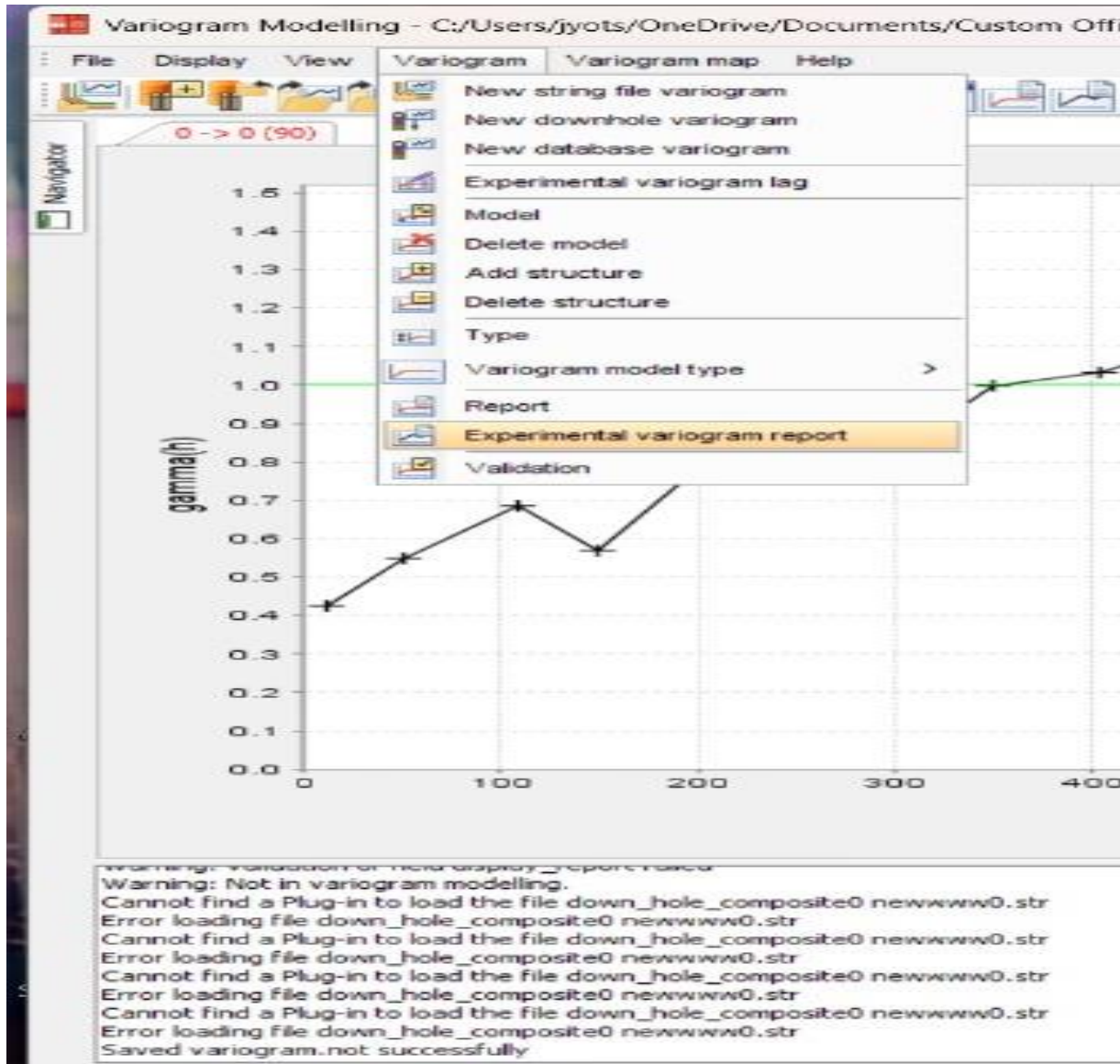
Maximum distance: fill in all other information according to the available data. >> APPLY



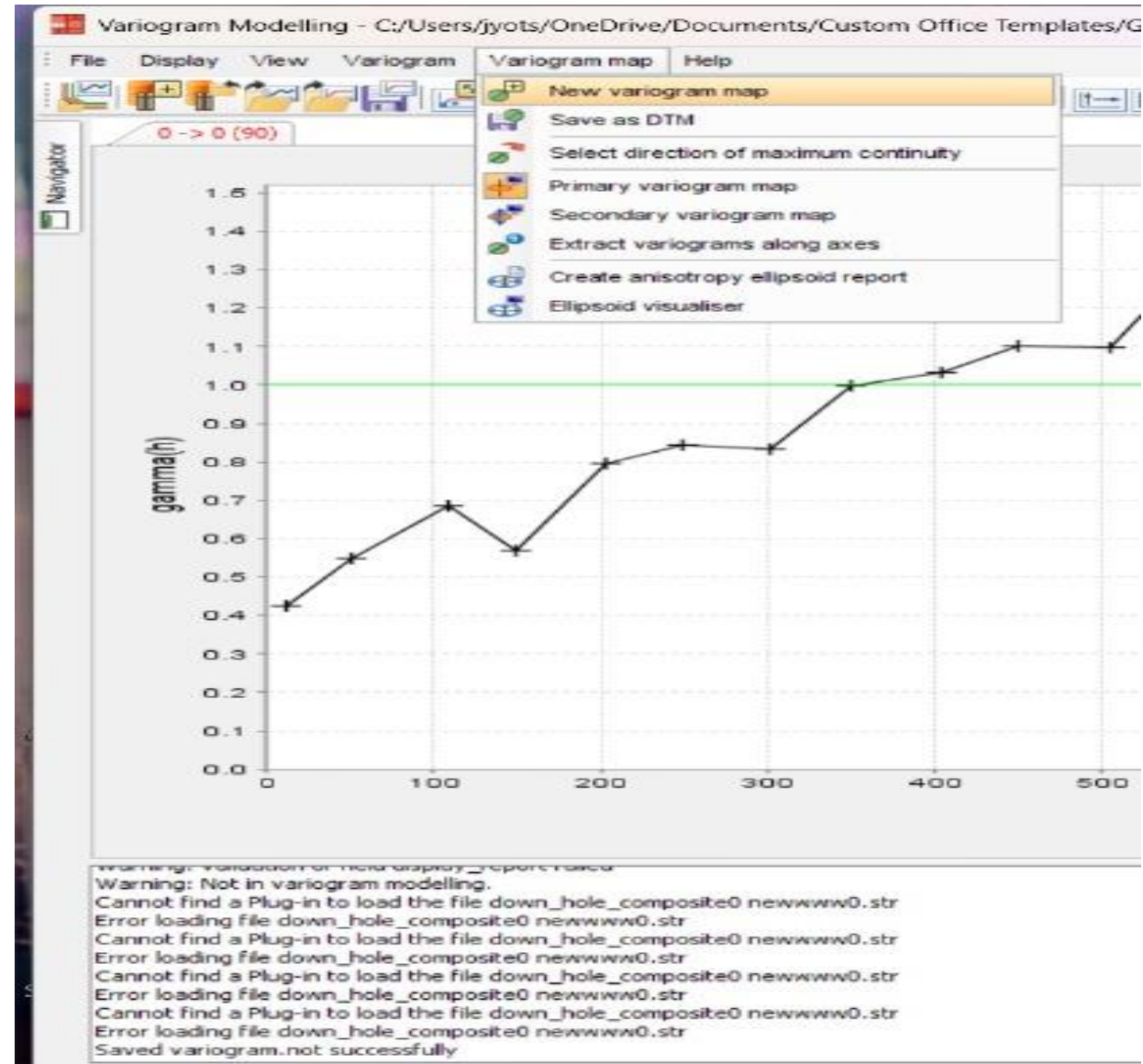
The Experimental Variogram will open like this



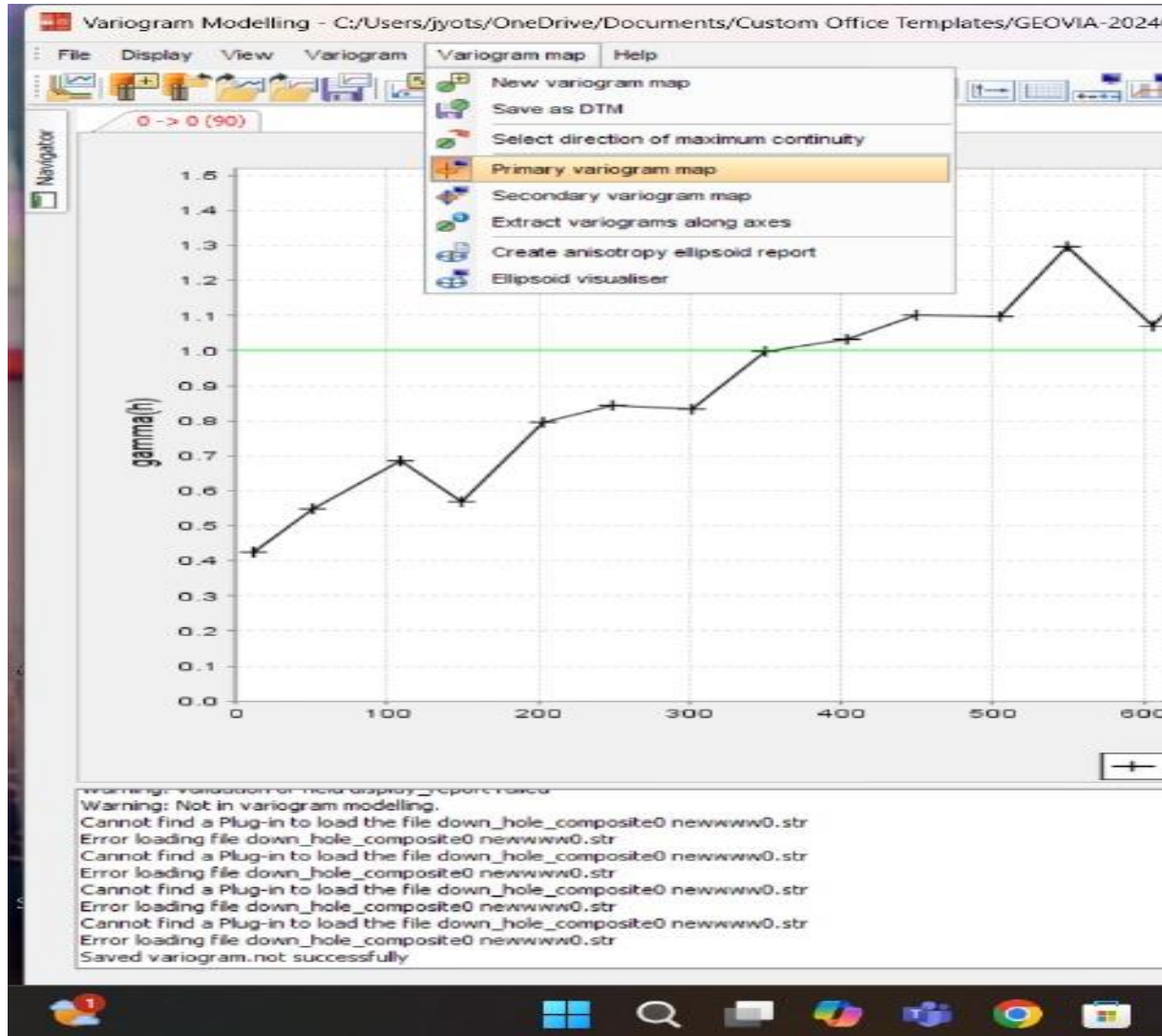
Variogram >> Experimental variogram Report >>
Enter File Name >> APPLY



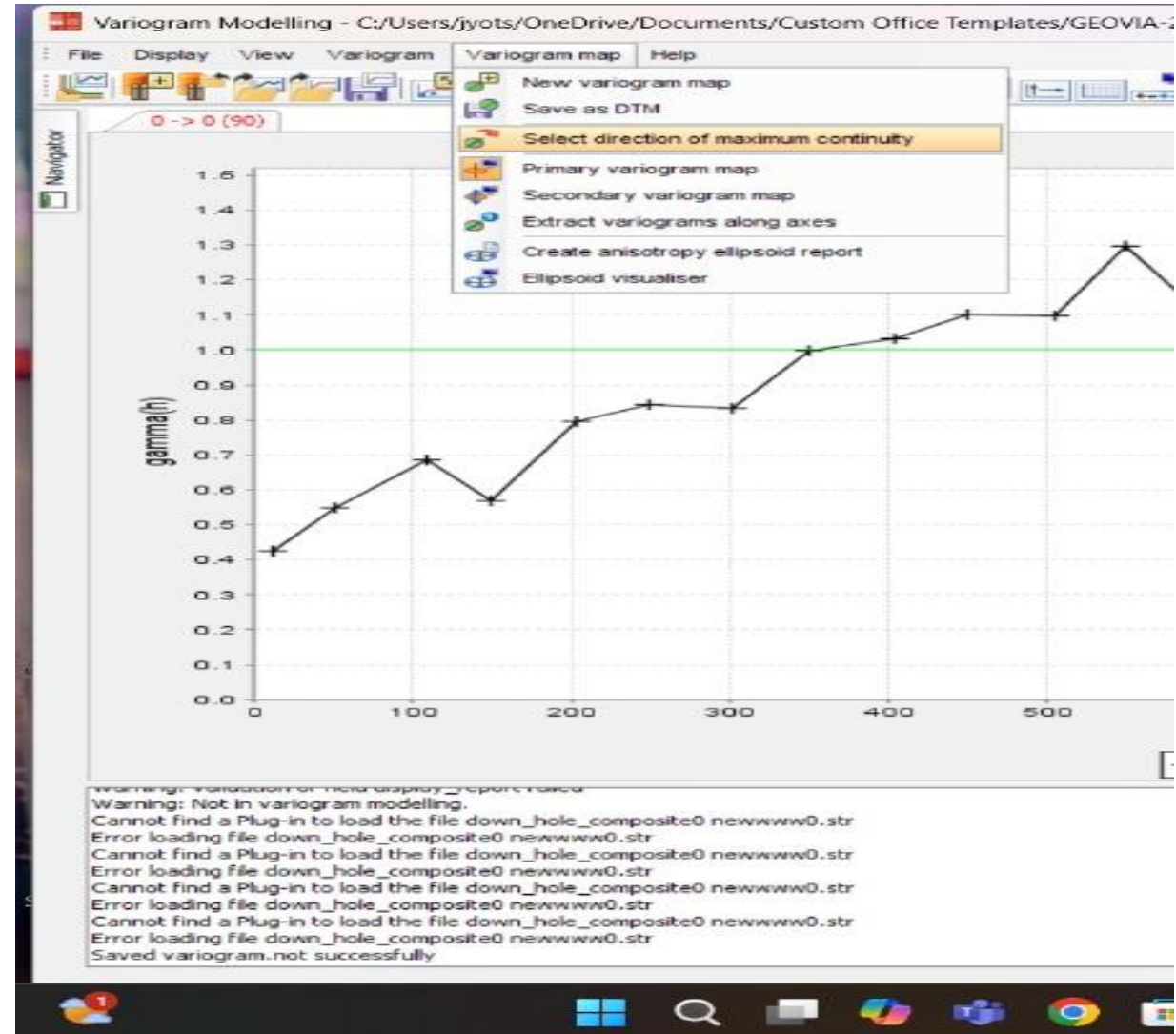
Variogram map >> New Variogram Map



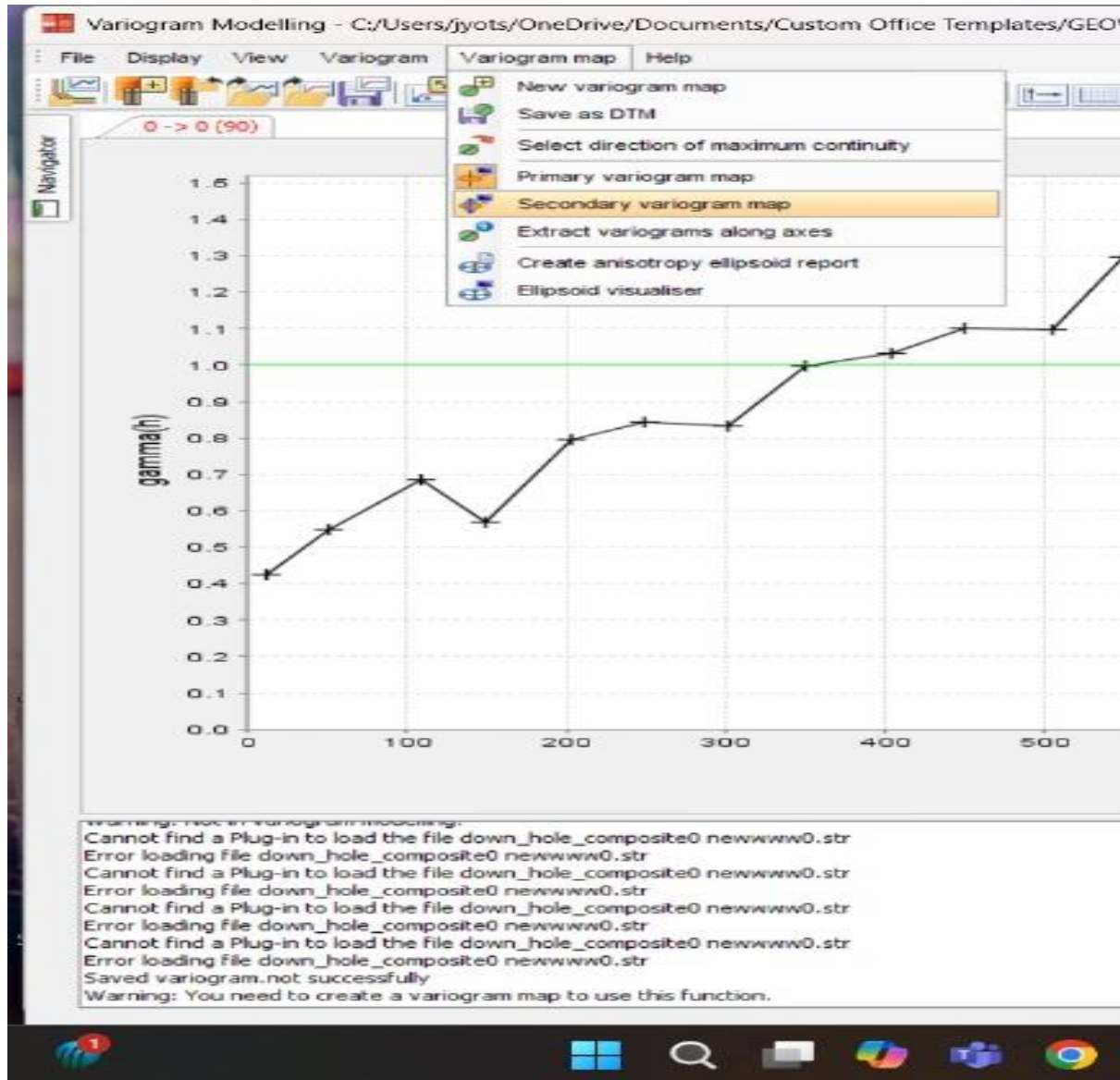
Variogram map >> Primary variogram map



Variogram map >> Select the direction of maximum continuity

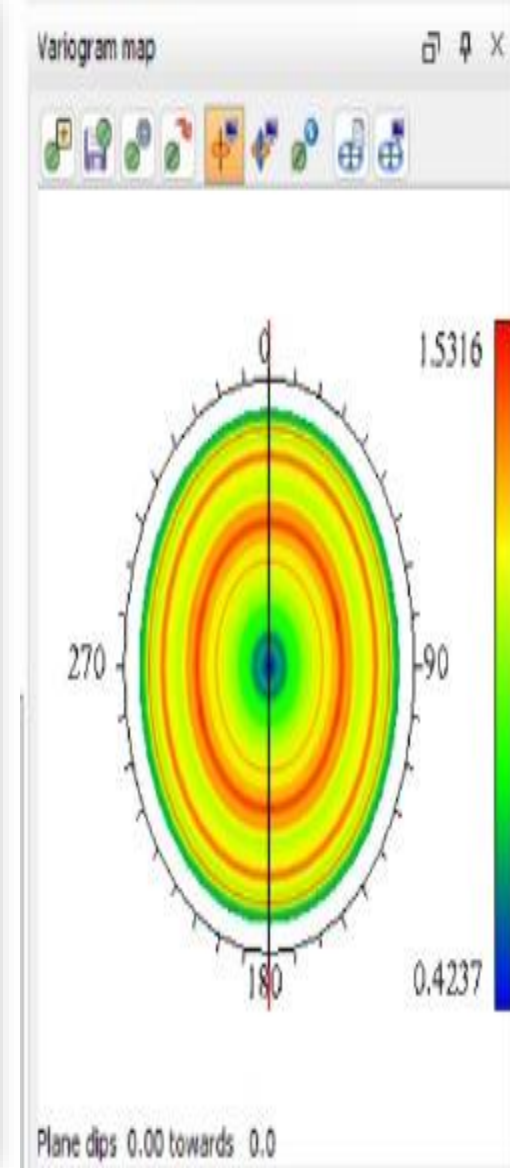
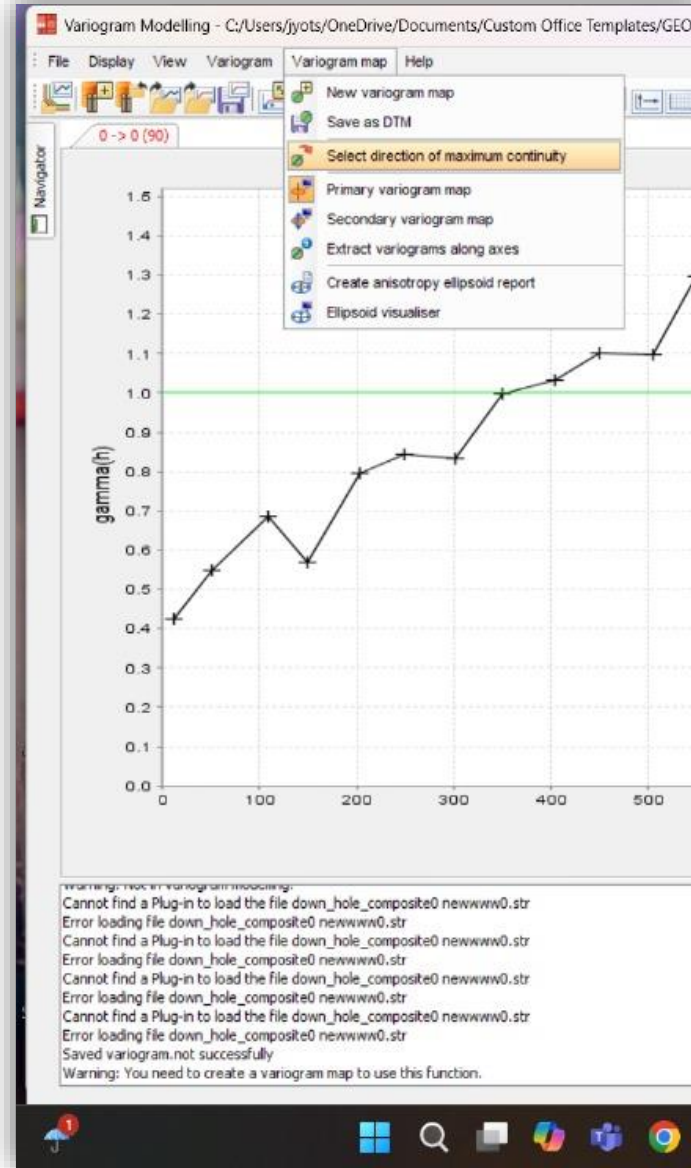


Variogram map >> Secondary variogram map

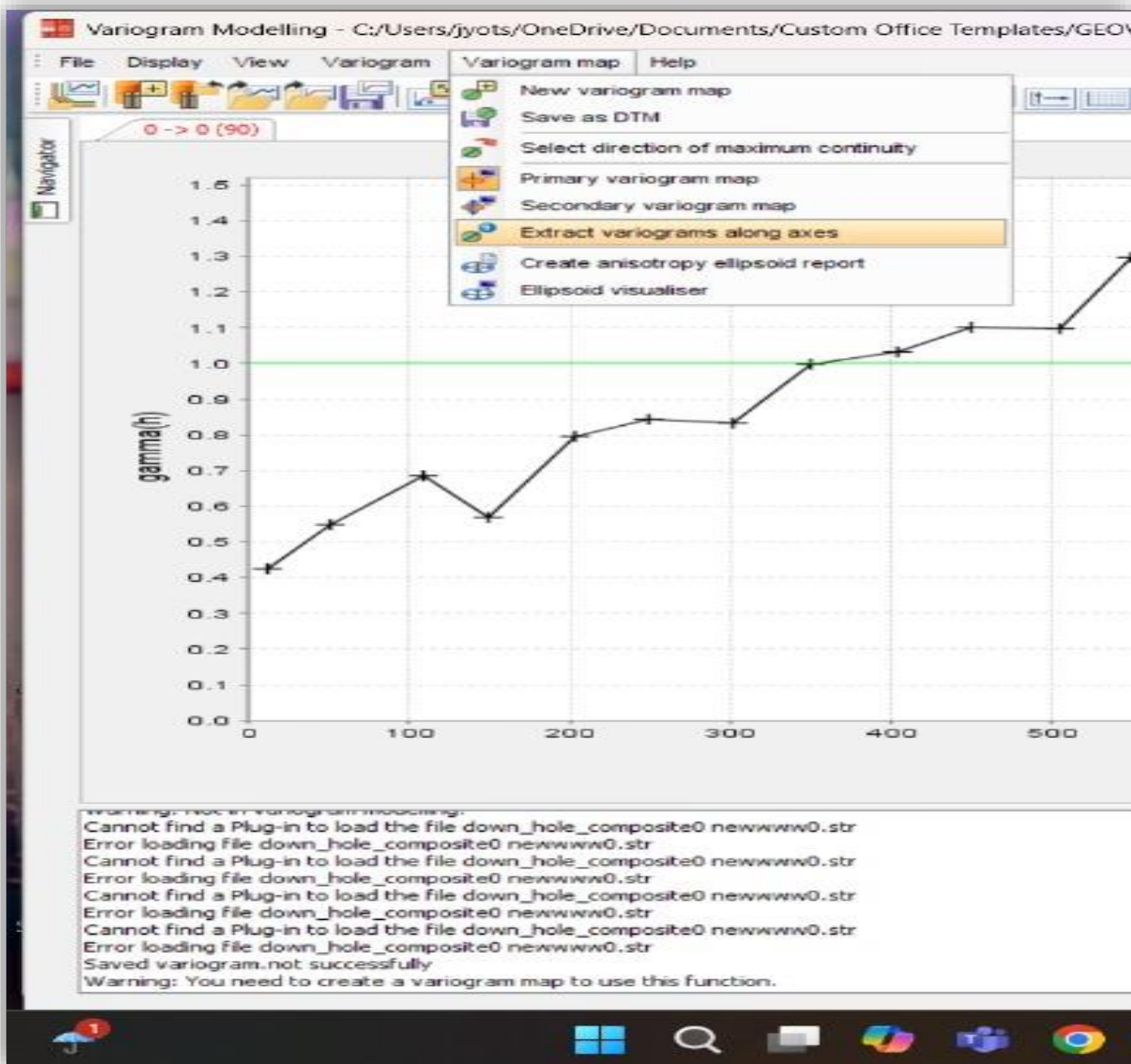


Variogram map >> Select the direction of maximum continuity

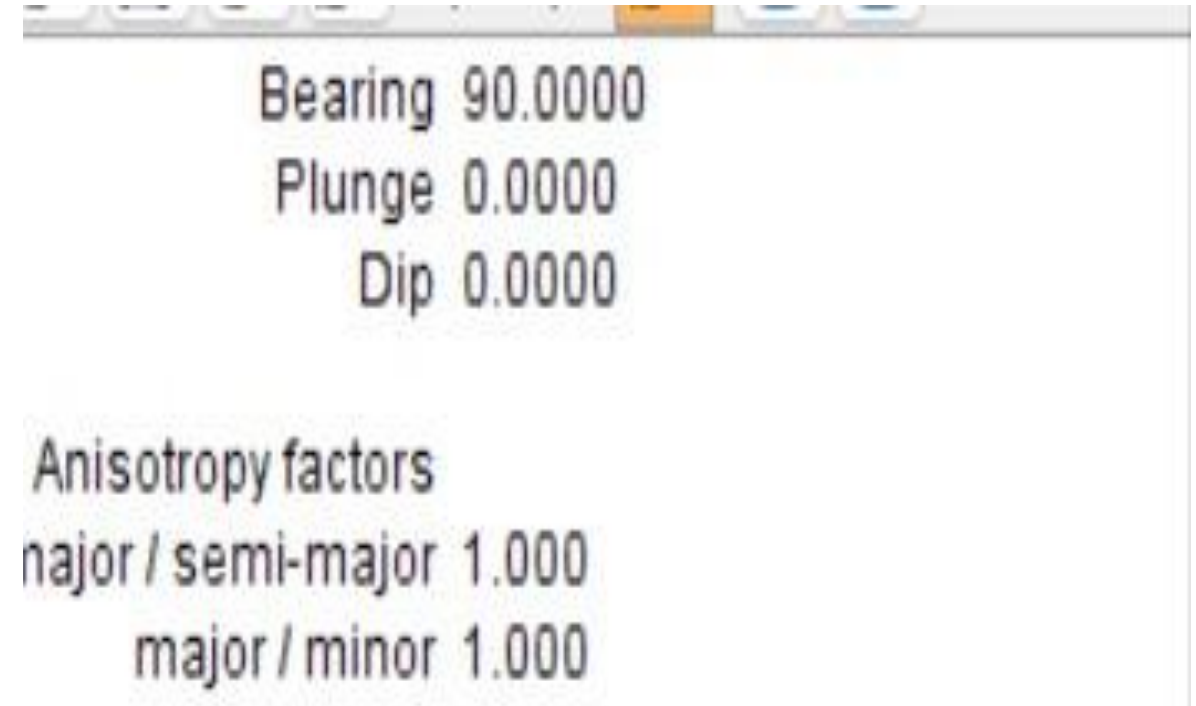
A Variogram map will be shown..



Variogram map >> Extract Variogram along axis

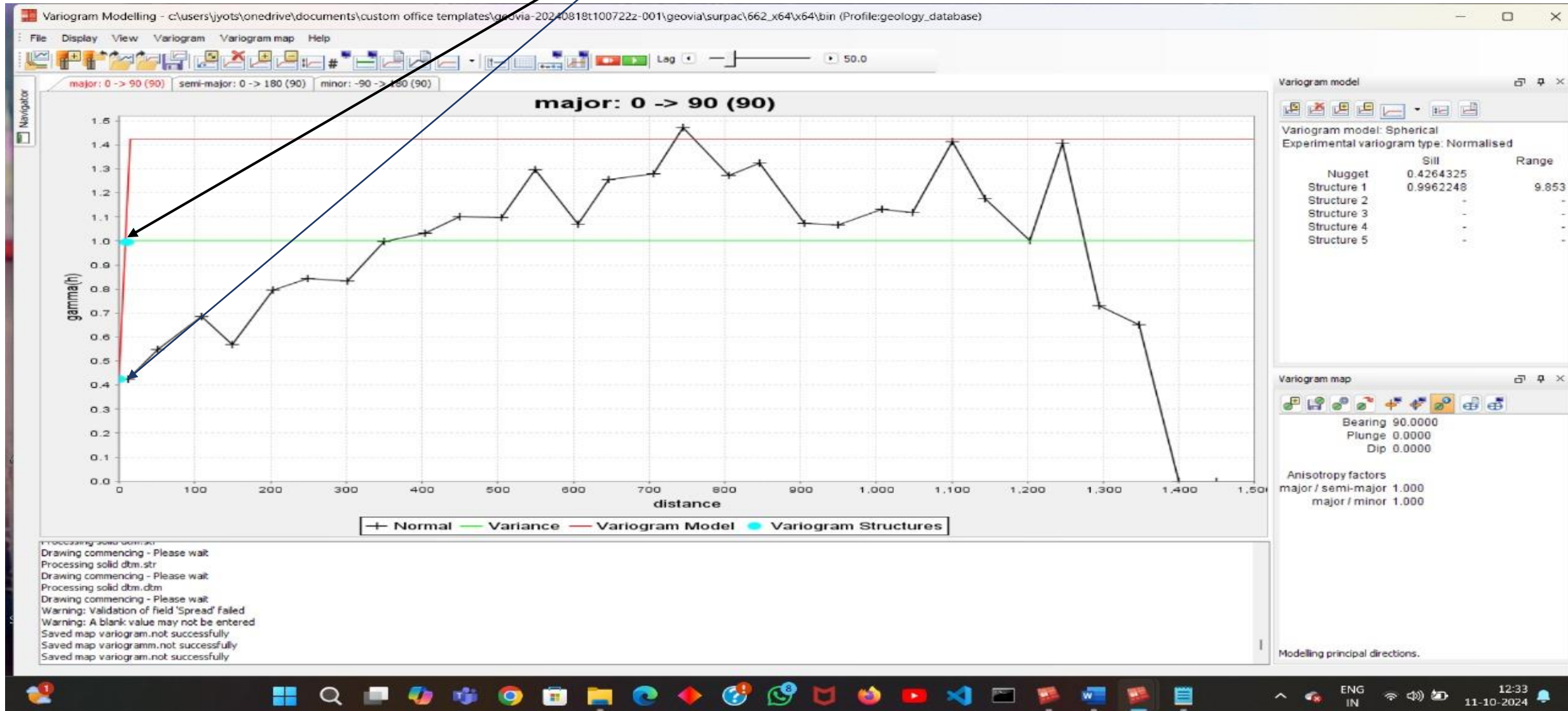


Anisotropy Report. We need this for further use

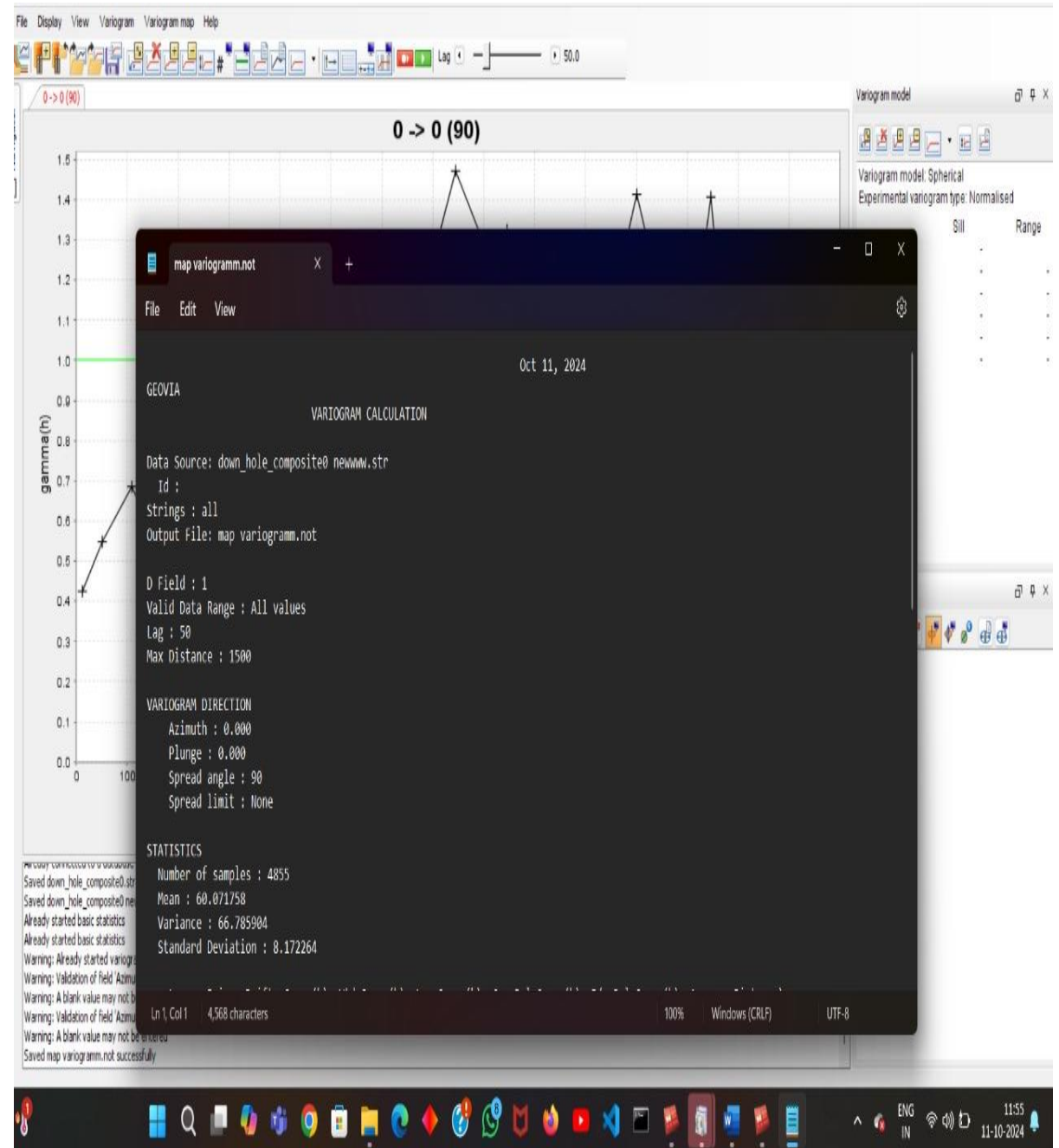
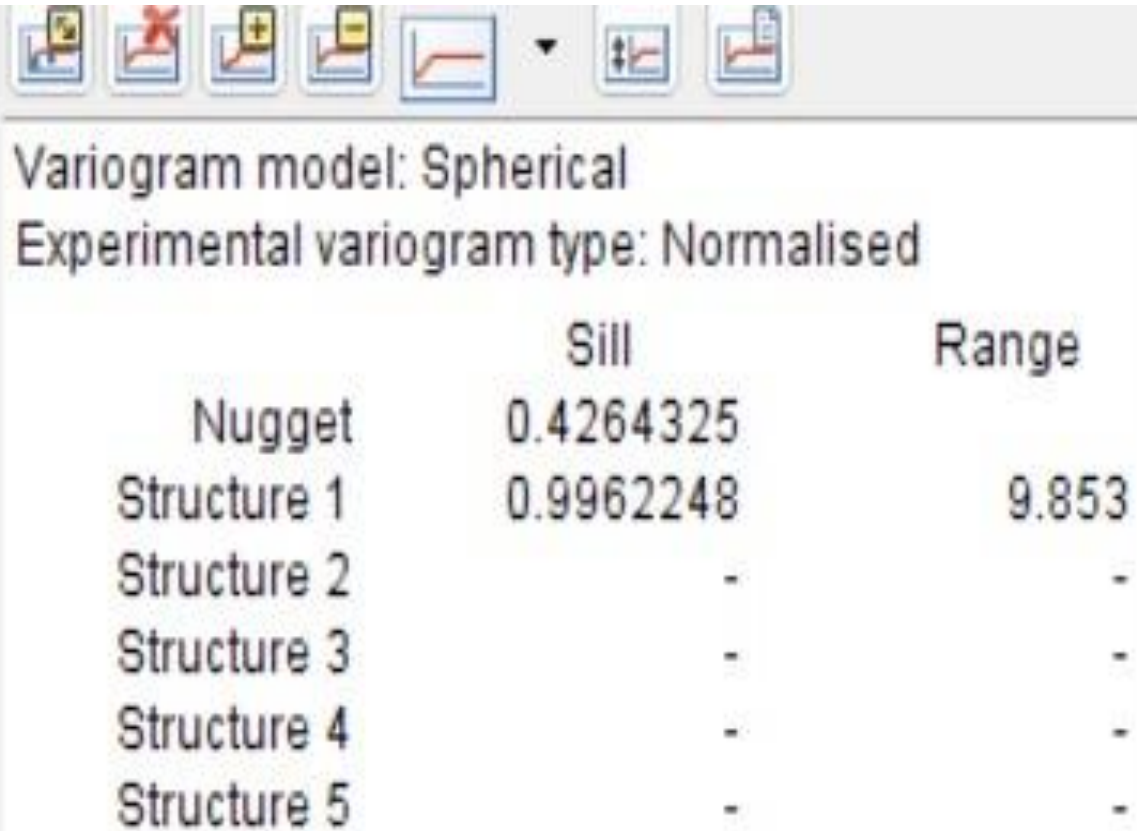


Go to Variogram >> Model

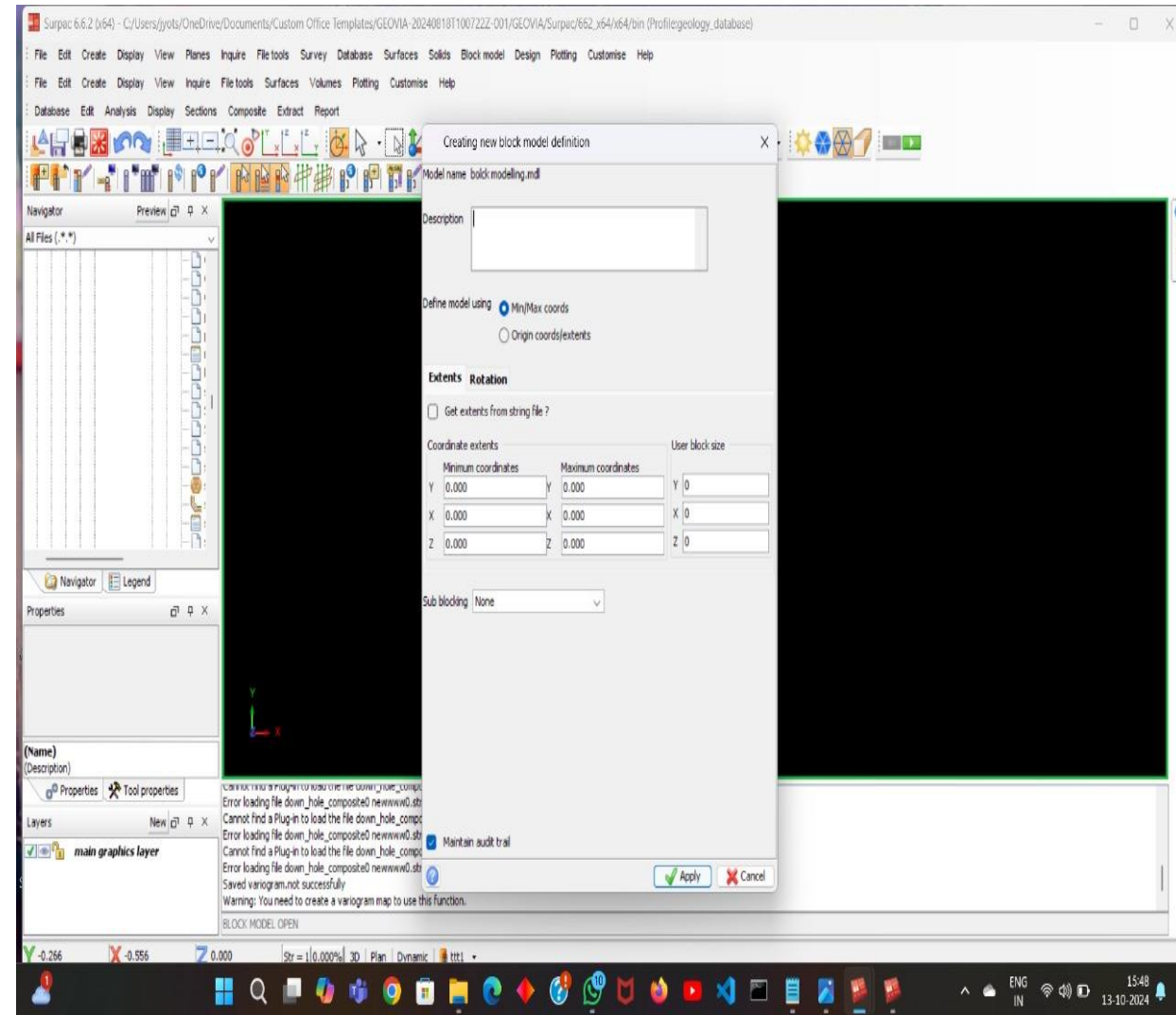
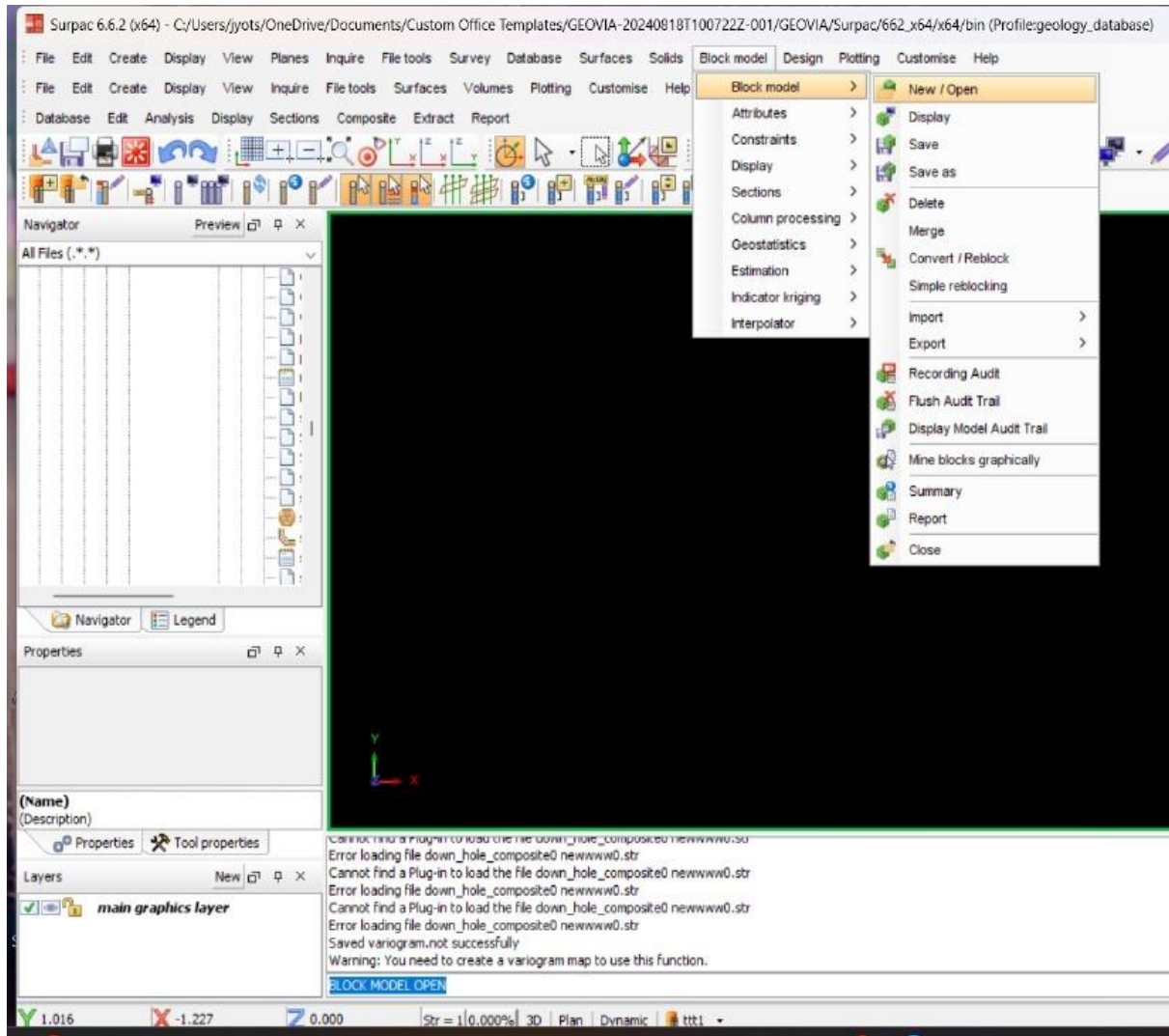
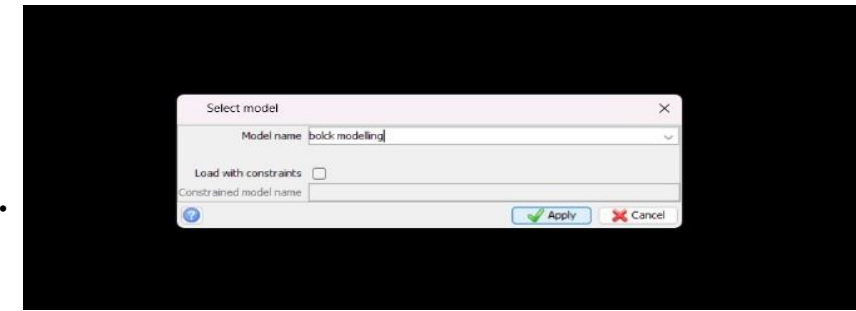
The variogram window will show the *sky-blue* points to adjust the perfect variogram map. Adjust those points and create a perfect variogram map.



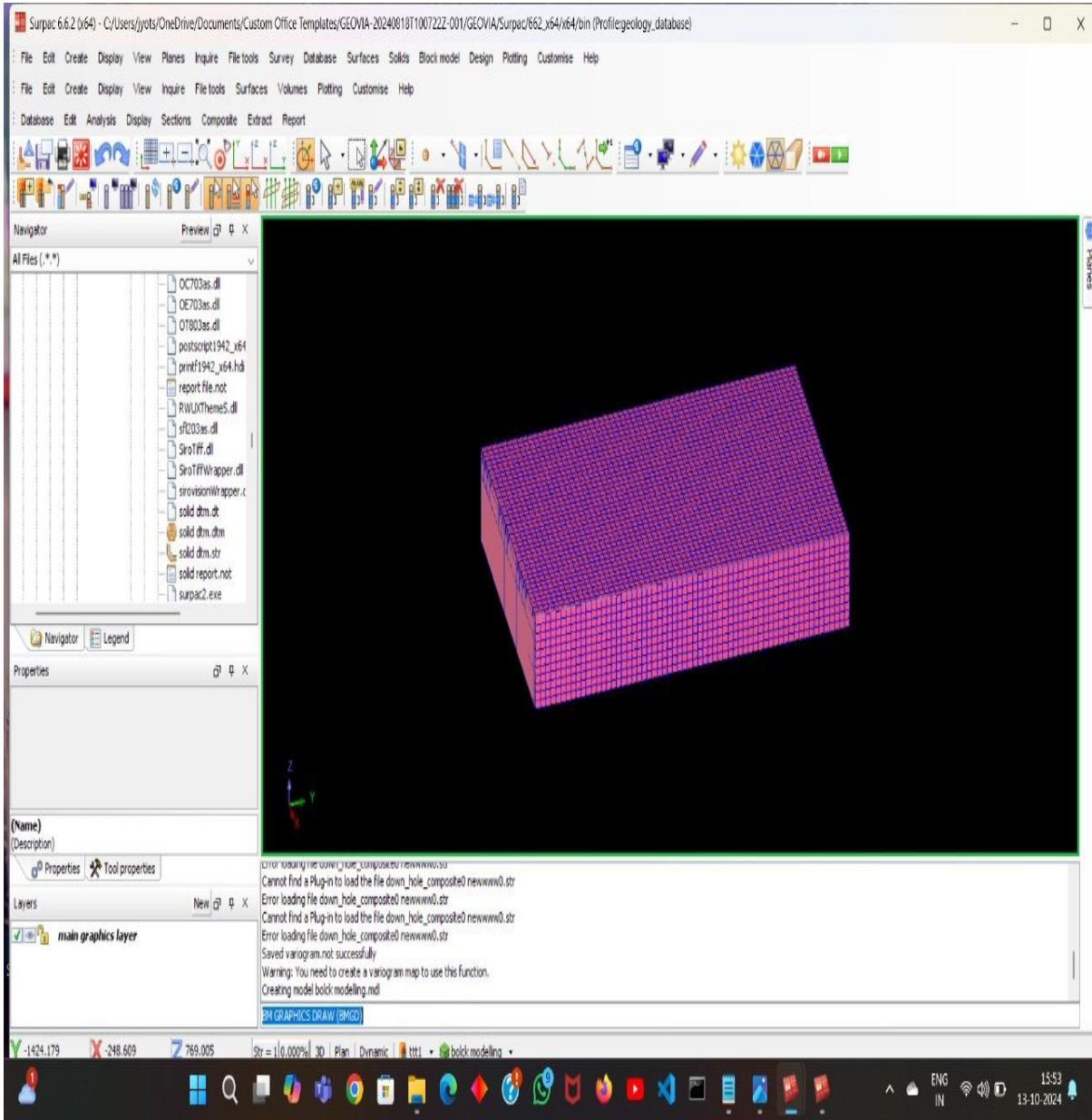
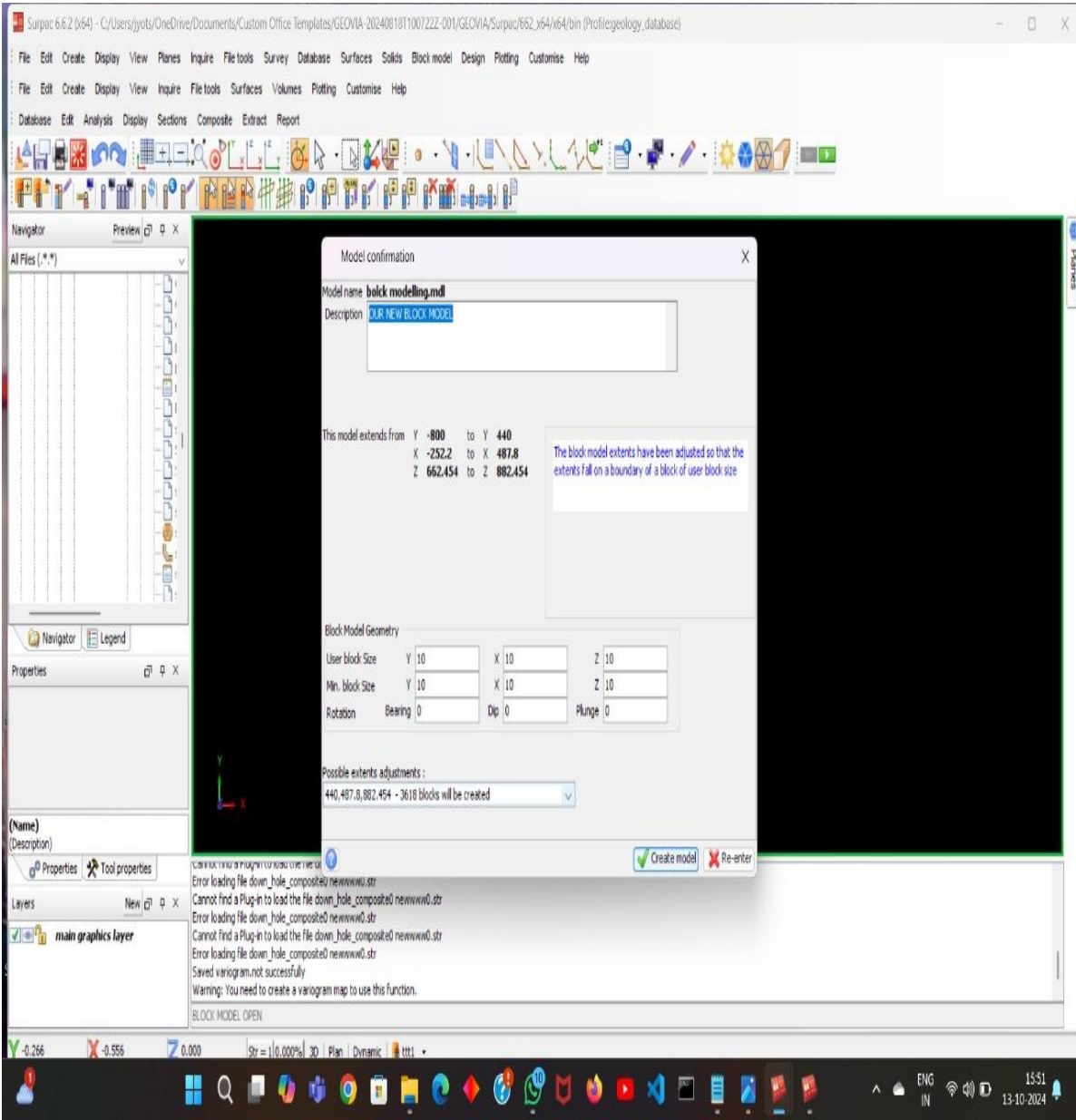
After adjustment, we will get nugget, sill, and range values



BLOCK MODELLING:- Go to block model >> New/Open.
Define a name for the block model, Click on Apply.
To get the X, Y and Z coordinates extents, Click on Get extents from string file.
Select the solid model string file and define the block sizes in X, Y and Z directions.
Click on Apply
A new block model will be created

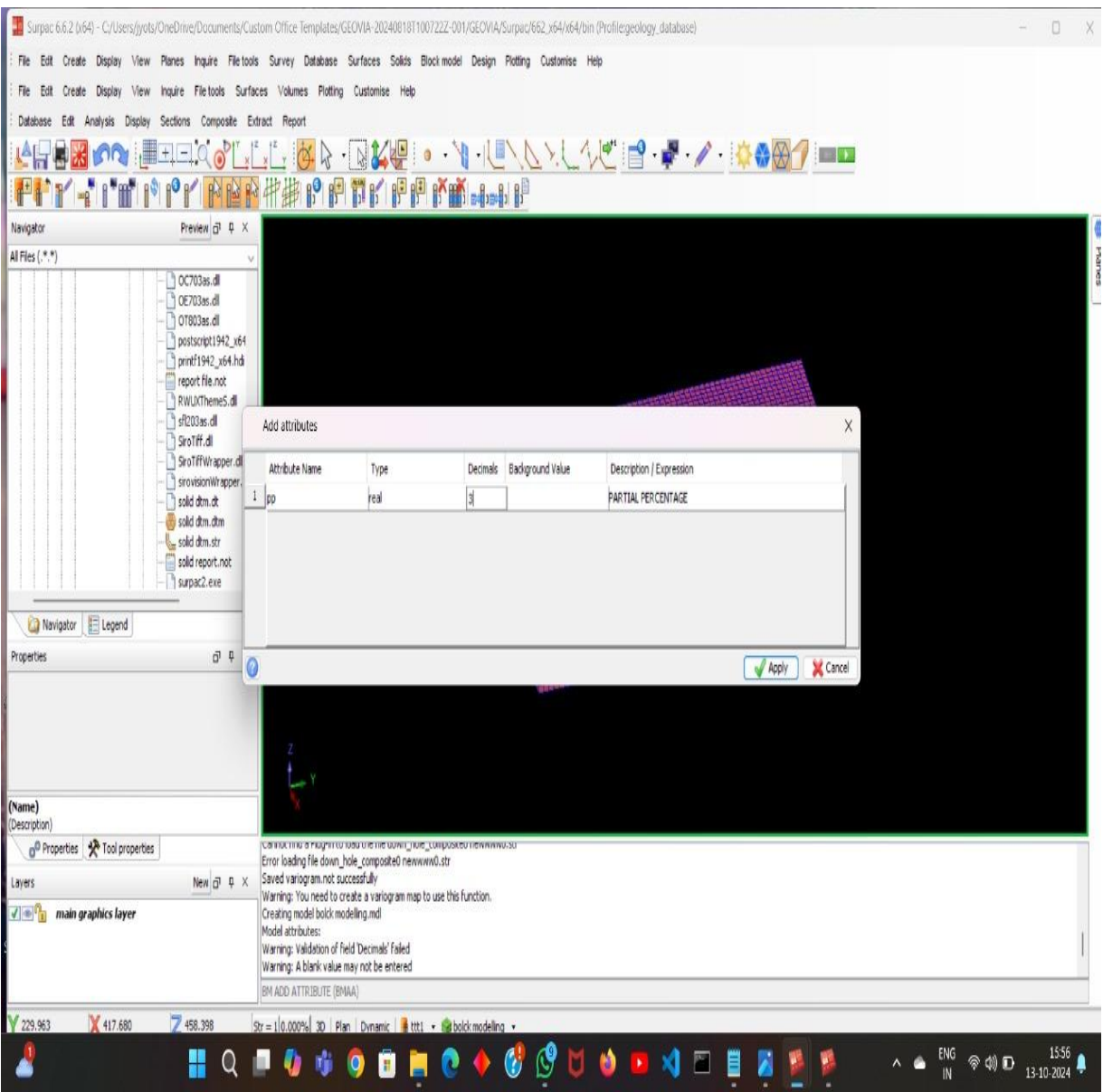
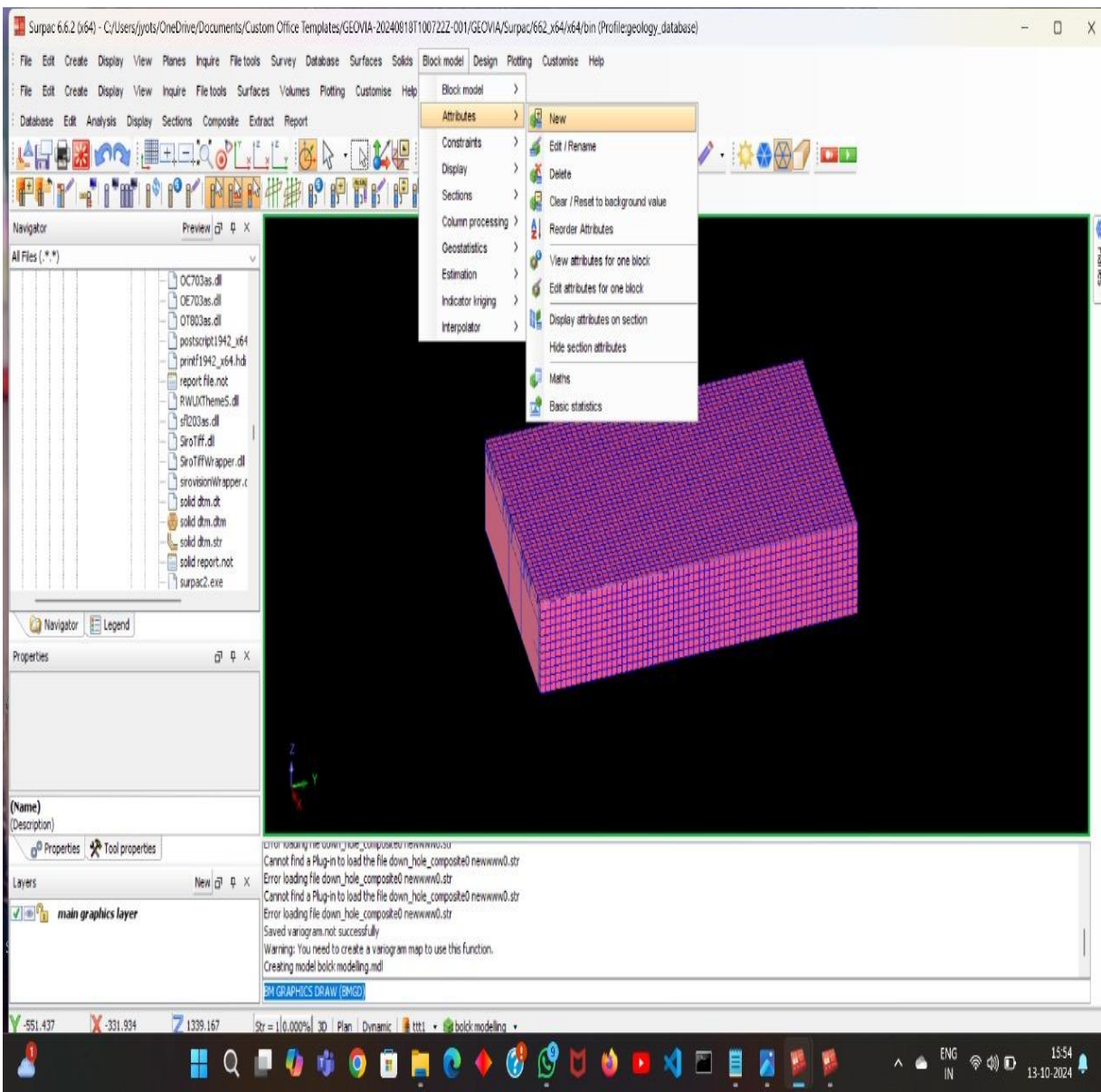


A NEW BLOCK MODEL WILL BE APPEAR LIKE THIS

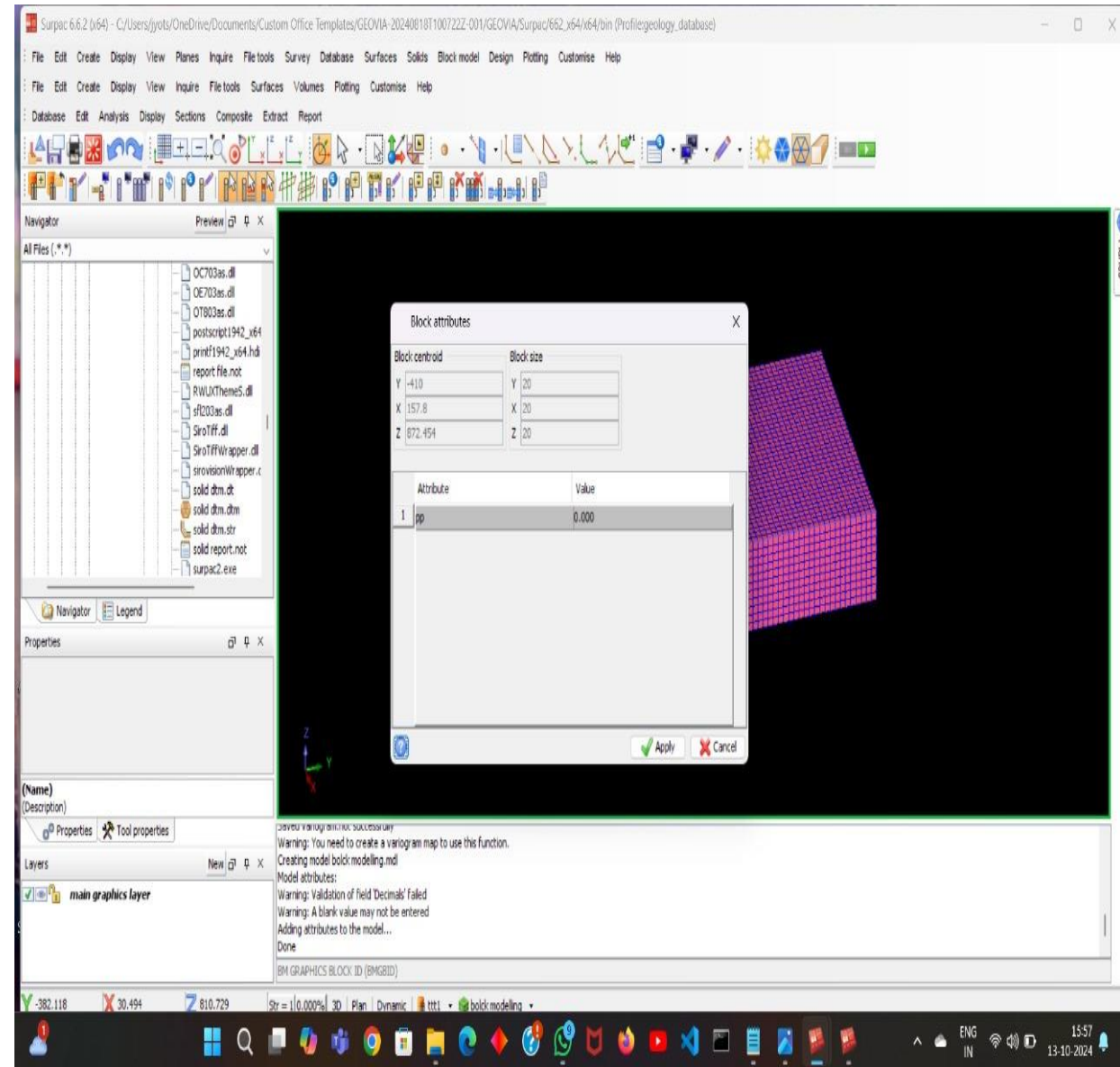
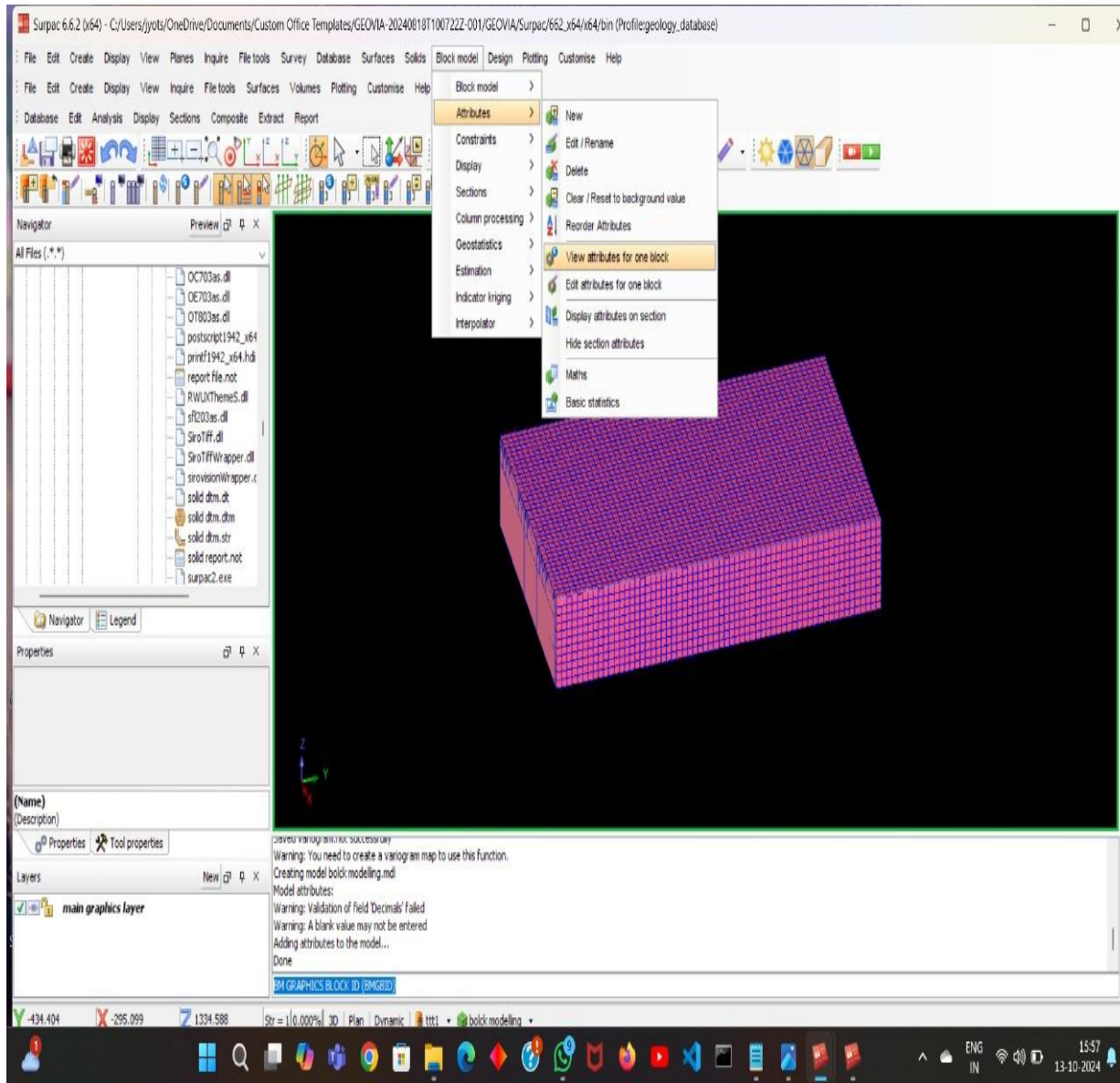


PARTIAL PERCENTAGE :- Go to Block model >> Attributes >> New

In next pop up click on apply

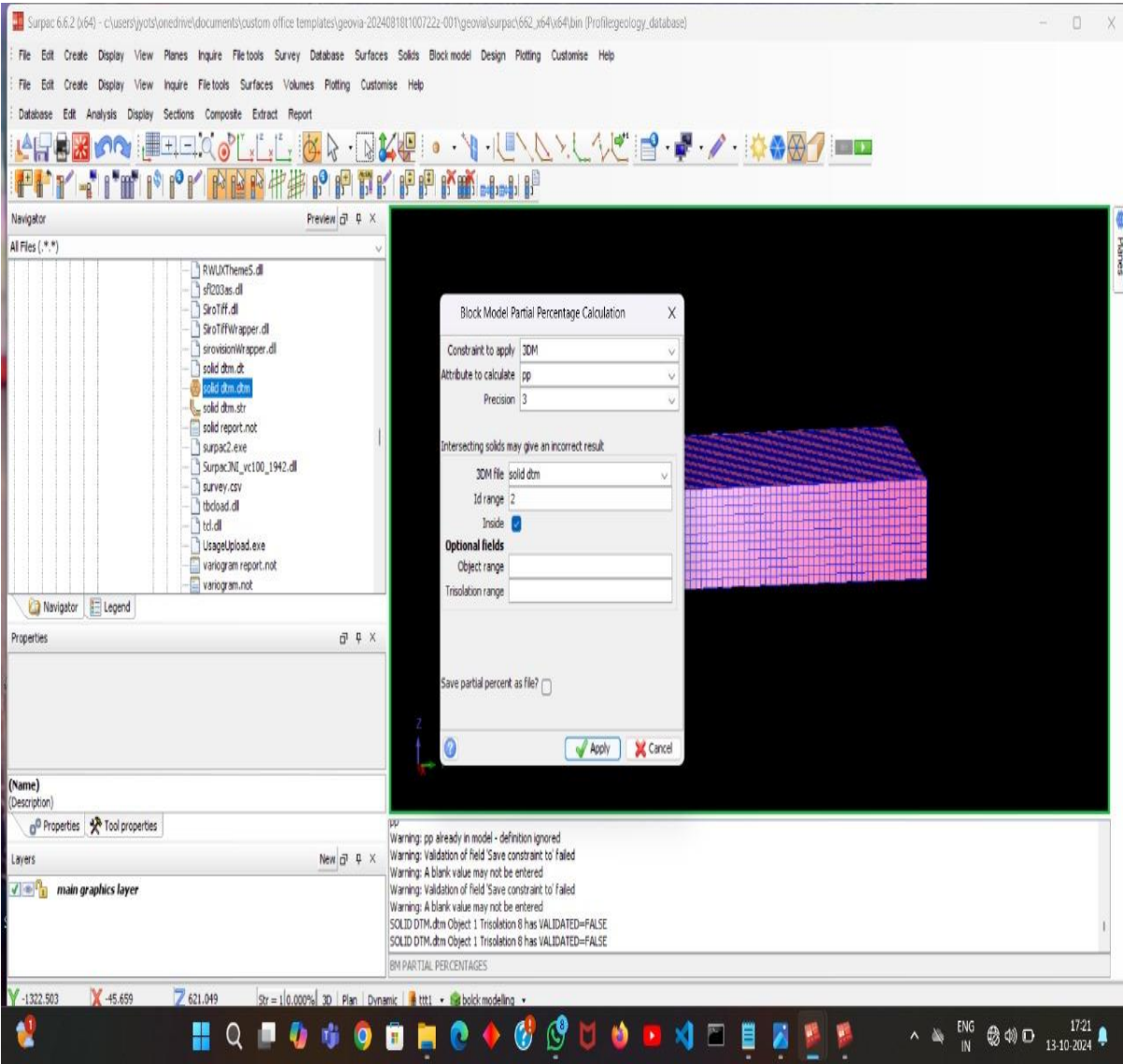
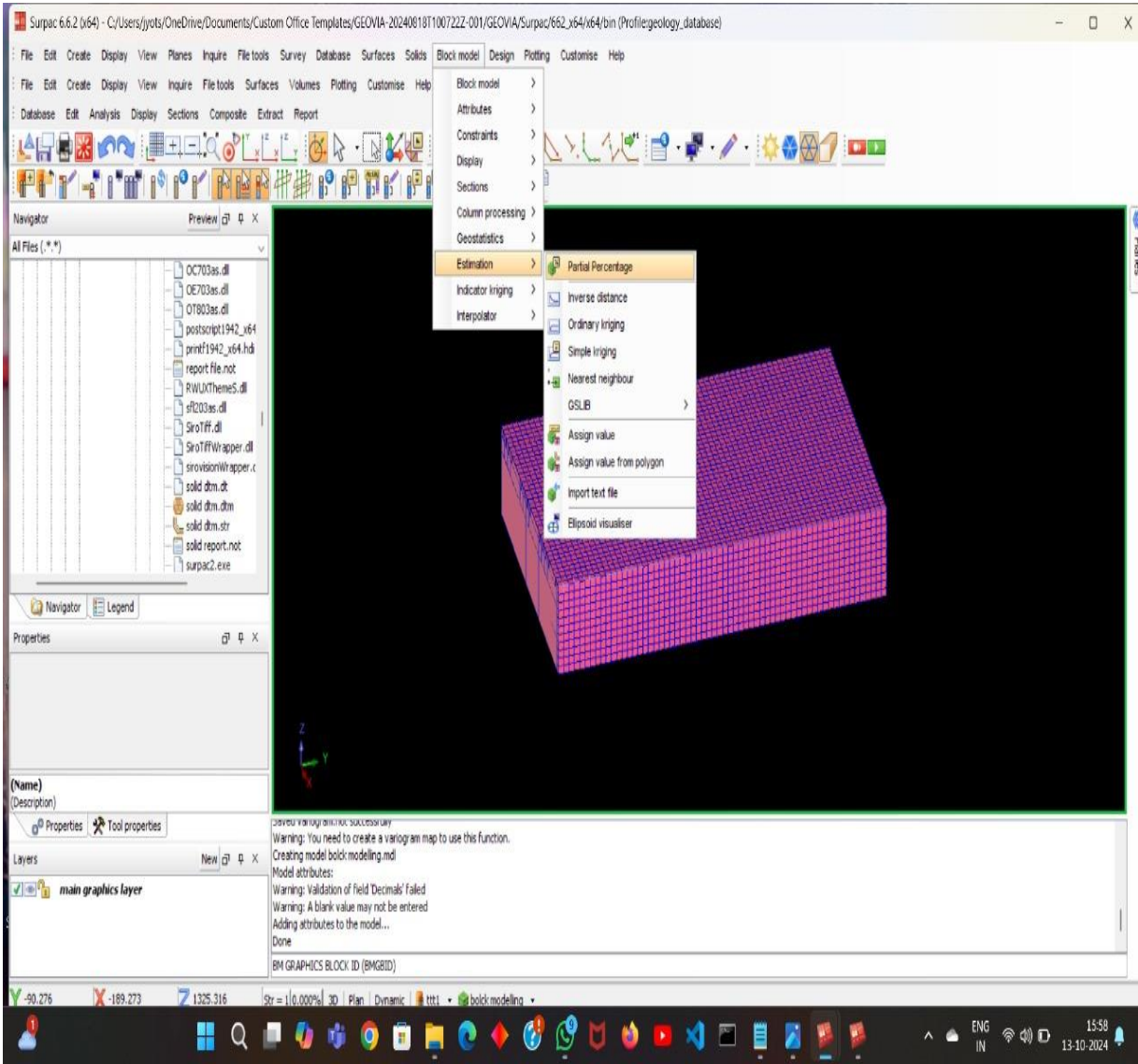


To check whether the attribute is added or not:- Go to Block model >> Attributes >> view attribute for one block >> Click on any block.
Attribute is because the value showing 0.

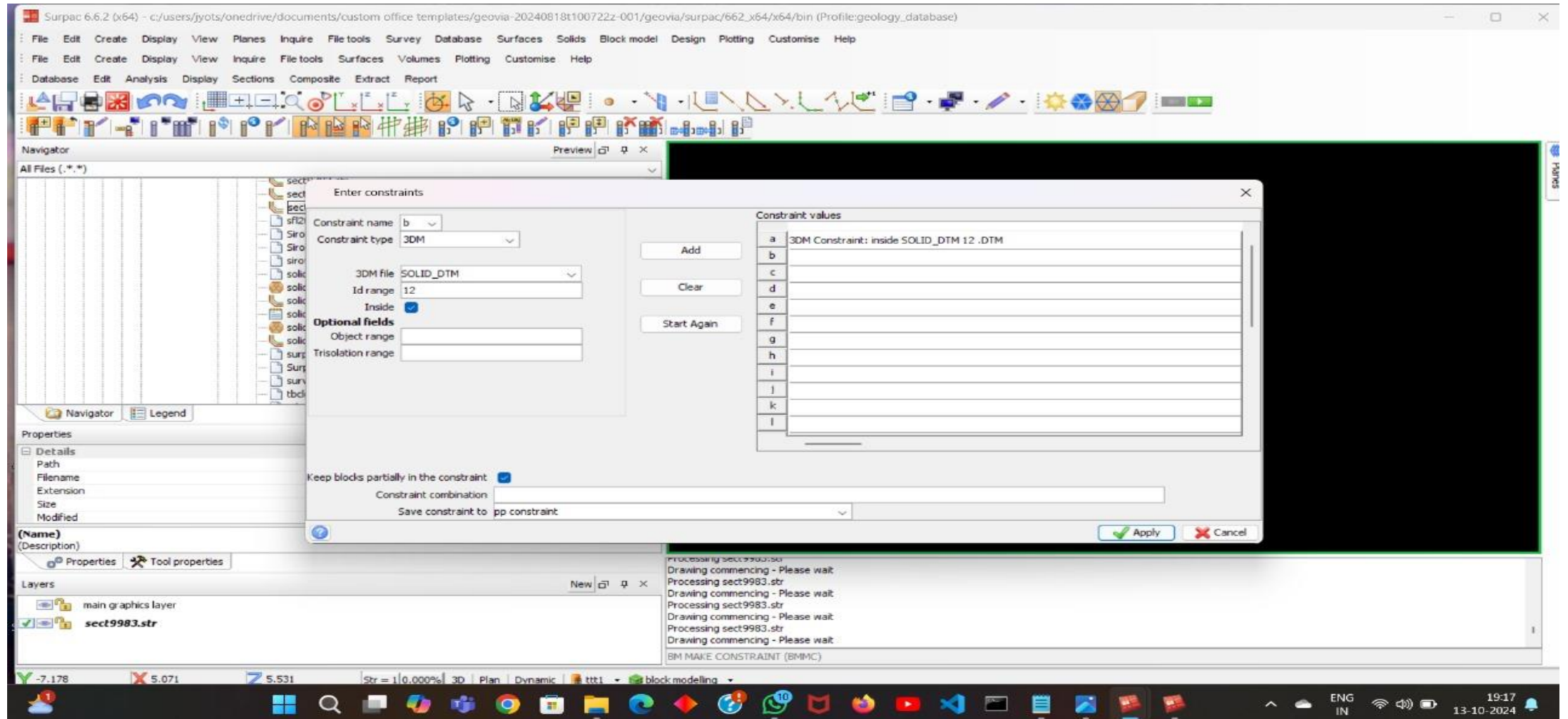


Go to Block model >> Estimation >> Partial percentage

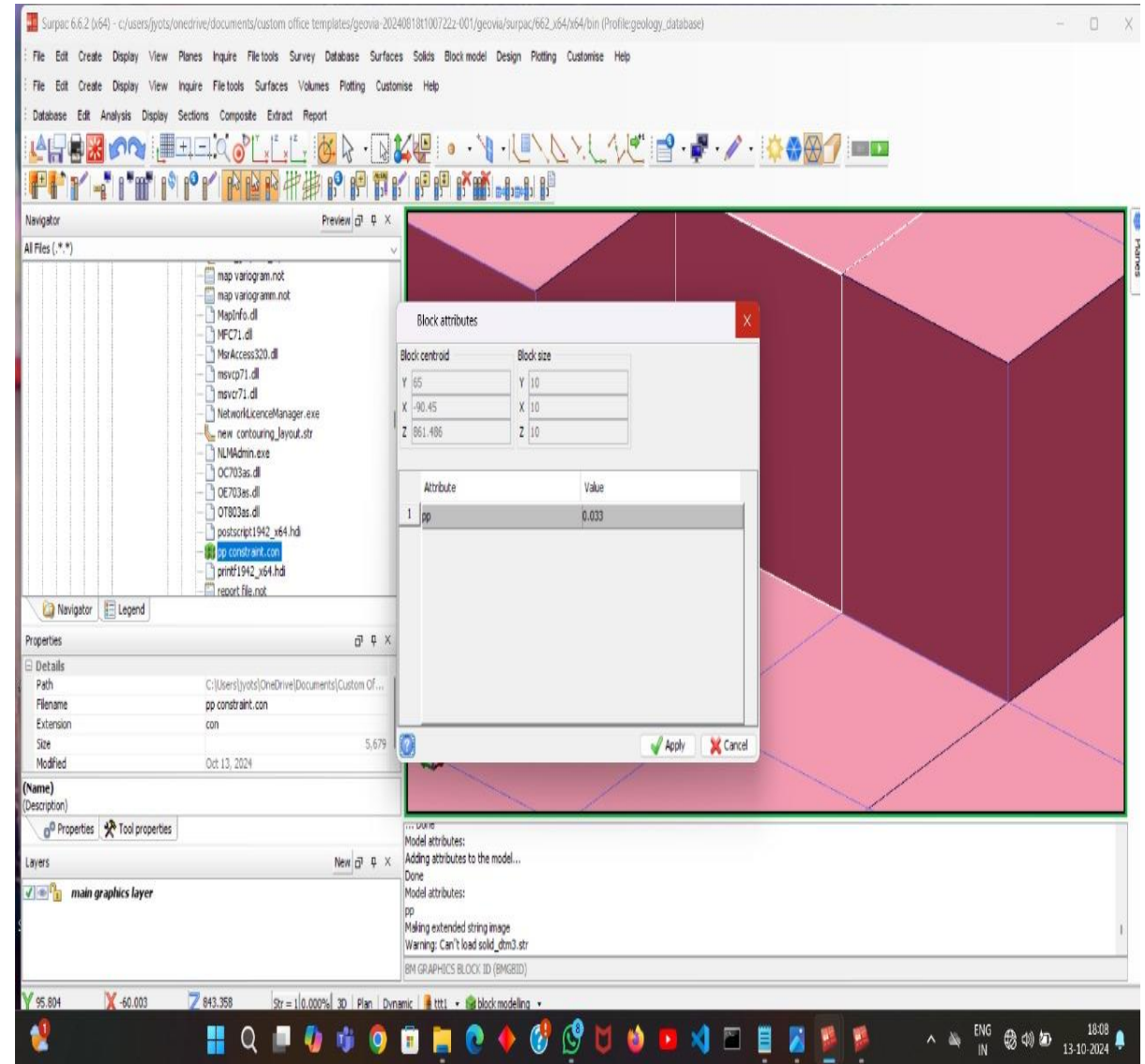
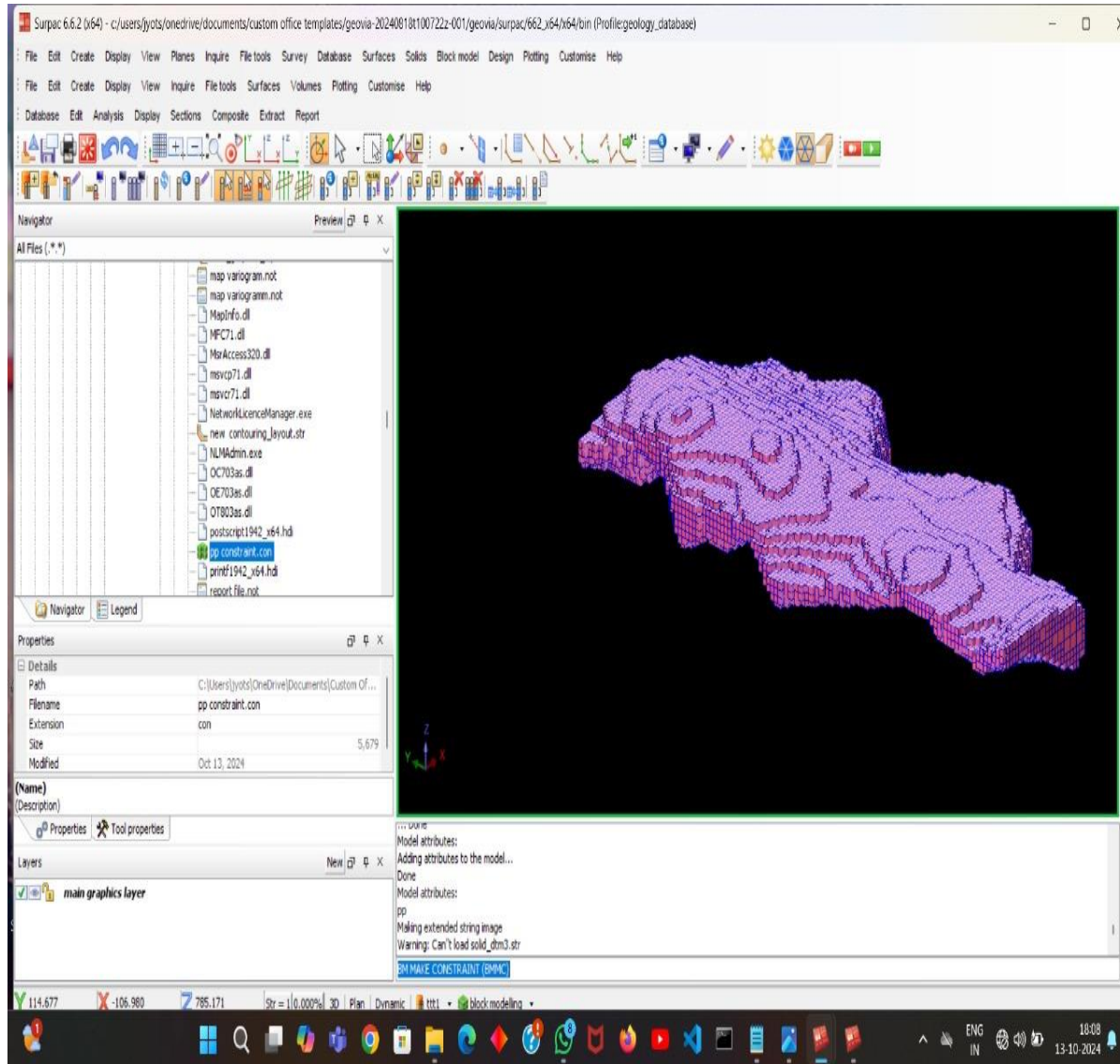
Select the solid model dtm in 3DM file .
Click on apply.



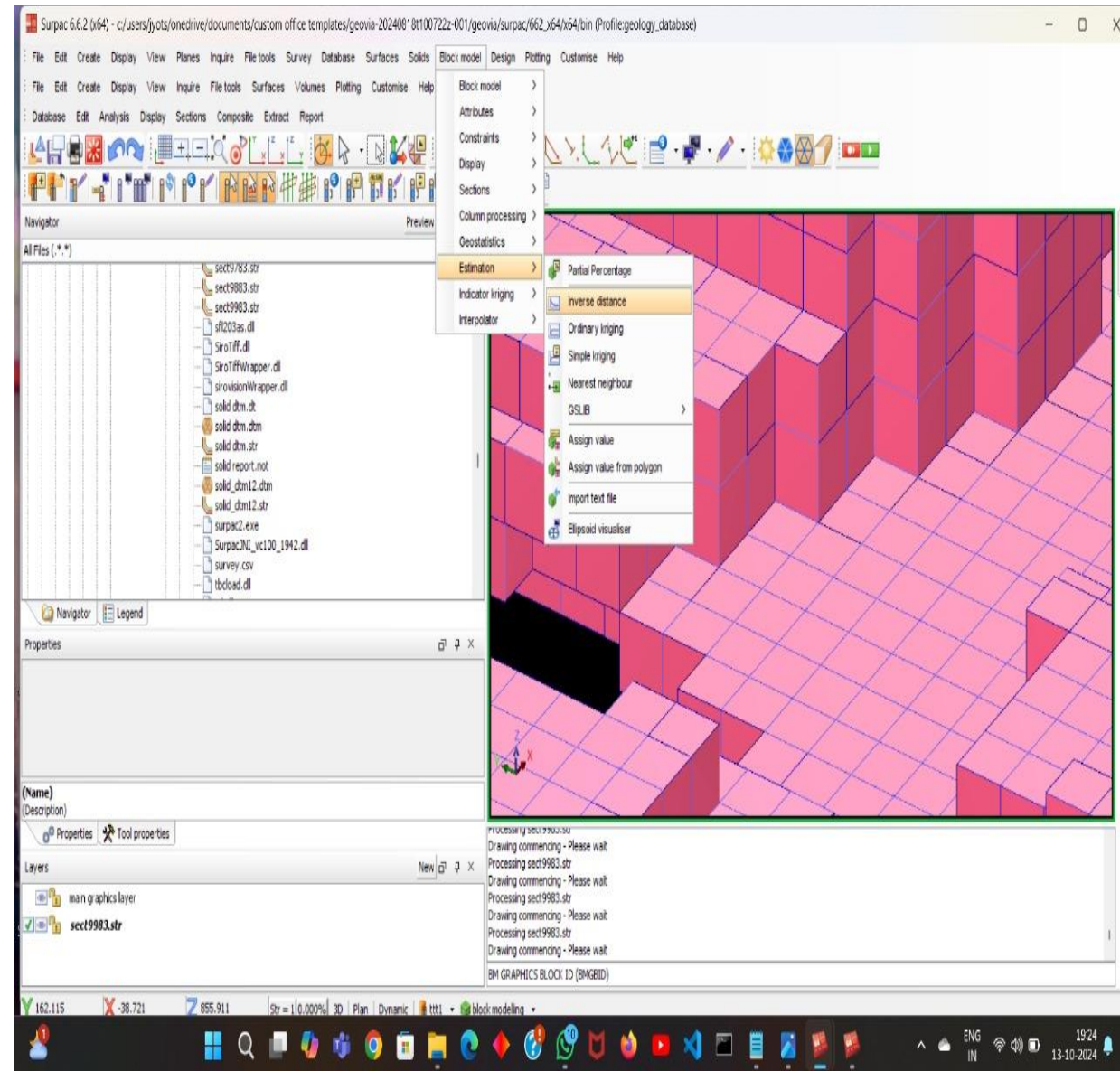
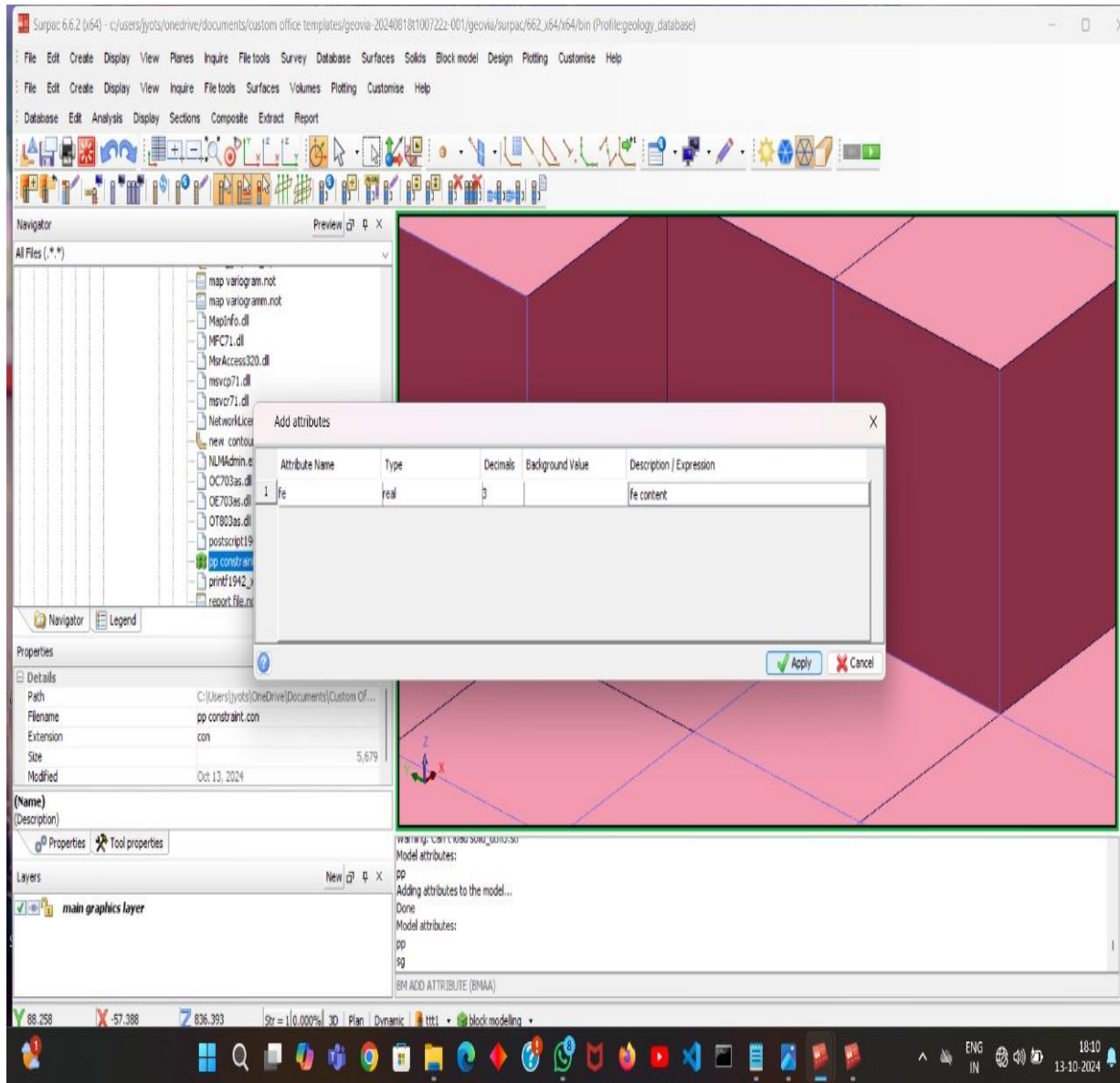
Go to block model >> constraints >> New constraint file
Select the constraint type as 3DM and select the solid model dtm in the 3DM file, click on add, click on apply.



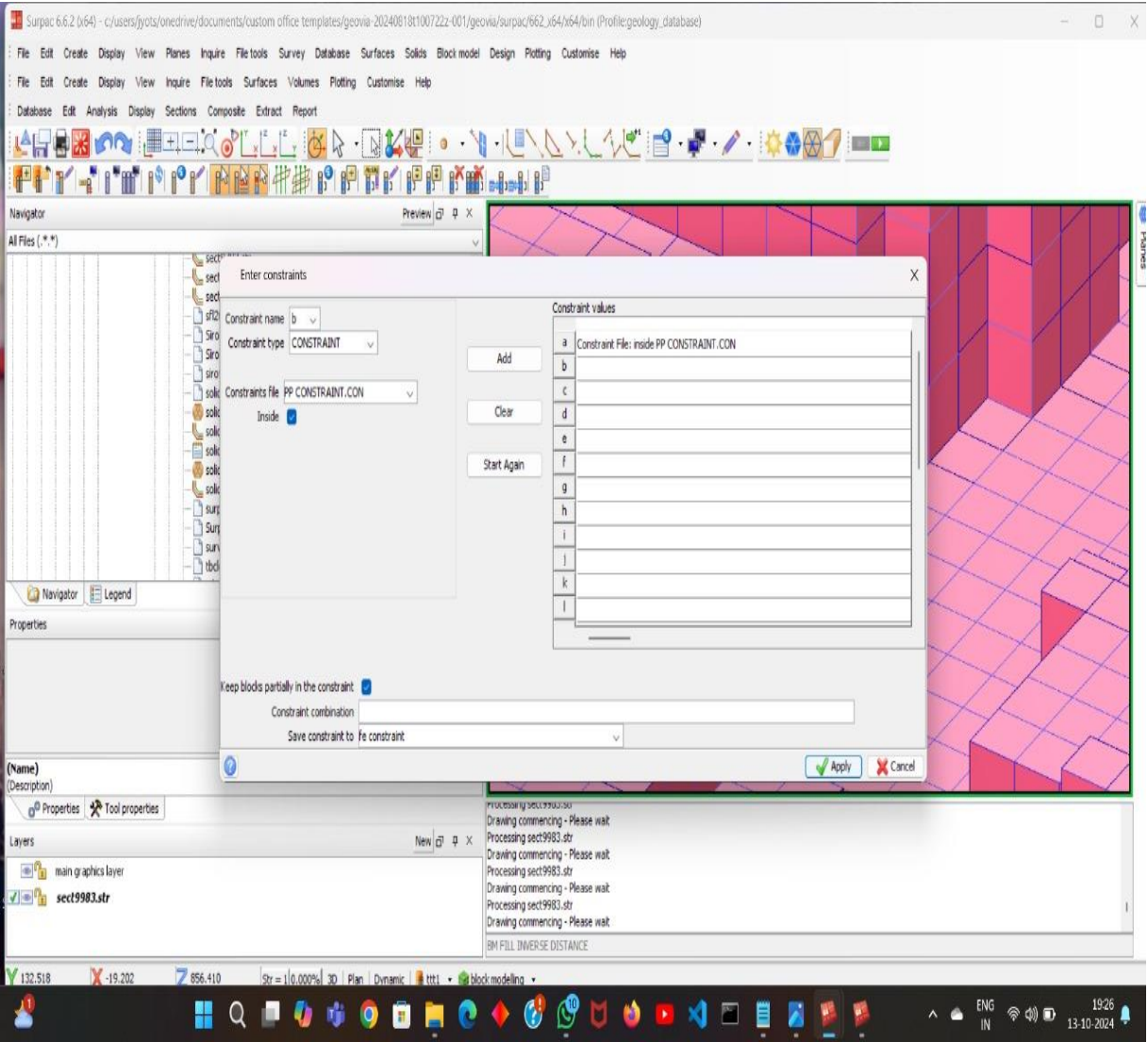
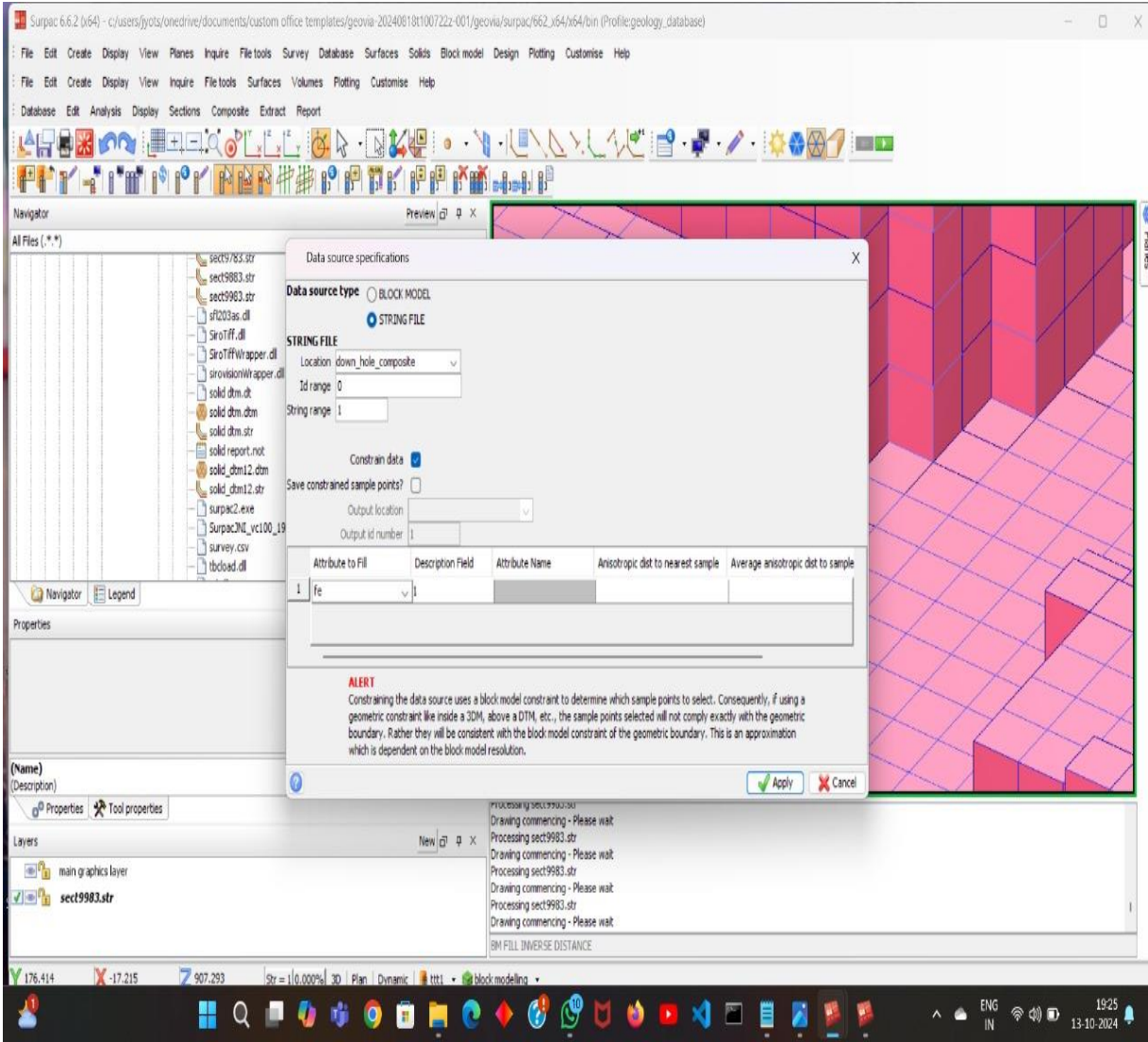
Go to Block model >> Attributes >> view attribute for one block >> Click on any block. It gives the value of Partial Percentage.



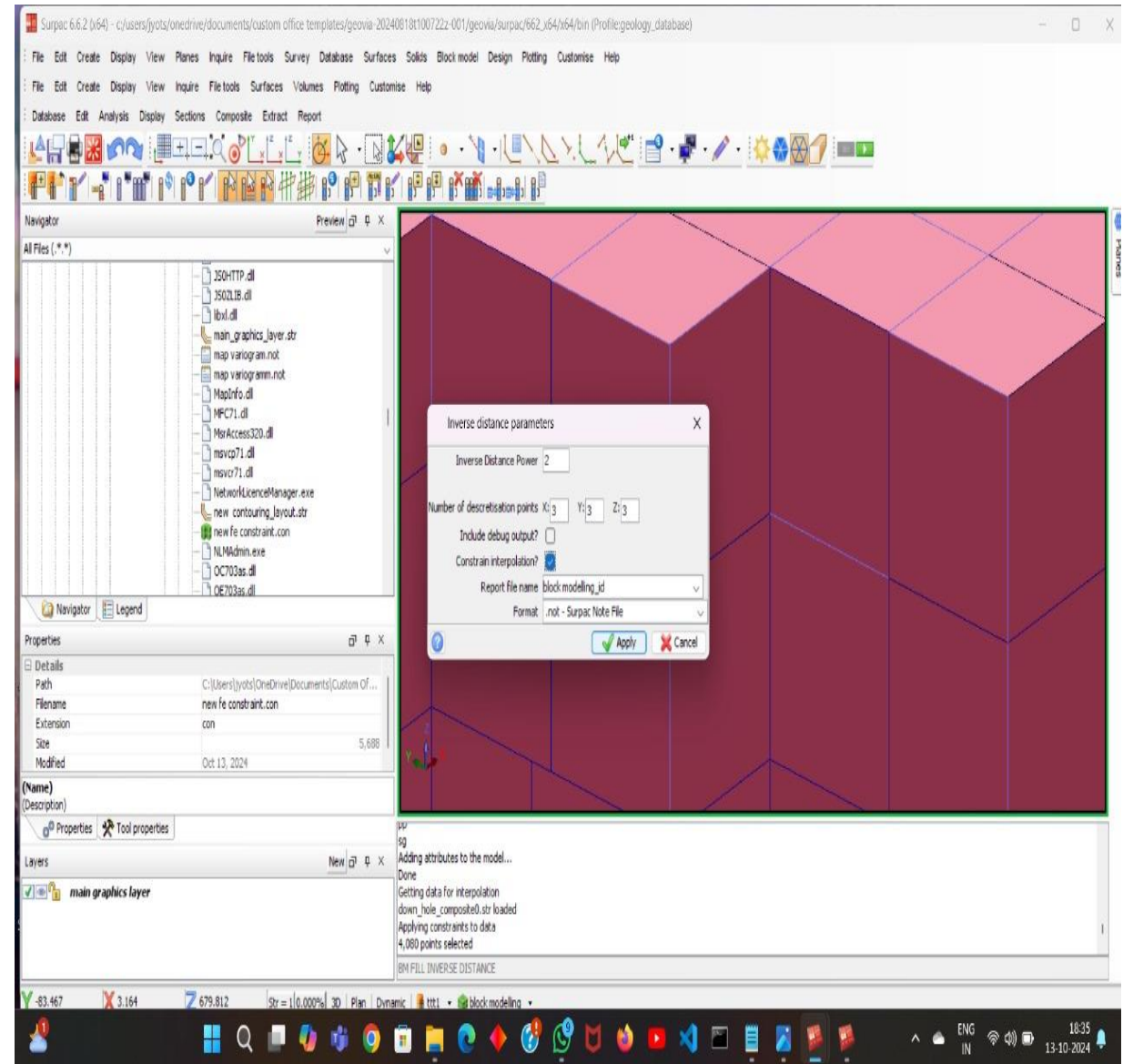
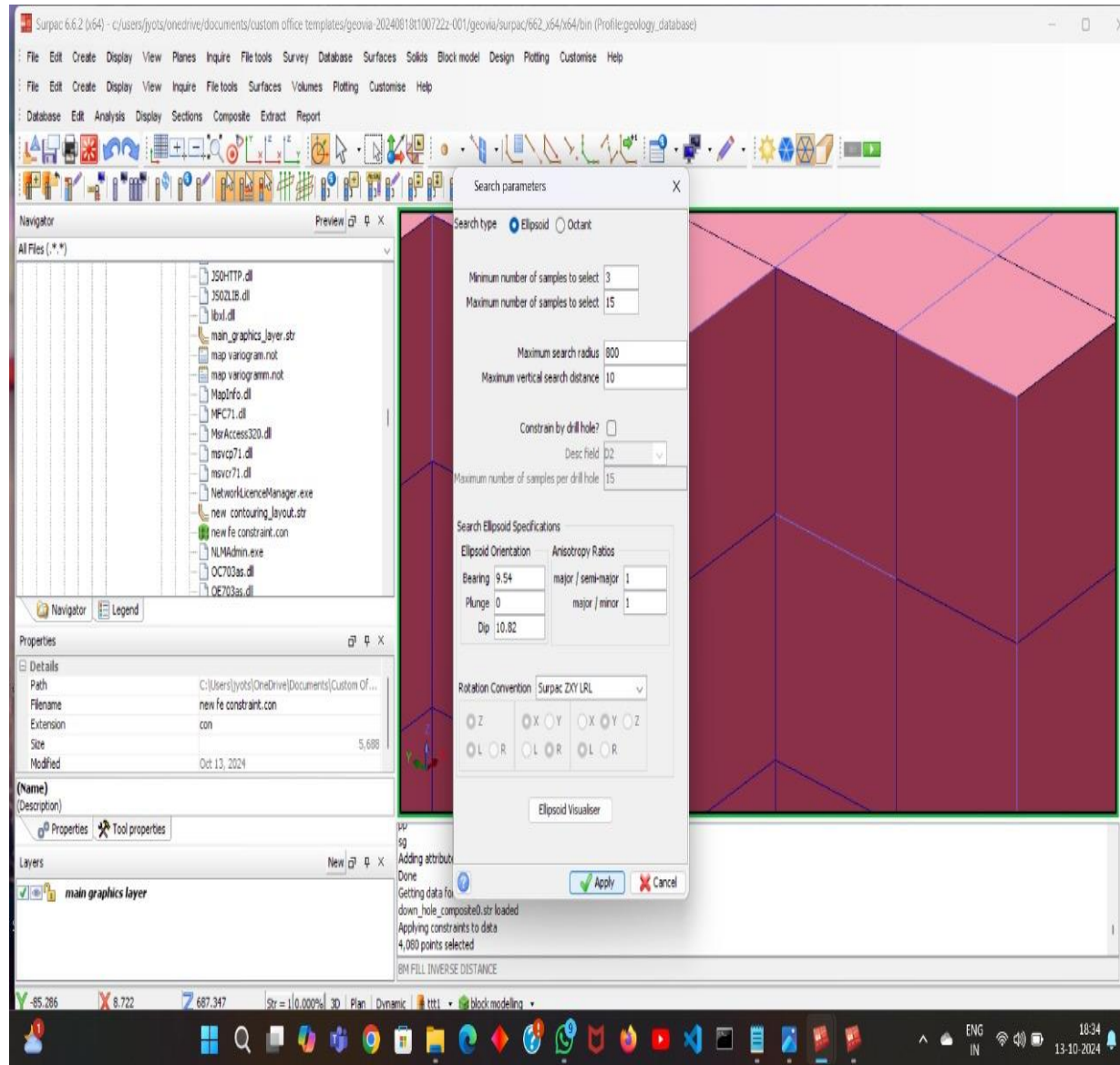
Fe attribute is used to assign the assay value to the block model.
Go to estimation >> Inverse distance >> Pop up>>Click On Apply



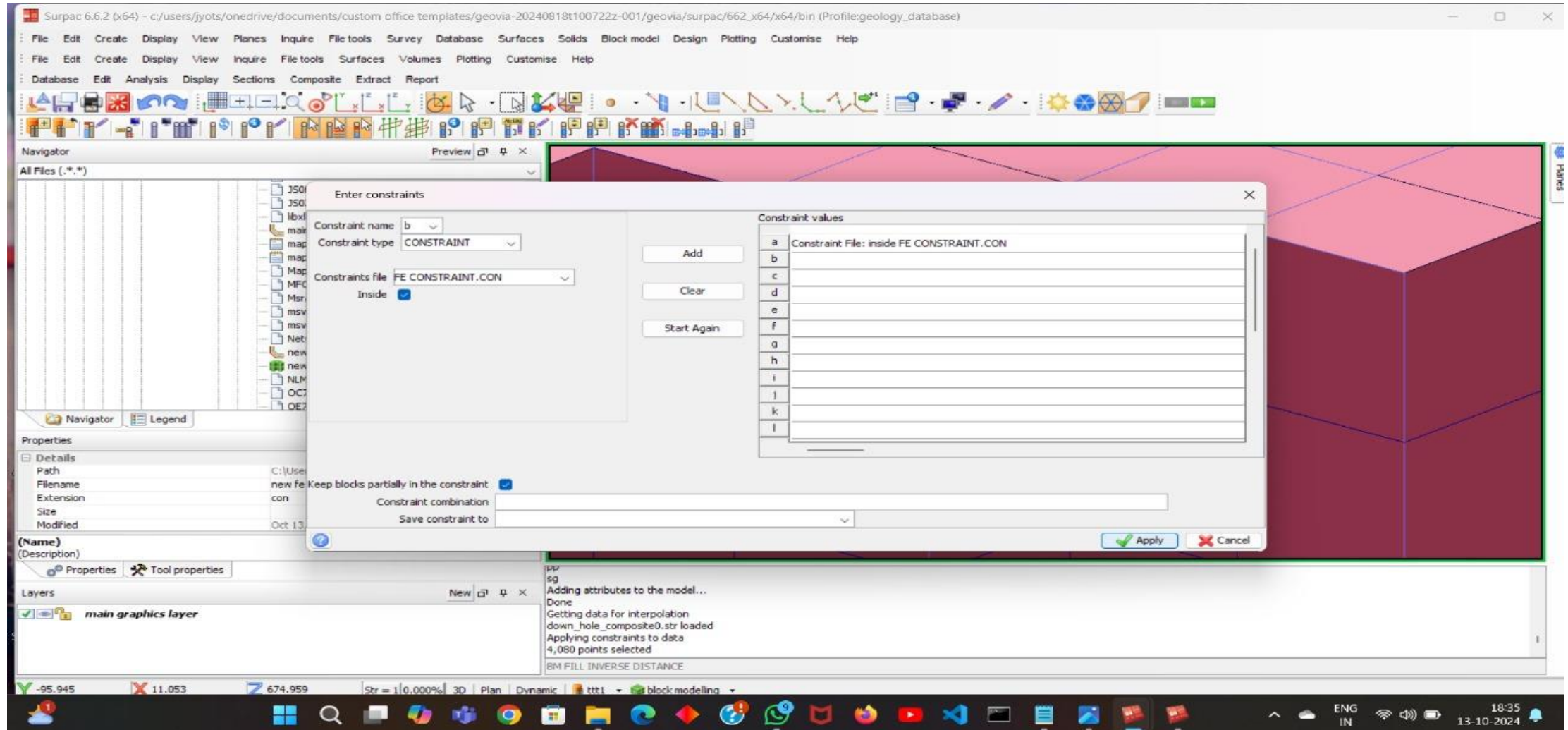
After, A new pop-up will appear to edit Constraints.
Select the pp constraint file for the constraint file, click on add, click on keep blocks partially in the constraint and save the constraint as fe constraint.
After editing save it as fe constraints then click on apply.



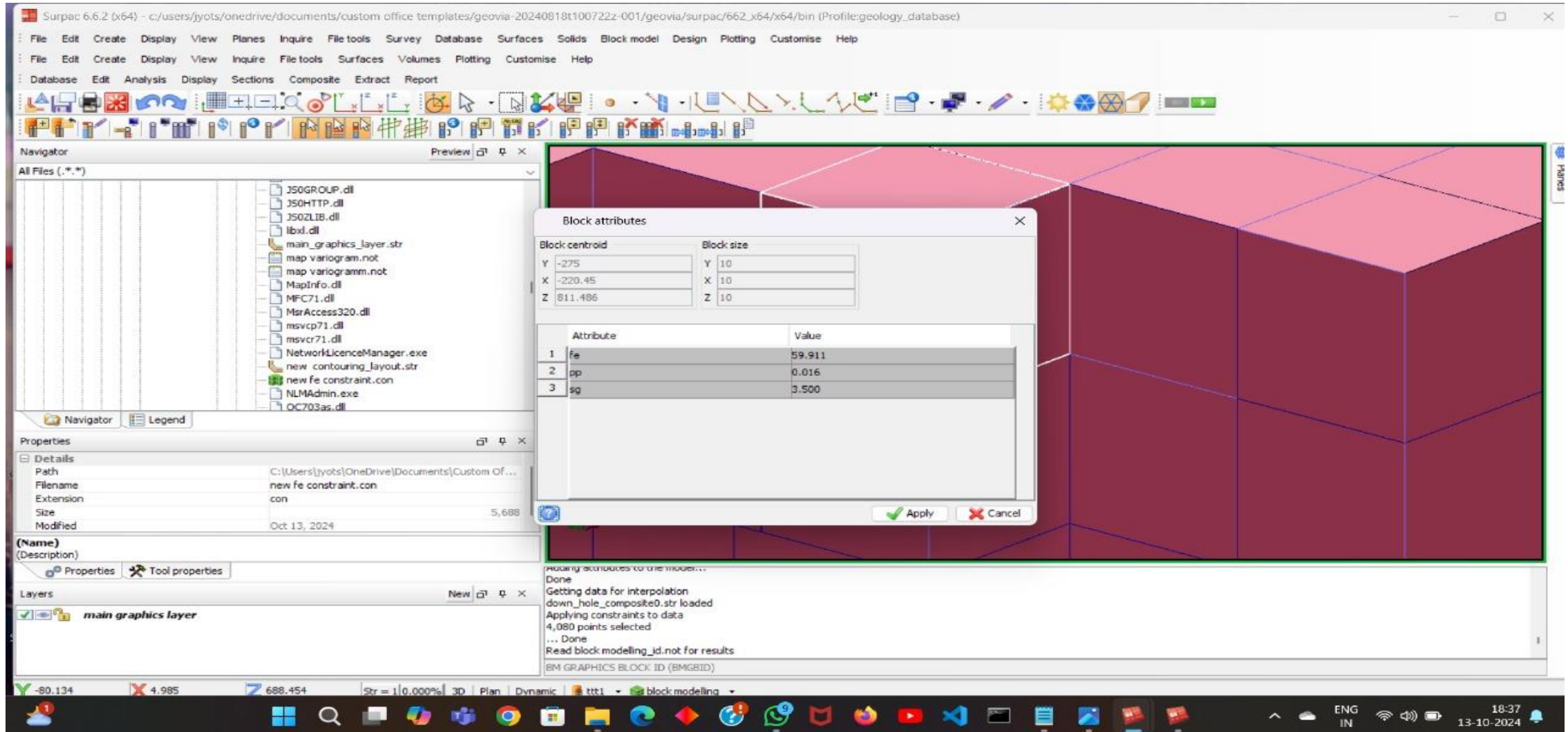
Enter the bearing and dip from the anisotropy report. Click on Apply



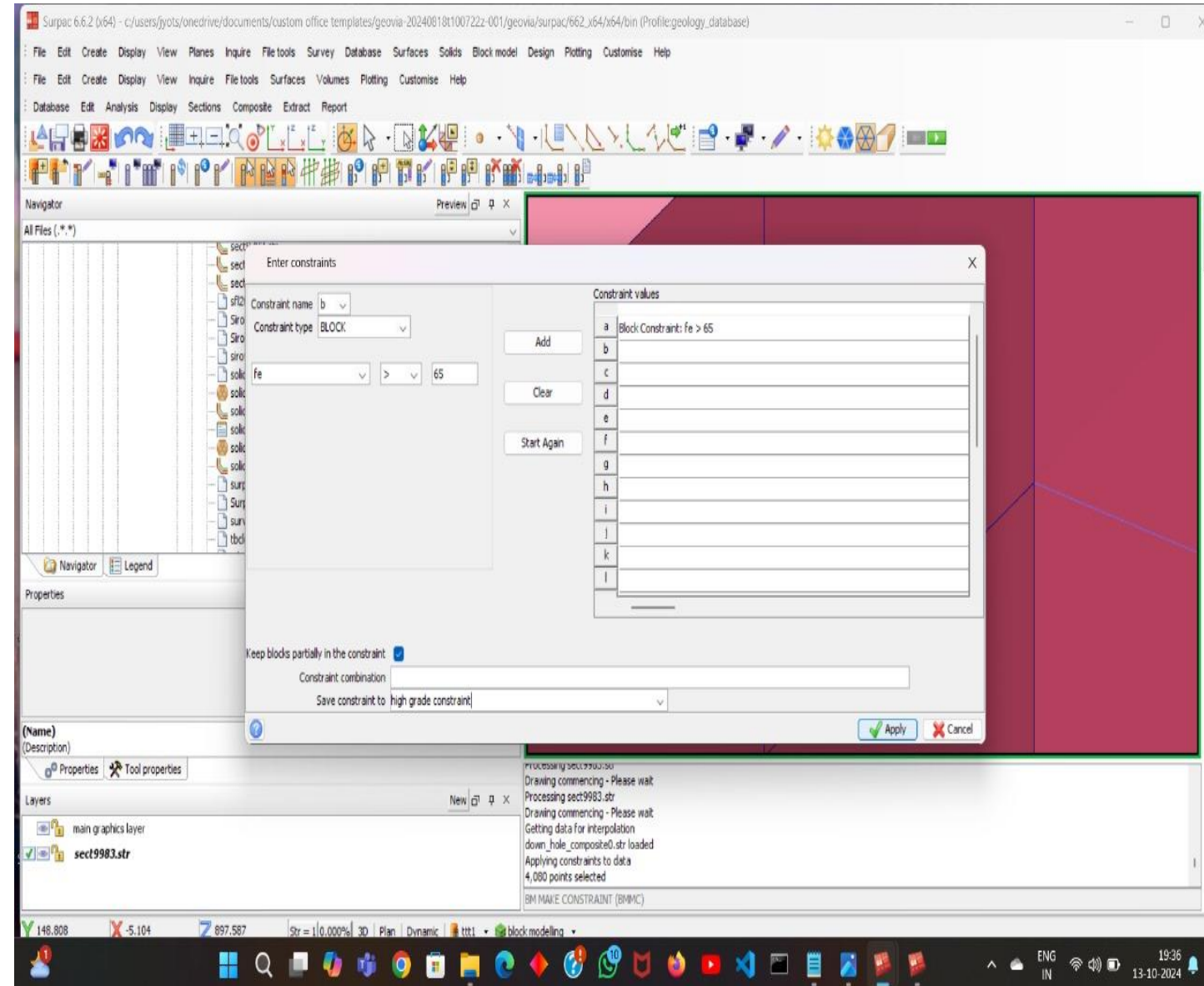
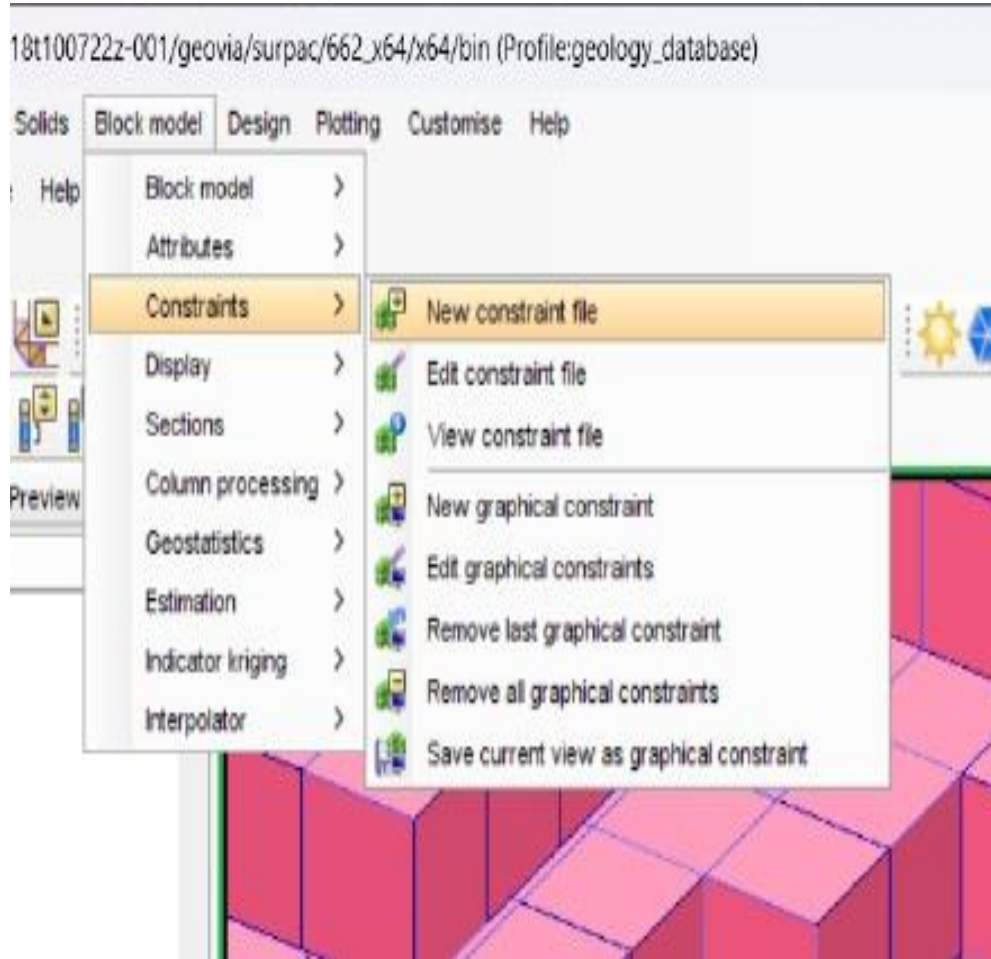
Select the fe constraint for the constraint file, click on add, and click on keep blocks partially in the constraint.
Click on Apply.



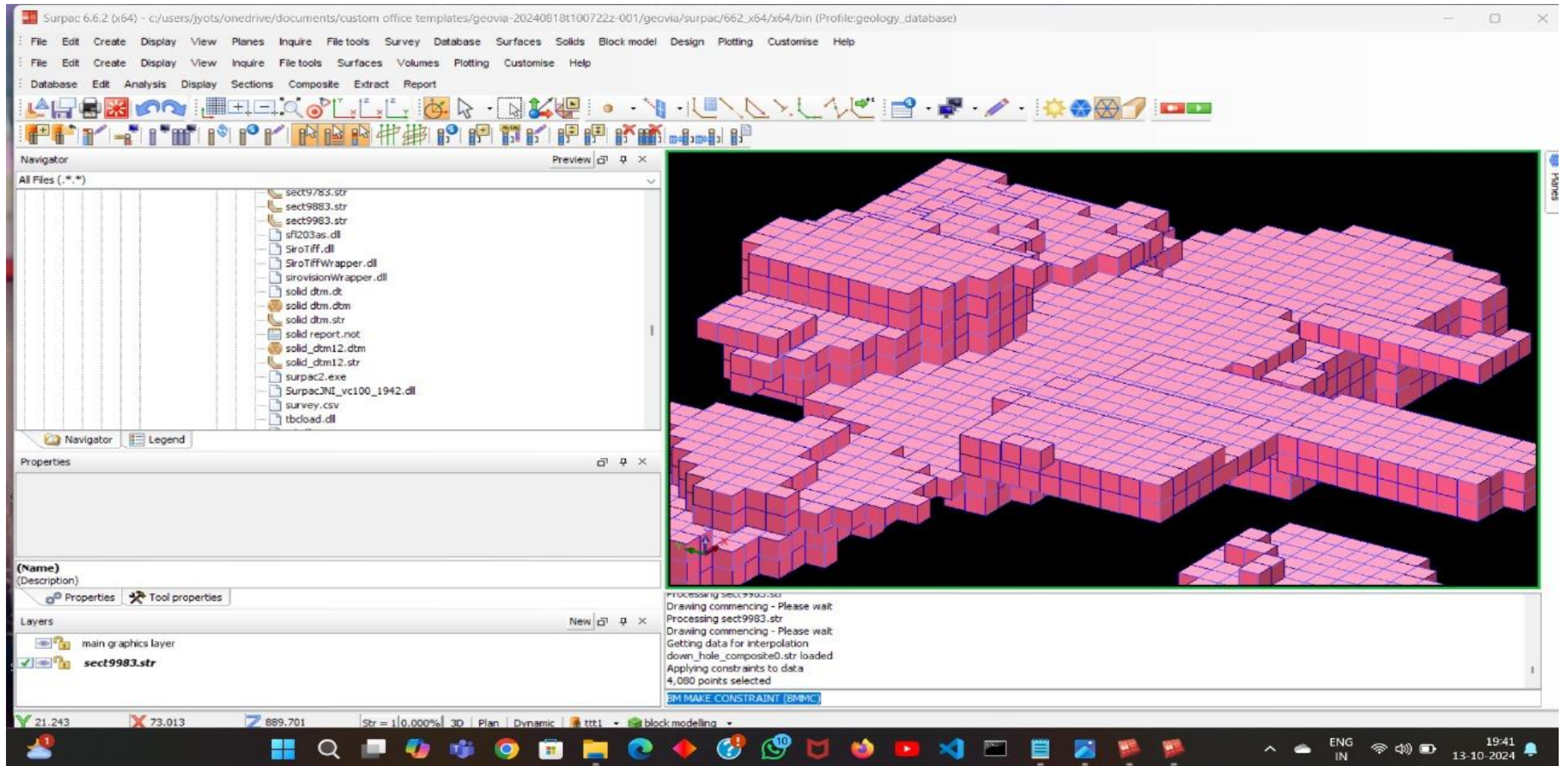
Go to Block model >> Attributes >> view attribute for one block >> Click on any block.



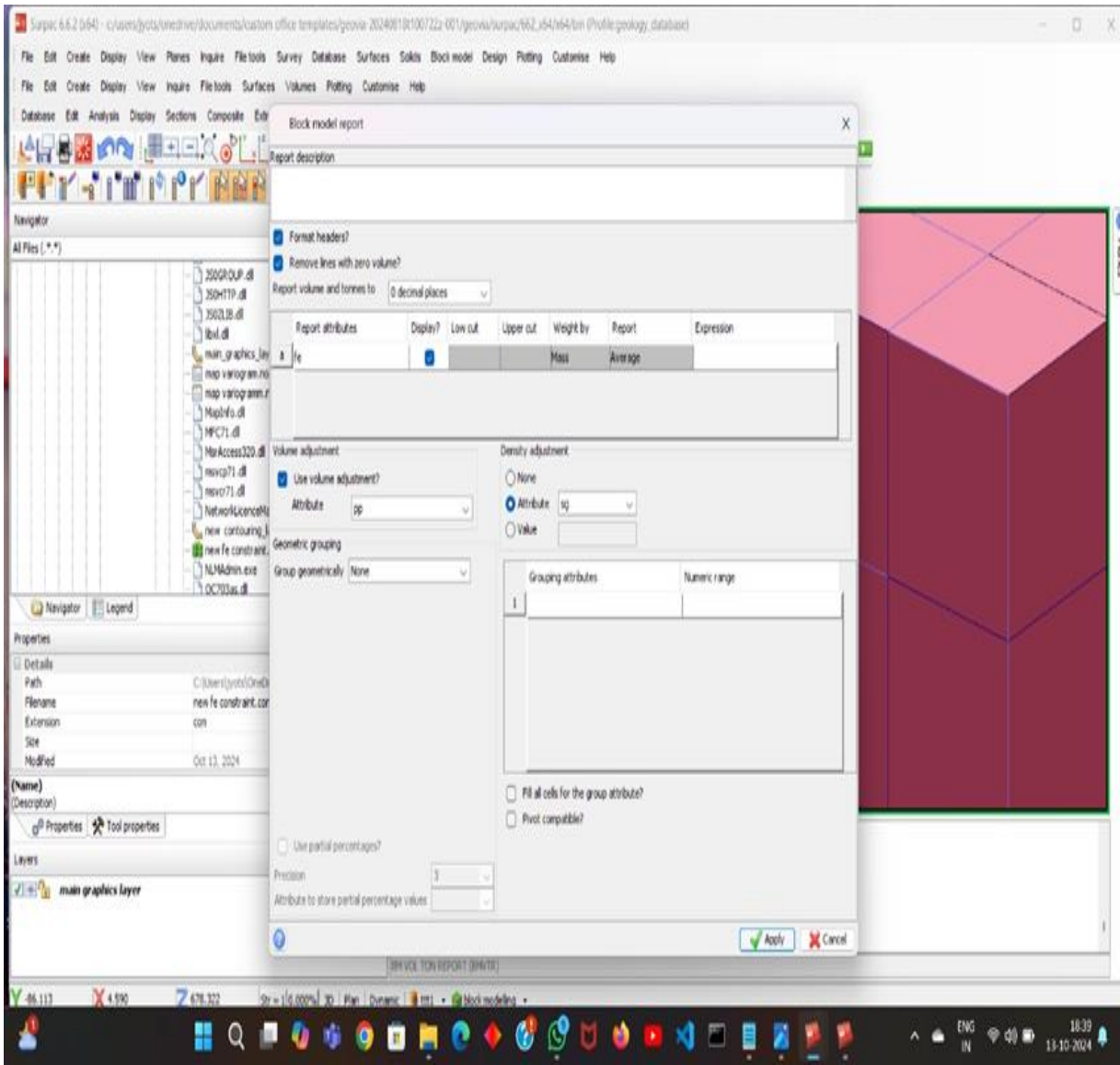
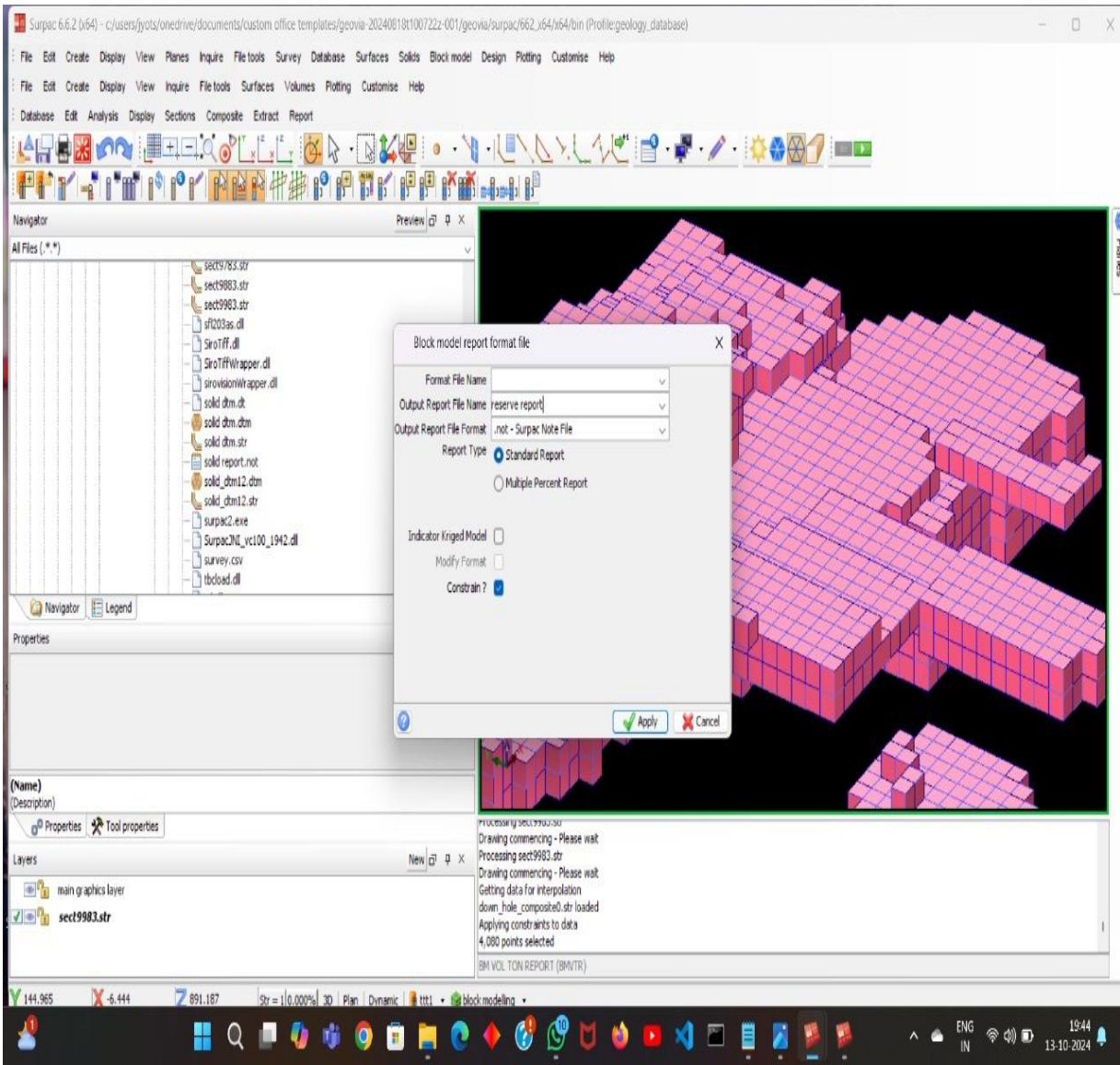
Block model >> constraints >> new constraint file >> Pop up
Select a block in constraint type, select fe grade >65, and Click on add.
Give a name to the output file as a high-grade ore constraint.
Click on Apply



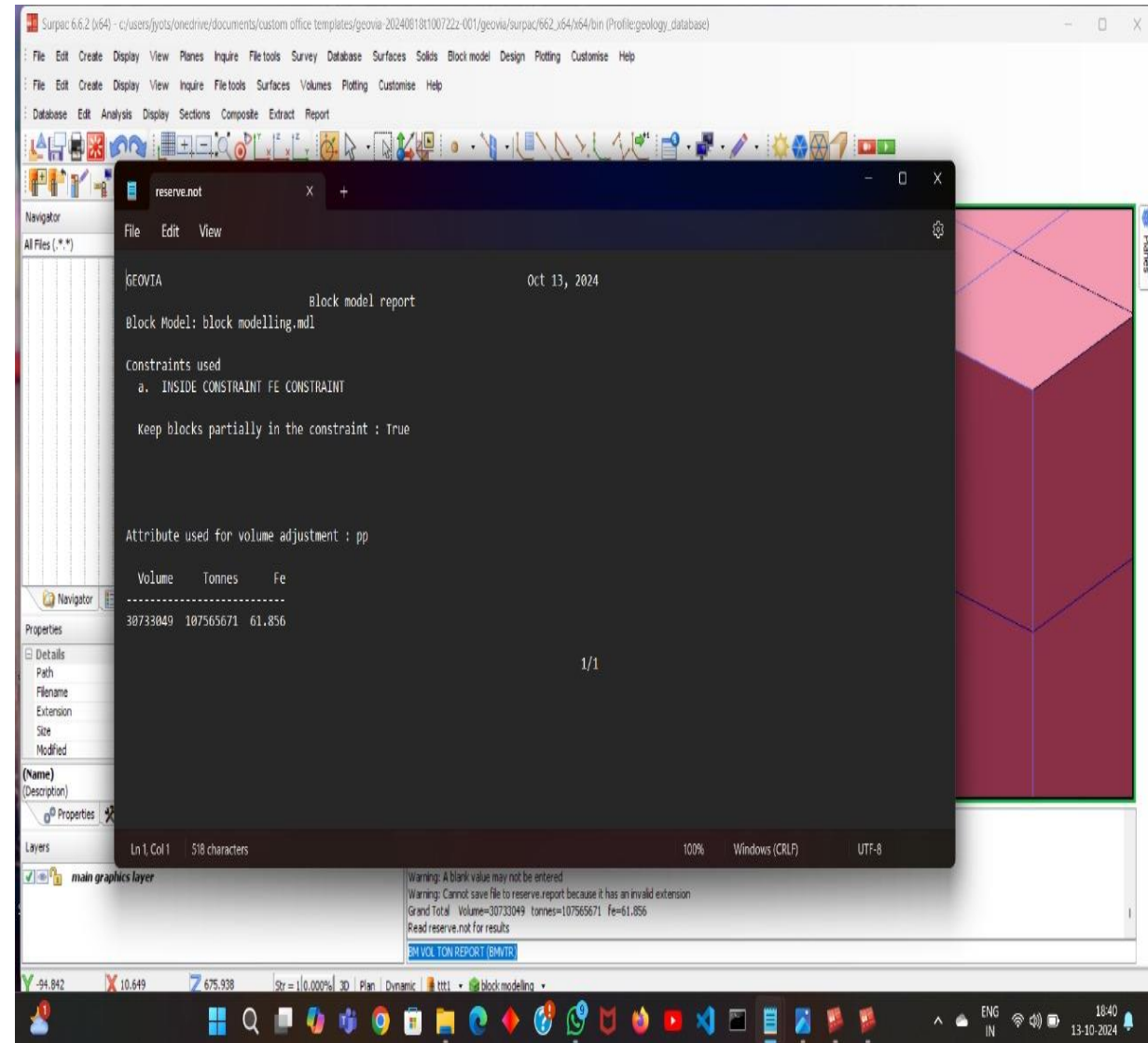
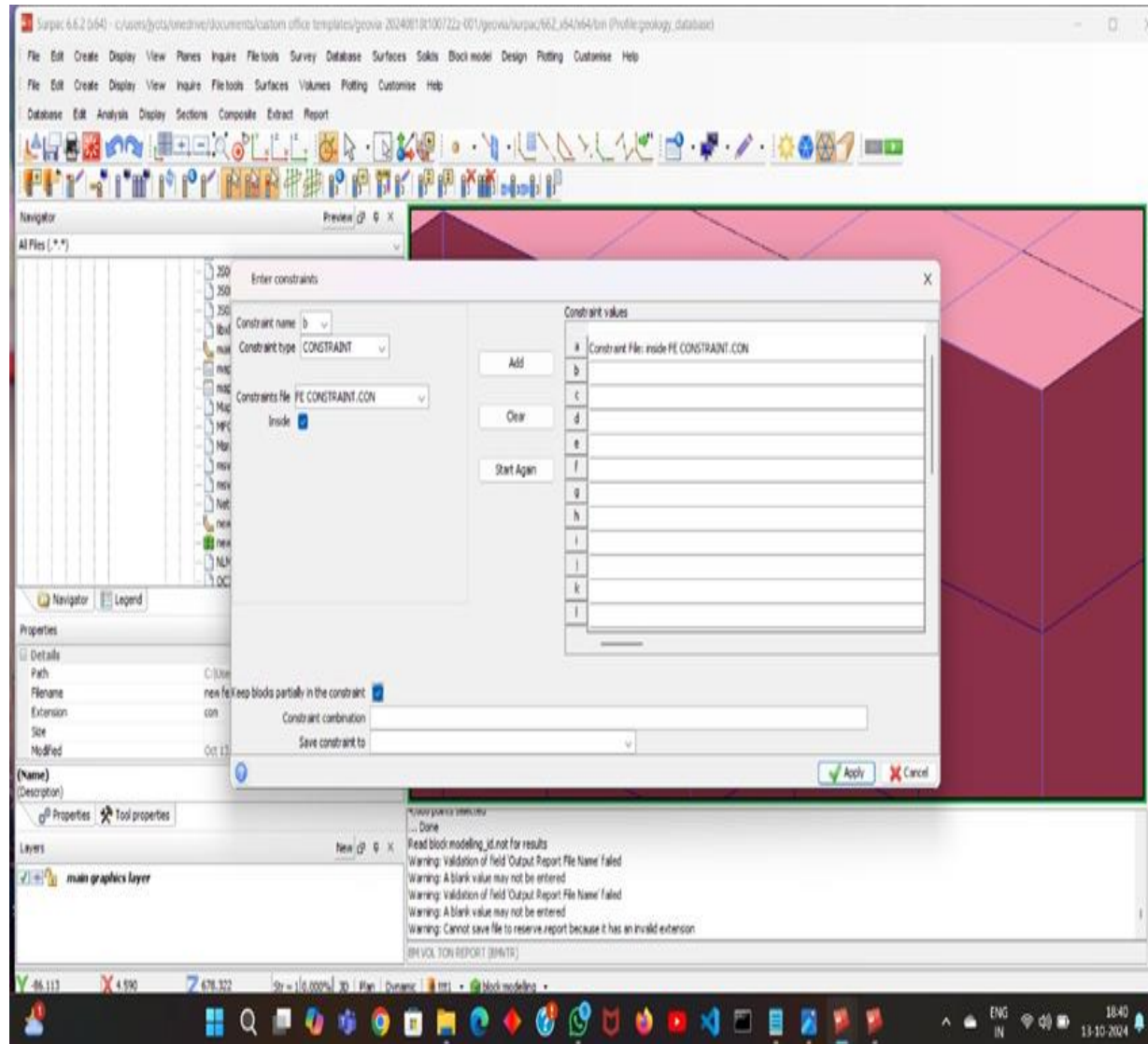
After Constraining with high-grade ore:



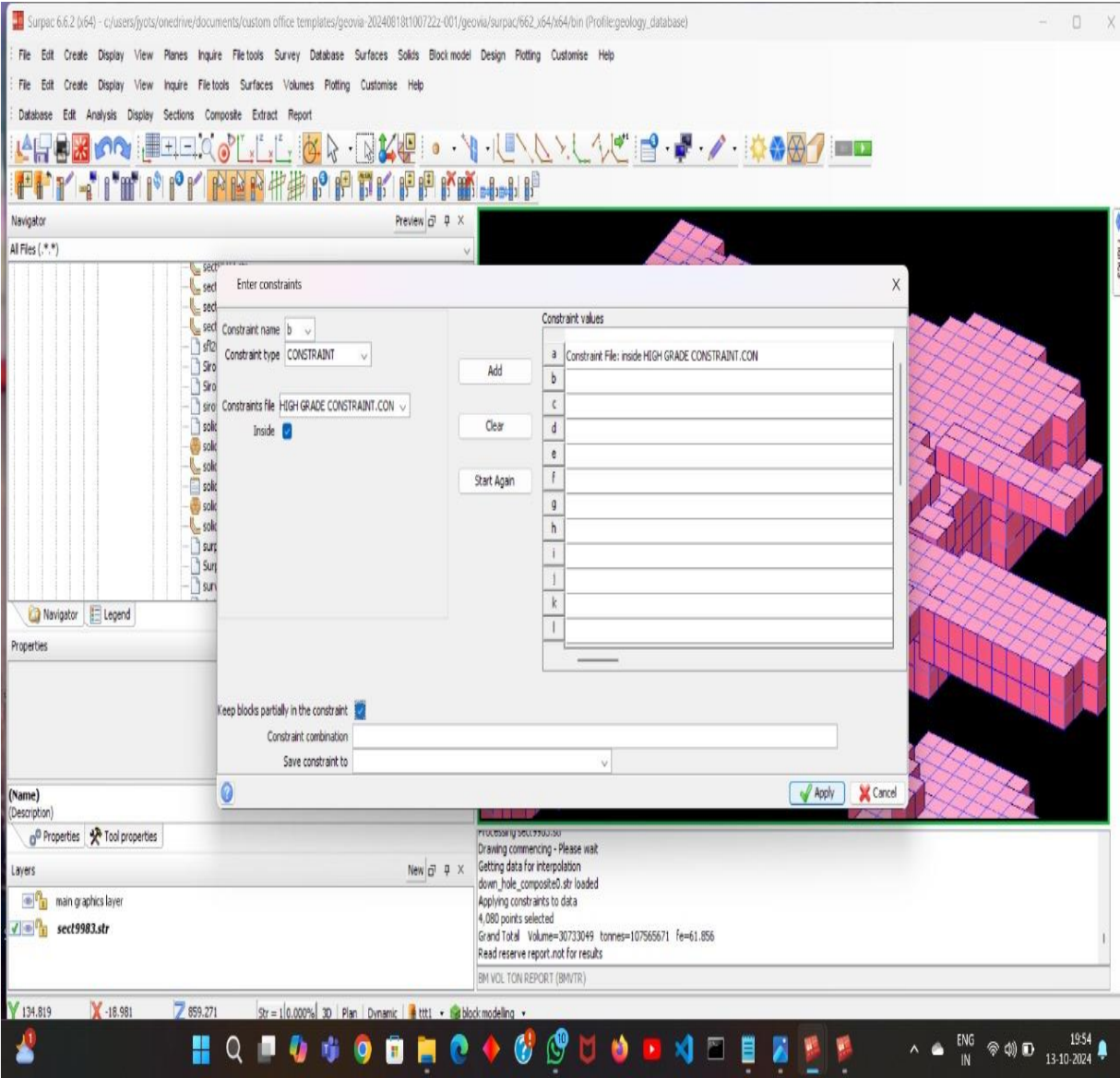
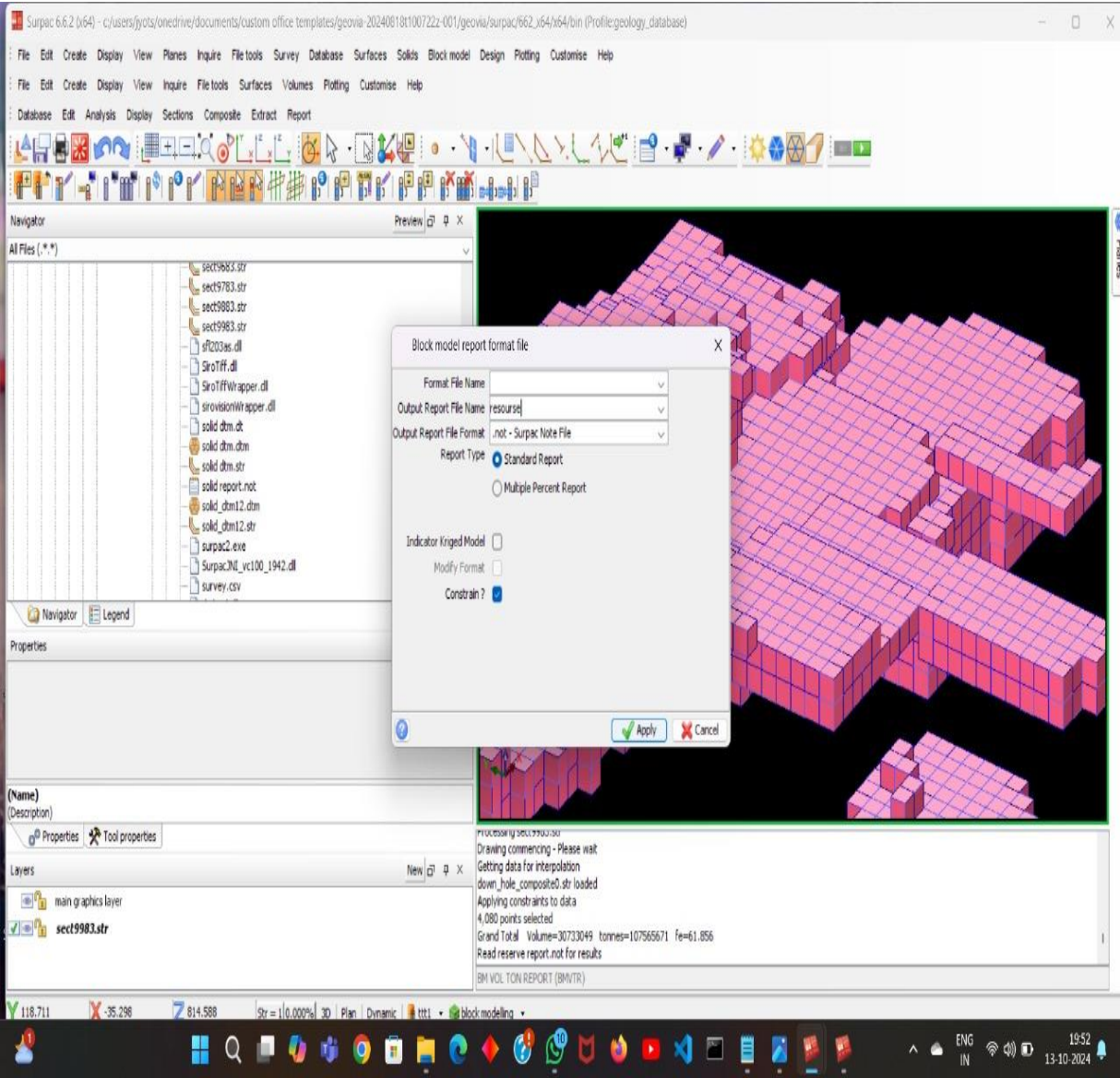
Go to Block model >> report>>Click on Apply
Select fe as report attributes, click on use volume adjustment and select pp attribute, density attribute will be the sg attribute.
Click on apply



Select the constraint type as a constraint and upload the fe constraint file
click on add >> Click on apply



Resource Report



This report will be your resource report because we considered only high-grade ore blocks in the constraint.

Surpac 6.6.2 (x64) - c:/users/jyots/onedrive/documents/custom office templates/geovia-20240818t100722z-001/geovia/surpac/662_x64/x64/bin (Profile:geology_database)

File Edit Create Display View Planes Inquire File tools Survey Database Surfaces Solids Block model Design Plotting Customise Help

File Edit Database

Database

Navigator

All Files (*.*)

Properties

(Name)
(Description)

Layers

main graphics layer

sect9983.str

21.570 20.362 802.197 Str = 1 0.000% 3D Plan Dynamic ttt1 block modelling

resourse.not

File Edit View

GEOVIA

Block model report

Block Model: block modelling.mdl

Constraints used

a. INSIDE CONSTRAINT HIGH GRADE CONSTRAINT

Keep blocks partially in the constraint : True

Attribute used for volume adjustment : pp

Volume	Tonnes	Fe
3695318	12933614	65.532

1/1

Ln 1, Col 1 520 characters 100% Windows (CRLF) UTF-8

down_hole_composite0.str loaded

Applying constraints to data

4,080 points selected

Grand Total Volume=30733049 tonnes=107565671 fe=61.856

Read reserve report.not for results

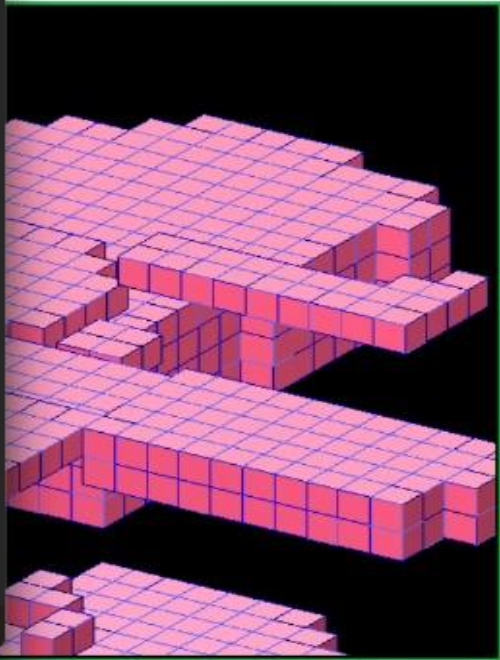
Grand Total Volume=3695318 tonnes=12933614 fe=65.532

Read resourse.not for results

BM VOL TON REPORT (BMVTR)

ENG IN

19:54 13-10-2024



Go to Block model >> Sections >> create

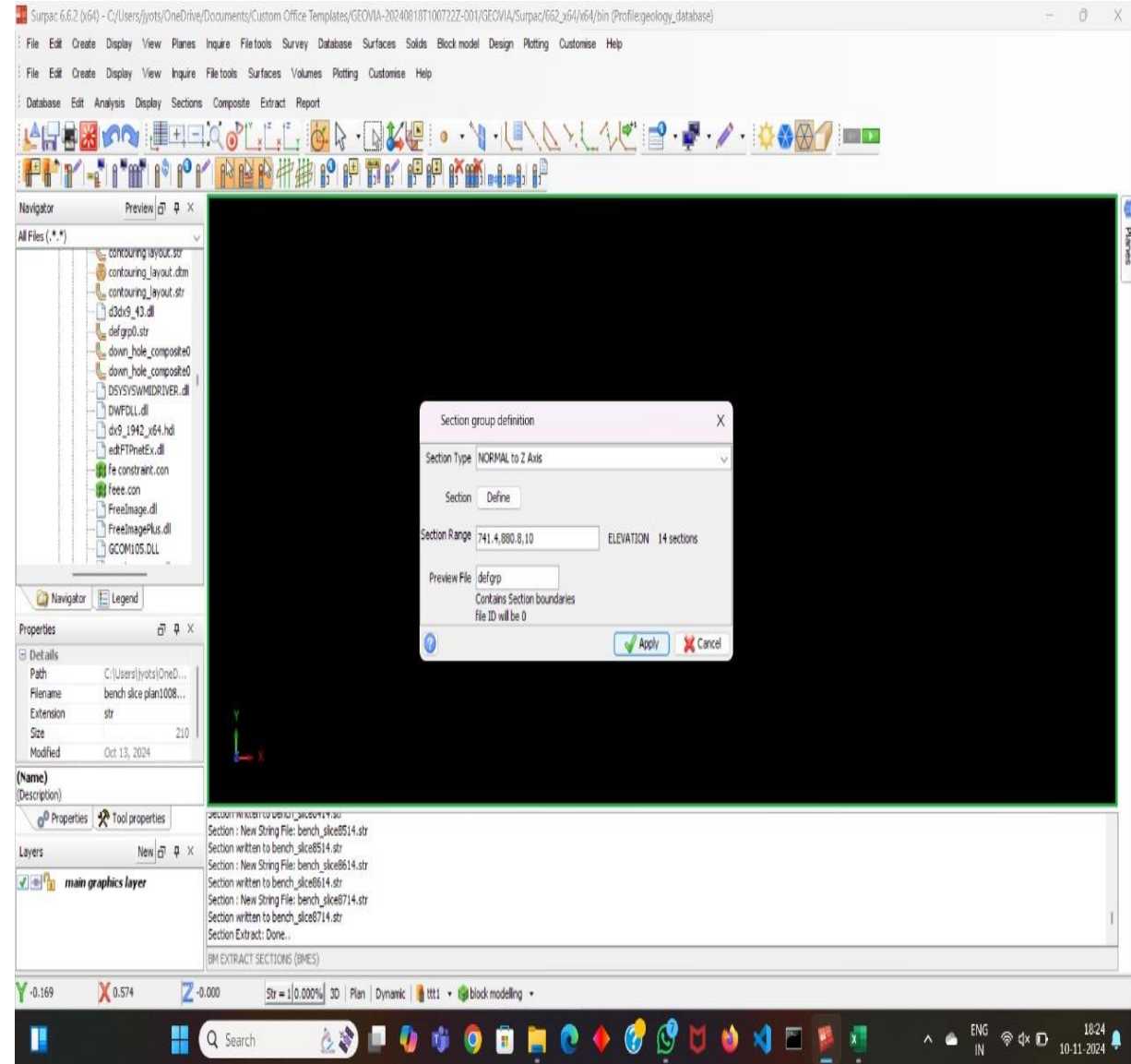
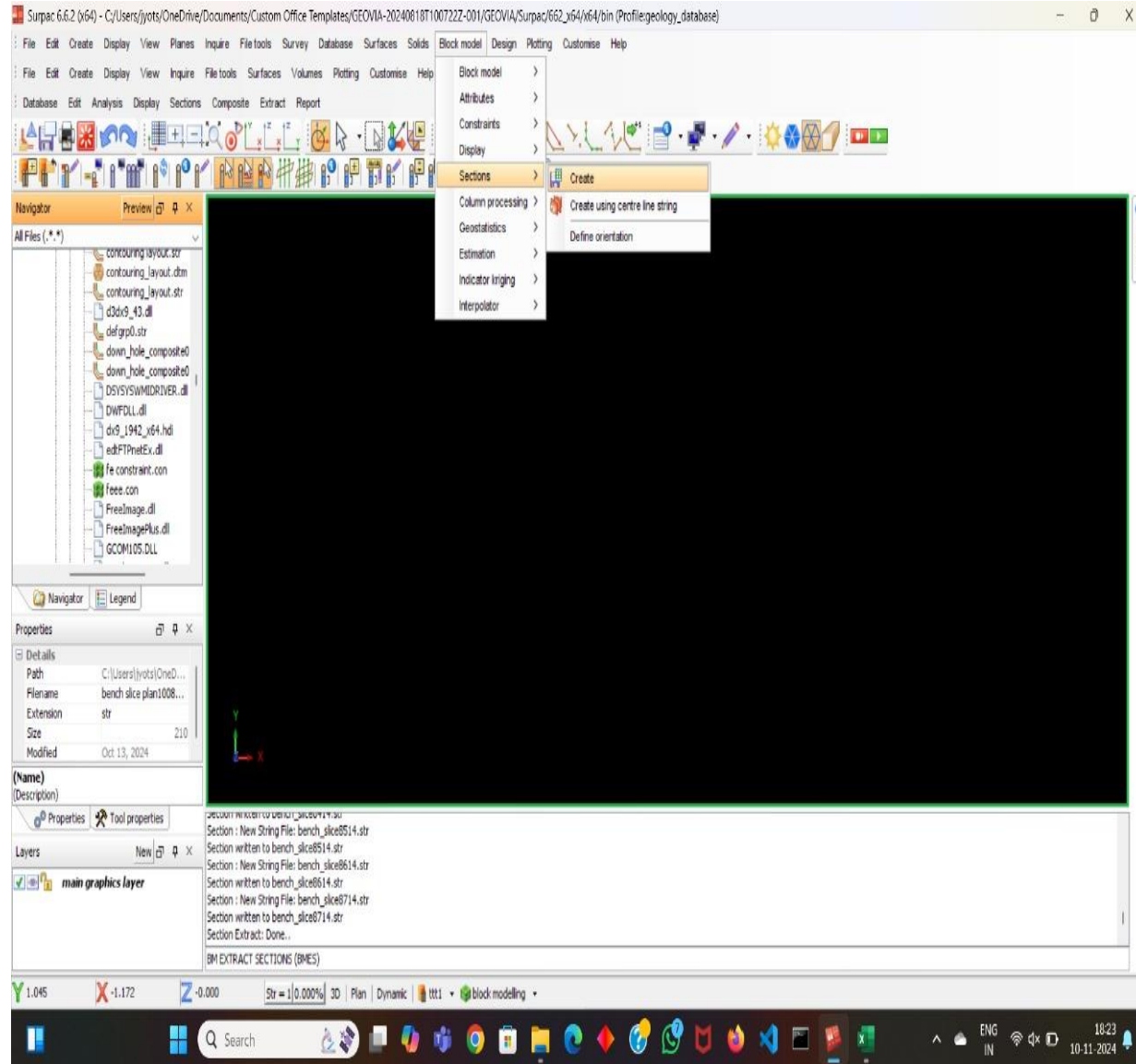
Select normal to z-axis in section type

In section range type your minimum and maximum extents of Z, using a comma.

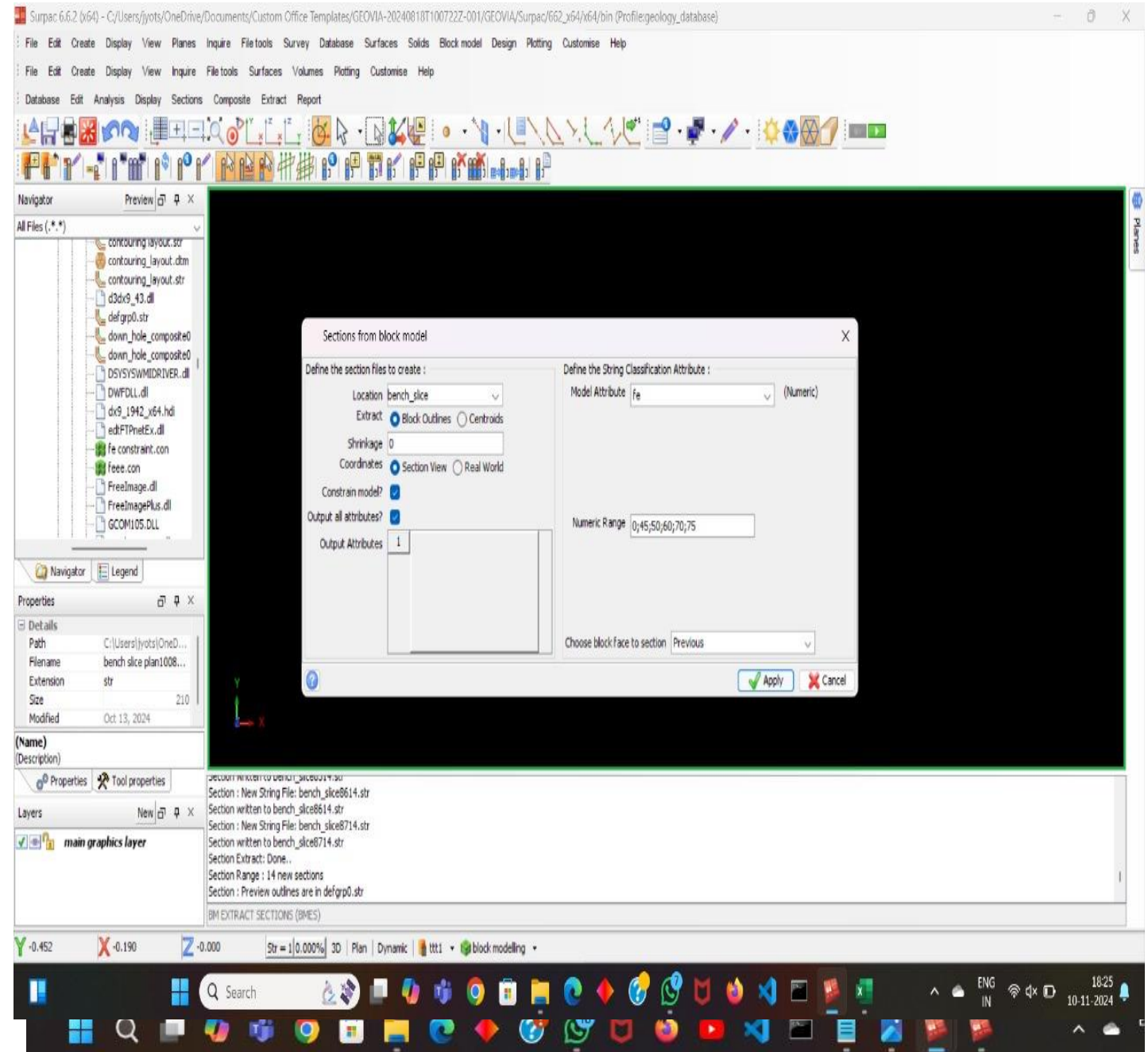
In section range type min z value, max z value, bench height

Click on Apply

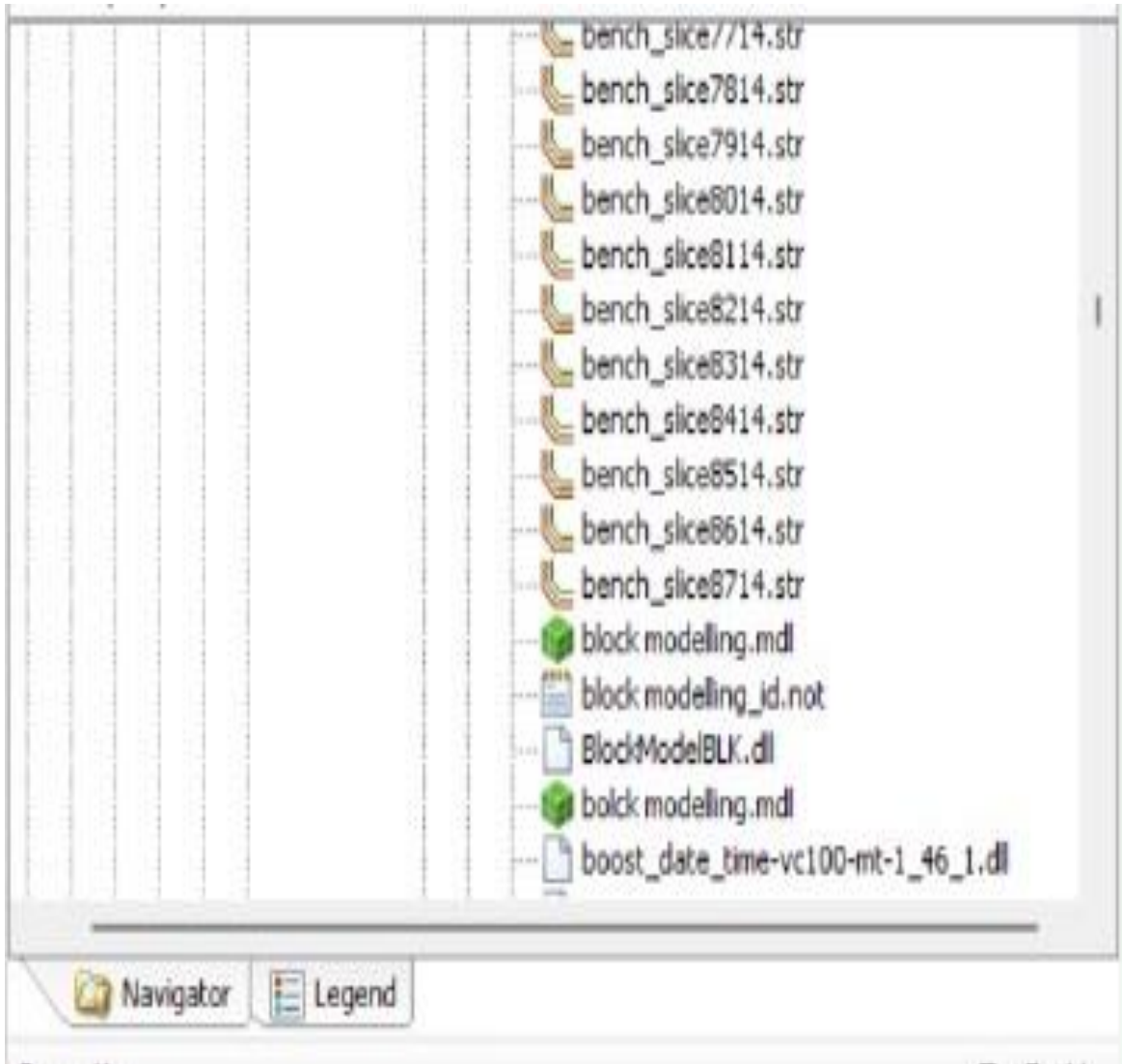
Extracting bench slices from block model:



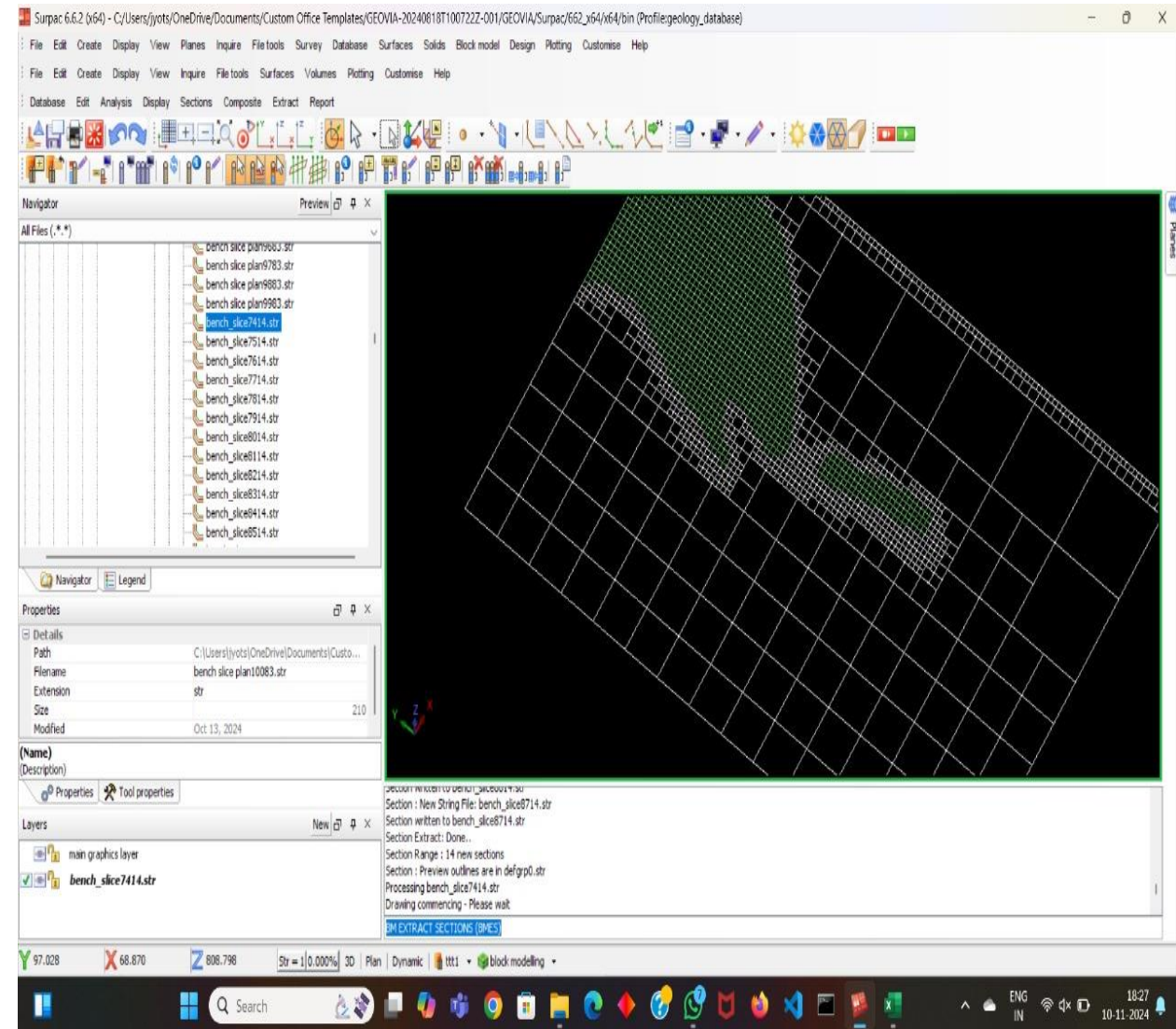
- Define the location of the section file as bench_slice
- Select Outlines as the Extract option.
- Select Model Attribute as Fe.
- Check on the constrained model, the model attribute is fe, and the numeric range will be 0;45;50;60;70;75
- Click on Apply



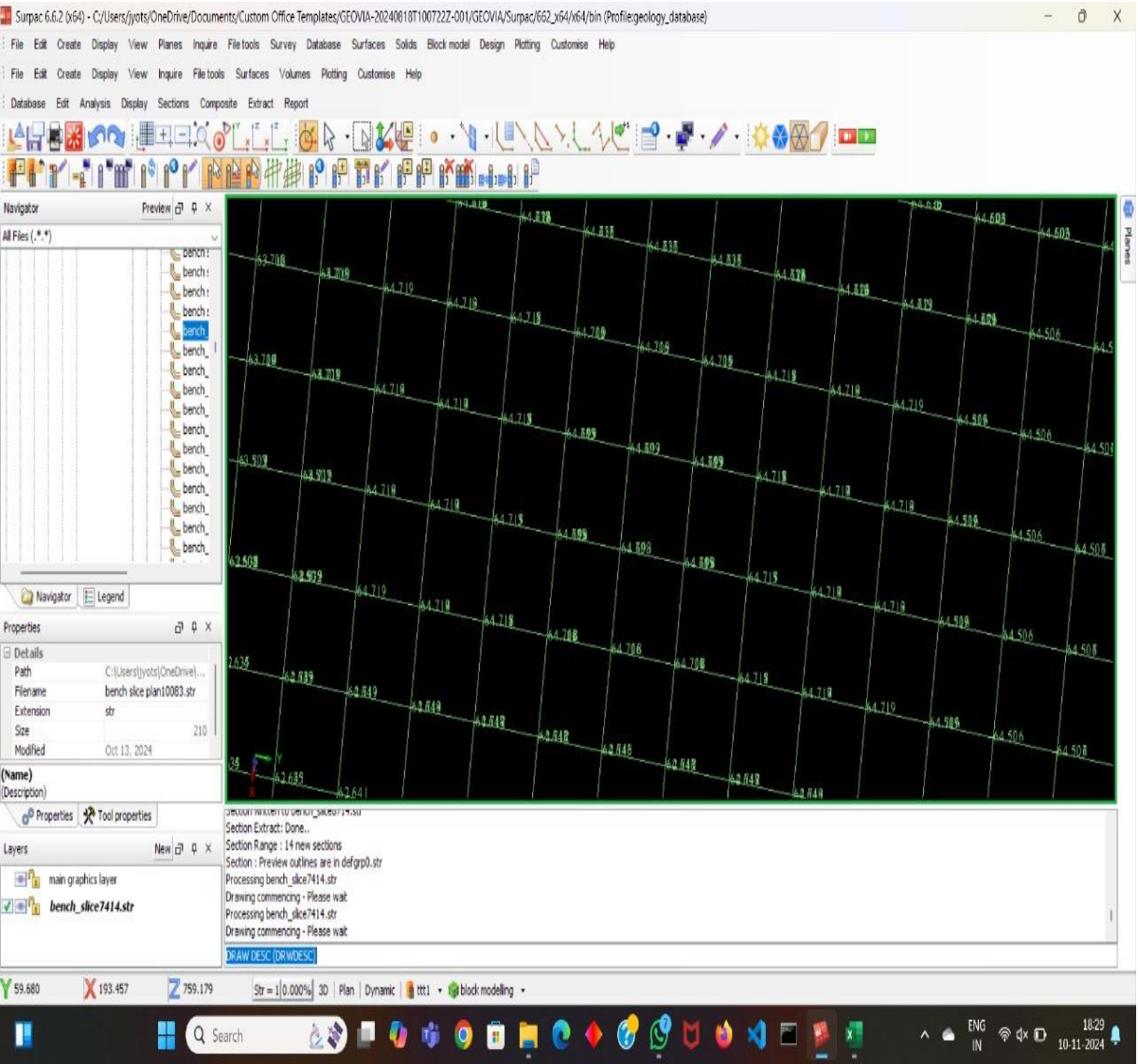
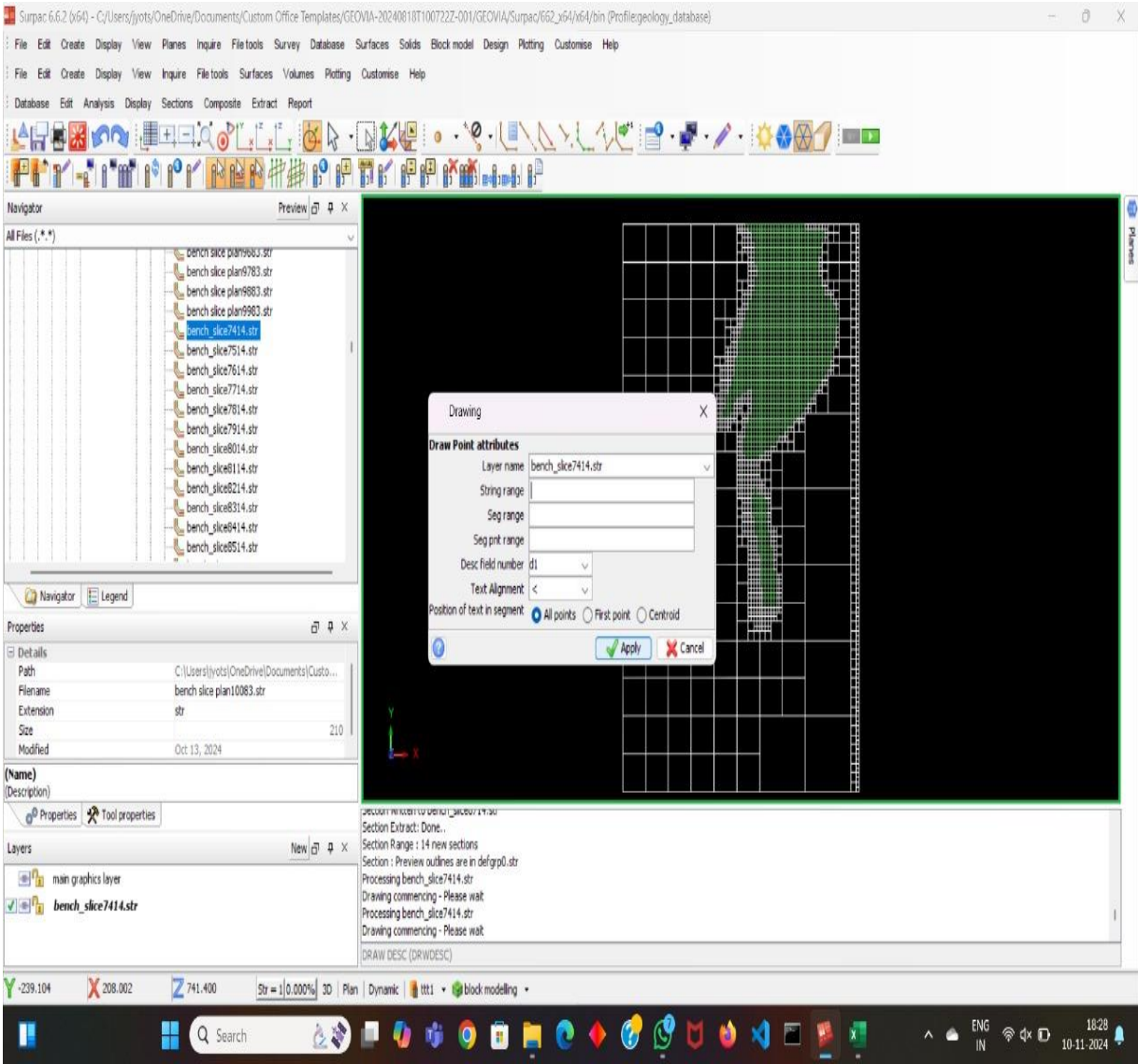
Bench slice plans are created as string files in your work directory.



- Bring the bottom slice plan to the main graphics layer and type the DRWDESC command in the command box.
- This command shows the assay value of each block in that slice plan.

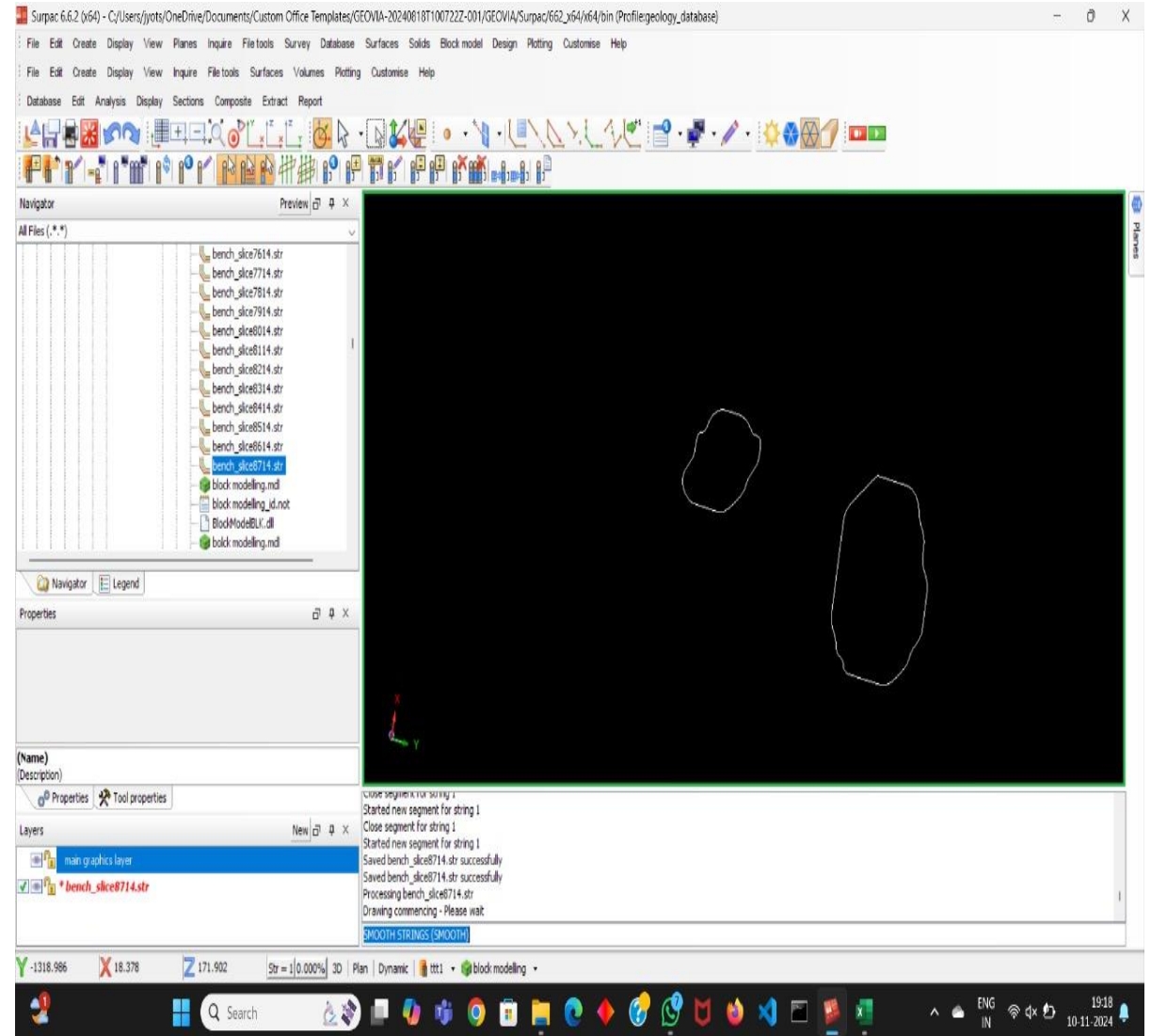
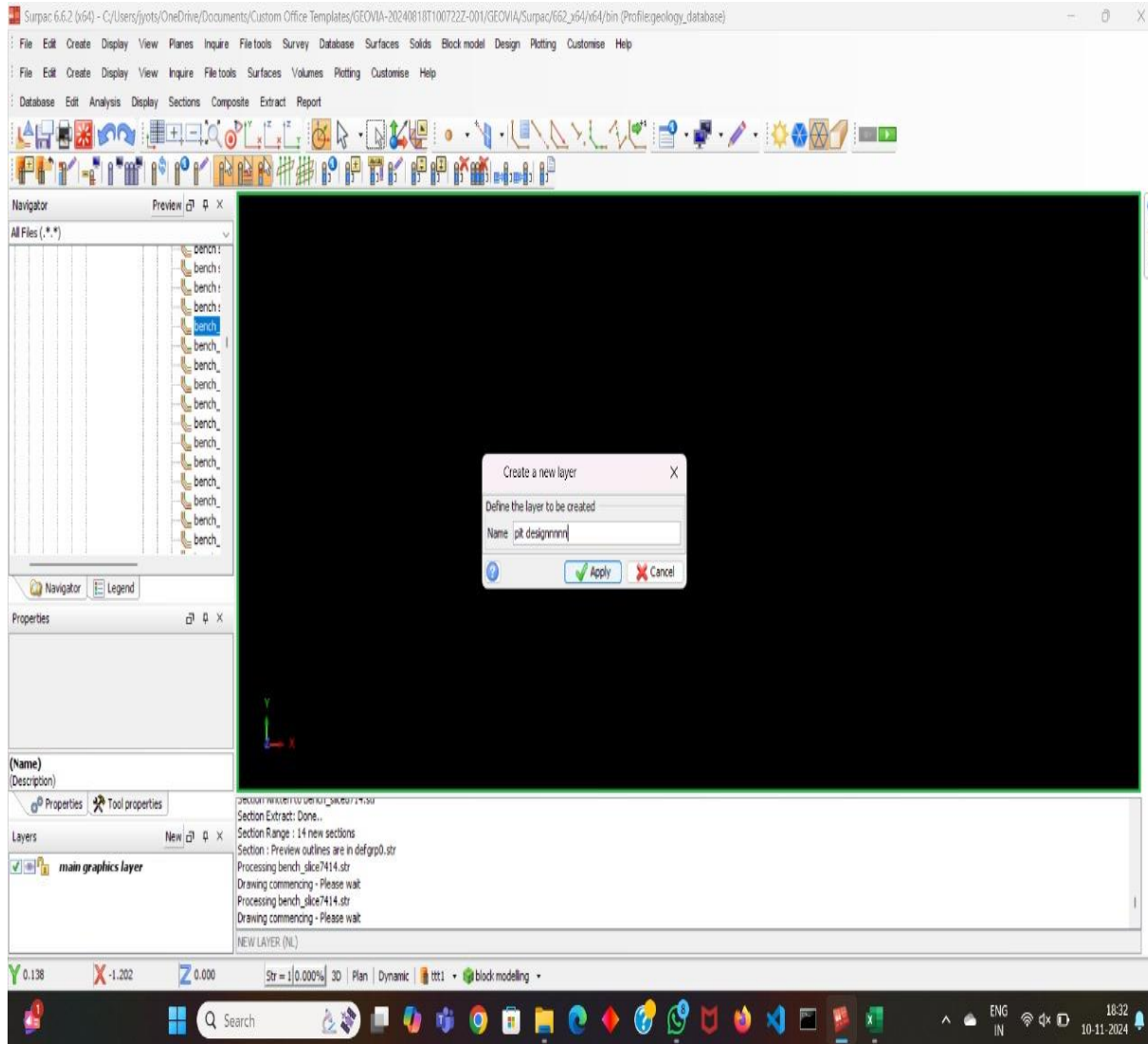


If the block shows 0, that means it is waste and we don't consider that waste blocks in pit design.
If all your slice plan doesn't show any assay value in the blocks, it means you didn't do the estimation and inverse distance properly.
You have to do it all over again, if you didn't find any coloured blocks in your slice plan.

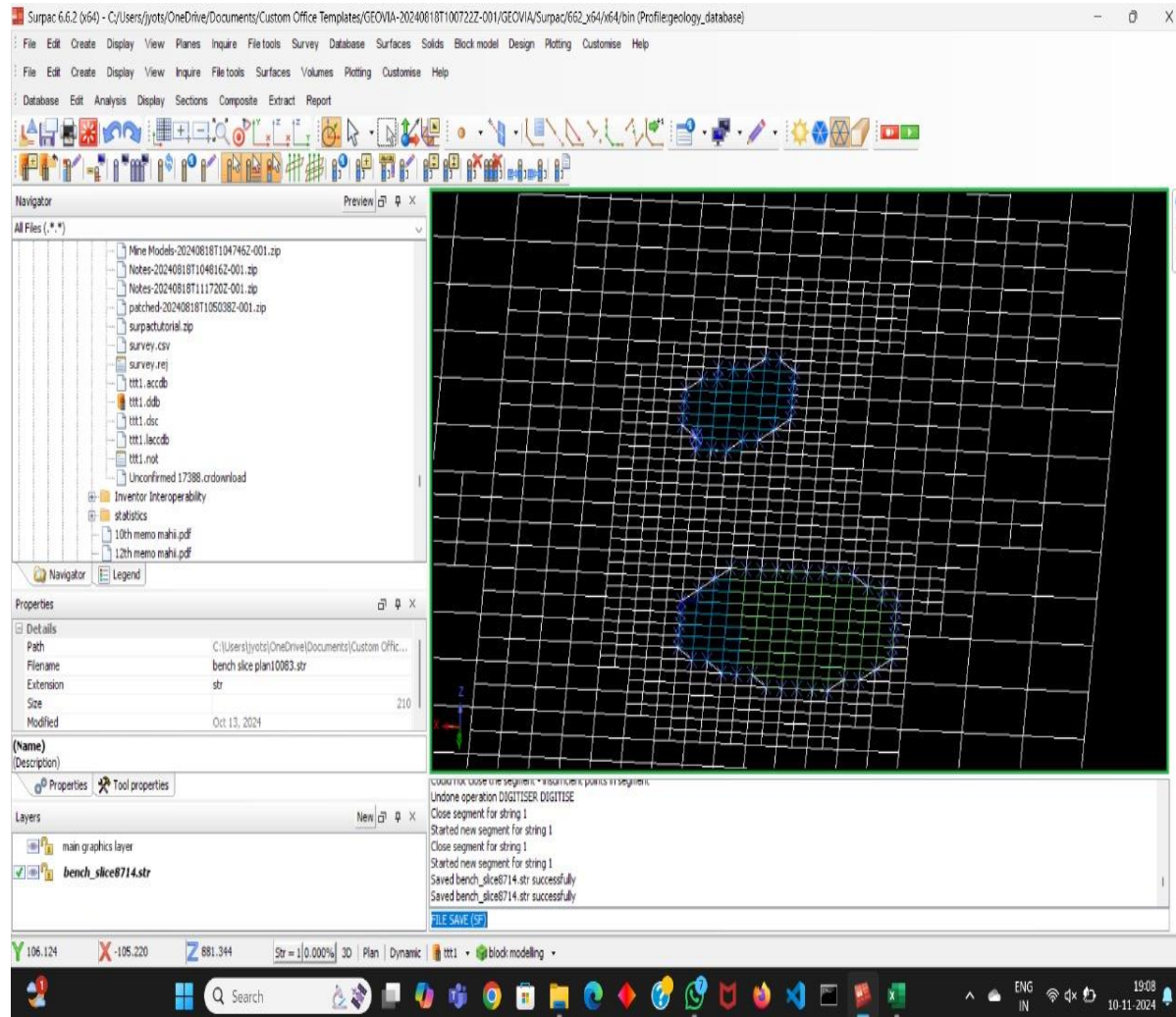


PIT DESIGN:-

Create a new layer and name that new layer as pit design .
Bring the bottom slice and digitise the slice with ore blocks.

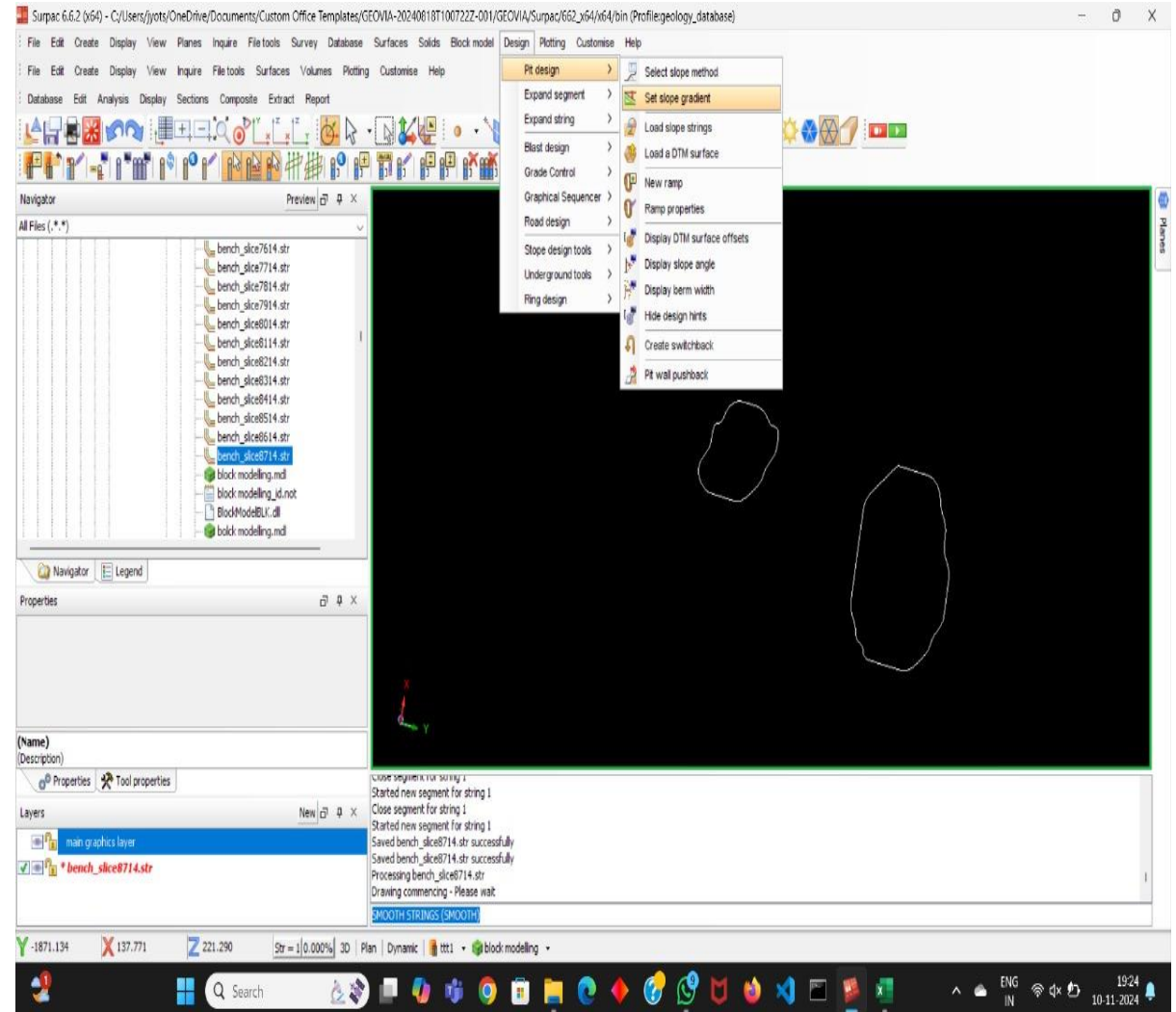


Blocks that have an assay value more than 0, will be displayed in different colours in the slice plan.
 Waste blocks will be displayed as white blocks.
 Only digitise the coloured blocks.
 Save the digitised layer
 Smooth the digitised segment, Edit>> Segment>> Smooth.

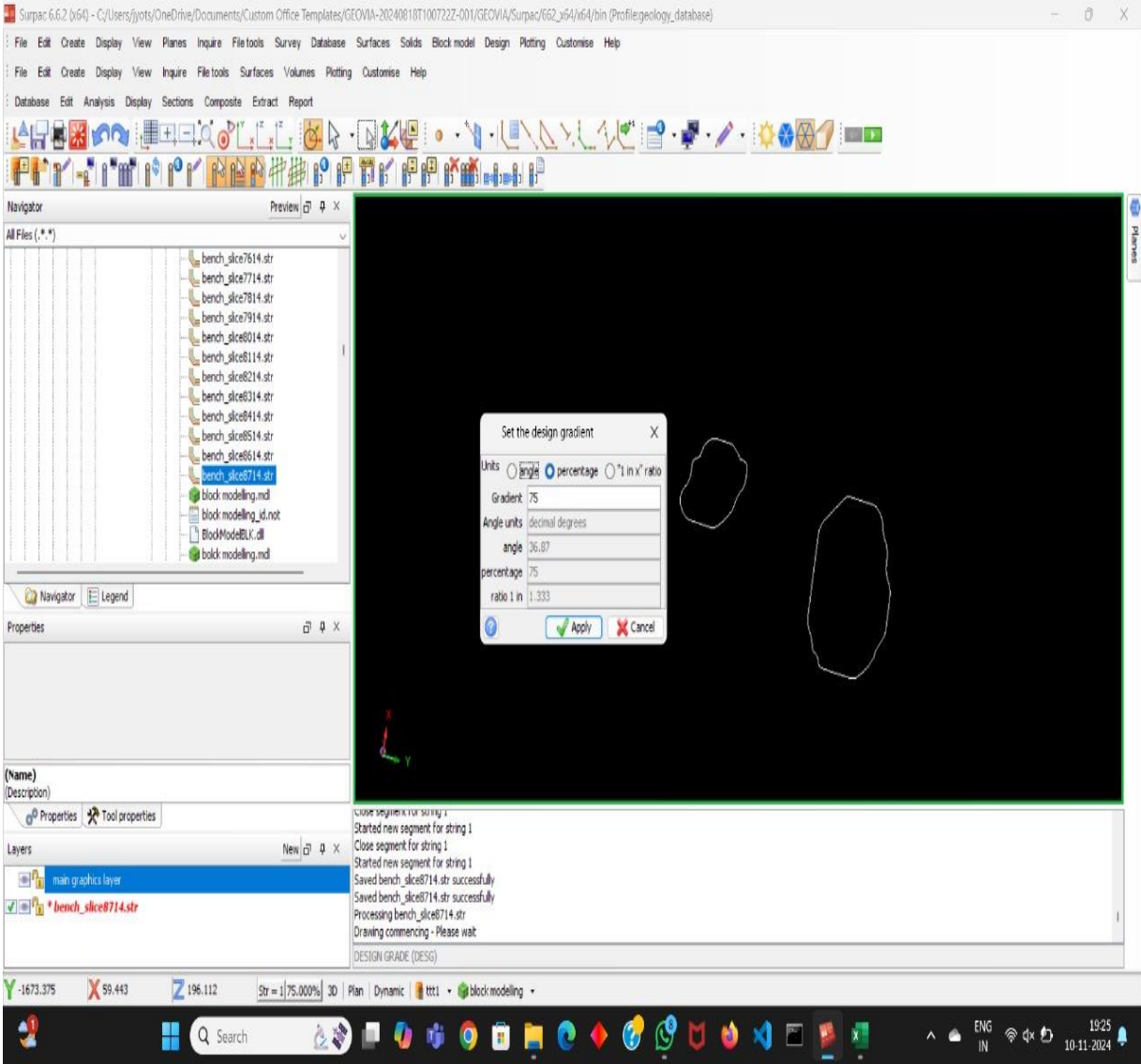


SLOPE GRADIENT

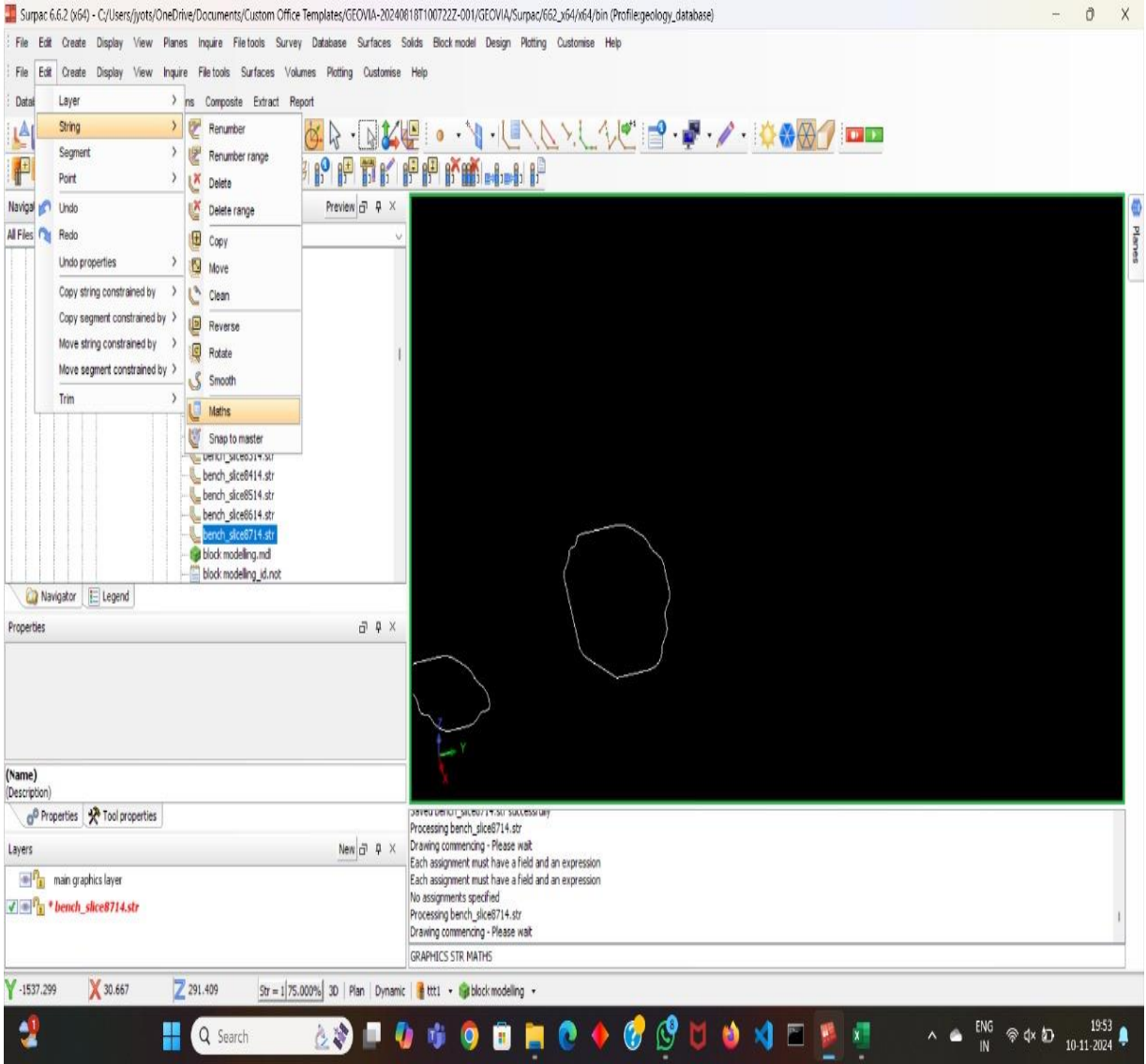
Go to pit design >> set slope gradient.
 Pop up



Units- angle, Gradient-75
Click on Apply.

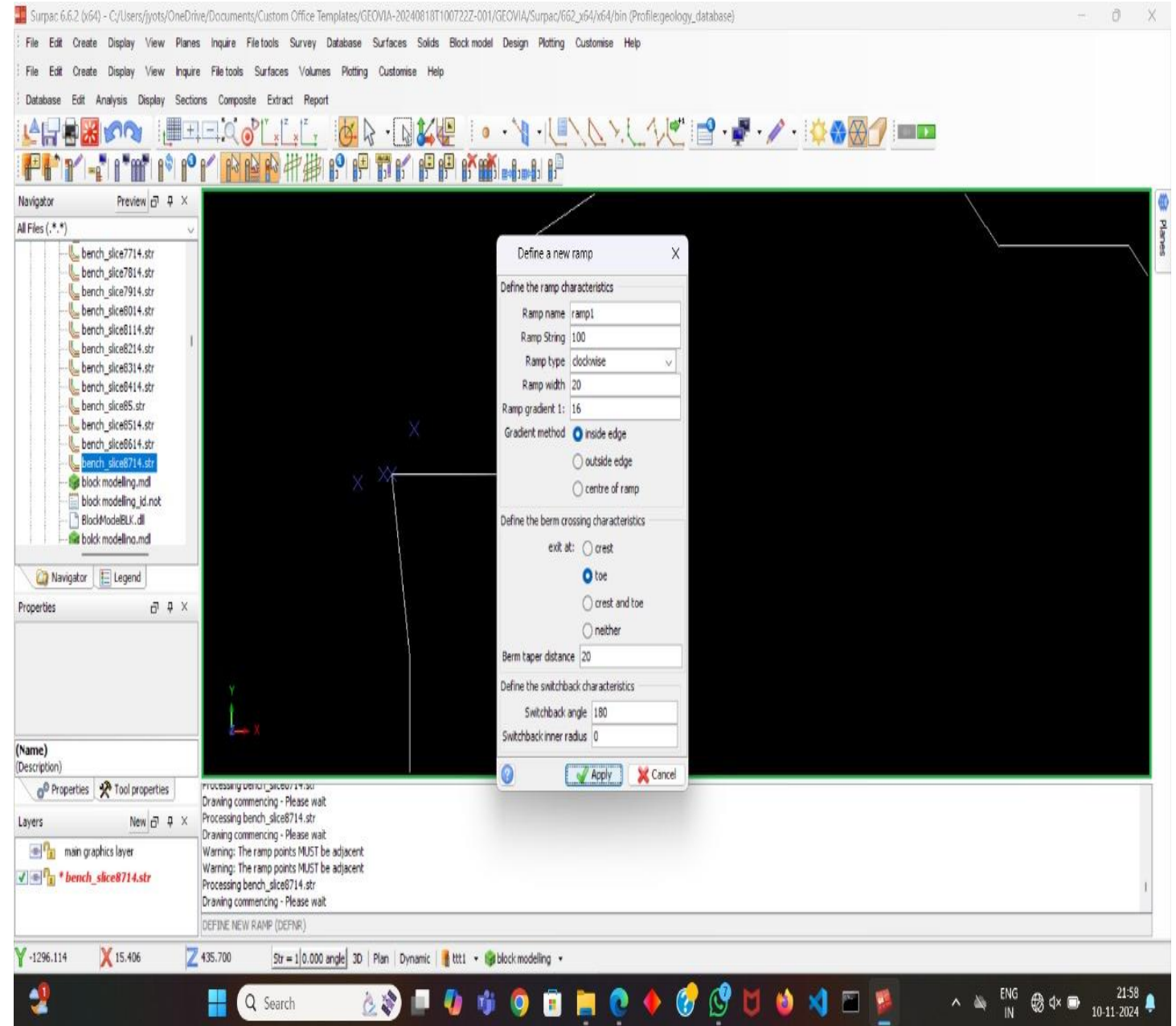


Go to string>> Maths >> Enter the elevation of
your slice plan .
Click on Apply.



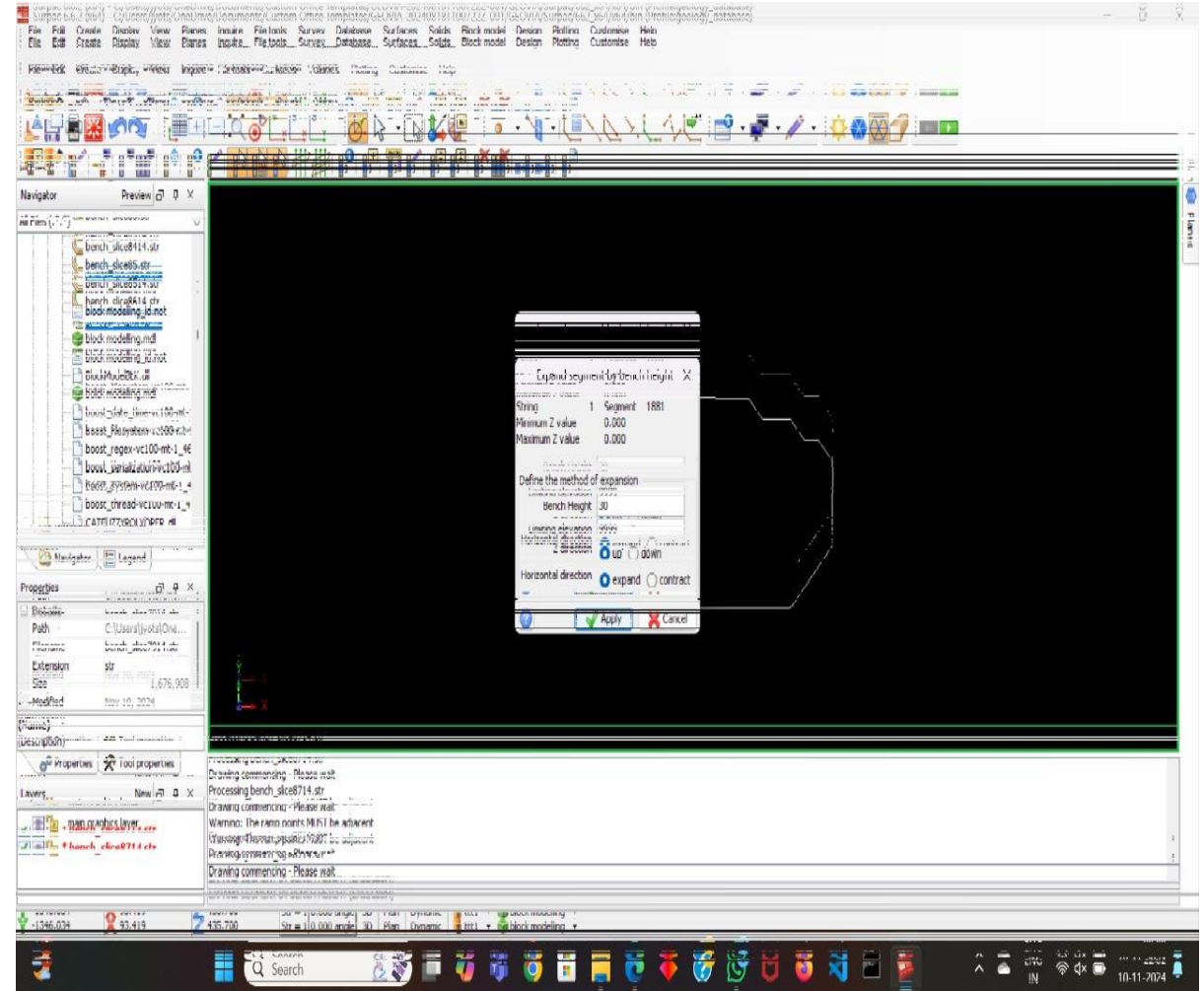
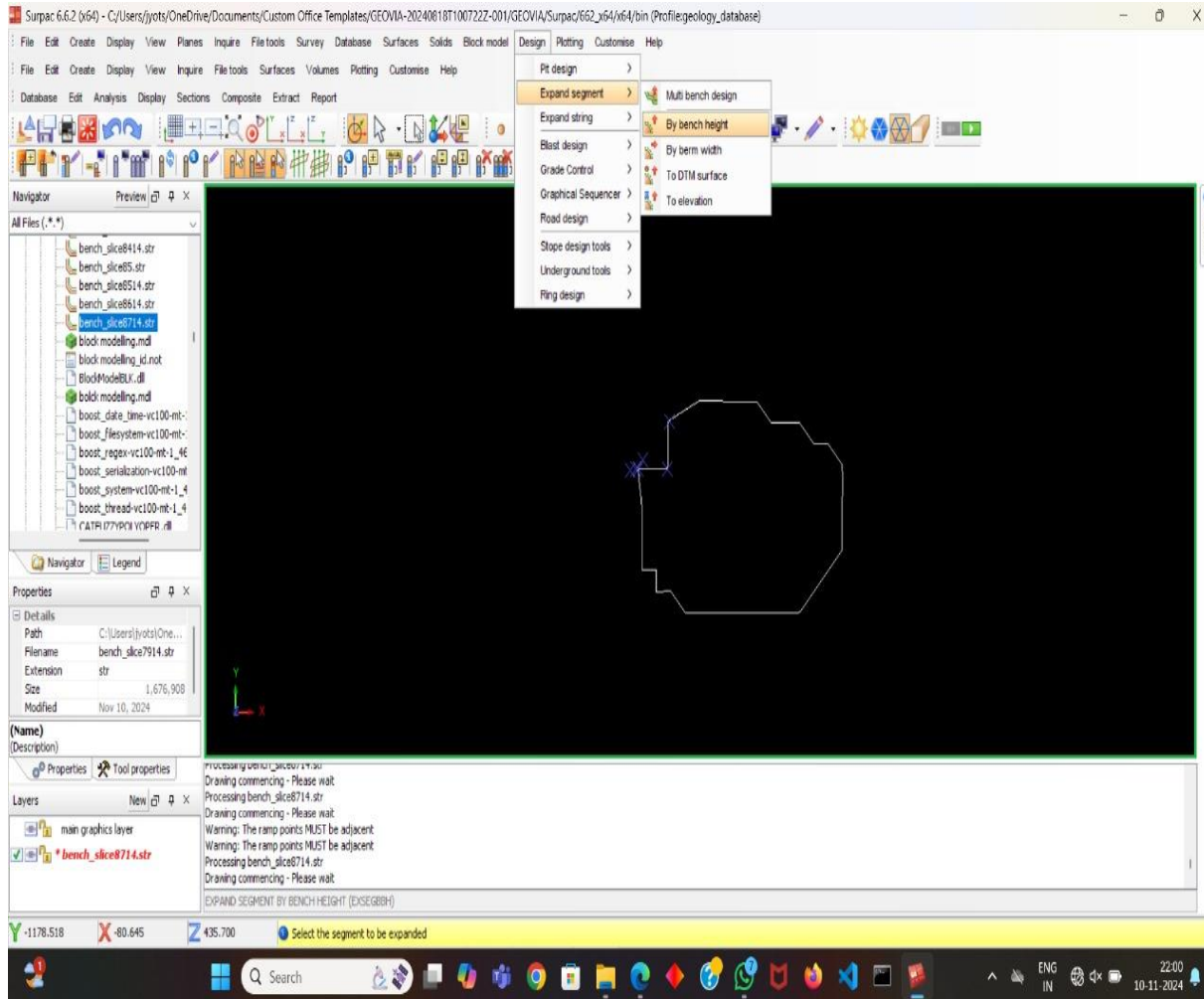
Creating a ramp:

- Go to pit design >> new ramp >> pop up.
- Give a name to the ramp
- Ramp string number should match the digitised string number.
- Choose the ramp type
Clockwise/anti-clockwise
- Ramp width 20m, ramp gradient will be 1 in 16
- Gradient method will be inside edge.
- Click on Apply.
- Select the first and second points of the ramp.
- A new ramp will be created.



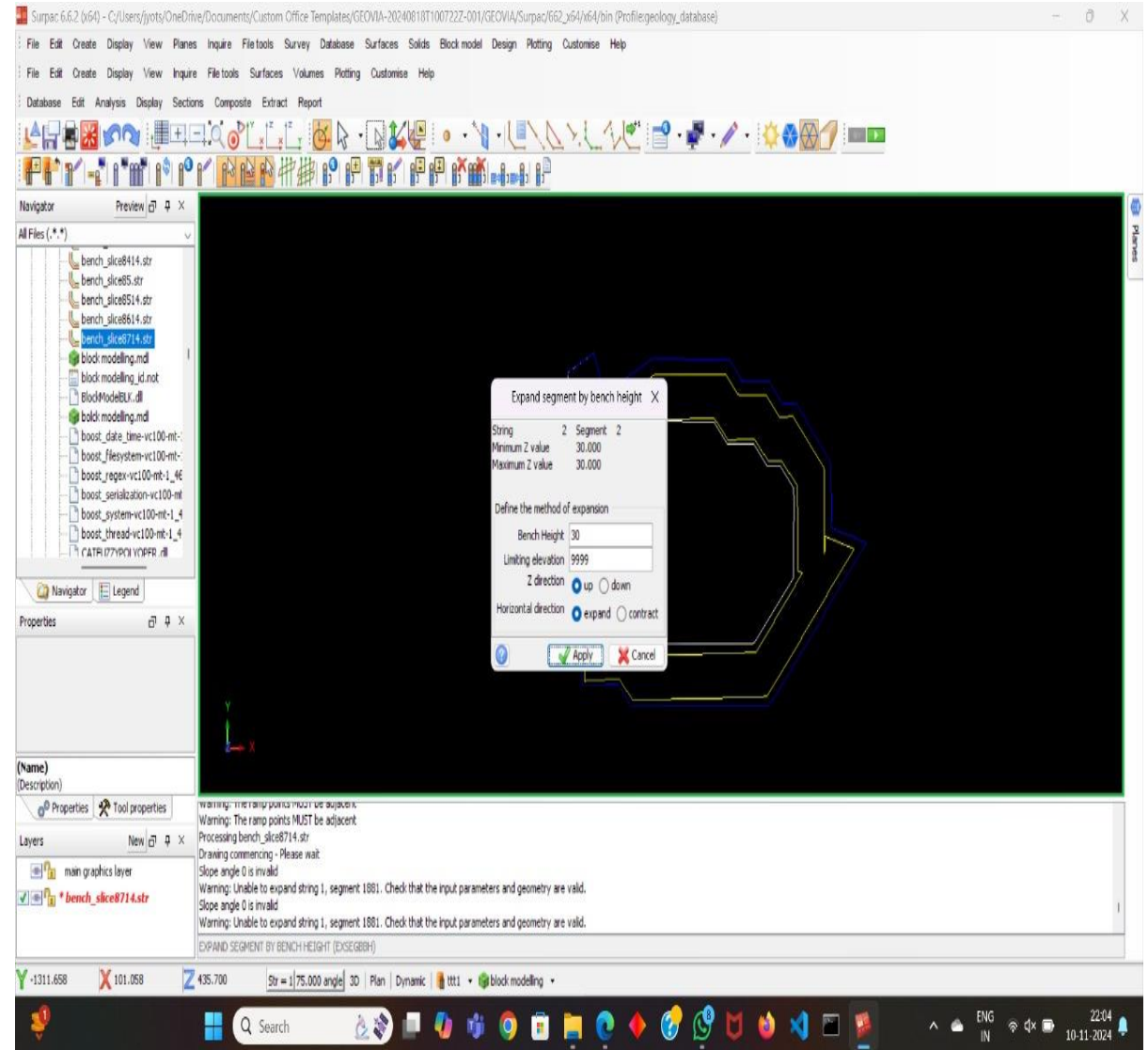
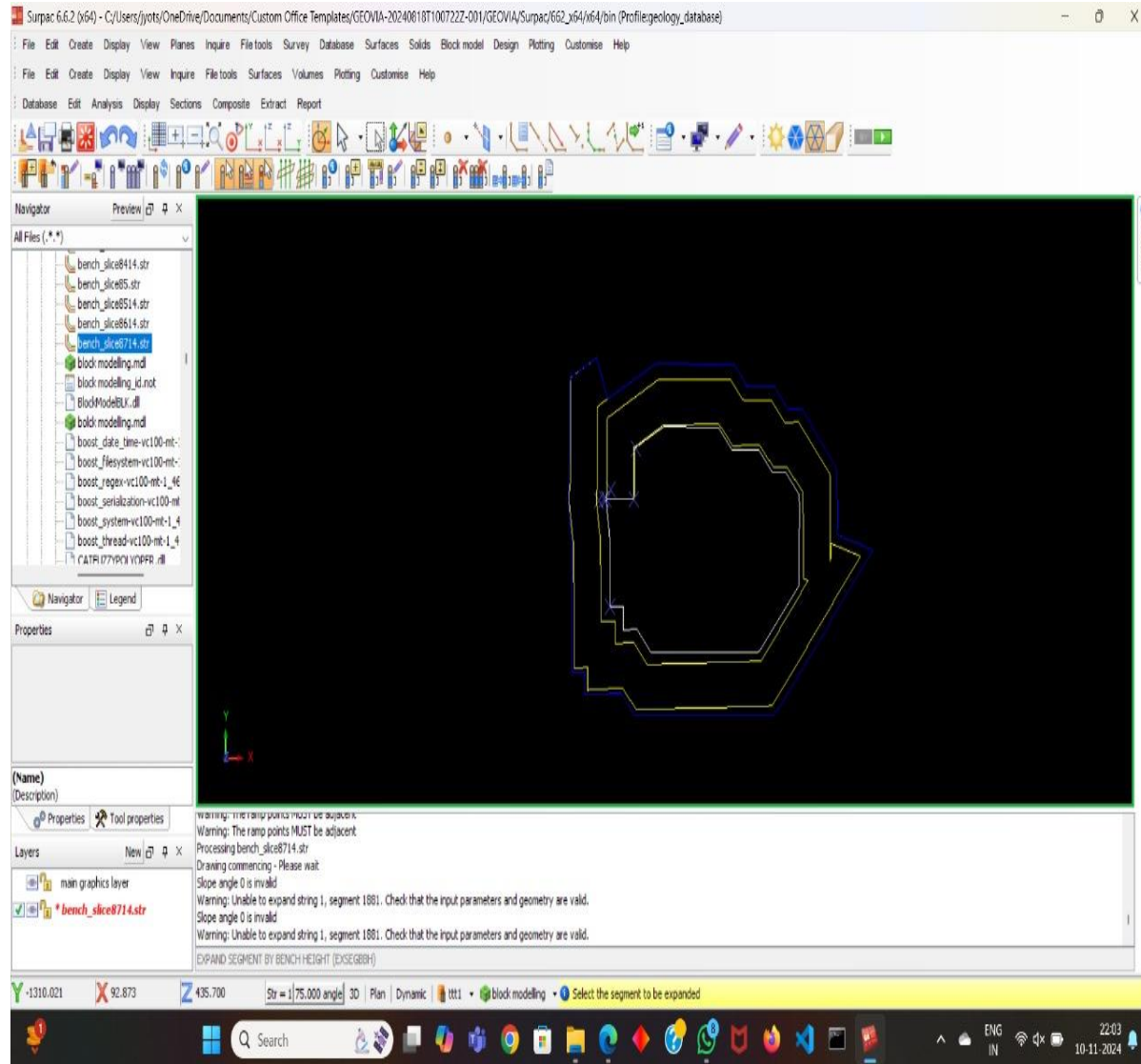
Expanding by bench height:

- The new ramp will not show up until you expand it by bench height.
- Go to pit design >> expand segment >> by bench height



Define the bench height:-

- Z direction should be upwards because we are doing the pit design from bottom to top.
- Click on expand
- Click on Apply



Expanding the pit by berm width: ➤
Go to pit design >> expand segment >>
by berm width

➤ Select the berm width method as fixed
width

➤ Define the berm width

➤ In horizontal direction Click on expand

➤ Click on apply.

➤ A new segment will be created

➤ Now design the next ramp in that
expanded segment to reach next bench.

➤ Follow the same procedure for all the
slice plans

➤ Digitise >> ramp design >> expanding by
bench height >> expanding by berm width
>> next bench

Expand Segment by Berm Width

String 1 Segment 1881
Minimum Z value 0
Maximum Z value 0

Define the method of expansion
Berm creation method always
Limiting elevation 999
Delta Z limit 0

Berm width method ☒ fixed width
☐ descriptions
☐ slope strings
☐ block model

Berm width attribute fe
Default berm width 5
Berm gradient (%) 0

Horizontal direction ☒ expand
☐ contract

Switchback characteristics

	Add switchback	Ramp name	Switchback angle	Switchback inner radius
1	☐	ramp1	180	0

Apply Cancel